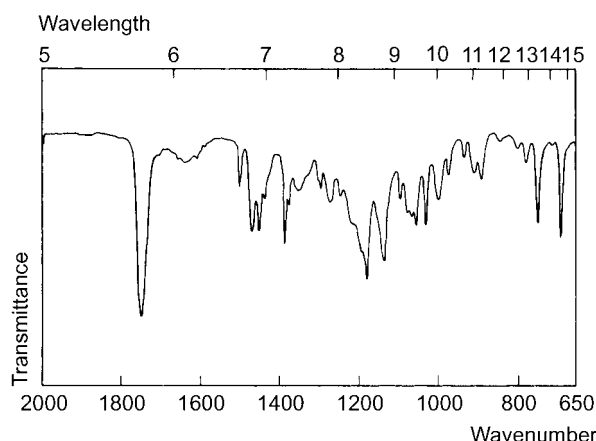




Mass Spectrum Principal ions at m/z 172, 187, 84, 57, 42, 188, 44, 43.

Quantification

Plasma GC FID. Limit of detection, 30 $\mu\text{g/L}$ [Fung *et al.* 1980].

Disposition in the Body Alphaprodine is a homologue of pethidine but its action is more rapid in onset and of shorter duration.

Therapeutic Concentration

After a rapid IV injection of 0.5 mg/kg to 6 subjects, plasma concentrations of about 0.8 mg/L were measured after 5 min, declining to 0.07 mg/L at 5 h [Fung *et al.* 1980].

Toxicity The estimated minimum lethal dose is 0.1 g. Prolonged use of alphaprodine is liable to produce dependence of the morphine type.

The following postmortem tissue concentrations were reported in an accidental fatality caused by the IV administration of 40 mg of alphaprodine: blood 0.62 mg/L, liver 0.97 $\mu\text{g/g}$, urine 0.30 mg/L; a blood-alcohol concentration of 1.7 mg/L was also reported [Griesemer, Nakamura 1973].

Half-life Plasma half-life, about 2 h.

Volume of Distribution About 2 L/kg.

Clearance Plasma clearance, about 10 mL/min/kg.

Dose 20 to 60 mg of alphaprodine hydrochloride by subcutaneous injection, or 10 to 30 mg intravenously.

Fung DL *et al.* (1980). A comparison of alphaprodine and meperidine pharmacokinetics. *J Clin Pharmacol* 20: 37–41.

Griesemer E, Nakamura GR (1973). *TIAFT Bull* 9(1–2)5.

Alpidem

Anxiolytic, Imidazopyridine

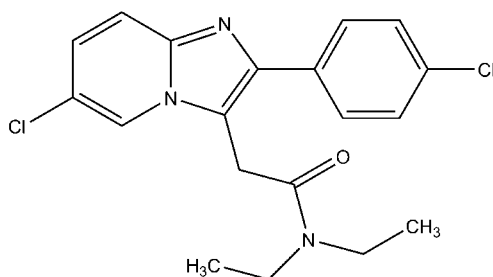
$\text{C}_{21}\text{H}_{23}\text{Cl}_2\text{N}_3\text{O}$ = 404.3

CAS—82626-01-5

IUPAC Name 2-[6-Chloro-2-(4-chlorophenyl)imidazo[3,2-a]pyridin-3-yl]-N,N-dipropylacetamide

Synonyms 6-Chloro-2-(4-chlorophenyl)-N,N-dipropylimidazo[1,2-a]pyridine-3-acetamide; SL-80.0342-00.

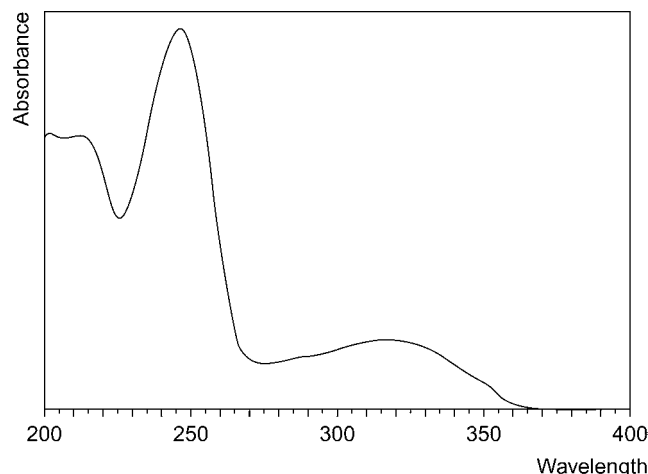
Proprietary Name Ananxyl



Chemical Properties White, fluffy, microcrystalline powder. Mp 140° to 141°. Sparingly soluble in water (20 mg/L). pK_a 4.25, Log P (octanol/water), 4.4. [Durand *et al.* 1992].

Ultraviolet Spectrum

Principal peak at 249 nm.



Quantification

Plasma GC Column: 5% phenylmethylsilicone (25 m $\times$ 0.32 mm i.d., 0.52 μm). Carrier gas: He, 2.1 mL/min. Temperature programme: 210° to 285° at 2.5°/min. NPD. Retention time: 1.77 (relative to prazepam). Limit of quantification not reported [Gaillard *et al.* 1993].

HPLC Column: C_8 (150 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: acetonitrile: methanol: 0.025 mol/L phosphate buffer (pH 4.5, 40:15:45), flow rate 1.5 mL/min. Fluorescence detection (λ_{ex} = 255 nm, λ_{em} = 423 nm). Retention time: 14 min. Limit of quantification, 2.5 $\mu\text{g/L}$ for alpidem; 1.5 $\mu\text{g/L}$ for metabolites [Flaminio *et al.* 1994]. Column: C_{18} (300 $\times$ 3.9 mm i.d., 4 μm). Mobile phase: methanol: tetrahydrofuran: phosphate buffer (pH 2.6, 65:5:30), flow rate 0.8 mL/min. DAD. Retention time: 11 min. Limit of quantification, not reported [Tracqui *et al.* 1993]. Column: C_{18} (250 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: acetonitrile: methanol: 0.025 mol/L potassium dihydrogen phosphate (pH 4.8; 45:10:45), flow rate 1.5 mL/min. Fluorescence detection (λ_{ex} = 255 nm, λ_{em} = 423 nm). Retention time: 24 min. Limit of detection, 1 $\mu\text{g/L}$ [Ascalone *et al.* 1987].

Disposition in the Body Alpidem undergoes first-pass metabolism, being metabolised to several active products via *N*-alkylation or oxidation (or a combination of both processes) of the propyl side chain and aromatic oxidation reactions. Elimination is mainly via the faeces and less than 0.1% of a dose is found in urine.

Therapeutic Concentration

A group of 21 healthy man volunteers was administered single oral doses of 25, 50, 100 or 200 mg alpidem. Mean peak plasma concentrations for the four doses were 39, 128, 237 and 270 $\mu\text{g/L}$, respectively, attained at 8, 3, 4, and 3 h, respectively [Jonkman *et al.* 1991].

A group of 10 healthy volunteers was administered single oral doses of 25, 50, or 100 mg alpidem. Mean peak plasma concentrations for the three doses were 24.7, 78.1 and 119.3 $\mu\text{g/L}$, respectively, reached after 2.2, 2.6 and 1.9 h [Saletu *et al.* 1986].

Toxicity Alpidem was withdrawn from the market following reports of hepatic dysfunction.

Bioavailability Approximately 32%.

Half-life Approximately 18 h (range 7 to 44 h); increases to 23 h in elderly subjects.

Volume of Distribution 8.7 L/kg.

Clearance 0.86 L/h/kg.

Protein Binding 99.1%.

Ascalone V *et al.* (1987). Determination of alpidem and its metabolites in human plasma by high-performance liquid chromatography and fluorimetric detection. *J Chromatogr* 414: 101–108.

Durand A *et al.* (1992). Comparative pharmacokinetic profile of two imidazopyridine drugs: zolpidem and alpidem. *Drug Metab Rev* 24: 239–266.

Flaminio L *et al.* (1994). Determination of alpidem, an imidazopyridine anxiolytic, and its metabolites by column-switching high-performance liquid chromatography with fluorescence detection. *J Chromatogr A* 668: 403–411.

Gaillard Y *et al.* (1993). Simultaneous screening and quantitation of alpidem, zolpidem, buspirone and benzodiazepines by dual-channel gas chromatography using electron-capture and nitrogen-phosphorus detection after solid-phase extraction. *J Chromatogr* 622: 197–208.

Jonkman JH *et al.* (1991). Clinical pharmacokinetics and tolerability of alpidem in healthy subjects given increasing single doses. *Eur J Clin Pharmacol* 41: 369–374.

Saletu B *et al.* (1986). Pharmacokinetic and dynamic studies with a new anxiolytic imidazo-pyridine alpidem utilizing pharmacology-EEG and psychometry. *Int Clin Psychopharmacol* 1: 145–164.

Tracqui A *et al.* (1993). High-performance liquid chromatographic assay with diode-array detection for toxicological screening of zopiclone, zolpidem, suriclone and alpidem in human plasma. *J Chromatogr* 616: 95–103.

Alprazolam

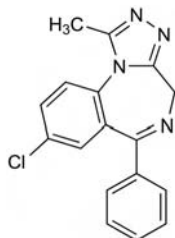
Anxiolytic, Benzodiazepine

$\text{C}_{17}\text{H}_{13}\text{ClN}_4$ = 308.8

CAS—28981-97-7

IUPAC Name 8-Chloro-1-methyl-6-phenyl-4*H*-[1,2,4]-triazolo[4,3-*a*][1,4]-benzodiazepine

Proprietary Names Alpralid; Alpraz; Alprox; Alzam; Anxirid; Apo-Alpraz; Apox; Azor; Calmax; Cassadan; Drimpam; Esparon; Gerax; Kalmor; Mialin; Novo-Alprazol; Nu-Alpraz; Panix; Ralozam; Valeans; Xanagis; Xanolam; Xanor; Zopax.



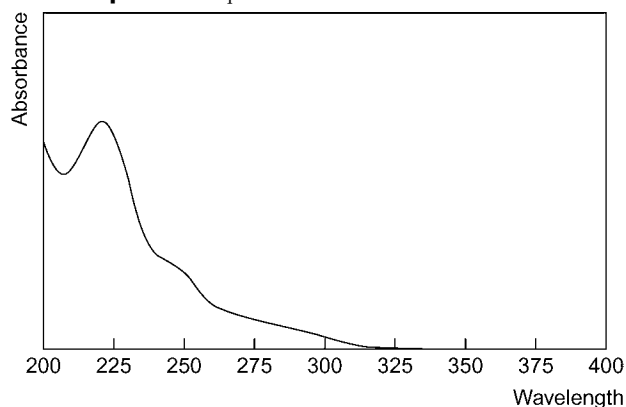
Chemical Properties A white crystalline powder. Mp 228° to 228.5°. Practically insoluble in water; sparingly soluble in alcohol and in acetone; freely soluble in chloroform and dichloromethane. pK_a 2.4. Log P 2.12. Extraction yield (chlorobutane), 0.95 [Demme *et al.* 2005].

Thin-layer Chromatography System TA— R_f 0.67; system TB— R_f 0.01; system TC— R_f 0.57; system TD— R_f 0.07; system TE— R_f 0.47; system TF— R_f 0.02; system TL— R_f 0.14; system TAD— R_f 0.40; system TAE— R_f 0.67; system TAF— R_f 0.66.

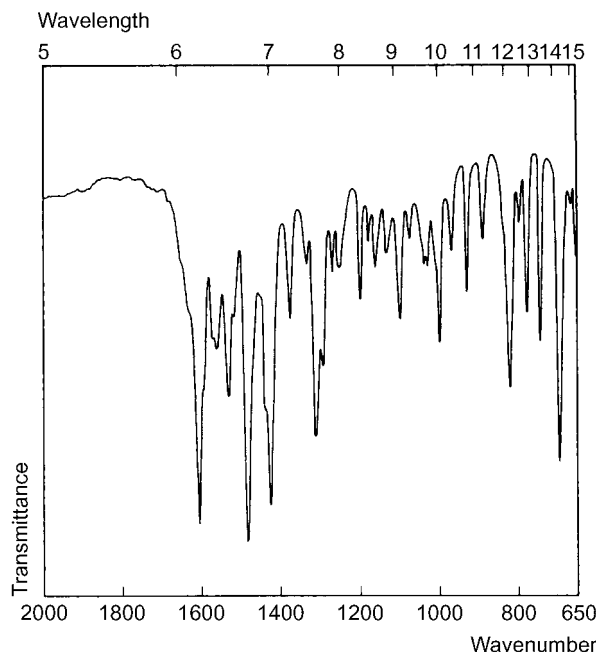
Gas Chromatography System GA—alprazolam RI 3100, M (α -OH-) RI 3045, M (OH-) RI 3245, M (4-OH-) not eluted; system GB—alprazolam RI 3108, M (OH-) not eluted; .

High Performance Liquid Chromatography System HI— k 4.70; system HK— k 2.79; system HX—RI 470; system HY—RI 379; system HZ—RT 4.0 min; system HAA—RT 17.0 min; system HAI—(relative to temazepam) RRT 1.47; system HAL—RT 6.3 min; system HAM— not detected; system HAX—RT 6.4 min; system HAY—RT 6.4 min; system HBH— k 3.35; system HBI— k 1.28.

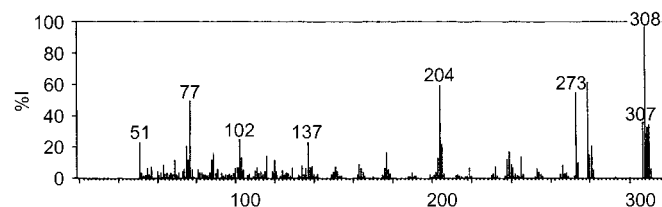
Ultraviolet Spectrum Aqueous acid—260 nm.



Infrared Spectrum Principal peaks at wavenumbers 1490, 1610, 697, 1316, 1540, 827 cm^{-1} (KBr disk). Three polymorphic forms may occur.



Mass Spectrum Principal ions at m/z 308, 279, 204, 273, 77, 307, 310, 309.



Quantification

Blood GC Column: SE-54 5% phenyl methyl silicone (25 m $\times$ 0.31 mm i.d., 0.17 μm). Carrier gas: He, 2–3 mL/min. Temperature programme: 200° for 1 min to 290° at 10°/min. NPD or ECD. Limit of detection not reported [Lillsunde, Seppälä 1990].

GC-MS Column: HP-1 methylsilicone (12.5 m $\times$ 0.2 mm i.d.). Carrier gas: He, 1.0 mL/min. Temperature programme: 70° to 280° at 20°/min. Retention time: 6.6 min. Limit of quantification, 18 $\mu\text{g/L}$ [Hall *et al.* 1995]. Column: Ultra-2 (25 m $\times$ 0.2 mm i.d., 0.33 μm). Carrier gas: He, 40 cm/s. Temperature programme: 150° for 1 min to 280° at 30°/min for 12 min. NCI. Limit of quantification, 4 $\mu\text{g/L}$ [Cairns *et al.* 1994].

HPLC Column: Acquity BEH Phenyl (100 $\times$ 2.1 mm i.d., 1.7 μm). Mobile phase: 0.1% formic acid in water : 0.1% formic acid in acetonitrile (80 : 20 for 0.25 min to 65 : 35 over 2.25 min for 2.5 min to 20 : 80 over 1 min to 80 : 20 over 0.01 min for 1.4 min), flow rate 0.5 mL/min. ESI, MRM acquisition mode. Limit of detection not reported [Gunn *et al.* 2010]. Column: Restek Allure C_{18} (150 $\times$ 3.2 mm i.d., 5 μm). Mobile phase: 5 mmol/L ammonium acetate : acetonitrile : methanol (90 : 5 : 5), flow rate 0.45 mL/min. DAD. Limit of quantification, 2 $\mu\text{g/L}$ [Dussy *et al.* 2006].

Plasma GC Column: CPSIL 8 CB (25 $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1 mL/min. Temperature programme: 150° to 230° at 40°/min for 2 min to 250° at 5°/min for 1 min to 300° at 15°/min for 3 min. NPD. Limit of detection, 0.01–0.48 $\mu\text{mol/L}$ [Reubsæet *et al.* 1998]. Column: C_8 Ultra-2 (25 m $\times$ 0.32 mm i.d., 0.52 μm). Carrier gas: He, 2.1 mL/min. Temperature programme: 210° to 285° at 2.5°/min. NPD and ECD. Relative retention time: 1.48. Limit of quantification, 2 $\mu\text{g/L}$ [Gaillard *et al.* 1993]. See Blood [Lillsunde, Seppälä 1990].

GC-MS Column: Restek-200 capillary (15 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: H_2 , 7 psi. Temperature programme: 190° for 1 min to 320° at 20°/min for 2 min. NCI. Limit of quantification, 1 $\mu\text{g/L}$ [Höld *et al.* 1996].

HPLC Column: C_{18} (250 $\times$ 4.6 mm). Mobile phase: methanol : water (65 : 35), flow rate 1.0 mL/min. UV detection ($\lambda = 230$ nm) [Liang *et al.* 2009]. Column: Kromasil C_8 (250 $\times$ 5 mm i.d., 5 μm). Mobile phase: methanol : acetonitrile : 0.05 mol/L ammonium acetate. DAD ($\lambda = 240$ nm). Limit of quantification, 0.07–1.57 mg/L, limit of detection, 0.02–0.47 mg/L [Uddin *et al.* 2008]. Column: Inertsil C_8 (250 $\times$ 4 mm i.d., 5 μm). Mobile phase: ammonium acetate : 0.05 mol/L methanol : acetonitrile (33 : 57 : 10). Limit of detection, 3.3–10.2 ng [Samanidou *et al.* 2007]. Column: reversed phase C_{18} . Mobile phase: methanol : water (60 : 40) containing 0.1% formic acid, flow rate 250 $\mu\text{L/min}$. Limit of detection, 0.05 $\mu\text{g/L}$ [Crouch *et al.* 1999]. Column: C_{18} ODS (150 $\times$ 4.6 mm i.d., 5 μm) at 30°. Mobile phase: methanol : water with 1% diethylamine (pH 6, 60 : 40), flow rate, 1.5 mL/min. UV detection ($\lambda = 240$ nm). Limit of detection, 200 $\mu\text{g/L}$ [Jenkins *et al.* 1997]. See also Akerman *et al.* [1996], and Rieck, Platt [1992].

LC-MS Column: XTerra MS C_{18} (150 $\times$ 2.1 mm i.d., 3.5 μm). Mobile phase: acetonitrile : water : 100 mmol/L ammonium formate (pH 3.0, 55 : 40 : 5), flow rate 0.125 mL/min. ESI, MRM acquisition mode. Retention time: 4.93 min. Limit of quantification, 20 $\mu\text{g/L}$, limit of detection, 10 $\mu\text{g/L}$ [Marin *et al.* 2008; Marin, McMillin 2010].

Serum HPLC See Blood [Dussy *et al.* 2006; Gunn *et al.* 2010]. See Plasma [Akerman *et al.* 1996]. Column: LiChrosorb RP 18 (100 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: methanol : water (65 : 35), flow rate 1.2 mL/min. UV detection ($\lambda = 230$ nm). Retention time: 3.1 min. Limit of detection, 10 mg/L [Goldnik *et al.* 1994]. Column: Supelco C_{18} (150 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: methanol : 0.001 mol/L phosphate buffer with 0.003 mol/L heptyltriethylammonium phosphate in water (pH 7.4, 40 : 60), flow rate 2 mL/min. Limit of detection, 1 $\mu\text{g/L}$ [Schmith *et al.* 1991].

LC-MS See Plasma [Marin, McMillin 2010]. Unison UK- C_{18} RP ODS (150 $\times$ 2 mm i.d., 3 μm). Mobile phase: 10 mmol/L aqueous ammonium formate containing 0.1% formic acid : methanol containing 0.1% formic acid (70 : 30 over 20 min to 20 : 80 over 5 min), flow rate 0.25 mL/min. MRM acquisition mode. Limit of detection, 0.3–11.4 $\mu\text{g/L}$ [Nakamura *et al.* 2009]. See Plasma [Marin *et al.* 2008].

Urine GC See Plasma [Reubsæet *et al.* 1998].

GC-MS Column: Column: BP1 polydimethylsiloxane (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.2 mL/min. Temperature programme: 65° for 1 min to 250° at 15°/min, for 8 min, to 300° at 10°/min for 2 min. SIM acquisition mode. Retention time: 21.49 min. Limit of detection, 1.0–1.7 $\mu\text{g/L}$ (SIM) and 13–30 $\mu\text{g/L}$ (scan mode) for alprazolam and other benzodiazepines [Borrey *et al.* 2001a; Borrey *et al.* 2001b]. Column: 5% phenylmethylsiloxane (25 m $\times$ 0.2 mm i.d., 0.33 μm). Carrier gas: He, 0.5 mL/min. Temperature programme: 240° to 260° at 25°/min to 300° at 30°/min. SIM acquisition mode. Limit of detection, 0.1 mg/L [Black *et al.* 1994]. Column: DB-5 bonded-phase (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 0.9 mL/min. Temperature programme: 130° for 1 min to 270° at 30°/min for 5 min to 300° at 40°/min for 6 min. EI ionisation, SIM acquisition mode. Limit of quantification, 16.5 $\mu\text{g/L}$, limit of detection, 6.8 $\mu\text{g/L}$ [Joern 1992].

HPLC See Plasma [Uddin *et al.* 2008]. Column: Inertsil C₈ (250 × 4 mm i.d., 5 µm). Mobile phase: ammonium acetate:0.05 mol/L methanol:acetonitrile (33:57:10). Limit of detection, 2.6–12.6 ng [Samanidou *et al.* 2007]. Column: Hypersil C₁₈ (100 × 4.6 mm i.d., 3 µm). Mobile phase: acetonitrile:0.04 mol/L phosphate buffer (pH 4.5; 35:65), flow rate, 1 mL/min. UV detection: DAD at 240 nm. Retention time: 3.48 min for α -hydroxyalprazolam and 4.98 min for alprazolam. Limit of detection, 9.2 µg/L [Segura *et al.* 2001].

LC-MS See Plasma [Marin *et al.* 2008; Marin, McMillin 2010]. Column: Symmetry C₁₈ (100 × 2.1 mm i.d., 3.5 µm). Mobile phase: acetonitrile:5 mmol/L ammonium acetate (pH 5.0, 10:90 to 30:70 at 5 min to 80:20 at 9.0 min to 10:90 at 9.1 min), flow rate 0.3 mL/min. Limit of quantification, 0.002 µmol/L [Hegstad *et al.* 2006].

Meconium LC-MS See Plasma [Marin *et al.* 2008; Marin, McMillin 2010].

Oral Fluid HPLC See Plasma [Uddin *et al.* 2008].

LC-MS Column: Atlantis dC₁₈ (50 × 2.1 mm i.d., 3.5 µm). Mobile phase: acetonitrile:5 mmol/L aqueous ammonium acetate (pH 5, 10:90 to 40:60 at 4 min to 90:10 at 4.1 min until 8 min to 10:90 at 8.1 min), flow rate 0.3 mL/min. ESI, MRM acquisition mode. Limit of quantification, 0.093 µg/L [oiestad *et al.* 2007].

Liver GC-MS See Blood [Cairns *et al.* 1994].

Disposition in the Body Alprazolam is readily absorbed after oral administration and undergoes extensive metabolism. The major metabolites are α -hydroxyalprazolam, 4-hydroxyalprazolam (both active) and a benzophenone derivative of alprazolam. Approximately 80% of a dose is excreted in the urine in 72 h, of which 11% is unchanged drug, 15% is α -hydroxyalprazolam and 9% is the benzophenone metabolite. Small amounts of desmethylalprazolam and 4-hydroxyalprazolam are also excreted in the urine. Approximately 7% of a dose is eliminated in the faeces.

Therapeutic Concentration The serum therapeutic concentration range is 0.005–0.05 mg/L; it has also been reported as 0.025–0.102 mg/L.

After a single oral dose of 1 mg alprazolam given to 10 subjects, peak plasma concentrations of 0.011–0.020 mg/L (mean, 0.015) were attained in ~1.75 h. Following daily oral doses of 0.5 mg 3 times a day to the same subjects, minimum steady-state plasma concentrations of 0.006–0.017 mg/L (mean, 0.011) and a mean maximum steady-state plasma concentration of 0.018 mg/L were reported [Smith *et al.* 1983].

In 227 patients treated for panic disorder with alprazolam (dosage titrated up to 6 mg daily), after 3 weeks of treatment, alprazolam plasma concentrations were significantly correlated with daily dosage: 70% of those with levels greater than 0.02 mg/L achieved complete remission of attacks compared with 30% of those with levels below 0.02 mg/L. At week 8, plasma alprazolam levels no longer correlated with efficacy [Greenblatt *et al.* 1993a].

Toxicity The serum toxic concentration range is 0.1–0.4 mg/L.

In a fatality attributed to acute alprazolam toxicity, the following postmortem tissue concentrations were reported for alprazolam and α -hydroxyalprazolam, respectively: heart blood 2.1 mg/L and not detected, subclavian blood 2.3 and 0.12 mg/L, urine 1.0 and 1.4 mg/L, bile 2.8 and 1.3 mg/L, vitreous humour 0.58 mg/L and not detected, liver 9.2 and 0.83 µg/g, kidney 3.8 µg/g and not detected, stomach contents 13 mg in 110 mL and not detected; the alprazolam concentration in the heart blood (2.1 mg/L) was the highest reported so far in the literature [Jenkins *et al.* 1997].

The following postmortem concentrations were reported in a 30-year-old female who ingested an unknown quantity of alprazolam and tramadol: alprazolam, α -hydroxyalprazolam and tramadol, respectively, in peripheral blood 0.21, not detected and 38.3 mg/L; bile 0.27, 0.12 and 44.0 mg/L, liver 0.23, <0.005 and 27.6 µg/g, and gastric contents (100 mL) 2.73, not detected and 130 mg [Michaud *et al.* 1999].

Half-life Plasma half-life, 11–15 h (mean, 12).

Volume of Distribution ~0.7 L/kg.

Distribution in Blood Plasma: whole blood ratio, 1.35

Clearance Plasma clearance, ~0.7 mL/min/kg.

Protein Binding ~70–80%.

Note For a review of alprazolam, see Dawson *et al.* [1984]; for its pharmacokinetics, see Greenblatt, Wright [1993] and Greenblatt *et al.* [1993b].

Dose 1 to 3 mg daily; up to 10 mg in the treatment of panic attacks.

Akerman KK *et al.* (1996). Analysis of low-dose benzodiazepines by HPLC with automated solid-phase extraction. *Clin Chem* 42: 1412–1416.

Black DA *et al.* (1994). Analysis of urinary benzodiazepines using solid-phase extraction and gas chromatography–mass spectrometry. *J Anal Toxicol* 18: 185–188.

Borrey D *et al.* (2001a). Sensitive gas chromatographic–mass spectrometric screening of acetylated benzodiazepines. *J Chromatogr A* 910: 105–118.

Borrey D *et al.* (2001b). Simultaneous determination of fifteen low-dosed benzodiazepines in human urine by solid-phase extraction and gas chromatography–mass spectrometry. *J Chromatogr B Biomed Sci Appl* 765: 187–197.

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Demme U *et al.* (2005). Systematic evaluation of 1-chlorobutane for liquid–liquid extraction of drugs. *Proceedings of the 12th TIAFT*, Seoul: 481–486.

Dussy FE *et al.* (2006). Quantification of benzodiazepines in whole blood and serum. *Int J Legal Med* 120: 323–330.

Gaillard Y *et al.* (1993). Simultaneous screening and quantitation of alpidem, zolpidem, buspirone and benzodiazepines by dual-channel gas chromatography using electron-capture and nitrogen–phosphorus detection after solid-phase extraction. *J Chromatogr* 622: 197–208.

Goldnik A *et al.* (1994). Determination of estazolam and alprazolam in serum by HPLC. *Acta Pol Pharm* 51: 311–312.

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Greenblatt DJ *et al.* (1993a). Plasma alprazolam concentrations: relation to efficacy and side effects in the treatment of panic disorder. *Arch Gen Psychiatry* 50: 715–722.

Greenblatt DJ *et al.* (1993b). Alprazolam pharmacokinetics, metabolism, and plasma levels: clinical implications. *J Clin Psychiatry* 54(Suppl): 4–11.

Gunn J *et al.* (2010). Simultaneous determination and quantification of 12 benzodiazepines in serum or whole blood using UPLC/MS/MS. *Meth Mol Biol* 603: 107–119.

Hall MA *et al.* (1995). High-performance liquid chromatography of alprazolam in postmortem blood using solid-phase extraction. *J Anal Toxicol* 19: 511–513.

Hegstad S *et al.* (2006). Determination of benzodiazepines in human urine using solid-phase extraction and high-performance liquid chromatography–electrospray ionization tandem mass spectrometry. *J Anal Toxicol* 30: 31–37.

Hödl KM *et al.* (1996). Determination of alprazolam and α -hydroxyalprazolam in human plasma by gas chromatography/negative-ion chemical ionization mass spectrometry. *J Mass Spectrom* 31: 1033–1038.

Jenkins AJ *et al.* (1997). A fatality due to alprazolam intoxication. *J Anal Toxicol* 21: 218–220.

Joern WA (1992). Confirmation of low concentrations of urinary benzodiazepines, including alprazolam and triazolam, by GC/MS: an extractive alkylation procedure. *J Anal Toxicol* 16: 363–367.

Liang X *et al.* (2009). [Simultaneous determination of 5 sedative hypnotics in human plasma by reversed phase high-performance liquid chromatography]. *Zhong Nan Da Xue Xue Bao Yi Xue Ban* 34: 689–692.

Lillsunde P, Seppälä T (1990). Simultaneous screening and quantitative analysis of benzodiazepines by dual-channel gas chromatography using electron-capture and nitrogen–phosphorus detection. *J Chromatogr* 533: 97–110.

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Michaud K *et al.* (1999). Fatal overdose of tramadol and alprazolam. *Forensic Sci Int* 105: 185–189.

Nakamura M *et al.* (2009). Simultaneous determination of benzodiazepines and their metabolites in human serum by liquid chromatography–tandem mass spectrometry using a high-resolution octadecyl silica column compatible with aqueous compounds. *Biomed Chromatogr* 23: 357–364.

oiestad EL *et al.* (2007). Drug screening of preserved oral fluid by liquid chromatography–tandem mass spectrometry. *Clin Chem* 53: 300–309.

Reubsaet KJ *et al.* (1998). Determination of benzodiazepines in human urine and plasma with solvent modified solid phase micro extraction and gas chromatography: rationalisation of method development using experimental design strategies. *J Pharm Biomed Anal* 18: 667–680.

Rieck W, Platt D (1992). High-performance liquid chromatographic method for the determination of alprazolam in plasma using the column-switching technique. *J Chromatogr* 578: 259–263.

Samanidou VF *et al.* (2007). Development of a validated HPLC method for the determination of four 1,4-benzodiazepines in human biological fluids. *J Sep Sci* 30: 679–687.

Schmith VD *et al.* (1991). New high-performance liquid chromatographic method for the determination of alprazolam and its metabolites in serum: instability of 4-hydroxyalprazolam. *J Chromatogr* 568: 253–260.

Segura M *et al.* (2001). Analytical methodology for the detection of benzodiazepine consumption in opioid-dependent subjects. *J Anal Toxicol* 25: 130–136.

Smith RB *et al.* (1983). Single- and multiple-dose pharmacokinetics of oral alprazolam in healthy smoking and nonsmoking men. *Clin Pharm* 2: 139–143.

Uddin MN *et al.* (2008). Validation of SPE-HPLC determination of 1,4-benzodiazepines and metabolites in blood plasma, urine, and saliva. *J Sep Sci* 31: 3704–3717.

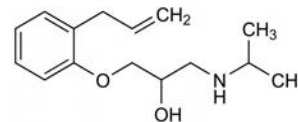
Alprenolol

β -Adrenoceptor Antagonist

C₁₅H₂₃NO₂ = 249.4

CAS—13655-52-2; 23846-70-0 (±)

IUPAC Name 1-(Propan-2-ylamino)-3-(2-prop-2-enylphenoxy)propan-2-ol



Chemical Properties Mp 57° to 59°, pK_a 9.5 (20°). Log P (octanol/water pH 7.0), 0.5. Extraction yield (chlorobutane), 1 [Demme *et al.* 2005].

Alprenolol Hydrochloride

C₁₅H₂₃NO₂·HCl = 285.8

CAS—13707-88-5; 13678-97-2 (±)

Proprietary Names Apillobal; Aptin(e); Betacard; Duriles; Gubernal; Regletin; Yobir.

Chemical Properties Colourless crystals or a white crystalline powder. Mp 108° to 111°. Soluble 1 in <1 of water, 1 in 2 of ethanol, and 1 in 3 of chloroform; soluble in acetone and dilute acetic acid; practically insoluble in ether.

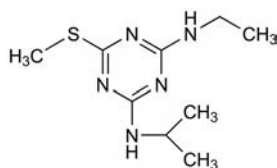
Colour Tests Liebermann's reagent—red; Mandelin's test—brown-violet; Marquis test—red; sulfuric acid—orange.

Thin-layer Chromatography System TA—R_f 0.52; system TB—R_f 0.11; system TC—R_f 0.12; system TE—R_f 0.49; system TAE—R_f 0.22; system TAG—R_f 0.11 (acidified iodoplatinate solution—positive).

Gas Chromatography System GA—RI 1820.

High Performance Liquid Chromatography System HA—k 1.2; system HY—RI 323; system HZ—RT 3.87 min.

Ultraviolet Spectrum Aqueous acid—270 (A₁¹ = 69b), 276 nm. No alkaline shift.



Chemical Properties Colourless crystals. Mp 84° to 86°. Very slightly soluble in water; soluble in chloroform and methanol.

Thin-layer Chromatography System TA— R_f 0.76; system TX— R_f 0.26; system TY— R_f 0.23 (dragendorff spray, positive).

Gas Chromatography System GA—RI 1879.

Infrared Spectrum Principal peaks at wavenumbers 1515, 1595, 1294, 1166, 806, 1140 cm^{-1} (KBr disk).

Amphetamine

Phenethylamine, Stimulant (Central), Sympathomimetic

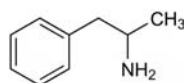
$\text{C}_9\text{H}_{13}\text{N}$ = 135.2

CAS—300-62-9

IUPAC Name 1-Phenylpropan-2-amine

Synonyms Amphetamine; desoxynorephedrine.

Proprietary Names *Elastonon; Fenopromin; Mydrial; Phenamine*. It is an ingredient of *Biphetmaine* and *Durophet*.



Chemical Properties A colourless, mobile, slowly volatile liquid. It absorbs carbon dioxide from the air forming a violent carbonate. Bp 200° to 203°. Soluble 1 in 50 of water; soluble in ethanol and ether; readily soluble in acids. pK_a 9.9 [Baselt 2008]. Log *P* (octanol/water), 1.8 [Baselt 2008]. Extraction yield (chlorobutane), 0.5 [Demme *et al.* 2005].

Amphetamine Hydrochloride

$\text{C}_9\text{H}_{13}\text{N} \cdot \text{HCl}$ = 141.7

IUPAC Name 1-Phenylpropan-2-amine hydrochloride

Amphetamine Phosphate

$\text{C}_9\text{H}_{13}\text{N} \cdot \text{H}_3\text{PO}_4$ = 233.2

CAS—139-10-6

IUPAC Name 1-Phenylpropan-2-amine; phosphoric acid

Synonyms Amphetamine phosphate; monobasic racemic amphetamine phosphate.

Proprietary Names *Dynaphenil; Monophos; Profetamine*.

Chemical Properties A white crystalline powder with no characteristic melting point; it sinters at ~150° and decomposes at ~300°. Freely soluble in water; slightly soluble in ethanol; practically insoluble in benzene, chloroform and ether.

Amphetamine Sulfate

$(\text{C}_9\text{H}_{13}\text{N})_2 \cdot \text{H}_2\text{SO}_4$ = 368.5

CAS—60-13-9

IUPAC Name 1-Phenylpropan-2-amine sulfuric acid

Synonym Amphetamine sulfate

Proprietary Names *Benzedrine; Benzpropamine; Centramina; Fabedrine; Fenamin; Phenaminium; Phenopromin; Sympamine*. It is an ingredient of *Adderall*, *Epipropine* and *Ortenal*.

Chemical Properties A white crystalline powder. Mp above 300°, with decomposition. Soluble 1 in 9 of water and 1 in 515 of ethanol; practically insoluble in chloroform and ether.

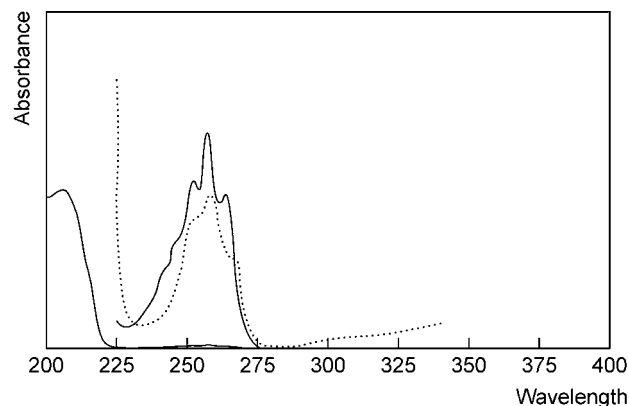
Colour Test Liebermann's reagent—red-orange; Marquis test—orange→brown; ninhydrin—pink-orange.

Thin-layer Chromatography System TA— R_f 0.43; system TB— R_f 0.20; system TC— R_f 0.09; system TE— R_f 0.43; system TL— R_f 0.18; system TAE— R_f 0.12; system TAF— R_f 0.75 (acidified iodoplatinate solution—positive; acidified potassium permanganate solution—positive; Dragendorff spray—positive; FPN reagent—pink; Marquis test—brown; ninhydrin—positive).

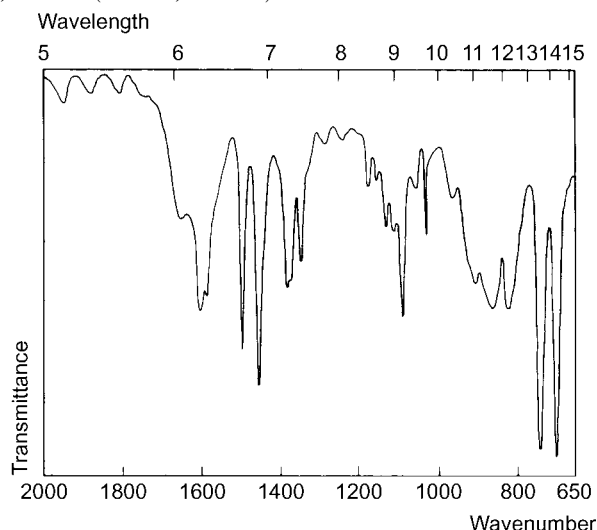
Gas Chromatography System GA—RI 1125, amphetamine-trifluoroacetic acid RI 1095, amphetamine-PFP RI 1330, amphetamine-TMS RI 1190, amphetamine-AC RI 1501, art (formyl) RI 1100, M (3-OH-)-PFP₂ RI 1520, M (3-OH-)-TMS₂ RI 1850, M (3-OH-)-AC₂ RI 1930, M (4-OH-)-AC₂ RI 1480, M (4-OH-)-AC₂ RI 1900, M (3,4-di-OH-)-AC₃ RI 2150, M (OH-methoxy-)-RI 1465, M (OH-methoxy-)-AC₂ RI 2065, M (desamino-oxo-OH-)-AC RI 1520, M (desamino-oxo-OH-methoxy-)-RI 1510, M (desamino-oxo-OH-methoxy-)-AC RI 1600, M (desamino-oxo-di-OH-)-AC₂ RI 1735; system GB—RI 1150art (formyl) RI 1142; system GC—RI 1536; system GF—RI 1315; system GAD—RT 7.5 min; system GAK—RT 4.9 min.

High Performance Liquid Chromatography System HA— k 0.9; system HB— k 8.48; system HC— k 0.98; system HX—RI 244; system HAA—RT 3.7 min; system HBC—RT 2.1 min; system HBD—RT 3.7 min.

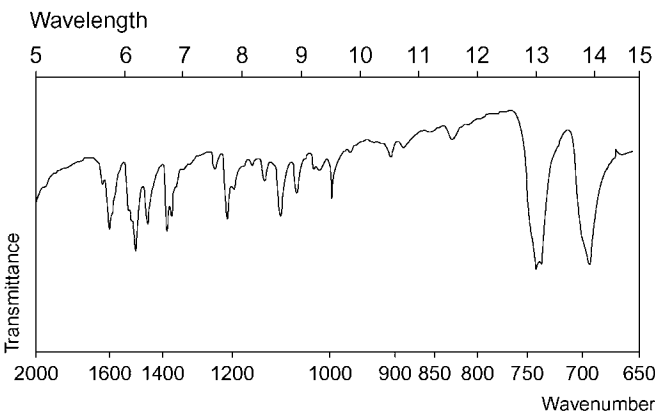
Ultraviolet Spectrum Aqueous acid—251 nm, ($A_1^1 = 14a$), 257 nm, 263 nm.



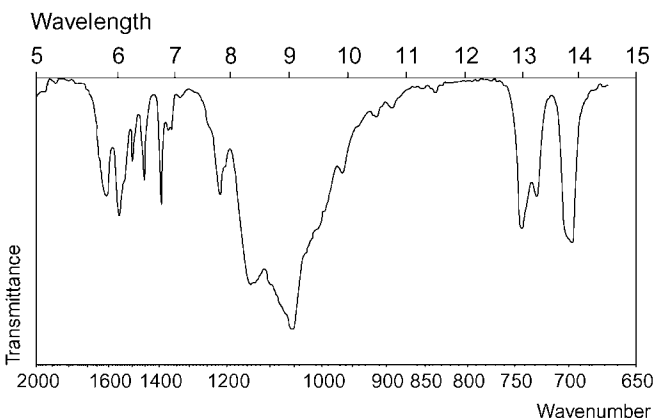
Infrared Spectrum Principal peaks at wave numbers 700, 740, 1495, 1090, 1605, 825 cm^{-1} (thin film, see below).



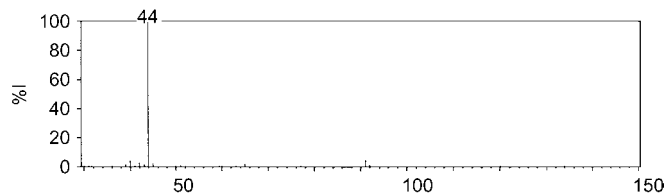
D-Amphetamine Hydrochloride



DL-Amphetamine Sulfate



Mass Spectrum Principal ions at m/z (amphetamine) 44, 91, 40, 42, 65, 45, 39, 43; (phenylacetone) 43, 91, 134, 92, 65, 39, 63, 135.



Quantification

Blood GC-MS Column: HP-5 MS (15 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He. Temperature programme: 60° for 1 min, to 220° at 20°/min. EI ionisation, SIM acquisition mode. Retention time: 4.66 min for amphetamine, 5.3 min for metamfetamine, 7.14 min for 3,4-methylenedioxymetamfetamine (MDMA) [Gunn *et al.* 2010]. Column: Supelco, PTE-5 (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 0.4 mL/min. Temperature programme: 60° for 3 min to 280° at 20°/min. EI ionisation, SIM acquisition mode. Limit of quantification, 0.5 ng/g for amfetamines and metamfetamine [Okajima *et al.* 2001]. Column: Supelco PTE-5 (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 60 kPa. Temperature programme: 60° for 3 min, to 280° at 20°/min. EI ionisation, SIM acquisition mode. Limit of detection, 10 ng/g [Namera *et al.* 2000]. Column: HP-5 MS capillary (30 m $\times$ 0.32 mm i.d.). EI ionisation, SIM acquisition mode. Limit of quantification, 10 μ g/L, limit of detection 0.5–8 μ g/L [Marquet *et al.* 1997]. Column: fused silica (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 8 psi. Temperature programme: 45° for 3.5 min to 270° at 25°/min. EI ionisation, SIM acquisition mode. Limit of detection, 0.01 μ g/g for amfetamine and metamfetamine [Nagasawa *et al.* 1996].

LC-MS Column: Varian Pursuit 3 C₁₈ (100 $\times$ 3 mm i.d., 3 μ m). Mobile phase: 2 mmol/L ammonium formate buffer: acetonitrile (pH 5.3, 92:8); methanol (97:3 for 3 min, to 80:20 over 4 min for 4 min to 70:30 over 6 min, to 10:90 over 13 min), flow rate 0.3 mL/min. MRM acquisition mode. Limit of quantification, 0.010 mg/g, limit of detection, 0.0008 mg/g [Bjork *et al.* 2010]. Column: Sphinx RP. Mobile phase: 10 mmol/L ammonium formate buffer: acetonitrile. Limit of quantification, 2.5 μ g/L, limit of detection, 0.05–0.5 μ g/L [Fernandez *et al.* 2009]. Column: Uptisphere ODB C₁₈ (150 $\times$ 2.1 mm i.d., 5 μ m). Mobile phase: acetonitrile:2 mmol/L formate buffer (pH 3.0, 20:80 for 8 min to 90:10 over 5 min to 20:80 over 7 min), flow rate 250 μ L/min. ESI, positive ion mode, MRM acquisition mode. Limit of quantification, 0.1 μ g/L [Chèze *et al.* 2007]. Column: BEH C₁₈ (50 $\times$ 2.1 mm). Mobile phase: aqueous pyrrolidine: methanol (52:48), flow rate 0.4 mL/min. Retention time: 1.32 min [Apollonio *et al.* 2006]. Column: Hypersil BDS-C₁₈ (100 $\times$ 2.1 mm i.d., 3 μ m). Mobile phase: acetonitrile:0.001% formic acid (6:94 for 5 min to 50:50 over 10 min to 6:94 over 5 min for 15 min), flow rate 0.3 mL/min. Sonic spray ionisation. Limit of detection, 5 μ g/L [Mortier *et al.* 2002].

Plasma GC Column: HP (25 m $\times$ 0.3 mm i.d., 1.05 μ m). Carrier gas: He, 2 mL/min. Temperature programme: 105° for 0.50 min to 300° at 5°/min for 15 min. ECD. Limit of detection, <1 μ g/L [Asghar *et al.* 2001]. Column: HP-5 (25 m $\times$ 0.32 mm i.d., 0.52 μ m). NPD. Limit of quantification, 1 μ g/L, limit of detection, 1 μ g/L [Cheung *et al.* 1997].

GC-MS See Blood [Gunn *et al.* 2010]. Column: HP-5 MS (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1.0 mL/min. Temperature programme: 80° to 280° at 20°/min for 1.5 min. EI ionisation, SIM acquisition mode. Limit of detection not reported [Kronstrand *et al.* 2003]. Column: SGE-BPX5 (15 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1.5 mL/min. Temperature programme: 100° for 1 min to 180° at 40°/min to 195° at 5°/min to 310° at 40°/min for 2 min. NICI, SIM acquisition mode. Limit of quantification, 0.049 μ g/L for (R)-(–)-amphetamine and 0.195 μ g/L for (S)-(+)-amphetamine [Leis *et al.* 2003]. Column: HP-5 MS (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 0.6 mL/min. Temperature programme: 100° to 250° at 10°/min to 310° at 30°/min for 1 min. EI ionisation, SIM acquisition mode. Limit of quantification, 5 μ g/L [Peters *et al.* 2003a]. Column: HP-5 MS (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° to 180° at 30°/min to 230° at 5°/min to 310° at 30°/min. NICI, SIM acquisition mode. Limit of detection not reported [Peters *et al.* 2003b].

HPLC Fluorescence detection. Limit of detection, 0.87 μ g/L for amphetamine and metamfetamine [Nakashima *et al.* 2003]. Column: MetaSil Basic (100 $\times$ 2 mm i.d., 3 μ m). Mobile phase: 0.1% formic acid: methanol (73:27), flow rate 0.25 mL/min. APCI, positive ion mode, MRM acquisition mode. Limit of quantification, 0.2 μ g/L [Slawson *et al.* 2002]. Column: ODS. Mobile phase: acetonitrile: citrate buffer (55:45). UV detection. Limit of detection, 20 μ g/L [Farrell, Jefferies 1983].

LC-MS Column: Alltima RP C₁₈ (250 $\times$ 4.6 mm i.d., 5 μ m). Mobile phase: acetonitrile:5 mmol/L formic acid (10:90 for 2 min to 25:75 in 16 min to 30:70 in 0.1 min to 100:0 over 12 min), flow rate 1.0 mL/min. Positive ion mode, MRM acquisition mode. Limit of quantification, 1.0 μ g/L, limit of detection, 0.6 μ g/L [Sergi *et al.* 2009]. Column: Atlantis dC₁₈ (100 $\times$ 2.1 mm i.d., 3 μ m). Mobile phase: acetonitrile: ammonium formate buffer (5:95 for 1 min, to 50:50 in 10 min for 1 min, to 5:95 in 1 min for 4 min), flow rate 0.2 mL/min. ESI, SIM acquisition mode. Limit of quantification, 2 μ g/L, limit of detection, 0.5–1 μ g/L [Concheiro *et al.* 2006]. Column: Hypersil BDS-C₁₈ (100 $\times$ 2.1 mm i.d., 3 μ m). Mobile phase: 10 mmol/L ammonium acetate: acetonitrile. ESI, positive ion mode, MRM acquisition mode. Limit of quantification, 0.5 μ g/L, limit of detection, 0.5 μ g/L [Wood *et al.* 2003].

Serum GC-MS See Blood [Gunn *et al.* 2010]. Column: DB-5 MS (30 m $\times$ 0.32 mm i.d., 1.0 μ m). Carrier gas: He. Temperature programme: 130° for 2 min, to 320° at 15°/min for 3 min. SIM acquisition mode. Limit of detection not reported [Kankaanpää *et al.* 2004]. Column: DB-5 MS (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1 mL/min. Temperature programme: 60° to 120° at 30°/min to 180° at 20°/min to 250° at 30°/min for 0.5 min. SIM acquisition mode. Limit of detection, 0.6 μ g/L [Lee *et al.* 2000]. Column: DB-5 MS (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Temperature programme: 80° for 1 min to 200° at 20°/min to 240° at 5°/min to 310° at 30°/min for 6 min. SIM acquisition mode. Limit of quantification, 4.9 μ g/L, limit of detection, 1.5 μ g/L [Weinmann *et al.* 2000].

HPLC Superspher 100 RP C₁₈ (125 $\times$ 3 mm i.d., 4 μ m). Mobile phase: acetonitrile:50 mmol/L ammonium formate buffer (25:75), flow rate 0.3 mL/min. APCI, SIM acquisition mode. Limit of detection, 2 μ g/L [Bogusz *et al.* 2000]. Column: Superspher Select B (125 $\times$ 3 mm i.d.). Mobile phase: 50 mmol/L ammonium formate buffer (pH 3.0): acetonitrile (60:40), flow rate 0.8 mL/min. UV detection (λ =250 nm). Limit of detection, 30 μ g/L [Bogusz *et al.* 1997]. Column: Superspher Select B (125 $\times$ 3 mm i.d.). Mobile phase: 50 mmol/L ammonium formate buffer (pH 3.0): acetonitrile (60:40), flow rate 0.8 mL/min. DAD. Limit of detection, 100 μ g/L [Bogusz *et al.* 1997].

Urine GC Column: HP-5 (300 $\times$ 32 mm i.d., 0.25 μ m). Carrier gas: He, 1.5 mL/min. Temperature programme: 140° to 300° at 20°/min. PDHID. Limit of detection, 0.5 mg/L [Casari, Andrews 2001]. See Plasma. Limit of quantification, 20 μ g/L, limit of detection, 1 μ g/L [Cheung *et al.* 1997]. NPD. Limit of quantification, 0.05 mg/L [Jonsson *et al.* 1996]. Column: DB-5 MS (150 $\times$ 2.5 mm i.d.). Carrier gas: He, 1 mL/min. Temperature programme: 70° to 200° at 15°/min. FTIR. Limit of quantification, 20 μ g/L, limit of detection, 10 μ g/L [Kalasinsky *et al.* 1993]. ECD. Limit of detection, 10 μ g/L [Terada *et al.* 1982]. See also Ugland *et al.* [1999], Ugland *et al.* [1997] and Van Bocxlaer *et al.* [1997].

GC-MS Column: DB-5 MS (5 m $\times$ 100 μ m i.d., 0.1 μ m). Carrier gas: He, 0.6 mL/min. Temperature programme: 85° to 115° at 62°/min for 0.8 min to 300° at 65°/min for 0.52 min. SIM acquisition mode. Limit of detection, 6.25 μ g/L [Marais, Laurens 2009]. Column: HP-5 MS (150 mm). Temperature programme: 100° for 1.8 min to 200° at 15 to 30°/min. SIM acquisition mode. Limit of detection, 62.5 μ g/L [Stout *et al.* 2002]. See Serum. Limit of quantification, 200 μ g/L [Kankaanpää *et al.* 2004]. Column: HP-5 MS (300 $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 0.8 mL/min. Temperature programme: 100° for 1 min to 300° at 10°/min for 3 min. EI ionisation, SIM acquisition mode. Limit of detection, 5 μ g/L [Nishida *et al.* 2003]. Column: HP-5 MS (300 $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1 mL/min. Temperature programme: 90° for 3 min to 280° at 20°/min for 3 min. SIM acquisition mode. Limit of quantification, 1.0 μ g/L, limit of detection, 0.3 μ g/L [Huang *et al.* 2002]. See also Centini *et al.* [1996], Dallakian *et al.* [1996], Dasgupta, Spies [1998], Hensley, Cody [1999], Jurado *et al.* [2000], Karacic, Skender [2000], McCambly *et al.* [1997], Myung *et al.* [1998], Namera *et al.* [2002], Pellegrini *et al.* [2002], Valentine *et al.* [1995], Wang *et al.* [2001] and Yashiki *et al.* [1995].

HPLC Column: Supelcosil LC-CN (33 $\times$ 4.6 mm i.d., 3 μ m). Mobile phase: acetonitrile:50 mmol/L ammonium acetate (15:85), flow rate 0.4 mL/min. ESI, SIM acquisition mode. Limit of quantification, 13 μ g/L [Wu *et al.* 2001]. Column: RP C₁₈ ODS (250 $\times$ 4 mm i.d.). Mobile phase: tetrahydrofuran: acetonitrile:0.1% TEA (15:15:70). UV detection (λ =260 nm). Limit of detection, 10 μ g [Veress 2000]. Column: ODS. Mobile phase: 0.1 mol/L Tris-hydrochloride buffer (pH 7.0): acetonitrile (45:55), flow rate 0.2 mL/min. Fluorescence detection [[Al-Dirbashi *et al.* 2000a]. UV detection. Limit of quantification, 47 μ g/L, limit of detection, 14 μ g/L [Fisher, Bourque 1993]. See Plasma. Limit of detection, 4 μ g/L [Farrell, Jefferies 1983]. See also Campins-Falcó [1995], Campins-Falcó *et al.* [1996], Al-Dirbashi *et al.* [1997], Al-Dirbashi *et al.* [1998], Katagi *et al.* [1996] and Molins Legua *et al.* [1995].

LC-MS See Blood. Limit of quantification, 25 μ g/L, limit of detection, 0.25–2.5 μ g/L [Fernandez *et al.* 2009]. Column: CAPCELL PAK C₁₈ MG-II (150 $\times$ 2.0 mm i.d., 5 μ m). Mobile phase: 5 mmol/L ammonium formate: acetonitrile (95:5 to 20:80 in 8 min for 1 min), flow rate 0.23 mL/min. ESI, MRM acquisition mode. Limit of quantification, 5.0 μ g/L, limit of detection, 0.87 μ g/L [Kim *et al.* 2008]. See Blood [Chèze *et al.* 2007]. Column: Symmetry Shield RP C₁₈ (150 $\times$ 2.1 mm i.d., 5 μ m). Mobile phase: acetonitrile:0.05% trifluoroacetic acid (10:90), flow rate 0.2 mL/min. MRM acquisition mode. Limit of quantification, 5 μ g/L, limit of detection, 1 μ g/L [Fuh *et al.* 2006]. Column: Symmetry Shield RP C₁₈ (50 $\times$ 2.1 mm i.d., 3.5 μ m). MRM acquisition mode. Limit of quantification, 9.1 μ g/L, limit of detection, 2.8 μ g/L [Wu, Fuh 2005].

Bile GC-MS See Vitreous Humour [Kiely *et al.* 2009].

LC-MS Column: Shiseido CAPCELL PAK SCX80 (250 $\times$ 1.5 mm i.d.). Mobile phase: 25 mmol/L ammonium acetate (pH 4.0): acetonitrile (3:7), flow rate 0.2 mL/min. ESI, SRM acquisition mode. Limit of detection not reported [Kuwayama *et al.* 2008].

CSF GC-MS SIM acquisition mode [Narasimhachari *et al.* 1979].

Meconium GC-MS Column: HP-5 MS (15 m $\times$ 0.25 mm i.d., 0.25 μ m). Temperature programme: 120° for 5 min, to 260° at 35°/min for 0.5 min. EI ionisation, SIM acquisition mode. Limit of quantification, 75 ng/g, limit of detection, 30 ng/g [Gunn *et al.* 2008]. DB-1 (10 m $\times$ 0.18 mm i.d., 0.4 μ m). Temperature programme: 120° for 1 min to 150° at 5°/min. Limit of quantification, 100 ng/g, limit of detection, 50 ng/g [Elseohly *et al.* 1999].

LC-MS Column: Synergi Hydro RP (150 $\times$ 2 mm i.d., 4 μ m). Mobile phase: 1 mmol/L ammonium formate, (pH 3.4): acetonitrile and 5% 1 mmol/L ammonium formate (100:0 to 70:30 over 8 min, to 65:35 over 2 min for 1 min, to

100:0 over 2.5 min), flow rate 0.250 mL/min. MRM acquisition mode. Limit of quantification, 5 ng/g, limit of detection, 1 ng/g [Gray *et al.* 2009]. Column: Phenomenex Synergi Hydro RP (50 × 2.0 mm i.d., 2.0 µm). Mobile phase: 10 mmol/L ammonium acetate with 0.1% formic acid:acetonitrile with 0.1% formic acid (88:12), flow rate 0.6 mL/min. Limit of quantification, 0.6 ng/g, limit of detection, 0.2 ng/g [Jones *et al.* 2009]. Column: Synergi Polar RP (150 × 2.0 mm i.d., 4 µm). Mobile phase: 10 mmol/L ammonium acetate containing 0.01% formic acid:acetonitrile (90:10 for 1 min to 50:50 over 13 min for 2 min to 10:90 over 2 min for 2 min to 90:10 over 2 min), flow rate 200 µL/min. MRM acquisition mode. Limit of quantification, 12.5 ng/g, limit of detection, 4 ng/g [Kelly *et al.* 2008]. Column: C₁₈. Mobile phase: 10 mmol/L ammonium bicarbonate:methanol. SIM acquisition mode. Limit of quantification, 0.005 µg/g [Pichini *et al.* 2004].

HPLC Column: Supelcosil LC-18 DB C₁₈ (150 × 4.6 mm i.d., 5 µm). Mobile phase: potassium dihydrogen phosphate:methanol:acetonitrile:tetrahydrofuran:TEA (600:100:25:7:1:5), flow rate 1 mL/min. DAD (λ = 204 nm). Limit of quantification, 0.5 µg/g [Franssen *et al.* 1994].

Milk HPLC Phenomenex Aqua C₁₈ (250 × 4.6 mm i.d., 5 µm). Mobile phase: acetonitrile:45 mmol/L phosphate buffer (12:88), flow rate 1.2 mL/min. UV detection (λ = 200 nm). Limit of quantification, 3 mg/L [Bartu *et al.* 2009].

Oral Fluid GC-MS Column: DB-5 (30 × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° for 1 min to 230° at 10°/min for 10 min. SIM acquisition mode. Limit of detection, 100 µg/L [Meng, Wang 2010]. Column: Varian (15 m × 0.25 mm i.d.). Limit of quantification, 3 µg/L [Cone *et al.* 2007]. Column: HP-5 MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1 mL/min. Temperature programme: 100° to 190° at 30°/min, to 230° at 5°/min to 245° at 30°/min to 260° at 5°/min to 310° at 30°/min. NCI mode, SIM acquisition mode. Limit of detection not reported [Peters *et al.* 2007]. Column: HP-5 MS (30 m × 0.32 mm i.d., 0.25 µm). Carrier gas: He, 1.5 mL/min. Temperature programme: 70° to 170° at 25°/min for 2 min to 195° at 5°/min for 0.5 min to 300° at 30°/min for 1 min. EI ionisation, SIM acquisition mode. Limit of quantification, 5.0 µg/L, limit of detection, 2.5 µg/L [Scheidweiler, Huestis 2006]. See Serum. Limit of quantification, 5 µg/L [Kankaanpää *et al.* 2004].

LC-MS ESI. Limit of quantification, 0.4–5 µg/L [Fritch *et al.* 2009]. See Plasma. Limit of quantification, 1.5 µg/L, limit of detection, 0.5 µg/L [Sergi *et al.* 2009]. Column: Pinnacle II C₁₈ (50 × 4.6 mm i.d., 5 µm). Mobile phase: 5 mmol/L ammonium acetate:methanol containing 0.1% formic acid, flow rate 1.2 mL/min. MRM acquisition mode. Limit of quantification, 10 µg/L, limit of detection, 10 µg/L [Kala *et al.* 2008]. Column: Atlantis dC₁₈ (100 × 2.1 mm i.d., 3 µm). Mobile phase: acetonitrile:ammonium formate buffer (pH 3.0, 5:95 for 1 min to 50:50 in 10 min for 1 min to 5:95 in 1 min), flow rate 0.2 mL/min. ESI, SIM acquisition mode. Limit of quantification, 1 µg/L, limit of detection, 0.5 µg/L [Concheiro *et al.* 2007]. See Plasma [Wood *et al.* 2003].

Stomach Contents GC-MS See Vitreous Humour [Kiely *et al.* 2009]. Column: DB-5 (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 0.9 mL/min. Limit of detection not reported [Takekawa *et al.* 2007]. Column: DB-5 fused silica (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 0.8 mL/min. Temperature programme: 70° for 1 min, to 100° at 30°/min to 270° at 10°/min. Limit of detection not reported [Meyer *et al.* 1997].

Vitreous Humour GC-MS Column: DB-5 MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.0 mL/min. Temperature programme: 80° to 120° at 40°/min to 240° at 10°/min for 5 min. Limit of detection not reported [Kiely *et al.* 2009].

HPLC Column: Hypersil BDS-C₁₈ (100 × 2.1 mm i.d., 3 µm). Fluorescence detection (λ_{ex} = 288 nm, λ_{em} = 324 nm). Limit of quantification, 2.0 µg/L, limit of detection, 0.8 µg/L [Clauwaert *et al.* 2000].

LC-MS Column: Hypersil BDS-C₁₈ (100 × 2.1 mm i.d., 3 µm). Limit of quantification, 2.0 µg/L, limit of detection, 0.8 µg/L [Clauwaert *et al.* 2000].

Bone and Bone Marrow GC-MS [Sato *et al.* 2000].

Brain GC-MS See Vitreous Humour [Kiely *et al.* 2009]. Column: HP-1 MS (3 m × 0.32 mm i.d., 0.25 µm). Carrier gas: He, 1.1 mL/min. Temperature programme: 70° for 1 min to 175° at 30°/min to 250° at 23°/min to 310° at 18°/min for 5 min. SIM acquisition mode. Limit of quantification, 50 ng/g, limit of detection, 50 ng/g [Lowe *et al.* 2006]. See Stomach Contents [Meyer *et al.* 1997].

Hair GC Column: CBJ-17. NPD. Limit of detection, 0.1 µg/g for amphetamine and 0.4 µg/g for metamfetamine [Koide *et al.* 1998].

GC-MS Column: DB-5 MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.1 mL/min. Temperature programme: 90° for 0.5 min to 150° at 25°/min for 1.5 min to 230° at 15°/min for 1 min to 300° at 40°/min for 1.3 min. EI ionisation, SIM acquisition mode. Limit of quantification, 0.1 µg/g, limit of detection, 0.027 µg/g [Kim *et al.* 2010a]. G&W Scientific Agilent Technologies 5% phenyl-MS (17 m × 0.2 mm i.d., 0.33 µm). Carrier gas: He, 20 psi. Temperature programme: 100° for 1 min to 140° at 8°/min to 310° at 30°/min for 2 min. Limit of quantification, 0.2 µg/g for (R)-(–) and (S)-(+)-enantiomers, limit of detection, 0.1 µg/g for (R)-(–) and (S)-(+)-enantiomers [Strano-Rossi *et al.* 2009]. Column: HP-5 MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° for 1 min to 300° at 10°/min for 3 min. Limit of detection, 1.6 µg/g [Nishida *et al.* 2006a]. See Plasma [Kronstrand *et al.* 2003]. See Urine. Limit of detection, 0.25 µg/g [Nishida *et al.* 2003]. See also Cirimele *et al.* [1995], Kintz *et al.* [1995], Liu *et al.* [2001], Musshoff *et al.* [2002], Rohrich, Kauert [1997] and Skender *et al.* [2002].

HPLC Column: SP-120-5-ODS-BP (250 × 4.6 mm i.d., 5 µm). Mobile phase: acetonitrile:methanol:water (30:40:30), flow rate 1.0 mL/min. Fluorescence detection (λ_{ex} = 325 nm, λ_{em} = 430 nm). Limit of detection, 11 ng/g [Kaddoumi

et al. 2004]. See Plasma. Mobile phase: acetonitrile:methanol:citrate buffer (45:20:37.5). Limit of detection, 0.12 ng/g for amphetamine and metamfetamine [Nakashima *et al.* 2003]. See Urine. Limit of quantification, 0.06 µg/g [Wu *et al.* 2001]. Column: Chiralcel OD-RH (150 × 2 mm i.d.). Mobile phase: phosphate-citrate buffer (pH 4.0) containing sodium hexafluorophosphate:acetonitrile (43:57), flow rate 0.1 mL/min. Fluorescence detection (λ_{ex} = 330 nm, λ_{em} = 440 nm). Limit of detection, 54.0 ng/g [Al-Darbashi *et al.* 2000b].

LC-MS See Blood. Limit of quantification, 14.7 ng/g [Chèze *et al.* 2007]. Column: Chiral DRUG (150 × 2 mm i.d.). Mobile phase: 2 mmol/L ammonium formate buffer (pH 5) in methanol:2 mmol/L ammonium formate buffer (pH 5.0, 40:60 for 20 min to 95:5 over 55 min), flow rate 0.1 mL/min. ESI, SIM acquisition mode. Limit of detection, 0.05 µg/g for (R)-(–) and (S)-(+)-enantiomers [Nishida *et al.* 2006b]. Column: Betasil C₈. Mobile phase: water:acetonitrile containing 0.1% formic acid (80:20). MRM acquisition mode. Limit of quantification, 25 ng/g, limit of detection, 25 ng/g [Cairns *et al.* 2004]. Column: LiChroCART Purospher 60 RP C₁₈ (125 × 4 mm i.d., 5 µm). Mobile phase: acetonitrile:water (100:0 to 60:40 at 15 min to 100:0 at 15.2 min for 5 min), flow rate 1 mL/min. APCL, SIM acquisition mode. Limit of detection, 0.10 µg/g [Stanaszek, Piekoszewski 2004]. Column: CAPCELL PAK SCX UG 80 (150 × 1.5 mm i.d., 5 µm). Mobile phase: 12.5 mmol/L ammonium acetate containing formic acid:acetonitrile (40:60), flow rate 0.15 mL/min. Positive ion mode, SIM acquisition mode. Limit of detection, 0.02 µg/g [Miki *et al.* 2003].

Kidney GC-MS See Stomach Contents [Meyer *et al.* 1997].

Liver GC-MS See Vitreous Humour [Kiely *et al.* 2009]. See Stomach Contents [Meyer *et al.* 1997].

Lung GC-MS [Sato *et al.* 2000].

Muscle GC-MS [Sato *et al.* 2000]. XTI-5 (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.7 mL/min. Temperature programme: 60° for 1 min to 290° at 20°/min. SIM acquisition mode. Limit of detection not reported [Hara *et al.* 1997].

Nail Clippings GC-MS Column: DB-5 MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1 mL/min. Temperature programme: 90° for 3 min to 170° at 15°/min for 2 min to 210° at 25°/min for 1.5 min to 230° at 20°/min for 0.5 min, to 300° at 40°/min for 0.3 min. EI ionisation, SIM acquisition mode. Limit of quantification, 0.063 µg/g, limit of detection, 0.019 µg/g [Kim *et al.* 2010b]. Column: DB-5 MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.1–1.4 mL/min. Temperature programme: 100° for 1 min to 150° at 20°/min for 1.5 min to 285° at 30°/min for 0.5 min to 300° at 30°/min for 2.5 min. EI ionisation, SIM acquisition mode. Limit of quantification, 0.1 µg/g, limit of detection, 0.016 µg/g [Kim *et al.* 2008]. HP-1 MS (30 m × 0.25 mm i.d., 0.25 µm). Temperature programme: 60° for 1 min to 300° at 20°/min for 1 min. Limit of detection, 0.2 µg/g [Lin *et al.* 2004]. See Hair. Limit of detection, 12.0 µg/g [Cirimele *et al.* 1995].

Placenta GC-MS Column: ZB-5 (15 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 0.8 mL/min. Temperature programme: 70° for 2 min, to 160° at 30°/min to 170° at 5°/min to 200° at 20°/min to 210° at 10°/min to 300° at 30°/min for 3 min. EI ionisation. Limit of quantification, 3.4 µg/L, limit of detection, 1.1 µg/L [Joya *et al.* 2010].

Disposition in the Body Amphetamine is readily absorbed after oral or rectal administration. It is rapidly distributed extravascularly and taken up, to some extent, by red blood cells. The main metabolic reaction is oxidative deamination to form phenylacetone, which is then oxidised to benzoic acid and conjugated with glycine to form hippuric acid; minor reactions include aromatic hydroxylation to form 4-hydroxyamphetamine (an active metabolite), β-hydroxylation, which is stereoselective for the (+)-isomer of amphetamine, to form norephedrine (phenylpropanolamine), and N-oxidation to form a hydroxylamine derivative. The products of aromatic hydroxylation and N-oxidation may be conjugated with sulfate or glucuronic acid.

Excretion of amphetamine is markedly dependent on urinary pH, being greatly increased in acid urine. After large doses, amphetamine may be detected in urine for several days. Under uncontrolled urinary pH conditions, ~30% of the dose is excreted unchanged in the urine in 24 h and a total of ~90% of the dose is excreted in 3–4 days. The amount excreted unchanged in 24 h may increase to 74% of the dose in acid urine and decrease to 1 to 4% in alkaline urine; under alkaline conditions, hippuric acid and benzoic acid account for ~50% of the urinary material. Under normal conditions, 16–28% is excreted as hippuric acid, ~4% as benzoylglucuronide, 2–4% as 4-hydroxyamphetamine and ~2% as norephedrine in 24 h; small amounts of conjugated 4-hydroxynorephedrine and phenylacetone are also excreted. No elimination in the faeces has been detected.

Amphetamine is a metabolite of benzphetamine, metamfetamine and selegiline.

Therapeutic Concentration After normal therapeutic doses the plasma concentration is usually <0.1 mg/L. However, continued use of amphetamine may cause addiction, and ingestion of 10 times the usual therapeutic dose is common among addicts; in such cases the plasma concentration may be up to 3 mg/L.

The IV administration of 160 mg amphetamine to a regular user resulted in a plasma concentration of 0.59 mg/L after 1 h [Anggård *et al.* 1970].

After a single oral dose containing 10 mg amphetamine to 4 subjects, peak plasma concentrations of ~0.02 mg/L were attained [Wan *et al.* 1978].

Toxicity The estimated minimum lethal dose in non-addicted adults is 200 mg. Toxic effects may be produced with blood concentrations of 0.2–3 mg/L, and fatalities with concentrations greater than 0.5 mg/L. Death from overdosage are comparatively rare.

In a fatality caused by IV administration of amphetamine, the following postmortem tissue concentrations were reported: blood 41 mg/L, liver 23 µg/g and urine 39 mg/L [Adjutantis *et al.* 1975].

Urine concentrations of >0.5–320 mg/L (mean, 76) were reported in 11 fatalities caused by amphetamine [Holmgren, Lindquist 1975].

In a 22-year-old man who died of cardiorespiratory arrest following amphetamine ingestion, the following postmortem tissue concentrations were reported for racemic, (R)- and (S)-amphetamine, respectively: blood 2.44, 1.26 and 1.18 mg/L; urine 33.4, 16.7 and 16.7 mg/L; liver 11.7, 6.07 and 5.64 µg/g; kidney 3.85, 2.00 and 1.85 µg/g; and brain 5.50, 2.95 and 2.55 µg/g [Meyer *et al.* 1997].

Half-life Plasma half-life, 4 to 8 h when the urine is acidic and approx. 12 h in subjects whose urinary pH values are uncontrolled.

Volume of Distribution $\approx$ 3–4 L/kg.

Distribution in Blood Plasma: whole blood, 1.0

Protein Binding 15–40%.

Saliva Plasma: saliva ratio, $\sim$ 0.35

Dose 20 to 100 mg of amphetamine sulfate daily has been used in the treatment of narcolepsy.

Note For a review of the toxicokinetics of amphetamines, see Kraemer, Maurer [2002]; for a study of the pharmacokinetic and pharmacodynamic drug interactions in the treatment of attention-deficit hyperactivity disorder, see Markowitz, Patrick [2001].

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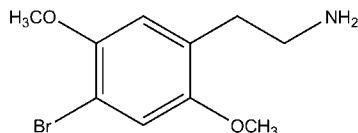
2C-B

5-HT_{2A} Receptor Antagonist, Entactogen, Hallucinogen, Phenethylamine
 $C_{10}H_{14}BrNO_2=260.1$
 CAS—66142-81-2

IUPAC Name 2-(4-Bromo-2,5-dimethoxy-phenyl)ethanamine

Synonyms BDMPEA; 4-bromo-2,5-dimethoxybenzeneethanamine; 4-bromo-2,5-dimethoxy- β -phenethylamine; 2-(4-bromo-2,5-dimethoxyphenyl)-1-aminoethane; α -desmethyl DOB; 2,5-dimethoxy-4-bromophenethylamine; MFT.

Street Names Bees; Bromo; Eroxo; Nexus; Performax; Venus; XTC.



Chemical Properties Log P (octanol/water), 2.03 [ACD 2007]. No indication of instability of processed samples over a time period of 32 h. When the samples were freeze-thawed, the ratio of the means was within 90 and 110% for all analytes at high concentrations [Habrdovala *et al.* 2005].

2C-B Hydrochloride

$C_{10}H_{14}BrNO_2 \cdot HCl=296.6$
 CAS—56281-37-9

Synonyms BDMPEA hydrochloride; 4-bromo-2,5-dimethoxybenzeneethanamine hydrochloride; 4-bromo-2,5-dimethoxy- β -phenethylamine hydrochloride; 2-(4-bromo-2,5-dimethoxyphenyl)-1-aminoethane hydrochloride; α -desmethyl DOB hydrochloride; 2,5-dimethoxy-4-bromophenethylamine hydrochloride; MFT hydrochloride.

Chemical Properties Pale pink crystals, fine white crystals when dry. Mp 237° to 239°. Soluble in water.

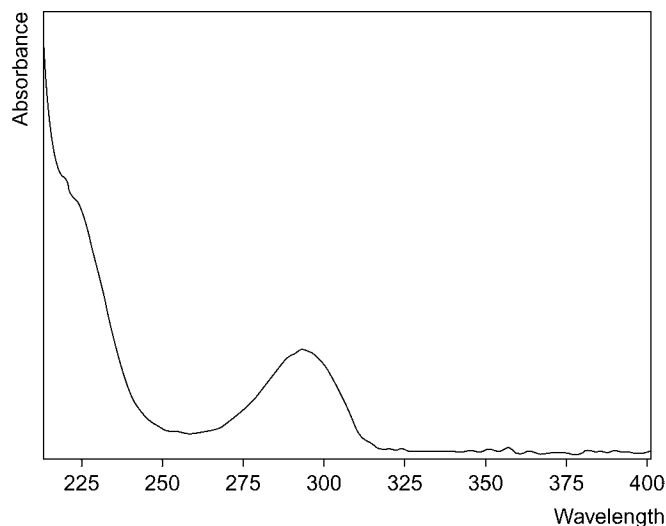
Colour Test Marquis test—yellow→green [Cole *et al.* 2002].

Gas Chromatography-Mass Spectrometry Column: HP-5MS cross-linked phenylmethylsiloxane (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1 mL/min. Temperature programme: 70° for 1 min to 200° at 20°/min for 8.84 min. EI ionisation at 70 eV. Limit of detection, 4.94 μ g/L [Tsai *et al.* 2006]. Column: HP Ultra-2 fused silica capillary (25 m $\times$ 0.2 mm i.d., 0.33 μ m). Carrier gas: He. Temperature programme: 55° for 1 min, to 190° at 20°/min to 305° at 10°/min for 11 min. EI ionisation at 70 eV. Retention time: 7.4 min [Giroud *et al.* 1998].

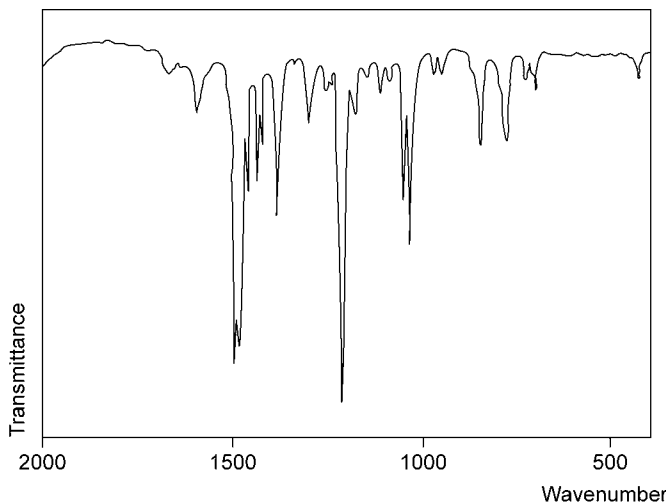
High Performance Liquid Chromatography Column: Spherisorb C₁₈ ODS2 (250 $\times$ 4.6 mm i.d., 5 μ m). Mobile phase: 0.05 mol/L ammonium acetate (pH 3.9) buffer (0.1% triethylamine): methanol (77:23 for 19 min to 65:35 for 16 min), flow rate 0.7 mL/min for 9 min, 1.2 mL/min for 12 min, and 1.4 mL/min for 14 min. DAD ($\lambda=210$ nm). Retention time: 27.6 min. Limit of detection, 105 μ g/L [Soares *et al.* 2004]. Column: Lichrospher RP-8 Select B (250 $\times$ 4 mm i.d., 5 μ m). Mobile phase: acetonitrile:10 mmol/L potassium phosphate buffer (pH 3.2; 5:95 to 60:40 after 20 min), flow rate 0.5 mL/min. DAD ($\lambda=210$ nm). Limit of detection, 1 μ g/L [Giroud *et al.*, 1998].

Capillary Electrophoresis Capillary: fused silica (total/effective length: 96/90 cm for sweeping MEKC and 100/94 cm for stacking MEKC, 50 μ m i.d.). Lamp: Xe. Fluorescence detection ($\lambda_{ex}=300$ nm, $\lambda_{em}=340$ nm). Limit of detection, 36.4 and 4.94 μ g/L for sweeping and stacking MEKC, respectively [Tsai *et al.* 2006].

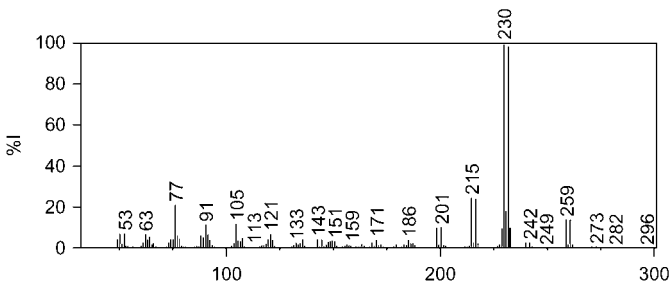
Ultraviolet Spectrum Methanol—203 nm, 296 nm.



Infrared Spectrum Principal peaks at wavenumbers 1250, 1500, 825, 740, 750 cm^{-1} .



Mass Spectrum Principal ions at m/z 230, 232, 215, 217, 77, 259, 261, 91; 242, 244, 229, 148, 301, 303, 77, 201 (AC); 244, 242, 229, 231, 148, 455, 457, 199 (HFB derivative) [Giroud *et al.* 1998].



Quantification

Blood GC-MS Column: Phenomex ZBI (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1 mL/min. Temperature programme: 100° for 1 min to 200° at 10°/min to 300° at 60°/min for 14 min. EI ionisation, SIM acquisition mode. Limit of detection, 5 to 10 μ g/L [Vorce, Sklerov 2004].

Plasma HPLC Column: Daisopak SP-120-5-ODS-BP (250 $\times$ 4.6 mm i.d., 5 μ m). Mobile phase: methanol-water-ethyl acetate (69:30:1): acetonitrile-water (70:30; 60:40 for 20 min, 5:95 at 26 min until 49 min to 60:40). Fluorescence detection ($\lambda_{ex}=325$ nm, $\lambda_{em}=430$ nm). Limit of detection, 312 ng/L [Kaddoumi *et al.* 2001].

GC-MS Column: HP-5MS capillary 5% phenylmethylsiloxane (30 m × 0.25 mm i.d., 250 nm). Carrier gas: He, 1.1 mL/min. Temperature programme: 80° for 0.5 min to 310° at 30°/min for 2 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of quantification, 10 µg/L [Habrdova *et al.* 2005].

Urine GC-MS See Blood [Vorce, Sklerov 2004]. Column: Supelco fused silica capillary (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 0.8 mL/min. Temperature programme: 80° for 3 min to 220° at 40°/min to 280° at 8°/min for 3 min. SIM acquisition mode. Limit of detection, 10 µg/L [Namera *et al.* 2002].

HPLC Column: Waters Spherisorb C₁₈ ODS2 (250 × 4.6 mm, 5 µm). Mobile phase: methanol: 0.05 mol/L ammonium acetate buffer containing 0.1% triethylamine (pH 3.9, 23 : 77 for 19 min to 35 : 65 for 16 min), flow rate 0.7 mL/min to 8 min to 1.2 mL/min to 19 min to 1.4 mL/min to 35 min. UV detection (λ = 210 nm). Limit of detection, 105 µg/mL [Soares *et al.* 2004].

LC-MS Direct injection, positive ion mode, MRM acquisition mode. Limit of detection, 50 µg/L [Nordgren *et al.* 2005]. Column: HypURITY Advance (30 × 2.1 mm, 3 µm). Mobile phase: 10 mmol/L ammonium acetate: methanol (95 : 5 at 0.1 min to 20 : 80 at 1 min to 95 : 5 for 5 min), flow rate 400 µL/min. APCL, positive ion mode. Retention time: 1.44 min. Limit of detection, 0.5 µg/L [Nordgren, Beck 2004].

Hair GC-MS Column: DB-5MS capillary (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.1 mL/min. Temperature programme: 90° for 3 min to 170° at 15°/min for 3 min to 210° at 25°/min for 1.5 min to 230° at 20°/min for 0.5 min to 300° at 35°/min for 0.5 min. EI ionisation mode at 70 eV, SIM acquisition mode. Limit of quantification, 20 pg/mg, limit of detection, 4 pg/mg [Kim *et al.* 2007].

Other HPLC Tablets. Column: Lichrospher RP-8 Select B (250 × 4 mm i.d., 5 µm). Mobile phase: acetonitrile: 10 mmol/L potassium phosphate buffer (pH 3.2; 5 : 95 to 60 : 40 at 20 min), flow rate 1.0 mL/min. UV detection (λ = 210 nm). Quantification, down to 1 ng/L [Giroud *et al.* 1998].

GC-MS Rat Urine. Column: HP-1 capillary cross-linked methylsilicone (12 m × 0.2 mm i.d., 330 nm). Carrier gas: He, 1 mL/min. Temperature programme: 100° to 310° at 30°/min. EI ionisation at 70 eV. Limit of detection not reported [Theobald *et al.* 2007]. Tablets. Column: HP Ultra-2 silica phenylmethylsilicone (25 m × 0.2 mm i.d., 0.33 µm). Carrier gas: He. Temperature programme: 55° to 190° at 20°/min to 305° at 10°/min for 11 min. Limit of detection, 1 mg/L [Giroud *et al.* 1998].

Disposition in the Body Metabolised by oxidative deamination and demethylation to form 2-(4-bromo-2,5-dimethoxyphenyl)-ethanol, 4-bromo-2,5-dimethoxyphenylacetic acid, 4-bromo-2,5-dimethoxybenzoic acid, and 2-(4-bromo-2-hydroxy-5-methoxyphenyl)ethanol [Carmo *et al.* 2005].

Toxicity There have been no published cases of fatal 2C-B toxicity. It is a partial agonist of serotonergic receptors producing euphoric and hallucinogenic effects (auditory and visual), which increase with dosage.

Note For further information, see Giroud *et al.* [1998] and Shulgin, Shulgin [1991].

Dose Light dose 5 to 15 mg; strong dose 20 to 50 mg [Cole *et al.* 2002].

ACD (2007) *ACD/LogP DB*. Toronto ON: Advanced Chemistry Development www.acdlabs.com (accessed November 2007).

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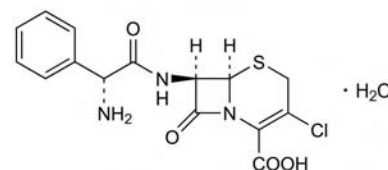
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IUPAC Name (6R,7R)-7-[[[(2R)-Aminophenylacetyl]amino]-3-chloro-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid monohydrate

Proprietary Names Alfatil; Cec; Cedor; Cedorbeta; Cefabioicin; Cefallone; Cefawolff; Cef-diolan; Distaclor; Infectocefl; Keflor; Kefolor; Keflid; Klocolor; Panacefl; Panoral; Sigacefl; Vercef.

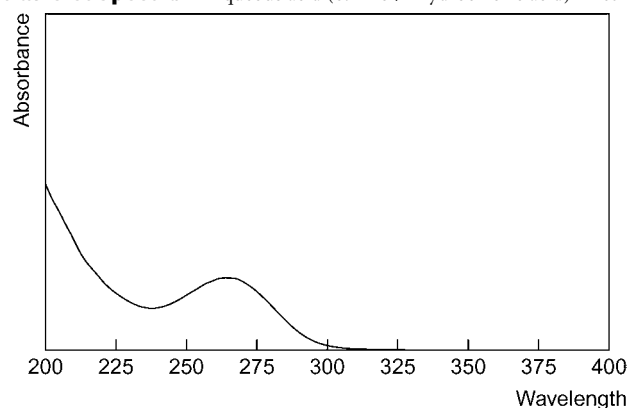


Chemical Properties A white to off-white, or slightly yellow, crystalline powder. It is soluble to slightly soluble in water (10 g/L); practically insoluble in benzene, chloroform, dichloromethane, and methanol (all <0.5 g/L). pK_a 1.5, 7.2 (water); 4.33, 7.34 (66% DMF). Log P (octanol/water), 0.35.

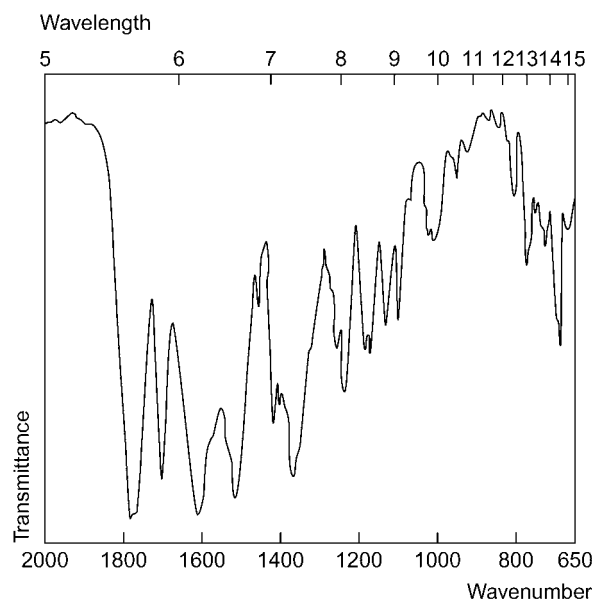
Thin-layer Chromatography System TE—R_f 0.00; system TF—R_f 0.00.

High Performance Liquid Chromatography System HX—RI 268.

Ultraviolet Spectrum Aqueous acid (0.1 mol/L hydrochloric acid)—265 nm.



Infrared Spectrum Principal peaks at wavenumbers 1775, 1693, 1600, 1365, 1500, 1560 cm⁻¹.



Quantification

Plasma HPLC UV detection (λ = 265 nm). Limit of detection, 0.5 mg/L [Kovach, Lantz 1991]. UV detection. UV detection (λ = 254 nm). Limit of detection, 0.5 mg/L [Signs *et al.* 1984].

Serum HPLC See Plasma [Kovach, Lantz 1991]. UV detection. Limit of detection, about 1 mg/L [Lindgren 1987]. UV detection (λ = 240 nm). Limit of detection, 1 mg/L [McAteer *et al.* 1987].

Urine HPLC See Plasma [Kovach, Lantz 1991].

Disposition in the Body Cefaclor is well absorbed after oral administration; food may delay absorption but the total amount absorbed is unchanged. The peak concentration, however, with food is 50 to 75% of that observed for the drug when administered in the fasting state. It is widely distributed in the body and crosses the placenta and low concentrations appear in breast milk. Cefaclor is rapidly excreted by the kidneys; up to 85%

Cefaclor

Antibacterial

C₁₅H₁₄ClN₃O₄S, H₂O = 385.8

CAS—53994-73-3 (anhydrous); 70356-03-5 (monohydrate)

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Codeine

Narcotic Analgesic

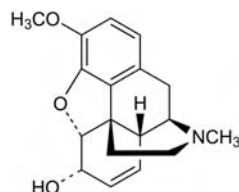
$C_{18}H_{21}NO_3 \cdot H_2O = 317.4$

CAS—76-57-3 (anhydrous); 6059-47-8 (monohydrate)

IUPAC Name (5 α ,6 α)-7,8-Didehydro-4,5-epoxy-3-methoxy-17-methylmorphinan-6-ol monohydrate

Synonyms Codeinum; methylmorphine; metilmorfina; morphine methyl ether.

Proprietary Names *Codicaps; Eulytan; Pneumogenol.*



Chemical Properties An alkaloid obtained from opium or prepared by methylation of morphine. Colourless crystals or white crystalline powder that effloresces slowly in dry air and is affected by light. Mp 154° to 156°. Soluble 1 in 120 of water, 1 in 15 of boiling water, 1 in 2 of ethanol, 1 in 0.5 of chloroform, 1 in 13 of benzene and 1 in 18 of ether; freely soluble in amyl alcohol, methanol and dilute acids; almost insoluble in petroleum ether or in solutions of alkali hydroxides. pK_a 8.2 (20°). Log *P* (octanol/water pH 7.4), 0.6. Extraction yield (chlorobutane), 0.8 [Demme *et al.* 2005].

Codeine Hydrochloride

C₁₈H₂₁NO₃·HCl·2H₂O = 371.9

CAS—1422-07-7 (anhydrous)

Chemical Properties A white crystalline powder. Mp ~280°, with some decomposition. Soluble 1 in 20 of water, 1 in 180 of ethanol and 1 in 800 of chloroform.

Codeine Phosphate

C₁₈H₂₁NO₃·H₃PO₄·1/2H₂O = 406.4

CAS—52-28-8 (anhydrous); 41444-62-6 (hemihydrate); 5913-76-8 (sesquihydrate)

Synonyms Codeine phosphate hemihydrate; codeini phosphas; methylmorphine phosphate.

Proprietary Names *Actacode; Bromophar; Bronchodine; Bronchosedal; Codant; Codedril; Codenfan; Codicompre; Evacode; Galcodine; Glottyl; Paderyl; Toularynx; Tricodoin; Tryasol; Tussoret.* Codeine phosphate is an ingredient of many proprietary preparations [Sweetman 2007].

Chemical Properties Small colourless crystals or white crystalline powder. Soluble 1 in 4 of water, 1 in 450 of ethanol and 1 in 125 of boiling ethanol; practically insoluble in chloroform and ether.

Codeine Sulfate

(C₁₈H₂₁NO₃)₂·H₂SO₄·3H₂O = 750.9

CAS—1420-53-7 (anhydrous); 6854-40-6 (trihydrate)

Synonym Codeine sulphate

Chemical Properties White crystals or crystalline powder. Soluble 1 in 30 of water and 1 in 1300 of ethanol; practically insoluble in chloroform and ether.

Colour Test Liebermann's reagent—black; Mandelin's test—green; Marquis test—violet.

Thin-layer Chromatography System TA—R_f 0.33; system TB—R_f 0.06; system TC—R_f 0.18; system TE—R_f 0.35; system TL—R_f 0.03; system TAE—R_f 0.21; system TAF—R_f 0.22; system TAJ—R_f 0.10; system TAK—R_f 0.00; system TAL—R_f 0.26 (acidified iodoplatinate solution—positive; Dragendorff Spray—positive; Marquis test—violet).

Gas Chromatography System GA—RI 2375, codeine-AC RI 2503, codeine-PPF RI 2430, codeine-trifluoroacetic acid RI 2280, codeine-TMS RI 2520, M (nor-) RI 2388, M (nor-) AC₂ RI 2945, M (morphine) RI 2445; system GB—RI 1511, codeine-AC RI 2645, codeine-TMS RI 2592, M (nor-) RI 2535, M (nor-) TMS₂ RI 2631; system GC—RI 2681, M (morphine) RI 2542; system GF—RI 2860; system GM—RRT 1.519, codeine-ACRI 1.449 (both relative to iprindole).

Gas Chromatography-Mass Spectrometry Column: HP-5 capillary 95% dimethylsiloxane, 5% diphenylsiloxane (25 m × 0.25 mm i.d., 0.25 μm). Temperature programme: 120° for 2 min to 280° at 20°/min for 10 min. Carrier gas: 1 mL/min. SIM acquisition mode (*m/z* 282, 395). Retention time: 13.18 min [Pragst *et al.* 1999].

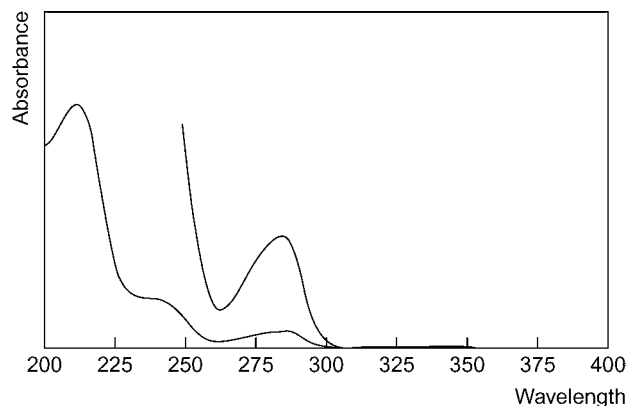
High Performance Liquid Chromatography system HA—*k* 4.8 (tailing peak), M (morphine) RI 3.8 (tailing peak), M (nor-) RI 3.1 (tailing peak); system HC—*k* 1.21, M (morphine) RI 1.30, M (nor-) 3.51; System HS—*k* 1.90, M (morphine) RI 5.16; system HX—RI 266; system HY—RI 237; system HZ—RT 1.9 min; system HAA—RT 5.0 min; system HAM—not detected; system HAX—RT 6.1 min; system HAY—RT 3.4 min.

Column: Basic C₈ (150 × 4.6 mm i.d., 3 μm). Mobile phase: acetonitrile: 5 mmol/L ammonium phosphate dibasic (pH 5.8, 8:92), flow rate 1.0 mL/min. Fluorescence detection, (λ_{ex} = 214 nm, λ_{em} = 345 nm). Retention time: 15 min [Weingarten *et al.* 1995].

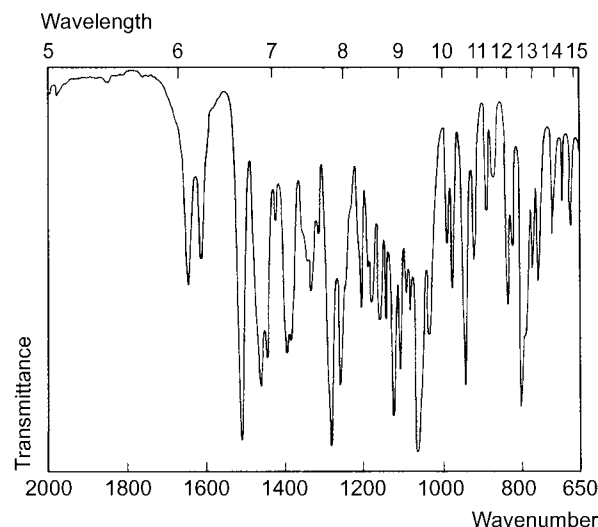
Column: μPorasil (300 × 3.9 mm i.d., 10 μm). Mobile phase: 5 mmol/L sodium acetate buffer (pH 3.75): acetonitrile (25:75), flow rate 1.2 mL/min. ECD. Retention time: 8.24 min (*k*, 3.76) [Liaw *et al.* 1998].

Column: Hypersil BDS-C₁₈ (100 × 46 mm i.d., 3 μm). Mobile phase: 10 mmol/L potassium dihydrogen phosphate (pH 3): acetonitrile (93:7), flow rate 1.0 mL/min. UV detection (λ = 212 nm). Retention time: 6.4 min for codeine, 5.5 min for norcodeine [Pascual, Sanagustin 1999].

Ultraviolet Spectrum Aqueous acid—285 nm (A₁¹ = 55a). No alkaline shift.



Infrared Spectrum Principal peaks at wave numbers 1052, 1268, 1500, 1111, 793, 934 cm⁻¹ (KBr disk).



Mass Spectrum Principal ions at *m/z* (codeine) 299, 42, 162, 124, 229, 59, 300, 69; (morphine) 285, 162, 42, 215, 286, 124, 44, 284; (norcodeine) 285, 81, 215, 148, 286, 164, 110, 115.

Quantification

Blood GC NPD. Limit of detection, 100 μg/L [Lee, Lee 1991]. Column: Borosilicate OV-1 (25 m × 0.36 mm i.d.). Carrier gas: He, 35 cm/s. Temperature: 220°. ECD. Limit of detection, 5 μg/L [Edlund 1981].

GC-MS Temperature programme: 150° for 0.5 min to 320° at 25°/min for 3 min. SIM acquisition mode. Retention time: 4.65 min. Limit of detection, 12.5 μg/L [Goldberger *et al.* 2010]. Column: DB-1 (15 m × 0.25 mm i.d., 0.25 μm). Carrier gas: He, 60 cm/s. Temperature programme: 140° for 1 min to 220° at 50°/min to 305° at 10°/min for 0.5 min. EI ionisation. Limit of quantification, 10 μg/L, limit of detection, 2 μg/L [Meatherall 2005]. Column: Chromabond DB-1 C₁₈ (30 m × 25 mm i.d., 0.25 μm). Carrier gas: He, 30 mL/min. Temperature programme: 100° to 300° at 20°/min. TIC, SIM acquisition mode. Limit of detection, <5 μg/L [Geier *et al.* 1996]. Column: OV1 (12 m × 0.2 mm i.d.). Temperature programme: 150° for 2 min to 220° at 40°/min for 6 min. MSD. Retention time: 8.2 min. Limit of detection, <1 μg/L [Muschhoff, Daldrop 1993]. Column: HP Ultra1 cross-linked methyl silicone (12 m × 0.2 mm i.d., 0.33 μm). Carrier gas: He, 2 mL/min. Temperature programme: 120° for 30 s to 220° at 40°/min to 244° at 4°/min to 300° at 40°/min. EI ionisation at 70 eV. Limit of detection, 0.05 μmol/L [Krogh *et al.* 1993]. See also Gjerde *et al.* [1991].

HPLC Column: Varian Pursuit 3 C₁₈ (100 × 3 mm i.d., 3 μm). Mobile phase: methanol: ammonium formate. Limit of quantification, 0.0005–0.01 mg/kg [Bjork *et al.* 2010]. Column: ODS-3 (150 × 3 mm i.d., 5 μm). Mobile phase: 1 mmol/L ammonium formate (pH 3.0): acetonitrile (95:5 for 5 min to 80:20 in 5 min for 5 min), flow rate 0.4 mL/min. ESI, SIM acquisition mode. Limit of quantification, 1 μg/L [Dienes-Nagy *et al.* 1999]. Column: Superspher RP-18 ODS (125 × 3 mm i.d., 4 μm). Mobile phase: acetonitrile: 50 mmol/L ammonium formate buffer (pH 3.0, 5:95), flow rate 0.6 mL/min for 4 min to 1.1 mL/min in 3 min for 10 min. APCI, SIM acquisition mode. Limit of detection, 2.5 μg/L [Bogusz *et al.* 1997]. Column: Nova-Pak phenyl (150 × 3.9 mm i.d., 5 μm). Mobile phase: acetonitrile: 10 mmol/L sodium dihydrogen orthophosphate (pH 6.6, 10:90), flow rate 1.2 mL/min. UV detection (λ = 210 nm). Retention time: 19.2 min. Limit of detection, 0.06 mg/L [Crump *et al.* 1994].

LC-MS Column: Nova-Pak CN (100 × 3.9 mm i.d., 4 µm). Mobile phase: acetonitrile: 2 mmol/L ammonium formate buffer (pH 3.0, 15:85). ESI, MRM acquisition mode. Limit of quantification, 2 µg/L, limit of detection, 1 µg/L [Coles *et al.* 2007].

Plasma GC Column: DB-17 fused silica capillary (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He. Temperature programme: 130° for 2 min to 300° at 8°/min. NPD. Retention time: 23.9 min. Limit of detection, 2.5 µg/L [Seno *et al.* 1995]. See Blood [Edlund 1981].

GC-MS See Blood [Geier *et al.* 1996]. Column: HP-1 (12 m × 0.25 mm i.d., 0.33 µm). Carrier gas: He, 5 psi. Temperature programme: 100° for 1 min to 250° at 20°/min to 257° at 2°/min. NICI. Limit of detection, <1 pg [Watson *et al.* 1995]. Limit of detection, 5 µg/L [Wang *et al.* 1994]. See Blood [Krogh *et al.* 1993].

HPLC Column: C₁₈ (150 × 2.1 mm i.d., 5 µm). Mobile phase: methanol: acetonitrile: 1% formic acid (70:10:20), flow rate 0.4 mL/min. SIM acquisition mode, *m/z* 300/165, positive ion mode. Limit of quantification, 0.2 µg/L [Liao *et al.* 2009]. Column: XTerra RP8 (250 × 4.6 mm i.d., 5 µm). Mobile phase: acetonitrile: 20 mmol/L phosphate buffer (pH 6.5, 10:90 to 2:98 at 5 min to 20:80 at 7 min to 35:65 at 10 min to 50:50 at 15 min for 5 min to 10:90 at 22 min), flow rate 0.7 mL/min to 0.8 mL/min at 7 min until 20 min to 0.7 mL/min. DAD (λ = 285 nm). Limit of detection, 0.05 mg/L [Fernandez *et al.* 2007]. Column: XTerra RP8 (250 × 4.6 mm i.d., 5 µm). Mobile phase: acetonitrile: phosphate buffer (pH 6.53, 10:90 to 15:85 at 5 min to 20:80 at 7 min to 35:65 at 10 min to 50:50 at 15 min for 5 min to 10:90 at 22 min), flow rate 0.7 mL/min to 0.8 mL/min at 7 min until 20 min to 0.7 mL/min at 22 min. DAD (λ = 285 nm). Limit of detection, 0.032 mg/L [Fernandez *et al.* 2006]. Column: Hypersil BDS-C₁₈ (100 × 4.6 mm i.d., 3 µm). Mobile phase: 10 mmol/L potassium dihydrogen phosphate (pH 3.0): acetonitrile (93:7), flow rate 1 mL/min. UV detection (λ = 212 nm). Retention time: 6.4 min. Limit of quantification, 1.56 µg/L, limit of detection, 0.47 µg/L [Pascual, Sanagustin 1999]. Column: MicroPorasil. Mobile phase: 5 mmol/L sodium acetate buffer (pH 3.75): acetonitrile (25:75), flow rate 1.2 mL/min. Electrochemical detection. Retention time: 8.24 min. Limit of detection, 0.14 µg/L [Liaw *et al.* 1998].

See also Chari *et al.* [1991], Freiermuth, Plasse [1997], He *et al.* [1998], Svensson *et al.* [1995], Vree, Verwey-van Wissen [1992] and Weingarten *et al.* [1995].

LC-MS See Blood [Coles *et al.* 2007].

Serum GC-MS Limit of quantification, 25 µg/L [Maresova *et al.* 2008]. Column: DB5-MS (30 m × 0.25 mm i.d., 0.25 µm). Temperature programme: 80° for 1 min to 200° at 20°/min to 240° at 5°/min to 310° at 30°/min for 6 min. SIM acquisition mode. Limit of quantification, 3.4 µg/L, limit of detection, 3.0 µg/L [Weinmann *et al.* 2000]. Column: DB-5 dimethylpolysiloxane (25 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 100 kPa. Temperature programme: 150° for 1 min to 250° at 35°/min for 4 min to 300° at 30°/min. NICI. Limit of quantification, 0.15 µg/L [Hofmann *et al.* 1999]. See Blood [Musshoff, Daldup 1993].

HPLC Column: Supelcosil LC-Si (250 × 2.1 mm i.d., 5 µm). Mobile phase: water: methanol: acetonitrile: formic acid (59.8:5.2:34.65:0.35), flow rate 230 µL/min. SIM acquisition mode. Limit of quantification, 4 µg/L [Zuccaro *et al.* 1997]. See Blood [Bogusz *et al.* 1997].

LC-MS See Blood [Coles *et al.* 2007].

Urine GC Column: EC-5 5% phenylpolysiloxane, 95% methylpolysiloxane (30 m × 0.32 mm i.d., 0.25 µm). Carrier gas: He, 2.0 mL/min. Temperature programme: 110° for 1 min to 260° at 32°/min. NPD. Limit of detection, 1.5 µg/L [Raikos *et al.* 2009]. Column: HP5MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.5 mL/min. Temperature programme: 200° for 0.5 min to 300° at 50°/min for 1.2 min. Retention time: 3.15 min. Limit of detection, 50 µg/L [Meadway *et al.* 2002]. See Blood [Seno *et al.* 1995]. Column: Chrompack (25 m × 0.32 mm i.d., 0.12 µm). Carrier gas: N₂, 1.3 mL/min. Temperature programme: 120° to 200° at 10°/min to 250° at 2°/min. NSD or FID. Limit of detection, 4 µg/L [Vuduc, Vernay 1990].

GC-MS See Blood [Goldberger *et al.* 2010]. See Serum [Hofmann *et al.* 1999]. Column: DB-1 (15 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 60 cm/min. Temperature programme: 140° for 1 min to 220° at 50°/min to 290° at 10°/min for 0.5 min. Limit of quantification, 25 µg/L, limit of detection, 10 µg/L [Meatherall 1999]. See CSF [Pragst *et al.* 1999]. Column: DB-5MS (15 m × 0.25 mm i.d.). Carrier gas: He, 1.4 mL/min. Temperature programme: 120° for 0.3 min to 235° at 30°/min for 2 min. Limit of detection, 6 µg/L [Paul *et al.* 1999]. See also Broussard *et al.* [1997], Cremese *et al.* [1998], O'Neal, Poklis [1997] and Wang *et al.* [1994].

HPLC Column: Nova-Pak phenyl (150 × 3.9 mm i.d., 5 µm). Mobile phase: acetonitrile: 10 mmol/L sodium dihydrogen phosphate (pH 6.6, 10:90), flow rate 1.2 mL/min. Dual ultraviolet spectrophotometric and electrochemical detection. Retention time: 19.2 min. Limit of detection, 0.05 mg/L [Gerostamoulos *et al.* 1993]. See Plasma [Chari *et al.* 1991; He *et al.* 1998; Svensson *et al.* 1995; Vree, Verwey-van Wissen 1992]. See Blood [Bogusz *et al.* 1997].

LC-MS See Blood [Coles *et al.* 2007].

Bile GC See Blood [Lee, Lee 1991].

HPLC See Blood. Limit of detection, 0.2 mg/L [Crump *et al.* 1994].

CSF GC-MS Column: HP-5 (25 m × 0.25 mm i.d., 0.25 µm). Carrier gas: 1 mL/min. Temperature programme: 120° for 2 min to 280° at 20°/min for 10 min. SIM acquisition mode (*m/z* 282, 395). Limit of quantification, 1 µg/L [Pragst *et al.* 1999].

HPLC See Blood [Bogusz *et al.* 1997]. See Plasma [Chari *et al.* 1991].

Meconium LC-MS See Blood [Coles *et al.* 2007].

Oral Fluid GC-MS Column: DB-5 (25 m × 0.2 mm i.d., 0.33 µm). Carrier gas: 1.5 mL/min. Temperature programme: 150° for 1.0 min to 245° at 20°/min for

8 min to 290° at 50°/min. EI ionisation. Limit of quantification, 2 µg/L, limit of detection, 2 µg/L [Jones *et al.* 2002]. See Plasma [Wang *et al.* 1994].

Vitreous Humour GC-MS See CSF [Pragst *et al.* 1999].

HPLC See Blood [Bogusz *et al.* 1997].

Hair GC-MS See Oral Fluid. Limit of quantification, 0.32 ng/mg, limit of detection, 0.03 ng/mg [Jones *et al.* 2002]. Column: 100% poly(dimethylsiloxane) (30 m × 0.245 mm i.d., 0.25 µm). Carrier gas: He, 35 cm/s. Temperature programme: 80° for 1 min to 120° at 50°/min to 300° at 20°/min for 11 min. SIM acquisition mode. Limit of detection not reported [Brewer *et al.* 2001]. Column: Restek Rtx-5 MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° for 1 min to 260° at 15°/min to 290° at 7°/min for 6 min. Limit of detection, 0.01 ng/mg [Pichini *et al.* 1999]. Column: DB-1 capillary (30 m × 0.32 mm i.d., 1.0 µm). Carrier gas: He, 8 psi. Temperature programme: 120° for 0.1 min to 270° at 20°/min to 310° at 10°/min for 1.4 min. PICI, SIM acquisition mode (*m/z* 372). Limit of quantification, 0.5 mg/g [Hold *et al.* 1998]. Column: DB-5 MS capillary (30 m × 0.25 mm i.d.). Carrier gas: He, 7 psi. Temperature programme: 175° for 1 min to 300° at 15°/min for 0.5 min. Limit of detection, 10 pg [Wilkins *et al.* 1995]. See also Kintz, Mangin [1995] and Wang *et al.* [1994].

HPLC Column: silica-based C₁₈ (150 × 4.6 mm i.d., 3 µm). Mobile phase: 20 µmol/L monobasic sodium phosphate: acetonitrile (90:10 to 50:50). Electrochemical detection. Retention time: 15.53 min. Limit of detection, 18 pg [Achilli *et al.* 1996].

LC-MS Column: Zorbax SB-Phenyl (2.1 × 100 mm i.d., 3.5 µm). Mobile phase: acetonitrile: 25 µmol/L formic acid (5:95). ESI, positive ion mode [Hegstad *et al.* 2008].

Note For a review of the analysis of opioids by GC-MS, see Wasels, Belleville [1994].

Disposition in the Body Codeine is well absorbed after oral administration, with peak plasma levels occurring after ~1 h; it is metabolised in the liver by *O*-demethylation involving P450 isozyme CYP2D6 to form morphine, *N*-demethylation to form norcodeine and conjugation to form glucuronides and sulfates of both unchanged drug and its metabolites. After an oral dose, ~86% is excreted in the urine in 24 h; of the excreted material, 40–70% is free or conjugated codeine, 5–15% is free or conjugated morphine, 10–20% is free or conjugated norcodeine and trace amounts may be free or conjugated normorphine; some of the dose is excreted in the bile and trace amounts are found in the faeces; unchanged drug accounts for 6–8% of the dose in urine in 24 h but this may increase to ~10% when the urinary pH is decreased. After an IM dose, 15–20% is excreted unchanged in acid urine in 24 h. Small amounts of hydrocodone, norhydrocodone, 6α-hydrocodol and 6β-hydrocodol have also been detected in urine.

Therapeutic Concentration

Thirteen healthy adult volunteers with a mean age of 36.5 years (range, 33–44) were administered 150 mg of the sustained-release formulation of codeine (Contin) every 12 h for 5 doses and an immediate-release codeine phosphate dose of 60 mg every 4 h for the first 3 doses followed by 30 mg every 4 h for 12 doses. After a single dose, the peak plasma concentration was 217.8 µg/L (range, 112.1–375.7) for the sustained-release formulation and 138.8 µg/L (range, 49.7–221.3) for the immediate release. These concentrations were observed at 2.3 h (range, 1.0–4.5) and 1.1 h (range, 0.5–2.0), respectively. The mean steady-state concentrations were 263.8 µg/L (range, 172.9–453.0) and 222.9 µg/L (range, 129.8–274.6), respectively, for the sustained-release and immediate-release formulations of codeine, and were attained at 3.2 h (range, 1.5–5.0) and 1.1 h (range, 0.5–1.5), respectively [Band *et al.* 1994].

Eight healthy male and female volunteers, aged 20–50 years, were administered an oral dose of 30 mg codeine phosphate (equivalent to 22 mg codeine). The mean peak plasma concentrations for codeine, codeine-6-phosphate and morphine 3-glucuronide, respectively, were 66.9 µg/L (range, 34.0–118.8), 968.0 µg/L (range, 693.0–1518.0) and 53.5 µg/L (range, 32.2–71.3). These concentrations were observed after 2 h [Vree, Verwey-van Wissen 1992].

In 11 women undergoing hysterectomy and using patient-controlled analgesia as a means of administering codeine, there was a nine-fold variation in the minimum plasma concentrations consistent with pain relief (40–350 µg/L). Two patients did not experience pain relief, one of whom had been classified as a poor metaboliser of dextromethorphan. In the 9 responding patients, the effective dose varied from 4.8–25.3 mg/h [Persson *et al.* 1995].

Toxicity The estimated minimum lethal dose is 800 mg, but codeine is much less toxic than morphine and death directly attributable to codeine is rare; in most fatalities involving codeine, other drugs and/or alcohol are also present. Drug addicts may use doses up to 10 times normal before showing toxic effects, while children may show toxicity with only 1/20th of the dose.

Of 107 drug-related deaths involving codeine, it was considered to be the major poison in six. The mean total concentration of codeine in femoral blood was 4.0 mg/L (range, 2.1–8.0), with the mean concentration of free codeine being 1.3 mg/L (range, 0.4–2.8). In the remaining 101 cases involving a combination of codeine and other drugs (most often paracetamol, diazepam, salicylates or ethanol), the mean total and free codeine blood concentrations were 1.8 mg/L (range, 0.04–26.00) and 0.82 mg/L (range, 0.02–9.00), respectively. In the absence of other contributing factors, total and free codeine concentrations of >2.0 and >0.4 mg/L, respectively, may be sufficient to cause death [Gerostamoulos *et al.* 1996].

In 39 deaths in which codeine was implicated, the postmortem blood concentrations for codeine ranged from <0.1–8.8 mg/L (mean, 1.2) and for morphine from <0.1–0.7 mg/L (mean, 0.16); in 22 of these deaths, the following concentrations ranges (mean) in other tissues and fluids were reported:

	Codeine	Morphine
Bile (mg/L)	0.5–43 (11)	<0.1–119 (33)
Kidney (µg/g)	<0.1–36.3 (7)	<0.1–12.4 (2)
Liver (µg/g)	<0.1–45 (5)	<0.1–6.4 (2)
Urine (mg/L)	3.2–229 (60)	<0.1–69.8 (20)

In all cases, other drugs were also present [Nakamura *et al.* 1976].

In 2 deaths involving the ingestion of large numbers of codeine tablets, postmortem blood concentrations of codeine of 15 and 48 mg/L were reported; urine concentrations were 155 and 370 mg/L; in both cases alcohol had also been ingested and in one case a low concentration of salicylate was also found in the blood [Peat, Sengupta 1977].

Bioavailability ~50%.

Half-life Plasma half-life, 2–4 h.

Volume of Distribution ~2.5–3.5 L/kg.

Clearance Plasma clearance, 10–15 mL/min/kg.

Protein Binding ~7–25%.

Dose For pain relief, 30 to 60 mg every 4 h to a usual maximum of 240 mg daily.

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Codeine N-oxide

Narcotic

$C_{18}H_{21}NO_4 = 315.4$

IUPAC Name Codeine N-oxide

Synonym Genocodeine

Colour Tests Ammonium molybdate test—green—blue (limit of detection, 0.1 µg); sulfuric acid–formaldehyde test—deep purple (limit of detection, 0.1 µg); Vitali's test—yellow/faint orange/purple—chocolate-brown (limit of detection, 0.1 µg).

Thin-layer Chromatography System T1— R_f 0.23 (location reagent acidified iodoplatinate spray, positive reaction).

Quantification

Other HPLC Elixirs. Column: µBondapak C_{18} (300 × 3.9 mm i.d.). Mobile phase: buffer (0.015 mol/L 1-butane sulfonic acid sodium salt, 0.015 mol/L potassium

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Diampromide

Narcotic

$C_{21}H_{28}N_2O = 324.5$

CAS—552-25-0

IUPAC Name *N*-[2-(Methylphenethylamino)propyl]propionanilide

Chemical Properties Extracted by organic solvents from aqueous alkaline solutions.

Colour Test Sulfuric acid–formaldehyde test—orange (limit of detection, 1.0 µg).

Thin-layer Chromatography System T1— R_f 0.42 (location reagent acidified iodoplatinate spray, positive reaction).

Ultraviolet Spectrum Aqueous acid (0.1 N sulfuric acid)—225, 257, 263 nm.

Diamthazole

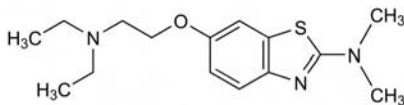
Antifungal

$C_{15}H_{23}N_3OS = 293.4$

CAS—95-27-2

IUPAC Name 6-[2-(Diethylamino)ethoxy]-*N,N*-dimethyl-2-benzothiazolamine

Synonyms Amycazol; dimazole.



Chemical Properties Log *P* (octanol/water), 2.5 (hydrochloride).

Diamthazole Hydrochloride

$C_{15}H_{23}N_3OS \cdot 2HCl = 366.4$

CAS—136-96-9

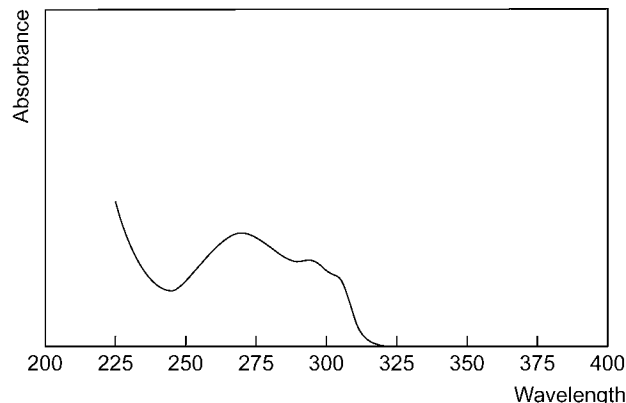
Proprietary Name Asterol

Chemical Properties A white, hygroscopic, crystalline powder. Mp 240° to 243°. Soluble in water, ethanol and methanol; very slightly soluble in chloroform and ether.

Colour Test Liebermann's reagent—green.

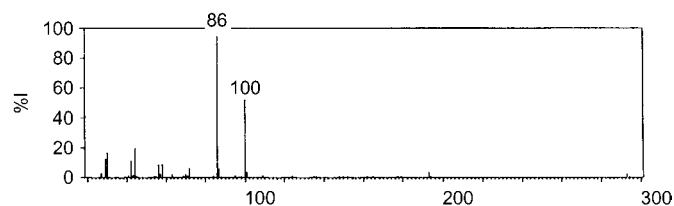
Thin-layer Chromatography System TA— R_f 0.52; system TB— R_f 0.30; system TC— R_f 0.30 (acidified iodoplatinate solution, positive).

Ultraviolet Spectrum Aqueous acid—270 ($A_1^1=370b$), 293 nm.



Infrared Spectrum Principal peaks at wavenumbers 1015, 1041, 667, 1548, 1602, 1190 cm^{-1} (KBr disk).

Mass Spectrum Principal ions at *m/z* 86, 100, 44, 30, 29, 42, 58, 56.



Use Diamthazole hydrochloride has been used in a concentration of 5%.

Diaveridine

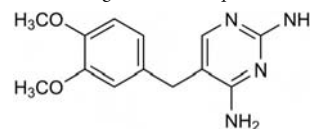
Coccidiostat (Veterinary)

$C_{13}H_{16}N_4O_2 = 260.3$

CAS—5355-16-8

IUPAC Name 5-[(3,4-Dimethoxyphenyl)methyl]-2,4-pyrimidinediamine

Proprietary Name It is an ingredient of *Saquadil*.



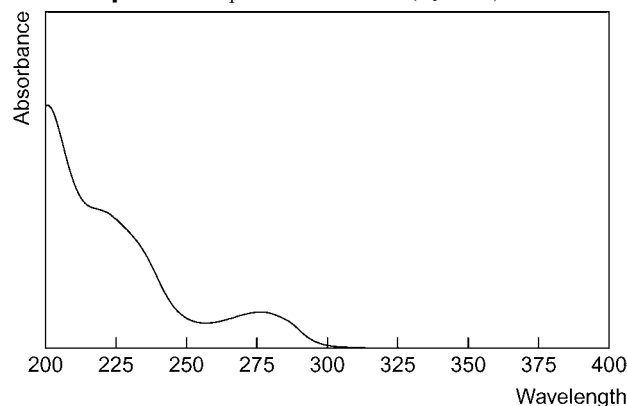
Chemical Properties A white or creamy-white crystalline powder. Mp 233°. Very slightly soluble in water and ethanol; soluble 1 in 600 of chloroform. Log *P* (octanol/water), 1.0.

Colour Tests Mandelin's test—green; Marquis test—grey→violet-brown.

Thin-layer Chromatography System TA— R_f 0.58 (acidified iodoplatinate solution, positive).

High Performance Liquid Chromatography System HAA—retention time 7.0 min.

Ultraviolet Spectrum Aqueous acid—276 nm ($A_1^1=300a$).



Infrared Spectrum Principal peaks at wavenumbers 1630, 1645, 1598, 1250, 1510, 1570 cm^{-1} (KBr disk).

Diazepam

Anxiolytic, Benzodiazepine, Tranquilliser

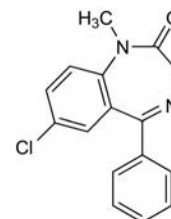
$C_{16}H_{13}ClN_2O = 284.8$

CAS—439-14-5

IUPAC Name 7-Chloro-1,3-dihydro-1-methyl-5-phenyl-3*H*-1,4-benzodiazepin-2-one

Synonyms Diazepamum; LA-III; NSC-77518; Ro-5-2807; Wy-3467.

Proprietary Names *Antenex*; *Anxicalm*; *Apozepam*; *Dialar*; *Diapam*; *Diazemuls*; *Diazep*; *Ducene*; *Faustan*; *Gewacalm*; *Hexalid*; *Lamra*; *Medipam*; *Novazam*; *Novodipam*; *Pro-Pam*; *Psychopax*; *Rimepam*; *Serenack*; *Stesolid*; *Tensium*; *Tranquase*; *Umbrium*; *Valaxona*; *Valclair*; *Valiquid*; *Valium*; *Vivol*.



Chemical Properties A white or yellow crystalline powder. Mp 125° to 126°. Slightly soluble in water; soluble 1 in 25 of ethanol, 1 in 2 of chloroform, and 1 in 39 of ether. pK_{a1} 3.5 [Krogh *et al.* 1997], pK_{a2} 3.3 (20°). Log *P* (octanol/water), 2.8

[Capella-Peiró *et al.* 2002], 3.08 [Mullett, Pawliszyn 2001], (octanol/water pH 7.4), 2.7. Extraction yield (chlorobutane), 0.95 [Demme *et al.* 2005].

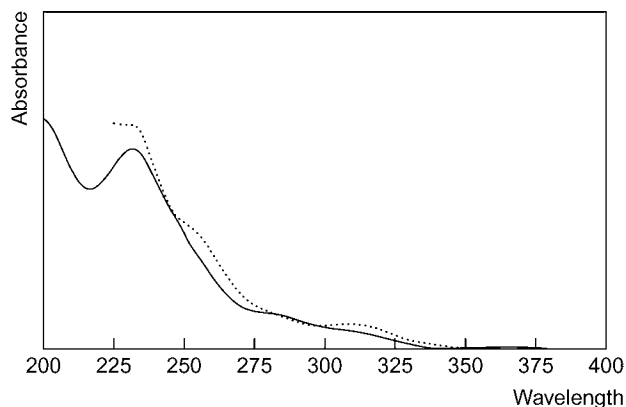
Colour Test Formaldehyde-sulfuric acid—orange.

Thin-layer Chromatography System TA— R_f 0.75; system TB— R_f 0.27; system TC— R_f 0.73; system TD— R_f 0.58; system TE— R_f 0.76; system TF— R_f 0.49; system TL— R_f 0.59; system TAD— R_f 0.72; system TAE— R_f 0.82; system TAF— R_f 0.85; system TAJ— R_f 0.67; system TAK— R_f 0.48; system TAL— R_f 0.96 (acidified iodoplatinate solution—positive; Dragendorff spray—positive; FPN reagent—yellow; Marquis test—yellow).

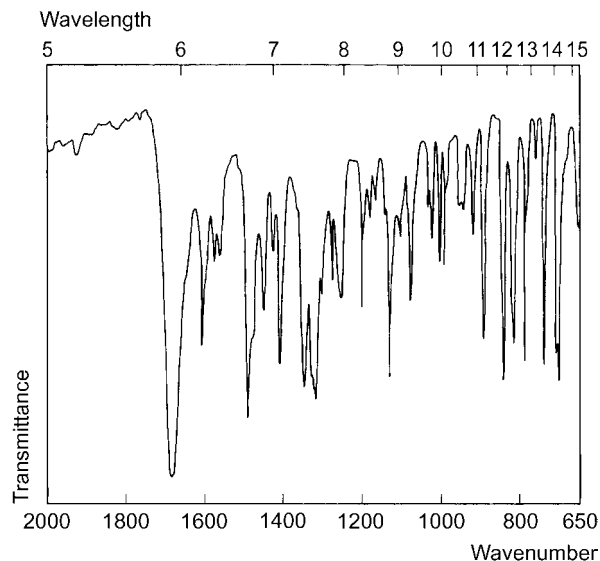
Gas Chromatography System GA—diazepam RI 2428, M (nordazepam [desmethyl-diazepam or nordiazepam]) RI 2490, M (oxazepam) RI 2325, M (temazepam) RI 2595; system GB—diazepam RI 2556, M (nordazepam) RI 2625, M (oxazepam) RI 2438, M (temazepam) RI 2727; system GF—RI 3045; system GG—diazepam RI 2940, M (nordazepam) RI 3041, M (oxazepam) RI 2803, M (temazepam) RI 3125. (Nordazepam, oxazepam and temazepam are metabolites of diazepam).

High Performance Liquid Chromatography System HA—diazepam k 0.1, M (nordazepam) k 0.2; system HI—diazepam k 9.47, M (nordazepam) k 8.00, M (oxazepam) k 4.62, M (temazepam) k 5.68; system HJ—diazepam k 2.29; system HK—diazepam k 2.49, M (nordazepam) k 1.99, M (oxazepam) k 0.73, M (temazepam) k 0.60; system HX—RI 528; system HY—RI 429; system HZ—RT 8.4 min; system HAA—RT 20.3 min; system HAF—RT 29.8 min; system HAL—RT 13.2 min; system HAM—diazepam RT 10.3 min, M (oxazepam) RT 3.4 and M (oxazepam) RT 4.4 min; system HAX—diazepam RT 7.7 min, M (nordazepam) RT 6.7 min, M (oxazepam) RT 6.0 min, M (temazepam) RT 8.9 min; system HAY—diazepam RT 8.8 min, M (nordazepam) RT 6.8 min, M (oxazepam) RT 4.5 min, M (temazepam) RT 6.7 min; system HAZ— k 2.25; system HBH—diazepam k 10.4, M (nordazepam) k 8.97, M (oxazepam) k 5.42, M (temazepam) k 6.80; system HBI—diazepam k 2.29, M (nordazepam) k 1.89, M (oxazepam) k 1.25, M (temazepam) k 1.49.

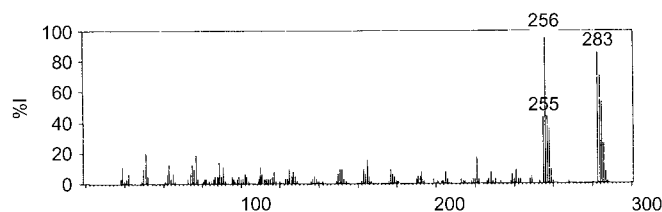
Ultraviolet Spectrum Aqueous acid—242 ($A_1^1=1020a$), 284, 366 nm.



Infrared Spectrum Principal peaks at wavenumbers 1681, 1313, 705, 840, 1125, 740 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at m/z 256, 283, 284, 285, 257, 255, 258, 286 (diazepam); 242, 269, 270, 241, 243, 271, 244, 272 (nordazepam); 257, 77, 268, 239, 205, 267, 233, 259 (oxazepam); 271, 273, 300, 272, 256, 77, 255, 257 (temazepam).



Quantification

Blood GC Columns: DB-1 and DB-1701 (15 m $\times$ 0.32 mm i.d., 0.25 μm). Carrier gas: He, 1 mL/min. Temperature programme: 120° for 1 min to 28° at 30°/min to 230° at 2°/min to 300° at 30°/min for 1 min or 120° for 1 min to 230° at 40°/min to 280° at 8°/min for 9 min. ECD. Limit of detection, 0.084 or 0.096 $\mu\text{mol/L}$ [Gjerde *et al.* 1992]. Column: SE-54 5% phenyl methyl silicone (25 m $\times$ 0.31 mm i.d., 0.17 μm). Carrier gas: He, 2–3 mL/min. Temperature programme: 200° for 1 min to 290° at 10°/min. NPD or ECD. Retention time: 5.57 min. Limit of detection not reported [Lillsunde, Seppälä 1990].

GC-MS Column: 100% methylsiloxane or 5% phenylsiloxane-95% methylsiloxane. Carrier gas: 1.0 mL/min. Temperature programme: 140° for 0.5 min to 320° at 25°/min. SIM acquisition mode. Limit of detection, 12.5 $\mu\text{g/L}$ [Goldberger *et al.* 2010]. Column: HP-5 MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1 mL/min. Temperature programme: 120° to 295° at 10°/min for 5 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of quantification and limit of detection, 117 and 38.6 $\mu\text{g/L}$, respectively [Papoutsis *et al.* 2010]. Column: 5% diphenyldimethylsiloxane DB-5MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 2 mL/min. Temperature programme: 150° for 0.6 min to 230° at 40°/min to 310° at 10°/min. EI ionisation, SIM acquisition mode. Limit of quantification, 25 $\mu\text{g/L}$, limit of detection, 25 $\mu\text{g/L}$ [Tiscione *et al.* 2008]. Column: DB-5 cross-linked 5% phenylmethylsiloxane (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 50° for 2 min to 180° at 30°/min to 280° at 5°/min for 19 min. Full scan mode. Retention time: 21.1 min. Limit of detection, 0.05 mg/L [Paterson *et al.* 2004]. Column: DB5-MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° to 230° at 8°/min to 313° at 4°/min. EI ionisation at 70 eV or CI in positive ion mode. Limit of detection, 10 $\mu\text{g/L}$ [Piray *et al.* 2002]. Column: HP-5 MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1 mL/min. Temperature programme: 120° to 295° at 10°/min for 5 min. EI ionisation [Papoutsis *et al.* 2010]. Column: BPX5 capillary (15 m $\times$ 0.32 mm i.d., 0.25 μm). SIM acquisition mode. Limit of detection, 0.2–20 $\mu\text{g/L}$ [Inoue *et al.* 2000].

HPLC Column: Chromolith Performance (RP-18e) (100 $\times$ 4.6 mm). Mobile phase: 35 mmol/L phosphate buffer (pH 2.1): acetonitrile (70:30), flow rate 2 mL/min. DAD ($\lambda=220$ nm). Limit of quantification, 30 $\mu\text{g/L}$ [Bugey, Staub 2004]. Column: Lichrospher select B (125 $\times$ 3 mm). Mobile phase: acetonitrile: 20 mmol/L potassium dihydrogen phosphate (pH 2.1, 30:70 to 35:65 at 30 min), flow rate 0.5 mL/min to 0.3 mL/min at 30 min. DAD ($\lambda=254$ nm). Limit of quantification, 30 $\mu\text{g/L}$, limit of detection, 10 $\mu\text{g/L}$ [El Mahjoub, Staub 2001a]. Column: Lichrospher Select B C_{18} (125 $\times$ 3 mm i.d., 5 μm). Mobile phase: 20 mmol/L phosphate buffer (pH 2.1): acetonitrile (65:35), flow rate 0.3 mL/min. UV detection ($\lambda=220$ nm). Limit of quantification, 5 $\mu\text{g/L}$, limit of detection, 2 $\mu\text{g/L}$ [El Mahjoub, Staub 2000a].

LC-MS Column: Acquity BEH Phenyl (100 $\times$ 2.1 mm i.d., 1.7 μm). Mobile phase: 0.1% formic acid in water: 0.1% formic acid in acetonitrile (80:20 for 0.25 min to 65:35 over 2.25 min for 2.5 min to 20:80 over 1 min to 80:20 over 0.01 min for 1.4 min), flow rate 0.5 mL/min. ESI, MRM acquisition mode. Limit of detection not reported [Gunn *et al.* 2010]. Column: Restek Allure C_{18} (150 $\times$ 3.2 mm i.d., 5 μm). Mobile phase: 5 mmol/L ammonium acetate: acetonitrile: methanol (90:5:5 to 50:25:25 at 7 min to 10:45:45 at 27 min for 3 min to 95:5:5 at 31 min), flow rate 0.45 mL/min. APCI. Limit of quantification, 2 $\mu\text{g/L}$ [Dussy *et al.* 2006]. Column: XTerra MS C_{18} (150 $\times$ 2.1 mm i.d., 3.5 μm). Mobile phase: 0.006 mol/L formic acid (pH 3.0): methanol (70:30 to 60:40 at 5 min to 50:50 in 25 min to 40:60 at 30 min for 5 min to 70:30 at 36 min for 9 min), flow rate 0.2 mL/min. Limit of quantification, 13.340 $\mu\text{g/L}$, limit of detection, 4.0 $\mu\text{g/L}$ [Smink *et al.* 2004].

Plasma GC NPD. Limit of detection, 20–115 nmol/L. [Ugland *et al.* 2000]. Column: CPSIL 8 CB (25 $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1 mL/min. Temperature programme: 150° to 230° at 40°/min for 2 min to 250° at 5°/min for 1 min to 300° at 15°/min for 3 min. NPD. Limit of detection, 0.01 $\mu\text{mol/L}$ [Reubsat *et al.* 1998]. Column: DB-1 methyl silicone (30 m $\times$ 0.2 mm i.d., 0.25 μm). Carrier gas: He, 1 mL/min. Temperature: 300°. NPD. Limit of detection, 100 nmol/L [Krogh *et al.* 1997]. Column: Ultra 2 5% phenyl methyl silicone (25 m $\times$ 0.32 mm i.d., 0.52 μm). Carrier gas: He, 2.1 mL/min. Temperature programme: 210° to 285° at 2.5°/min. NPD. Retention time: 0.722. Limit of quantification, 2 $\mu\text{g/L}$ [Gaillard *et al.* 1993].

GC-MS Column: HP-1 capillary (12 m $\times$ 0.2 mm i.d., 330 nm). Carrier gas: He, 0.6 mL/min. Temperature programme: 100° for 2 min to 310° at 40°/min for 2.5 min. EI ionisation, SIM acquisition mode. Retention time: 6.68 min. Limit of quantification, 0.05 mg/L [Peters *et al.* 2005].

HPLC Column: LC-18 DB (250 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: 5 $\mu\text{mol/L}$ potassium dihydrogen phosphate buffer solution (pH 6.0): methanol: diethyl ether (55:40:5), flow rate 0.8 mL/min. UV detection ($\lambda=245$ nm). Limit of quantification, 30 and 50 $\mu\text{g/L}$ [Borges *et al.* 2009]. Column: C_{18} (250 $\times$ 4.6 mm i.d.). Mobile phase: methanol: water (65:35), flow rate 1.0 mL/min. UV detection ($\lambda=230$ nm) [Liang *et al.* 2009]. Column: Chromolith Performance RP-18e (100 $\times$ 4.6 mm i.d.).

Mobile phase: 10 µmol/L phosphate buffer (pH 2.5) methanol:acetonitrile (63:10:27), flow rate 2 mL/min. Limit of quantification, 2 µg/L [Rouini *et al.* 2008]. Column: Kromasil C₈ (250 × 5 mm i.d., 5 µm). Mobile phase: methanol:acetonitrile:0.05 mol/L ammonium acetate. Limit of quantification, 0.07–1.57 mg/L, limit of detection, 0.02–0.47 mg/L [Uddin *et al.* 2008].

See also Atta-Politou *et al.* [1999], Azzam *et al.* [1998], El Mahjoub, Staub [2000b], Kamali [1993], Lacroix *et al.* [1993], Muchoh *et al.* [2001], Mullett, Pawliszyn [2001], Samanidou *et al.* [2007], Iwase *et al.* [1994].

LC-MS Column: XTerra MS C₁₈ (150 × 2.1 mm i.d., 3.5 µm). Mobile phase: acetonitrile: water: 100 mmol/L ammonium formate (pH 3.0, 55:40:5), flow rate 0.125 mL/min. ESI, MRM acquisition mode. Retention time: 7.75 min. Limit of quantification, 20 µg/L, limit of detection, 10 µg/L [Marin *et al.* 2008; Marin, McMillin 2010]. Column: XTerraMS C₈ (100 × 2.1 mm, i.d., 3.5 µm). Mobile phase: 2 mmol/L ammonium formate buffer (pH 3): acetonitrile (99:1 to 98:2 from 2 to 3 min to 5:95 at 3.1 min until 7.5 min to 99:1 at 8 min for 3 min), flow rate 0.2 mL/min. ESI, positive ion mode, MRM acquisition mode. Retention time: 7 to 7.5 min. Limit of quantification, 1 µg/L, limit of detection, 0.1 µg/L [Abbata *et al.* 2008]. Column: Merck LiChroCART (125 × 2 mm i.d.). Mobile phase: 5 mmol/L ammonium formate: acetonitrile (60:40 for 5.5 min to 10:90 in 8 min to 60:40 in 9.5 min for 10 min). APCI, SIM acquisition mode. Limit of quantification, 0.1 mg/L, limit of detection, 0.01 mg/L [Kratzsch *et al.* 2004].

Serum GC-MS Column: HP-5 MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.0 mL/min. Temperature programme: 60° for 1 min to 280° at 10°/min for 20 min. EI ionisation at 70 eV. Retention time: 23.1 min. Limit of quantification, 0.025 mg/L, limit of detection, <0.025 mg/L [Maresova *et al.* 2008]. Column: SPB-1 (30 m × 0.75 mm i.d., 1.0 µm). Carrier gas: He, 0.2 bar. Temperature programme: 230° for 2 min to 280° at 7°/min for 2 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of quantification, 3.0 ng/L, limit of detection, 2.0 ng/L [Duthel *et al.* 1992].

HPLC See Blood [Dussy *et al.* 2006]. Column: Symmetry Shield RP8 (250 × 4.6 mm i.d., 5 µm). Mobile phase: acetonitrile:0.1 mol/L potassium dihydrogen phosphate (40:60), flow rate 0.9 mL/min. UV detection (λ = 230 nm). Limit of detection, 1.0 µg/L [He *et al.* 2005]. Column: Kromasil C₁₈ (150 × 4.6 mm i.d., 5 µm). Mobile phase: butanol:0.06 mol/L SDS (pH 7, 5:95), flow rate 1.0 mL/min. UV detection (λ = 230 nm). Limit of detection, 10 µg/L [Capella-Peiró *et al.* 2002]. Column: Supelcosil C₁₈ (50 × 4.6 mm i.d., 5 µm). Mobile phase: water: methanol (52:48), flow rate 1.0 mL/min. UV detection (λ = 230 nm). Limit of detection, 46 µg/L [Mullett, Pawliszyn 2002]. Column: Supelcosil C₁₈ (50 × 4.6 mm i.d., 5 µm). Mobile phase: water: methanol (54:46), flow rate 1.0 mL/min. UV detection (λ = 230 nm). Limit of quantification, 150.0 µg/L, limit of detection, 45.0 µg/L [Mullett, Pawliszyn 2001]. See Blood [El Mahjoub, Staub 2000a]. Column: µBondapak C₁₈ (300 × 3.9 mm i.d., 10 µm). Mobile phase: flow rate 1.5 mL/min. DAD (λ = 254 nm). Limit of quantification, 127.6 µg/L, limit of detection, 35.7 µg/L [Ahrens *et al.* 2000]. Column: TSK gel Super-ODS (100 × 4.6 mm i.d., 2 µm) or Hypersil ODS-C₁₈ (100 × 4.6 mm i.d., 5 µm). Mobile phase: methanol:5 µmol/L sodium dihydrogen phosphate (pH 6, 45:55), flow rate 0.65 mL/min. UV detection (λ = 254 nm) [Tanaka *et al.* 1996]. Column: Supelcosil C₈ DB 5 (150 × 4.6 mm i.d., 5 µm). Mobile phase: acetonitrile: water:0.5 mol/L monobasic potassium phosphate (370:600:30), flow rate 1.65 mL/min. UV detection (λ = 219 nm). Limit of quantification, 1 µg/L [Welk 1996]. Column: Chromspher C₈ (100 × 3.0 mm i.d., 5 µm). Mobile phase: methanol: water (20:80 for 2 min to 30:70 at 2.2 min until 4 min to 40:60 at 4.4 min until 4.5 min to 43:57 at 5 min to 45:55 at 6 min to 52:48 at 7 min to 58:42 at 9.5 min to 75:25 at 10.5 min for 5 min to 20:80 at 15.3 min until 19 min), flow rate 0.7 mL/min. DAD (λ = 450 nm). Limit of quantification, 30 µg/L, limit of detection, 10 µg/L [Lambert *et al.* 1995]. See also Goldnik *et al.* [1993].

LC-MS See Plasma [Marin *et al.* 2008; Marin, McMillin 2010]. Unison UK-C₁₈ RP ODS (150 × 2 mm i.d., 3 µm). Mobile phase: 10 mmol/L aqueous ammonium formate containing 0.1% formic acid: methanol containing 0.1% formic acid (70:30 over 20 min to 20:80 over 5 min), flow rate 0.25 mL/min. MRM acquisition mode. Limit of quantification, 6.8 µg/L, limit of detection, 2.0 µg/L [Nakamura *et al.* 2009]. Column: LiChrospher 60-RP select B (100 × 2.0 mm i.d., 5 µm). Mobile phase: methanol: water: acetonitrile (1:1:1), flow rate 100 µL/min. APCI, SRM acquisition mode. Limit of quantification, 2 µg/L [Kleinschnitz *et al.* 1996].

Urine GC See Plasma [Ugland *et al.* 2000]. Column: DB-17 (30 m × 0.32 mm i.d., 0.25 µm). Carrier gas: He, 80 kPa. Temperature programme: 150° for 1 min to 230° at 10°/min for 5 min to 300° at 10°/min for 9 min. ECD. Limit of detection, 160 µg/L [Guan *et al.* 1999]. See Plasma. Limit of detection, 0.01–0.45 µmol/L [Reubsat *et al.* 1998]. Column: DB-17 (15 m). Carrier gas: N₂. Temperature: 225°. ECD. Retention time: 2.07 min. Limit of detection, 0.13 ng [Beischlag, Inaba 1992].

HPLC See Plasma [Uddin *et al.* 2008]. Column: Inertsil C₈ (250 × 4 mm i.d., 5 µm). Mobile phase: ammonium acetate:0.05 mol/L methanol:acetonitrile (33:57:10). UV detection (λ = 240 nm). Limit of detection, 2.6–12.6 ng [Samanidou *et al.* 2007]. Column: Hypersil C₁₈ (100 × 4.6 mm i.d., 3 µm). Mobile phase: 0.04 mol/L phosphate (pH 4)-0.4% octylamine:acetonitrile (73:27, 70:30, 65:35 or 60:40), flow rate 1 mL/min. DAD (λ = 240 nm). Retention time: 17.9 min. Limit of detection not reported [Segura *et al.* 2001]. Hypersil BDS RP₁₈ (250 × 4.0 mm i.d., 5.0 µm). Mobile phase: methanol:acetonitrile:0.05 mol/L potassium dihydrogen phosphate (pH 3.5, 50:10:40), flow rate 1.2 mL/min. UV detection (λ = 232 nm). Retention time: 11.6 min. Limit of quantification, 10 µg/L [Azzam *et al.* 1998]. See Serum [Kleinschnitz *et al.* 1996]. Column: LiChrospher 100 RP-18(e) (250 × 4 mm i.d., 5 µm). Mobile phase: water:

methanol: triethylamine (pH 5.5, 70:30:0.1), flow rate 0.7 mL/min. UV detection (λ = 240 nm). Limit of detection, 2 µg/L [Chiba *et al.* 1995].

GC-MS See Blood [Goldberger *et al.* 2010].

LC-MS Column: Hypurity C₈ (150 × 3 mm i.d.). Mobile phase: 4 mmol/L ammonium acetate (pH 6.8) in methanol-water (5:95):1% propan-2-ol, 0.05% formic acid in methanol (100:0 for 1 min to 0:100 at 3 min for 1.5 min to 100:0 over 0.1 min for 1.4 min). TIS, MRM acquisition mode. Limit of quantification, 5 µg/L, limit of detection, 2.5 µg/L [Glover, Allen 2010]. See Plasma [Marin *et al.* 2008; Marin, McMillin 2010]. Column: Shodex MSpak GF-310 4B (50 × 4.6 mm i.d., 6 µm). Mobile phase: 10 mmol/L ammonium acetate:acetonitrile (100:0 for 3 min to 0:100 from 3.01 to 6 min to 100:0 from 6.01 to 10 min for 5 min), flow rate 0.9 mL/min for 3 min to 0.3 mL/min until 10 min to 0.9 mL/min. ESI, positive ion mode. Limit of quantification, 0.5 µg/L, limit of detection, 0.1 µg/L [Umezawa *et al.* 2008]. Column: Symmetry C₁₈ (100 × 2.1 mm i.d., 3.5 µm). Mobile phase: acetonitrile:5 mmol/L ammonium acetate (pH 5, 10:90 to 30:70 at 5 min to 80:20 at 9 min to 10:90 at 9.1 min), flow rate 0.3 mL/min. ESI, MRM acquisition mode. Limit of quantification, 0.01 µmol/L [Hegstad *et al.* 2006]. See Serum [Kleinschnitz *et al.* 1996].

CE-MS Running buffer: methanol: water (50:50) containing 0.1% formic acid, flow rate 0.5 mL/min. ESI, TOF. Limit of quantification, 12.5 µg/L, limit of detection, 1.1 µg/L [Blas, McCord 2008].

Meconium LC-MS See Plasma [Marin *et al.* 2008; Marin, McMillin 2010].

Oral Fluid HPLC See Plasma [Uddin *et al.* 2008].

Hair GC-MS Column: HP-Ultra 2 capillary (12 m × 0.2 mm i.d., 0.33 µm). Carrier gas: He, 2 mL/min. Temperature programme: 70° for 2 min to 220° at 25°/min to 255° at 5°/min to 300° for 7 min. EI ionisation at 70 eV. Retention time: 13.1 min. Limit of detection, 0.01 ng/mg. [Yegles *et al.* 1997].

HPLC Column: Lichrospher select-B (125 × 3 mm i.d., 5 µm). Mobile phase: acetonitrile:20 mmol/L potassium dihydrogen phosphate (pH 2.1, 30:70 to 35:65 in 30 min), flow rate 0.5 mL/min to 0.3 mL/min at 30 min. UV detection (λ = 254 nm). Limit of quantification, 0.3–0.45 ng/mg, limit of detection, 0.2 ng/mg [El Mahjoub, Staub 2001b].

LC-MS Mobile phase: 3 mmol/L ammonium formate and 0.001% formic acid in water:acetonitrile (65:35 to 20:80 after 13 min to 10:90 at 13.5 min until 16.5 min to 65:35 until 20 min), flow rate 0.3 mL/min. ESI, positive ion mode. Limit of quantification, 0.22 ng/30 mg, limit of detection, 0.13 ng/30 mg [Miller *et al.* 2006].

Disposition in the Body Diazepam is rapidly and completely absorbed after oral administration, with peak plasma level occurring within ~30–90 min. The main metabolic reactions are *N*-demethylation, 3-hydroxylation and glucuronic acid conjugation. The major active metabolite is desmethyldiazepam (nordazepam), which accumulates during chronic dosing; other metabolites include oxazepam and temazepam, both of which are active. The formation of nordazepam and oxazepam is catalysed by the P450 isozymes CYP2C19 and CYP3A. Only small traces of unchanged diazepam are excreted in the urine and the relative amounts of metabolites are variable and appear to be dose dependent. Approximately 70% of a dose is excreted in the urine, mainly as oxazepam glucuronide and conjugated nordazepam, together with smaller amounts of conjugated temazepam. Approximately 10% of the dose may be eliminated in the faeces. Diazepam and its metabolites cross the blood–brain barrier and the placenta; they are also found in breast milk.

Diazepam is a metabolite of ketazolam and medazepam.

Therapeutic Concentration In plasma, usually in the range 0.1–2.5 mg/L. After discontinuation of chronic therapy, concentrations of nordazepam may be substantially higher than diazepam and both unchanged drug and metabolite are still detectable 7 days after cessation of dosing.

Following a single oral dose of 10 mg to 4 subjects, peak blood concentrations of 0.14–0.19 mg/L (mean, 0.15) were attained in 1–1.5 h; average peak concentrations of 0.03 mg/L of nordazepam were attained after 24 h [Kaplan *et al.* 1973].

After chronic daily oral dosing of 5 mg twice daily to 15 subjects, steady-state plasma concentrations were: diazepam 0.09–0.37 mg/L (mean, 0.23), nordazepam 0.13–0.46 mg/L (mean, 0.29), oxazepam 0.01–0.03 mg/L (mean, 0.02), temazepam 0.01–0.05 mg/L (mean, 0.03) [Zingales 1973].

After intranasal administration of diazepam to nine healthy subjects (20 mg/mL solution; 100 µL into one nostril ≡ 2 mg dose), a mean serum plasma concentration of 0.039 mg/L was attained in 18 min; this equated to 32.9% of the concentration that was attained 10 min after the same dose given IV to the same subjects [Gizurason *et al.* 1999].

Toxicity Toxic effects may be produced by blood concentrations >1.5 mg/L; fatalities caused by diazepam alone are rare but may occur at blood concentrations >5 mg/L.

In a review of 914 drug-related deaths in which diazepam was involved, it was found to be the sole cause of death in only 2 cases; postmortem concentrations of diazepam in these 2 cases were blood 5 and 19 mg/L, liver 13 µg/g in the first case [Finkle *et al.* 1979].

In a fatality from diazepam and alcohol ingestion, postmortem tissue diazepam concentrations were: blood 1.3 mg/L, bile 4.5 mg/L, brain 2.4 µg/g, kidney 11.7 µg/g, liver 11.4 µg/g, urine 6.6 mg/L [Simon 1976].

Half-life Plasma half-life, diazepam 20–100 h (mean, 48); nordazepam, ~40–100 h but there is considerable intersubject variation (see Nordazepam monograph). The plasma half-life appears to be increased in elderly subjects and neonates, and in subjects with liver disease; sex differences have also been suggested.

Volume of Distribution Diazepam and nordazepam, 0.5–2.5 L/kg; increased in elderly subjects.

Clearance Plasma clearance, diazepam ~0.3–0.5 mL/min/kg, nordazepam ~0.1–0.3 mL/min/kg.

Distribution in Blood Plasma : whole blood ratio, diazepam 1.8, nordazepam 1.7.

Protein Binding Diazepam 98–99%, nordazepam ~97%.

Note For reviews of the clinical pharmacokinetics of diazepam, see Friedman *et al.* [1992] and Mandelli *et al.* [1978].

Dose Usually 5 to 30 mg daily.

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Diazoxide

Antihypertensive, Hyperglycaemic

$C_6H_7ClN_2O_2S = 230.7$

CAS—364-98-7

IUPAC Name 7-Chloro-3-methyl-2H-1,2,4-benzothiadiazine 1,1-dioxide

Synonym SRG-95213

Proprietary Names Eudemine; Hyperstat; Hypertonalum; Proglidem; Proglycem.

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γ -Hydroxybutyrate

Anaesthetic

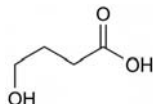
$C_4H_8O_3 = 104.1$

CAS—591-81-1

IUPAC Name 4-Hydroxybutanoic

Synonyms BRN-1720582; gamma hydroxybutyric acid; GHB; 4-hydroxybutyric acid.

Street Names Georgia home boy; grievous bodily harm; liquid ecstasy; scoop; fantasy.



Chemical Properties Crystals. It is water soluble and produces solutions that are mildly saline. Extraction yield (chlorobutane), 0 [Demme *et al.* 2005].

Sodium Gamma Hydroxybutyrate

Synonyms NSC-84223; sodium gamma hydroxybutyric acid; sodium oxybate; sodium oxybutyrate; Wy3478.

Proprietary Names Alcover; Gamma-OH; Somatomax PM; Somsanit; Xyrem.

Sodium-4-hydroxybutyrate

$C_4H_8NaO_3 = 127.1$

CAS—502-85-2

Chemical Properties A colourless, odourless powder, capsules or liquid.

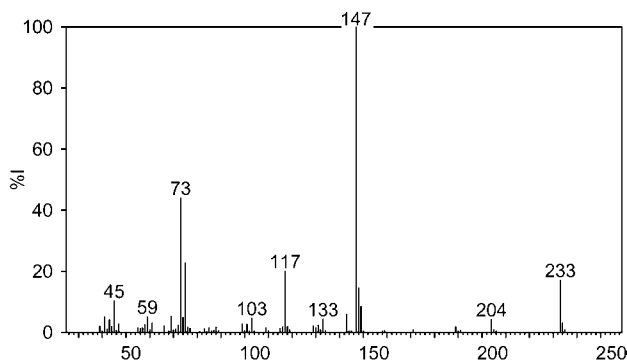
Colour Test Specific tests using mixtures of several common reagents are given in the Colour Tests chapter.

Gas Chromatography System GAO— γ -hydroxybutyrate (GHB)-TMS₂ RT 5.6 min, γ -butyrolactone (GBL) RT 4.0 min.

High Performance Liquid Chromatography System HBG—GHB-TMS₂ RT 3.5 min, GBL RT 4.0 min.

Column: μ Bondapak C₁₈ (300 $\times$ 3.9 mm i.d., 10 μ m). Mobile phase: 10 mmol/L potassium dihydrogen phosphate buffer (pH 3.0): methanol (70:30), flow rate 1.0 mL/min. Retention time: GHB 3.45 min, GBL 3.96 min [Mesmer, Satzger 1998].

Mass Spectrum Principal ions at m/z 147, 73, 75, 233, 117, 148, 77, 59 (GHB-TMS₂).



Quantification

Blood GC Column: DB-624 (30 m $\times$ 0.25 mm i.d., 1.4 μ m). Carrier gas: N₂, 34 cm/s. Temperature programme: 50° for 3 min to 150° at 20°/min for 7 min. FID. Limit of detection, 0.5 mg/L [LeBeau *et al.* 2000].

GC-MS See GC. EI ionisation, full scan mode. Retention time: 10.9 min. Limit of detection, 0.5 mg/L [LeBeau *et al.* 2000].

Plasma GC-MS Column: ULTRA-1 (12 m $\times$ 0.2 mm i.d., 0.33 μ m). Carrier gas: He, 0.2 kg/cm². Temperature programme: 50° for 0.6 min to 275° at 15°/min. EI

ionisation, SIM acquisition mode. Limit of detection, 0.2 mg/L [Ferrara *et al.* 1993]. Column: HP ULTRA 1 bonded phase (12 m $\times$ 0.2 mm i.d., 0.3 μ m). EI ionisation, SIM acquisition mode. Limit of detection, 1 mg/L [Ferrara *et al.* 1992].

Urine GC See Blood [LeBeau *et al.* 2000].

GC-MS Column: HP-1 (25 m $\times$ 0.32 mm i.d., 0.5 μ m). Carrier gas: He, 1.0 mL/min. Temperature programme: 50° for 1 min to 300° at 10°/min. Limit of detection, 2 mg/L [Kavanagh *et al.* 2001]. Column: HP-1 100% polydimethylsiloxane (12 m $\times$ 0.2 mm i.d., 0.33 μ m). Temperature programme: 60° for 2 min to 260° at 30°/min for 1 min. Carrier gas: He, 0.5 mL/min. I.S.: GHB-d₆. SIM acquisition mode, m/z 233, 234, 235 for GHB and 239, 240, 241 for IS. Retention time: GHB 4.43 min, IS 4.41 min. Limit of detection, 2 mg/L [Elián 2000]. See Blood [LeBeau *et al.* 2000]. Column: HP-1MS cross-linked methylsiloxane (12 m $\times$ 0.25 mm i.d., 0.25 μ m). Temperature programme: 65° for 0.5 min to 105° at 15°/min to 300° at 25°/min. Carrier gas: He, 1.6 mL/min. SIM acquisition mode, m/z as above. I.S.: GHB-d₆. Retention time: GHB 3.7 min. Limit of detection, 5 mg/L [McCusker *et al.* 1999]. See Plasma. Limit of detection, 0.1 mg/L [Ferrara *et al.* 1993]. See Plasma. Limit of detection, 0.2 mg/L [Ferrara *et al.* 1992].

Hair LC-MS Limit of quantitation 0.4 ng/mL, limit of detection, 0.2 ng/mL [Stout *et al.* 2010].

Disposition in the Body GHB is readily absorbed after oral administration and rapidly metabolised in the liver by oxidative enzymes. Conversion to GBL can occur and elimination is rapid, via the kidneys, with urine recovery virtually complete within 8 h of administration. Only negligible amounts of the parent drug are recovered unchanged in urine (<5%); the drug is not detectable after 12 h. GHB crosses the blood-brain barrier and the placental barrier. Endogenous concentrations of GHB are thought to be up to 10 mg/L in urine and 4 mg/L in antemortem plasma of individuals not suffering from the genetic disorder gamma hydroxybutyric aciduria. Endogenous concentrations appear to be increased postmortem, particularly in blood.

Therapeutic Concentration

Eight healthy non-smoker males, aged between 22 and 26 years, were administered with 12.5, 25 and 50 mg/kg single doses of GHB after an overnight fast. Peak plasma concentrations reached 23 $\pm$ 9, 23 $\pm$ 11 and 20 $\pm$ 7 mg/L, respectively, achieved within 25 $\pm$ 20, 30 $\pm$ 20, 45 $\pm$ 30 min, respectively [Palatini *et al.* 1993].

Toxicity Doses greater than 50 mg/kg can lead to a decrease in cardiac output, respiratory depression/arrest, seizure-like activity and, possibly, death. An intoxicating dose is 15 mg/kg and a dose of 4 g has been associated with death. It has also been reported that doses between 5 and 10 g can be toxic or fatal. Effects usually occur within 20 min of taking a dose and are also associated with a steep dose-response curve that may result in accidental overdosage.

An 18-year-old male was admitted to hospital unresponsive and in respiratory failure, he died 30 min later. Toxicological postmortem analysis showed a 309.4 mg/L blood concentration of GHB along with alcohol, cocaine, cocaethylene and benzoyllecgonine [Mozayani *et al.* 1998].

A 47-year-old Caucasian male was found dead on his living room floor. At postmortem GHB concentrations in his blood, vitreous and urine were 538, 652, and 2927 mg/L, respectively [Win, Baselt 2000].

A 22-year-old female who had taken a suspected single dose of liquid GHB revealed a postmortem femoral blood concentration of 330 mg/L and a frontal cortex brain tissue concentration of 221 ng/mg. GHB was also detected in her hair. No other drugs were detected [Kalasinsky *et al.* 2001].

A 24-year-old woman ingested 4 mL fantasy – the South Australian street name for GHB. Her blood concentration was 1200 mg/L [Strickland *et al.* 2005].

Half-life Terminal half-life, 0.3–1.0 h.

Volume of Distribution 0.5 L/kg.

Clearance Oral, 9.0 mL/min/kg (range, 4.7–15.3).

Protein Binding None.

Dose Usual dose required for general anaesthesia is 50 mg/kg, for treatment of amnesia; as an analgesic dose is 10 to 20 mg.

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Kalasinsky KS *et al.* (2001). Blood, brain, and hair GHB concentrations following fatal ingestion. *J Forensic Sci* 46: 728–730.

Kavanagh PV *et al.* (2001). The urinary excretion of gamma-hydroxybutyric acid in man. *J Pharm Pharmacol* 53: 399–402.

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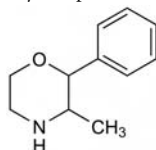
McCusker RR *et al.* (1999). Analysis of gamma-hydroxybutyrate (GHB) in urine by gas chromatography-mass spectrometry. *J Anal Toxicol* 23: 301–305.

Mesmer MZ, Satzger RD (1998). Determination of gamma-hydroxybutyrate (GHB) and gamma-butyrolactone (GBL) by HPLC/UV-VIS spectrophotometry and HPLC/thermospray mass spectrometry. *J Forensic Sci* 43: 489–492.

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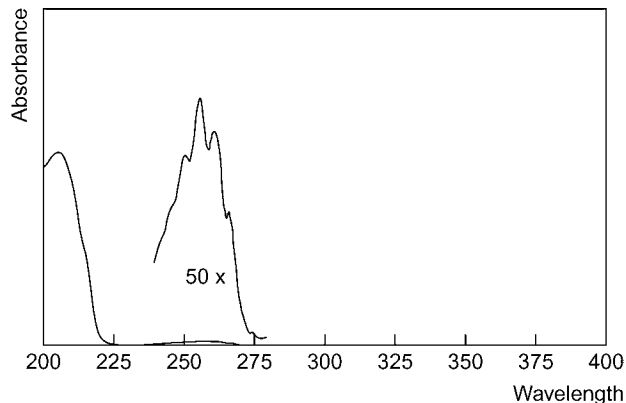
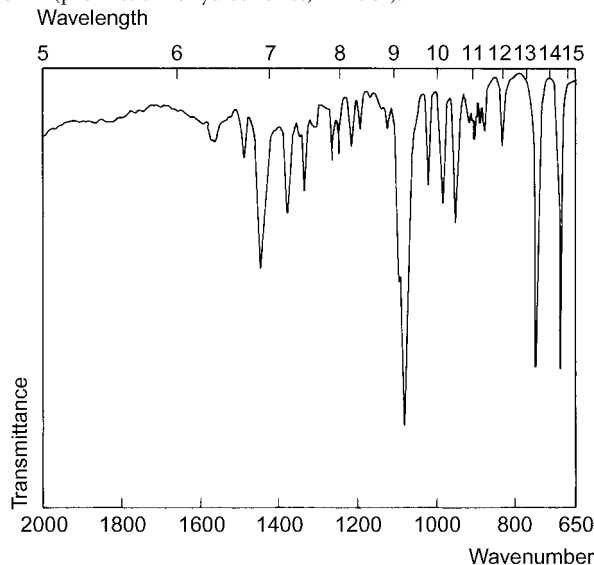
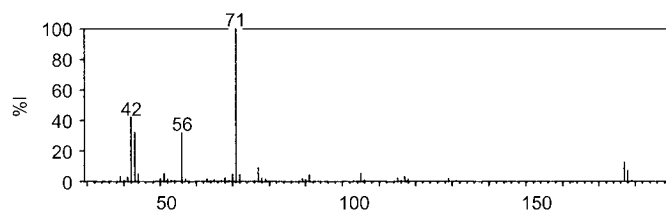
Stout PA *et al.* (2010). Quantitative analysis of gamma-hydroxybutyrate at endogenous concentrations in hair using liquid chromatography tandem mass spectrometry. *J Forensic Sci* 55: 531–537.

IUPAC Name 3-Methyl-2-phenylmorpholine**Chemical Properties** A liquid. pK_a 8.4 (25°). Log *P* (octanol/water), 1.5.**Phenmetrazine Hydrochloride** $C_{11}H_{15}NO \cdot HCl = 213.7$

CAS—1707-14-8

Synonym Oxazimédrine**Proprietary Name** Preludin**Chemical Properties** A white crystalline powder. Mp 182°. Soluble 1 in 0.4 of water, 1 in 2 of ethanol and 1 in 2 of chloroform; sparingly soluble in ether.**Phenmetrazine Teoclate** $C_{11}H_{15}NO \cdot C_7H_7ClN_4O_2 = 391.9$

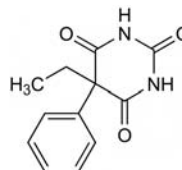
CAS—13931-75-4

Synonyms Phenmetrazine 8-chlorotheophyllinate; phenmetrazine theoclate.**Chemical Properties** A white powder. Soluble in water and ethanol; slightly soluble in acetone and ether.**Colour Test** Liebermann's reagent—red-orange.**Thin-layer Chromatography** System TA— R_f 0.50; system TB— R_f 0.14; system TC— R_f 0.27; system TE— R_f 0.46; system TL— R_f 0.14; system TAE— R_f 0.34; system TAF— R_f 0.45; system TAJ— R_f 0.20; system TAK— R_f 0.08; system TAL— R_f 0.60 (acidified iodoplatinate solution, positive; acidified potassium permanganate solution, positive).**Gas Chromatography** System GA—phenmetrazine RI 1432, phenmetrazine-TFA RI 1530, phenmetrazine-TMS RI 1620, phenmetrazine-AC RI 1810, M (OH-) isomer 1 RI 1830, M (OH-) isomer 2 RI 1865, M (OH-methoxy-) RI 1900, M (OH-) isomer 1-AC₂ RI 2150, M (OH-) isomer 2-AC₂ RI 2200, M (OH-methoxy-) AC₂ RI 2320; system GB—phenmetrazine RI 1483; system GC—RI 1873.**High Performance Liquid Chromatography** System HA— k 1.7; system HX—RI 258; system HY—RI 241.**Ultraviolet Spectrum** Aqueous acid—250, 256 ($A_1^1 = 13.3a$), 261, 263, 267 nm.**Infrared Spectrum** Principal peaks at wavenumbers 1083, 757, 695, 965, 990, 1030 cm^{-1} (phenmetrazine hydrochloride, KBr disk).**Mass Spectrum** Principal ions at m/z 71, 42, 56, 43, 177, 77, 178, 105.**Quantification****Serum** GC AFID. Phendimetrazine and phenmetrazine. Limit of detection, 10 $\mu g/L$ [Long *et al.* 1982].**Urine** GC NPD. Phenmetrazine, phentermine and amfetamines [Jonsson *et al.* 1996]. See Serum [Long *et al.* 1982].**GC-MS** Limit of detection, 0.2–0.5 mg/L [Dasgupta, Mahle 1997; Dasgupta *et al.* 1998]. Limit of quantification, 50 $\mu g/L$ [Kronstrand *et al.* 1996].**Disposition in the Body** Readily absorbed after oral administration. About 70% of a dose is excreted in the urine in 24 h with about 19% as unchanged drug, 19% as the lactam (5-methyl-3-oxo-6-phenylmorpholine), 12% as free 4-hydroxyphenmetrazine, 10% as conjugated 4-hydroxyphenmetrazine and about 5% as the *N*-oxide.

Phenmetrazine is a metabolite of phendimetrazine.

Therapeutic ConcentrationAfter a single oral dose of 75 mg given to 5 subjects, peak plasma concentrations of 0.10–0.24 mg/L (mean 0.16) were attained in about 2 h. Following a single oral dose of 75 mg of a slow-release preparation, average peak plasma concentrations of about 0.07 mg/L were reported at 5 h; concentrations remained almost constant over the period 5–24 h [Quinn *et al.* 1967].**Toxicity** Estimated minimum lethal dose for children up to 2 years of age, 200 mg. Large doses and prolonged treatment may lead to severe mental depression and dependence of the amphetamine type.In a fatality following IV overdosage, the following postmortem distribution was reported: blood 4 mg/L, bile 7 mg/L, liver 5 $\mu g/g$, urine 24 mg/L [Norheim 1973].The following postmortem tissue concentrations were reported in 12 fatalities: blood 0.1–4.9 mg/L (mean, 1.1), bile 0.2–23 mg/L (mean, 5.5), brain 0.1–15 $\mu g/g$ (mean, 2.9), kidney 0.1–8 $\mu g/g$ (mean, 1.5), liver 0.1–20 $\mu g/g$ (mean, 3.1), urine 0.1–90 mg/L (mean, 21) [Gottschalk, Baselt 2000].**Half-life** Plasma half-life, ≈ 8 h.**Dose** 25 to 75 mg of phenmetrazine hydrochloride daily.Dasgupta A, Mahle CE (1997). Determination of phenmetrazine in urine by gas chromatography-mass spectrometry after liquid-liquid extraction and derivatization with perfluorooctanoyl chloride. *J Forensic Sci* 42: 937–941.Dasgupta A *et al.* (1998). A convenient derivatization method for gas chromatography/mass spectrometric determination of phenmetrazine in urine using 2,2,2-trichloroethyl chloroformate. *J Forensic Sci* 43: 630–635.Dasgupta A *et al.* (1998). Gas chromatography-electron ionization and chemical ionization mass spectrometric analysis of urinary phenmetrazine after derivatization with 4-carboxyhexafluorobutyl chloride—a new derivative. *J Forensic Sci* 43: 636–640.Gottschalk L, per Baselt RC (2000). *Disposition of toxic chemicals and drugs in man*, 5th edn. Davis, California: Biomedical Publications, 687–688.Jonsson J *et al.* (1996). A convenient derivatization method for the determination of amphetamine and related drugs in urine. *J Forensic Sci* 41: 148–151.Kronstrand R *et al.* (1996). Determination of Phenmetrazine in urine by gas chromatography-mass spectrometry. *J Anal Toxicol* 20: 277–280.Long G *et al.* (1982). *Drug Dev Ind Pharm* 8: 203–213.Norheim G (1973). A fatal case of phenmetrazine poisoning. *J Forensic Sci Soc* 13: 287–289.Quinn GP *et al.* (1967). The effect of formulation on phenmetrazine plasma levels in man studied by a sensitive analytic method. *Clin Pharmacol Ther* 8: 369–373.**Phenobarbital***Hypnotic, Sedative, Barbiturate* $C_{12}H_{12}N_2O_3 = 232.2$

CAS—50-06-6

IUPAC Name 5-Ethyl-5-phenyl-1,3-diazinane-2,4,6-trione**Synonyms** 5-Ethyl-5-phenyl-2,4,6-(1*H*,3*H*,5*H*)-pyrimidinetrione; fenobarbital; phenemalum; phenobarbitalum; phenobarbitone; phenylethylbarbituric acid; phenylethylmalonylurea.**Proprietary Names** Aparoxal; Comizial; Edhanol; Fenemal; Fenocriz; Gardenal (e); Kaneuron; Luminal(e); Luminaletas; Luminalette; Luminaletten; Neurobiol; Phenaemal; Phenaemaletten; Solfoton. It is an ingredient of many proprietary preparations [Sweetman 2002].

Chemical Properties Colourless crystals or a white crystalline powder which may exhibit polymorphism. Mp 174° to 178°. Soluble 1 in 1000 of water, 1 in 8 of ethanol, 1 in 40 of chloroform, 1 in 13 of ether and 1 in about 700 of benzene. pK_a 7.4 (25°). Log P (octanol/water), 1.5.

Phenobarbital Sodium

$C_{12}H_{11}N_2NaO_3 = 254.2$
CAS—57-30-7

Synonyms Phenemalnatium; phenobarbitone sodium; sodium phenylethylbarbiturate; soluble phenobarbitone.

Proprietary Names *Garnenal(e)*; *Luminal(e)*. It is an ingredient of *Garoin*.

Chemical Properties A white hygroscopic powder, granules, or flakes. Mp about 175°. Soluble 1 in about 1 of water and 1 in about 10 of ethanol; practically insoluble in chloroform and ether.

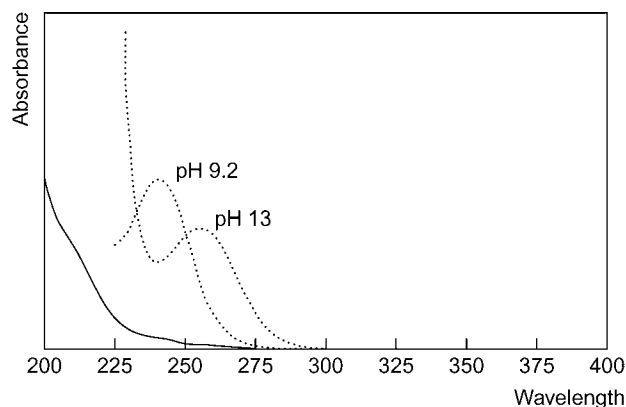
Colour Tests Koppanyi–Zwicker test—violet; Liebermann's reagent—red-orange; mercurous nitrate—black.

Thin-layer Chromatography System TD— R_f 0.47; system TE— R_f 0.28; system TF— R_f 0.65; system TH— R_f 0.38; system TAD— R_f 0.53; system TAE— R_f 0.85 (mercuric chloride–diphenylcarbazone reagent, positive; mercurous nitrate spray, black; Zwicker's reagent, pink).

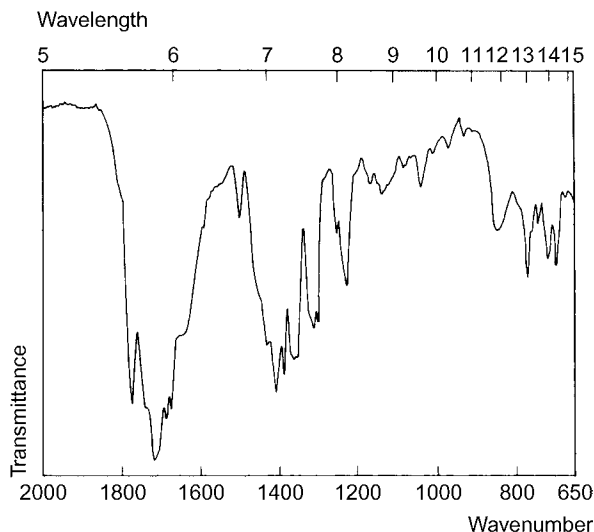
Gas Chromatography System GA—phenobarbital RI 1953, M (4-OH-) RI 2295, phenobarbital-Me₂ RI 1855, M (4-OH-)-Me₃ RI 2200; system GB—phenobarbital RI 2031, M (4-OH-) RI 2378; system GE—RRT 0.74 (relative to phenytoin); system GF—RI 2960; system GAJ—RRT 1.150 (relative to methylphenobarbital).

High Performance Liquid Chromatography System HE— k 2.76; system HG— k 3.09; system HH— k 1.23; system HX—RI 379; system HY—RI 335; system HZ—retention time 3.0 min; system HAA—retention time 14.0 min; system HAK—not detected; system HAM—retention time 2.8 min; system HAX—retention time 5.6 min; system HAY—retention time 4.7 min; system HAZ—phenobarbital k 0.39, M (4-OH-) k 0.14.

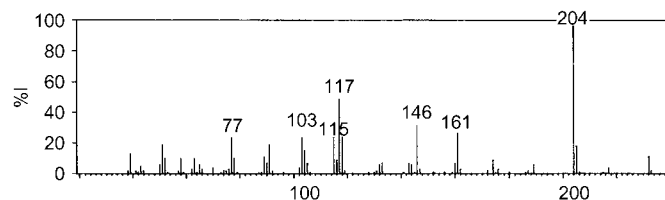
Ultraviolet Spectrum Borax buffer 0.05 mol/L (pH 9.2)—239 nm ($A_1^1=452a$); 1 mol/L sodium hydroxide (pH 13)—254 nm ($A_1^1=342b$).



Infrared Spectrum Principal peaks at wavenumbers 1712, 1684, 1670, 1770, 1310, 1300 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at m/z 204, 117, 146, 161, 77, 103, 115, 118; 4-hydroxyphenobarbital 219, 248, 148, 220, 120, 218, 133, 64.



Quantification See also under Amobarbital.

Blood GC-MS Phenobarbital and other anticonvulsants [Speed *et al.* 2000].

Plasma GC-MS Phenobarbital and 4-hydroxyphenobarbital. Limit of detection, 100 $\mu g/L$ [Patel *et al.* 1980].

HPLC Phenobarbital and other anticonvulsants [Matar *et al.* 1999]. UV detection. Phenobarbital and other anticonvulsants [Johansen *et al.* 1995]. UV detection. Phenobarbital and other anticonvulsants [Romanyshyn *et al.* 1994]. UV detection. Phenobarbital and other anticonvulsants. Limit of detection, about 600 $\mu g/L$ for phenobarbital [Christofides, Fry 1980].

HPLC-MS Phenobarbital and other sedatives [Kanazawa *et al.* 1998].

Polarisation Fluoroimmunoassay Limit of detection, 800 $\mu g/L$ [Sidki *et al.* 1982].

Serum GC Thermionic specific detection. Phenobarbital and other anticonvulsants, Limits of quantification, 50 to 200 $\mu g/L$ [Queiroz *et al.* 2002]. FID. Limit of detection, 500 $\mu g/L$ [Di Corcia *et al.* 1982].

HPLC UV detection. Phenobarbital and other anticonvulsants. Limit of quantification for phenobarbital, about 1 $\mu g/L$ [Levert *et al.* 2002]. UV detection. Phenobarbital and other anticonvulsants [Kouno *et al.* 1997]. UV detection. Phenobarbital and other anticonvulsants [Kouno *et al.* 1993]. Photodiode-array detection. Phenobarbital, other anticonvulsants, and their metabolites [Liu *et al.* 1993]. Phenobarbital, phenytoin, and theophylline, comparison with immunoassay [Hannak *et al.* 1992]. See Plasma [Christofides, Fry 1980].

Polarisation Fluoroimmunoassay Phenobarbital and other anticonvulsants [Steijns *et al.* 2002]. See Plasma [Sidki *et al.* 1982].

Urine GC-MS See Plasma [Patel *et al.* 1980].

HPLC UV detection. Phenobarbital and its metabolites [Paibir, Soine 1997]. See Serum [Liu *et al.* 1993].

Saliva HPLC See Serum [Liu *et al.* 1993].

Hair GC-MS Phenobarbital, phenytoin and their metabolites [Saisho *et al.* 2001; Goulle *et al.* 1995].

Liver GC-MS See Blood [Speed *et al.* 2000].

Biological Samples HPLC Phenobarbital and other barbiturates [Tanaka *et al.* 1997].

Disposition in the Body Readily absorbed after oral administration. The major metabolites are *N*-glucopyranosylphenobarbital and 4-hydroxyphenobarbital, together with its glucuronide conjugate. Other metabolites include two dihydrodiol compounds and hydroxymethylphenylbarbituric acid. During chronic dosing, up to about 25% of a dose is excreted in the urine in 24 h as unchanged drug and up to about 17% as total 4-hydroxyphenobarbital, about half of which is the glucuronide conjugate. Urinary excretion of unchanged drug is increased when the urine is alkaline or when the urinary volume is increased; the excretion of conjugated 4-hydroxyphenobarbital is reduced in patients with liver disease. After a single dose, about 80 to 90% is excreted in the urine in 16 days; the *N*-glucoside accounts for about 30% of a single dose. Phenobarbital crosses the placenta and is found in breast milk.

Phenobarbital is a metabolite of methylphenobarbital and primidone.

Therapeutic Concentration In plasma, usually in the range 2 to 30 mg/L . However, there is considerable intersubject variation and there does not appear to be any correlation between plasma concentration and clinical effect. Concentrations in CSF reach about 50% of those in plasma.

Following a single oral dose of 100 mg to 6 subjects, peak serum concentrations of 2.1 to 3.8 mg/L (mean, 2.9) were attained in 0.5 to 4 h (mean, 1.5) [Wilensky *et al.* 1982].

Following daily oral doses of 100 mg to 10 subjects for 15 days, serum concentrations of 10.6 to 22.1 mg/L (mean, 17) were reported 12 h after a dose [Luoma *et al.* 1982].

Toxicity The estimated minimum lethal dose is 1.5 g although recovery has occurred after ingestion of as much as 16 g . Toxic effects have been associated with blood concentrations of 4 to 17 to 90 mg/L and fatalities with concentrations of 4 to 45 to 120 mg/L . However, a degree of tolerance may develop in chronic dosing.

Blood concentrations ranged from 10 to 300 mg/L (mean 86) in 113 fatalities attributed to phenobarbital overdose [Parker *et al.* 1970].

In 20 fatalities caused by phenobarbital overdose, the following postmortem concentrations were reported: blood 15 to 540 mg/L (mean, 113), brain 0 to 833 $\mu g/g$ (mean, 135), kidney 17 to 867 $\mu g/g$ (mean, 152), liver 24 to 1450 $\mu g/g$ (mean, 236), spleen 0 to 850 $\mu g/g$ (mean, 137) [Sunshine, Hackett 1957].

Bioavailability Almost 100%.

Half-life Plasma half-life, about 90 to 100 h in adults, prolonged in neonates and reduced to about 65 to 70 h in children; increased in subjects with liver disease.

Volume of Distribution About 0.5 L/kg .

Clearance Plasma clearance, about 0.06 $mL/min/kg$.

Distribution in Blood Plasma : whole blood ratio, 0.93.

Saliva Plasma : saliva ratio, about 3.

Protein Binding About 50%.

Dose 60 to 180 mg daily.

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- Gouille JP *et al.* (1995). Phenobarbital in hair and drug monitoring. *Forensic Sci Int* 70: 191–202.
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- Johansen K *et al.* (1995). Automated analysis of free and total concentrations of three antiepileptic drugs in plasma with on-line dialysis and high-performance liquid chromatography. *J Chromatogr B Biomed Appl* 669: 281–288.
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- Lever H *et al.* (2002). Simultaneous determination of four antiepileptic drugs in serum by high-performance liquid chromatography. *Biomed Chromatogr* 16: 19–24.
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Phenol

Disinfectant, Anaesthetic (Topical)

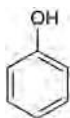
C₆H₅O = 94.11

CAS—108-95-2

IUPAC Name Phenol

Synonyms Carbolic acid; fenol; hydroxybenzene; oxybenzene; phenyl hydrate; phenylic acid.

Proprietary Names Chloraseptic; Phenaseptic; Septosol.



Chemical Properties Colourless or faintly pink, deliquescent crystals or crystalline masses, becoming pink on storage. Mp 40° to 41°. Commercial products may contain impurities which raise the bp to about 182°. Soluble 1 in about 15 of water, 6 in 1 of ethanol, 2 in 1 of chloroform and 5 in 1 of ether; soluble in glycerol. pK_a 10.0 (25°). Log P (octanol/water), 1.5.

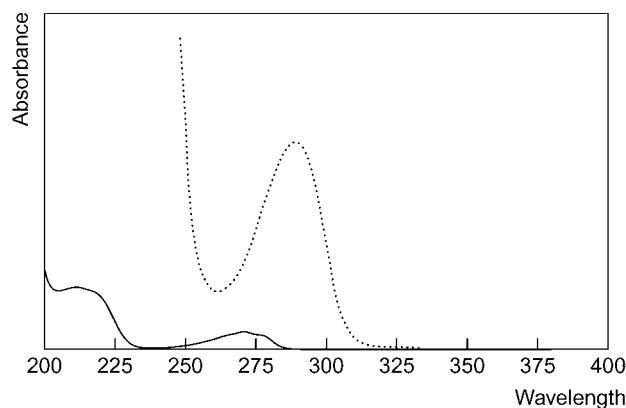
Caution Phenol is caustic to the skin.

Colour Tests p-Dimethylaminobenzaldehyde—orange/violet; ferric chloride—blue; Folin-Ciocalteu reagent—blue; Liebermann's reagent—green; potassium dichromate (method 1)—yellow—brown (2 min).

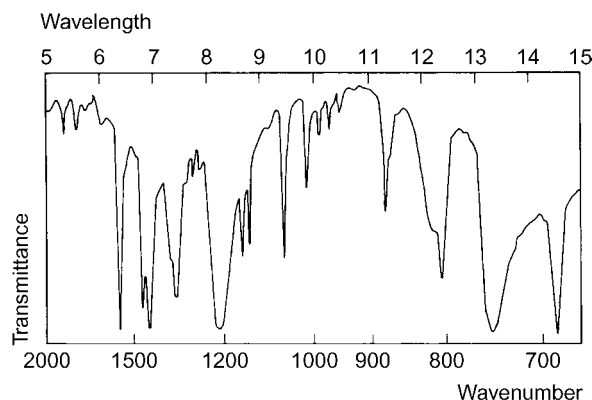
Gas Chromatography System GA—RI 981.

High Performance Liquid Chromatography System HAA—retention time 13.4 min.

Ultraviolet Spectrum Aqueous acid—270 nm (A₁¹=172b); aqueous alkali—287 nm (A₁¹=281a).



Infrared Spectrum Principal peaks at wavenumbers 690, 755, 1235, 1595, 1497, 811 cm⁻¹ (Nujol mull).



Mass Spectrum Principal ions at m/z 94, 66, 39, 65, 40, 55, 38, 95.

Quantification

Plasma GC-MS [Harrison *et al.* 1991].

Serum GC-MS Limit of detection, 1 mg/L [Hart, Dasgupta 1997].

Urine GC-MS Phenol and other phenolic compounds. Limit of detection, 0.1 to 0.2 mg/L [Bieniek 1996]. Limit of detection, 1 mg/L [Ahmed, Hale 1994]. FID [Rick *et al.* 1982]. FID [Baldwin *et al.* 1981].

HPLC Phenol and p-cresol. Fluorescence detection. Limits of detection, 0.17 pmol and 0.25 pmol per injection respectively [Tsuruta *et al.* 1996]. UV detection. Phenol and p-cresol [Birkett *et al.* 1995]. Fluorescence detection. Limit of detection, 10 ppb [Jen, Tsai 1994]. Fluorimetric detection. Phenol, hydroquinone, and catechol [Lee 1993]. UV detection. Limit of detection, <500 µg/L [Eadsforth, Coveney 1984].

Faeces HPLC See Urine [Birkett *et al.* 1995].

Disposition in the Body Rapidly absorbed from the gastrointestinal tract and readily penetrates the skin. It is metabolised by conjugation to yield phenyl glucuronide and phenyl sulfate; small amounts are oxidised to catechol and quinol which are also conjugated. Oxidation of these metabolites to quinones may tint the urine green. Acid-labile phenol conjugates are present endogenously in serum at concentrations of about 0.1 mg/L. Phenol is a metabolite of benzene. Endogenous phenol concentrations in urine average about 5 to 10 mg/L.

Toxicity Phenol denatures and precipitates cellular proteins and thus may cause rapid poisoning. The minimum lethal dose by mouth is about 1 g. Severe and even fatal poisoning may also arise from absorption of phenol from open wounds or through the intact skin. The maximum permissible atmospheric concentration is 5 ppm.

In a fatality due to the deliberate ingestion of a mixture containing phenol, the following postmortem tissue concentrations were reported: blood, phenol 56 mg/L, alcohol 300 mg/L; liver, phenol 74 µg/g. In a second fatality due to accidental percutaneous absorption, postmortem blood concentrations of 27 mg/L of phenol and 300 mg/L of alcohol were reported [Soares, Tift 1982].

In a fatality resulting from the ingestion of phenol, the following postmortem concentrations were reported: blood 130 mg/L, urine 47 mg/L, bile 187 mg/L, brain 486 µg/g, kidney 331 µg/g, muscle 204 µg/g, liver 228 µg/g, stomach content 668 mg [Lo Dico *et al.* 1989].

In a fatality due to accidental dermal exposure to phenol, the following postmortem concentrations were reported: blood 4.7 mg/L, unhydrolysed liver 3.3 µg/g, hydrolysed liver 7.1 µg/g [Lewin, Cleary 1982].

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Asparaginase

Antineoplastic

CAS—9015-68-3

Synonyms L-Asnase; L-asparaginase; L-asparagine amidohydrolase; colaspase; E. C.3.5.1.1; Mk-965; NSC-109229.

Proprietary Names Crasnitin; Crasnitine; Elspar; Erwinase (Crisantaspase-recombinant); Kidrolase; Laspar; Leucogen; Leunase; Oncaspar (Pegaspargase-recombinant); Paronal.

Chemical Properties Commonly obtained from *E. coli* and *Erwinia* sp. Asparaginase from *E. coli* has a molecular weight of about 136000. A white crystalline powder. It is freely soluble in water where it appears to be globular in shape; practically insoluble in methanol, in acetone and in chloroform.

Quantification

Serum TLC Reaction with ^{14}C -asparagine and detection with scintillation counter. Limit of detection, 0.05 U/mL [Clausen, Christensen 1986].

Disposition in the Body It is sequestered slowly by the reticulo-endothelial system. It is found in the lymph at about 20% of the concentration in plasma. There is virtually no diffusion into the CSF. Little is excreted in the urine. Preparations derived from *E. coli* and *Erwinia* differ in their pharmacokinetic and pharmacodynamic properties.

Toxicity Allergic reactions are frequent and can lead to death.

Half-life Plasma, 8 to 30 h (IV administration); up to 49 h (intramuscular administration).

Protein Binding 30%.

Dose Up to 1000 units/kg intravenously daily for 10 days.

Clausen N, Christensen E (1986). Determination of L-asparaginase activity in serum by thin-layer chromatography: application to the treatment of acute lymphoblastic leukemia. *Clin Chim Acta* 161: 111–116.

Aspirin

Analgesic, Antiplatelet, Antipyretic, NSAID

$\text{C}_9\text{H}_8\text{O}_4 = 180.2$

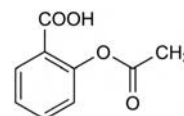
CAS—50-78-2

IUPAC Name 2-Acetoxybenzoic acid

Synonyms 2-Acetoxybenzoic acid; acetylsalicylic acid; salicylic acid acetate.

Proprietary Names Adprin-B; Angettes; Ascriptin; Aspergum; Asprimox; Aspro; Asatard; Beechams Lemon Tablets; Bufferin; Buffex; Caprin; Cardiprin; Cemerit; Claragine; Disprin; Easprin; Ecotrin; Empirin; Encaprin; Enprin; Gencardia;

Genprin; Halfprin; Micropirin; Magnaprin; Nu-Seals; Platet; PostMI; Rhonal; Solprin; ZORprin. Aspirin is an ingredient of many proprietary preparations [Sweetman 2007].



Chemical Properties Colourless or white crystals or white crystalline powder or granules. It is stable in dry air but gradually hydrolyses in contact with moisture to acetic and salicylic acids. Mp 135° . Soluble 1 in 300 of water, 1 in 5 of ethanol, 1 in 17 of chloroform, and 1 in 10 to 15 of ether; soluble in solutions of acetates and citrates and, with decomposition, in solutions of alkali hydroxides and carbonates. pK_a 3.5 (25°). Log *P* (octanol/pH 7.4), -1.1 .

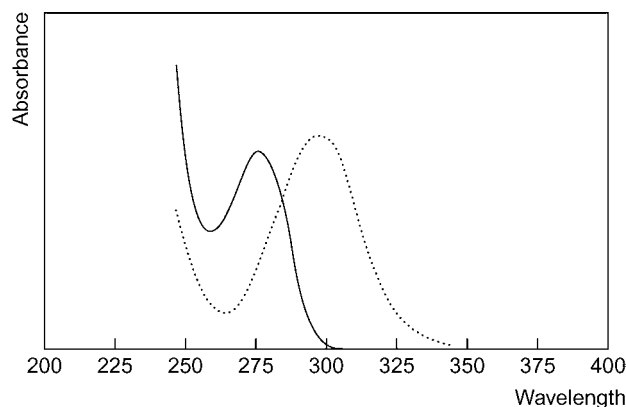
Colour Test McNally's test—red (after hydrolysis).

Thin-layer Chromatography System TA— R_f 0.90; system TD—aspirin R_f 0.18, salicylic acid R_f 0.07, salicylic acid R_f 0.00; system TE—aspirin R_f 0.09, salicylic acid R_f 0.10, salicylic acid R_f 0.00; system TF—aspirin R_f 0.30, salicylic acid R_f 0.01, salicylic acid R_f 0.00; system TAD—aspirin R_f 0.31, salicylic acid R_f 0.24; system TAE—aspirin R_f 0.78, salicylic acid R_f 0.86; system TAJ— R_f 0.40; system TAK— R_f 0.65; system TAL— R_f 0.90.

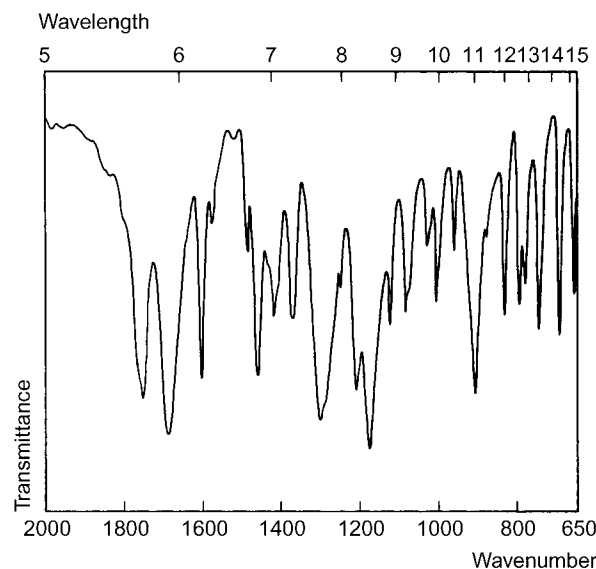
Gas Chromatography System GA—aspirin RI 1545, M (aspirin-Me) RI 1394, salicylic acid RI 1307, M (salicylic acid-Me) RI 1195; system GB—salicylic acid RI 1340, M (salicylic acid-Me) RI 1228.

High Performance Liquid Chromatography System HD—aspirin k 0.5, salicylic acid k 0.7; system HW—aspirin k 2.70, salicylic acid k 4.60; system HX—aspirin RI 350; system HY—aspirin RI 318; system HZ—aspirin RT 2.7 min; system HAX—salicylic acid RT 5.2 min; system HAY—salicylic acid RT 4.4 min.

Ultraviolet Spectrum Aqueous acid—230 ($A_1^1 = 466a$), 278 nm ($A_1^1 = 68a$); aqueous alkali—231 ($A_1^1 = 409b$), 298 nm ($A_1^1 = 190b$).



Infrared Spectrum Principal peaks at wavenumbers 1183, 1688, 1305, 1755, 925, 1219 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at m/z 120, 43, 138, 92, 121, 39, 64, 63 (aspirin); 120, 92, 138, 64, 39, 63, 121, 65 (salicylic acid); 121, 120, 69, 92, 195, 39, 93, 45 (salicylic acid).

Quantification

Plasma GC Column: 5% OV-17 on CQ 80/100 mesh (1.8 m × 4 mm i.d.). Carrier gas: N₂, 50 mL/min. Temperature programme: 160° to 200° at 2°/min. FID. Limit of detection, 200–500 µg/L for aspirin, salicylic acid and salicylamide [Rance *et al.* 1975].

GC-MS Column: reversed phase Luna C₁₈ (50 × 2.0 mm i.d., 3 µm). Mobile phase: acetonitrile and water containing 0.1% formic acid (8:2), flow rate 0.2 mL/min. MRM acquisition mode, negative ion mode. Limit of quantification, 5 µg/L for aspirin and 50 µg/L for salicylic acid [Bae *et al.* 2008]. Column: Column: DB-1701 (15 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 25 kPa. Temperature programme: 100° for 2 min to 300° at 25°/min. NICI or EI ionisation. Limit of quantification, 200 ng/L [Tsikas *et al.* 1998].

HPLC Column: MC-SAX (10 × 4 mm i.d., 50 µm). Mobile phase: water-acetonitrile (1000:10) containing 0.2% trifluoroacetic acid: water-acetonitrile (100:900) containing 0.1% trifluoroacetic acid (100:0 to 70:30 for 18 min for 2 min), flow rate 1.0 mL/min. UV detection (λ = 235 nm). Limit of quantification, 60 µg/L for aspirin and salicylic acid [Yamamoto *et al.* 2007]. Column: C₈ Nucleosil (250 × 4.6 mm i.d., 5 µm). Mobile phase: water:methanol:acetonitrile: orthophosphoric acid (650:200:150:1), flow rate 1 mL/min. UV detection (λ = 225 nm). Limit of quantification, 0.1 mg/L, limit of detection, 0.04 mg/L for aspirin and salicylic acid [McMahon, Kelly 1998]. Column: LiChrospher 100 RP-18. Mobile phase: water: phosphate buffer (pH 2.5): acetonitrile (35:40:25), flow rate 1 mL/min. Limit of detection, 0.1 mg/L for aspirin and salicylic acid [Pirola *et al.* 1998]. Column: Alltech ODS (150 × 4.6 mm i.d., 5 µm). Mobile phase: 30 mmol/L sodium citrate: 30 mmol/L sodium acetate (pH 4.8), flow rate, 1.0 mL/min. UV detection (λ = 305 nm). Limit of detection, <5 µmol/L for salicylic acid [Coudray *et al.* 1996]. Column: Nova-Pak C₁₈ (150 × 4 mm i.d., 4 µm). Mobile phase: water: 85% orthophosphoric acid: acetonitrile (740:0.9:180). UV detection (λ = 237 nm). Limit of quantification, 100 µg/L for aspirin and salicylic acid [Kees *et al.* 1996]. See Urine [Krivosiková *et al.* 1996]. See Oral Fluid [Legaz *et al.* 1992]. See also Buskin *et al.* [1982], Gaspari, Locatelli [1987], Owen *et al.* [1987] and Rumble *et al.* [1981].

Serum HPLC Column: ACE C₁₈ (150 × 4.6 mm i.d., 5 µm). Mobile phase: acetonitrile: 0.1% aqueous orthophosphoric acid (40:60 to 45:55 at 3.5 min to 70:30 at 5.5 min until 8 min to 40:60 at 9 min for 1 min), flow rate 1 mL/min for 3.5 min to 2 mL/min at 5.5 min to 3 mL/min at 7 min. Fluorescence detection (λ_{ex} = 290 nm, λ_{em} = 445 nm). Limit of detection, <0.05 pmol for aspirin, salicylic acid, piroxicam and mefenamic acid [Ibrahim *et al.* 2007]. Column: ODS. Mobile phase: methanol: 0.7 mmol/L phosphoric acid (pH 2.5, 50:50). UV detection. Limit of detection, 114 µg/L for aspirin and 38 µg/L for salicylic acid [Ohwaki *et al.* 2007].

Urine HPLC Column: LiChroCART (250 × 4 mm i.d., 5 µm). Mobile phase: acetonitrile: methanol: 0.05% trifluoroacetic acid in water (96:3:1 to 70:29:1 at 30 min to 50:49:1 at 60 min), flow rate 1.0 mL/min to 1.2 mL/min at 30 min to 1.4 mL/min at 60 min. DAD (λ = 200–450 nm). Limit of quantification, 0.46 mg/L, limit of detection, 0.14 mg/L [Baranowska *et al.* 2009]. Column: Separon SGX C₁₈ (150 × 3.3 mm i.d.). Mobile phase: water: 85% phosphoric acid: butanol: tetrabutylammonium hydroxide: methanol (134:1:1:1:63), flow rate, 0.9 mL/min. UV detection (λ = 305 nm). Limit of detection, 0.2 µmol/L salicylic acid and salicyluric acid [Krivosiková *et al.* 1996]. Column: µBondapak Phenyl (300 × 4.6 mm i.d., 10 µm). Mobile phase: methanol: acetonitrile: TEA: 10 mmol/L potassium dihydrogen phosphate (18:110:1:871, pH 2.8), flow rate 1.0 mL/min. UV detection (λ = 237 nm). Limit of detection, 2 mg/L for salicyl acyl glucuronide, salicyl phenolic acid glucuronide and salicyluric phenolic glucuronide; 1 mg/L for gentisic acid; and 0.5 mg/L for salicylic acid and salicyluric acid [Shen *et al.* 1991].

Oral Fluid HPLC Column: Spherisorb ODS-2 (200 × 4.6 mm i.d., 5 µm). Mobile phase: methanol: water: acetic acid (400:98:2), flow rate 1.0 mL/min. UV detection (λ = 237 nm). Limit of detection, 100 g/L [Legaz *et al.* 1992].

Skin HPLC See Plasma. Limit of detection, 0.1 µg/cm² for aspirin and salicylic acid [Pirola *et al.* 1998].

Note For a review of the analysis of salicylates, see Hori *et al.* [2003].

Disposition in the Body Aspirin is readily absorbed after oral administration and rapidly hydrolysed to salicylic acid, which is the active agent. Salicylic acid is conjugated with glucuronic acid and glycine to form acyl and ether glucuronides and salicyluric acid; some hydroxylation also occurs to give dihydroxy and trihydroxy derivatives of salicylic acid. Aspirin is excreted almost entirely in the urine with ~50–80% of a dose as salicyluric acid, 10–30% as salicyl O-glucuronide, 5% as salicyl ester glucuronide, and 5–10% as free salicylic acid, together with small amounts of gentisic acid, gentisuric acid, and unchanged drug; salicylates are reabsorbed by the renal tubules from acid urine and thus alkaline diuresis increases the rate of salicylate elimination; ~85% of a dose is excreted as free salicylic acid if the urine is made alkaline.

Aspirin is a metabolite of aloxiprin and benorilate.

Therapeutic Concentration In plasma, salicylic acid is usually in the range 20–100 mg/L for analgesia and 150–300 mg/L for anti-inflammatory effect.

After a single oral dose of 900 mg given to 5 subjects, peak plasma aspirin concentrations of 16–50 mg/L (mean, 37) were attained in ~14 min, and peak plasma salicylic acid concentrations of 47–113 mg/L (mean, 78) were reported at 0.5–1 h [Cham *et al.* 1980].

Following daily oral doses of 3.9 g to 8 subjects for 8 days, steady-state plasma salicylic acid concentrations of 105–227 mg/L (mean, 173) were reported; by the 36th day of treatment, the steady-state plasma concentrations had declined to 45–208 mg/L (mean, 129) [Rumble *et al.* 1980].

Toxicity The estimated minimum lethal dose is 15 g. Plasma concentrations of salicylic acid greater than 300 mg/L are likely to produce toxic reactions and

concentrations greater than 500 mg/L are associated with moderate to severe intoxication. The maximum permissible atmospheric concentration is 5 mg/m³.

In a review of 62 fatalities attributed to salicylate poisoning, the following postmortem tissue concentrations were reported: blood 61–7320 mg/L (mean, 661; 52 cases), brain 22–700 µg/g (mean, 218; 20 cases), kidney 19–1200 µg/g (mean, 408; 21 cases), liver 13–1000 µg/g (mean, 437; 24 cases), spleen 103–1230 µg/g (mean, 421; 16 cases), urine 180–1350 mg/L (mean, 593; 15 cases) [Rehling 1967].

A 43-year-old female attempted suicide by self-administration, in the form of an enema, of ~700 aspirin tablets dissolved in water. The initial salicylate concentration in the serum was 590 mg/L and rose to 900 mg/L 12 h later; after haemodialysis, serum salicylate concentration fell to 160 mg/L but the patient remained in a coma for more than a year. The patient's poor outcome was attributed to retention of aspirin products in the rectal vault plus poor recognition of the delayed absorption properties of rectally administered aspirin [Watson, Tagupa 1994].

In 3 cases of severe salicylate poisoning, haemofiltration treatment reduced serum salicylate from 8.5–3.5 mmol/L after 3 h in 1 patient and in another patient from 6.2–4 mmol/L after 4 h and to 1.4 mmol/L after a further 7 h. The third patient who had persistent salicylism (serum level of 3 mmol/L), serum salicylate levels were decreased with haemofiltration [Wrathall *et al.* 2001].

An 86-year-old woman who took an overdose of tablets each containing 300 mg aspirin had respiratory alkalosis and a blood salicylate concentration of 850 mg/L. As the patient was elderly and frail, she was treated with urinary alkalisation and she showed rapid clinical improvement [Basavarajaiah *et al.* 2004].

A 14-year-old girl who had ingested 120 tablets each containing 81 mg aspirin had a salicylate level of 10 mg/L 4 h after ingestion and she denied having any adverse symptoms. She was given one dose of activated charcoal. At 35 h she had dizziness, tinnitus and epigastric discomfort and a salicylate level of 460 mg/L. The delayed salicylate toxicity was treated with a second dose of activated charcoal and IV bicarbonate and potassium [Rivera *et al.* 2004].

A 52-year-old woman who died after ingesting ~300 tablets each containing 325 mg aspirin had the following post-mortem salicylate levels: 1.1 g/L (heart blood), 1.3 g/L (femoral blood), 0.3–0.4 mg/g (brain), 0.9–1.4 mg/g (lung), 0.6–0.8 mg/g (liver) and 0.9 g/L (kidney) [Ihama *et al.* 1997].

A 41-year-old man died after ingesting ~200 aspirin tablets (65 g aspirin). Blood levels of salicylate ~12 h after ingestion were 475 to 557 mg/L. Postmortem levels were 762 mg/L (heart blood) and 215 mg/L (femoral blood). Lethal levels were also found in all tested organs (e.g. 503 mg/L (liver) and 251 mg/L (brain) [Wollersen *et al.* 2007].

A 74-year-old patient admitted to intensive care because of impaired consciousness developed irreversible asystole ~6 h later and was found to have a plasma salicylate concentration of 876 mg/L [Galbois *et al.* 2009].

A 35-year-old man admitted to hospital 7.5 h after ingesting 400 tablets each containing 325 mg aspirin was treated with activated charcoal and IV hydration with sodium bicarbonate. 2 h after admission his salicylate concentration was 916 mg/L. He was treated with haemodialysis and 22 h after presentation, the salicylate level was 884 mg/L and he died 40 h after the overdose [Minns *et al.* 2010].

Half-life Plasma half-life, aspirin ~17 min, salicylic acid dose dependent (2–4 h after doses of less than 3 g, increasing to ~19 h after large doses).

Volume of Distribution Aspirin ~0.15 L/kg, salicylic acid ~0.1–0.2 L/kg (dose dependent).

Protein Binding Salicylic acid ~90% at concentrations below 100 µg/mL, decreasing to 50% at concentrations above 400 µg/mL.

Note For reviews of the pharmacokinetics of salicylates, see Levy [1978] and Mandelli, Tognoni [1980]. For reviews of salicylate poisoning, see Pearlman, Gambhir [2009] and Reingardiene, Lazauskas [2006].

Dose Usually 1.2 to 4 g daily; doses of up to 8 g daily are given in acute rheumatic disorders. Low-dose aspirin for antiplatelet effects; initially 300 mg then 75 to 150 mg daily.

Bae SK *et al.* (2008). Determination of acetylsalicylic acid and its major metabolite, salicylic acid, in human plasma using liquid chromatography–tandem mass spectrometry: application to pharmacokinetic study of Astrix in Korean healthy volunteers. *Biomed Chromatogr* 22: 590–595.

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Cadralazine

Antihypertensive

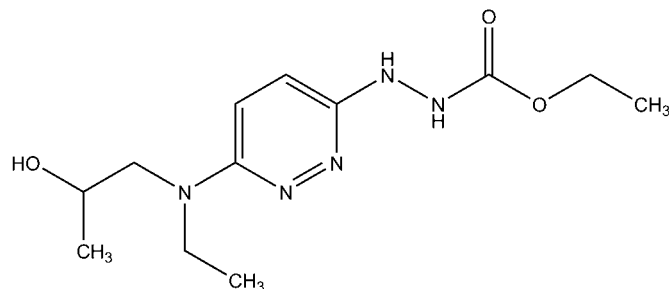
$C_{12}H_{21}N_5O_3 = 283.3$

CAS—64241-34-5

IUPAC Name Ethyl N-[[6-(ethyl-(2-hydroxypropyl)amino)pyridazin-3-yl]amino]carbamate

Synonyms 3-(2-Carboethoxyhydrazino)-6-[N-(2-hydroxypropyl)ethylamino]-pyridazine; DC-826; 2-[6-[ethyl(2-hydroxy-propyl)amino]-3-pyridazinyl]hydrazinecarboxylic acid ethyl ester; ISF-2469.

Proprietary Names *Cadral*; *Cadraten*; *Cadrikan*; *Presmode*.



Chemical Properties Crystals. Mp 160° to 162°. pK_a 6.0.

Dose 10 mg orally once a day.

Caffeine

Xanthine Stimulant

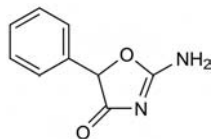
$C_8H_{10}N_4O_2 = 194.2$

CAS—58-08-2

IUPAC Name 1,3,7-Trimethylpurine-2,6-dione

Synonyms 1,3,7-Trimethyl-2,6-dioxopurine; 1,3,7-trimethylxanthine; 3,7-dihydro-1,3,7-trimethyl-1H-purine-2,6-dione; anhydrous caffeine; coffeine; guaranine; methyltheobromine; methylxanthine theophylline.

Proprietary Names *Caffedrine*; *Enerjets*; *Keep Alert*; *NoDoz*; *Pro-Plus*; *Quick Pep*; *Spray-n-Wake*; *Stay Alert*; *Tirend*; *Vivarin*; *Wake-Up Tablets*. Caffeine is an ingredient of many proprietary preparations [Sweetman 2007].



Chemical Properties Silky white crystals, usually matted together, or a white crystalline powder. It sublimes at 178°. Mp 238°. When crystallised from water, caffeine contains 1 molecule of water of crystallisation, but it is anhydrous when crystallised from ethanol, chloroform or ether. It is decomposed by strong solutions of caustic alkalis. Freely soluble in pyrrole, in tetrahydrofuran containing approx. 4% water; soluble in ethyl acetate; soluble 1 in 46 of water, 1 in 5.5 of water at 80°, 1 in 1.5 of boiling water, 1 in 66 of alcohol, 1 in 22 of alcohol at 60°, 1 in 50 of acetone, 1 in 5.5 of chloroform; 1 in 530 of ether, 1 in 100 of benzene, 1 in 22 of boiling benzene; slightly soluble in petroleum ether. Solubility in water is increased by alkali benzoates, cinnamates, citrates or salicylates. pK_a 10.4 (40°) [Dean *et al.* 1985], 14.0 (25°). Log *P* (octanol/water), −0.07 [Hansch *et al.* 1995]. Extraction yield (chlorobutane), 0.3 [Demme *et al.* 2005].

Caffeine Citrate

$C_8H_{10}N_4O_2 \cdot C_6H_8O_7 = 386.3$

CAS—69-22-7

Synonyms Caffeina citrate; citrated caffeine; coffeinum citricum; cofeincitrat.

Proprietary Names *Cafcit*. An ingredient of *Askit*.

Chemical Properties A mixture of caffeine and citric acid containing 47–50% of anhydrous caffeine. A white, crystalline powder that is decomposed by water. Soluble 1 in 4 of hot water, dissociating on further dilution with the separation of caffeine on cooling, which redissolves in ~1 in 32 of water; soluble 1 in 25 of ethanol. Log *P* (octanol/water), 0.16.

Caffeine Hydrate

$C_8H_{10}N_4O_2 \cdot H_2O = 212.2$

CAS—5743-12-4

Synonyms Caféine monohydraté; caffeine monohydrate; coffein monohydrat; coffeinum monohydricum.

Proprietary Names An ingredient of *Migril* and *Pardale*.

Chemical Properties Silky white crystals, usually matted together, or a white crystalline powder containing 8.5% water. It effloresces in dry air and loses its water of crystallisation when heated, becoming anhydrous at 80°. It sublimes at ~180°. Mp 234° to 239°. Soluble 1 in 60 of water and 1 in 110 of ethanol; soluble in chloroform with the separation of water; slightly soluble in ether.

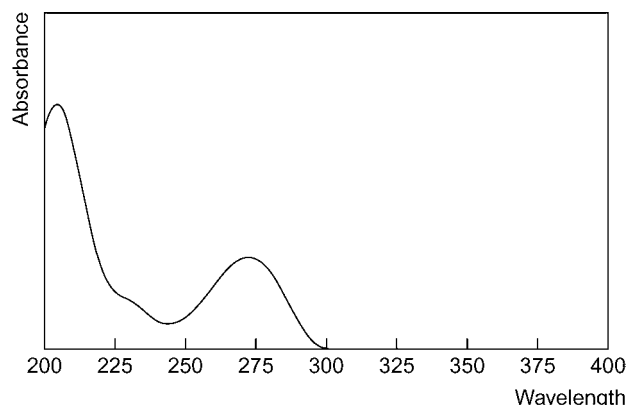
Colour Test Amalic acid test—orange/violet.

Thin-layer Chromatography System TA— R_f 0.52; system TB— R_f 0.03; system TC— R_f 0.58; system TD— R_f 0.15; system TE— R_f 0.52; system TF— R_f 0.10; system TL— R_f 0.25; system TAD— R_f 0.55; system TAE— R_f 0.59; system TAF— R_f 0.55; system TAJ— R_f 0.54; system TAK— R_f 0.18; system TAL— R_f 0.81 (acidified iodoplatinate solution—positive; Dragendorff spray—positive).

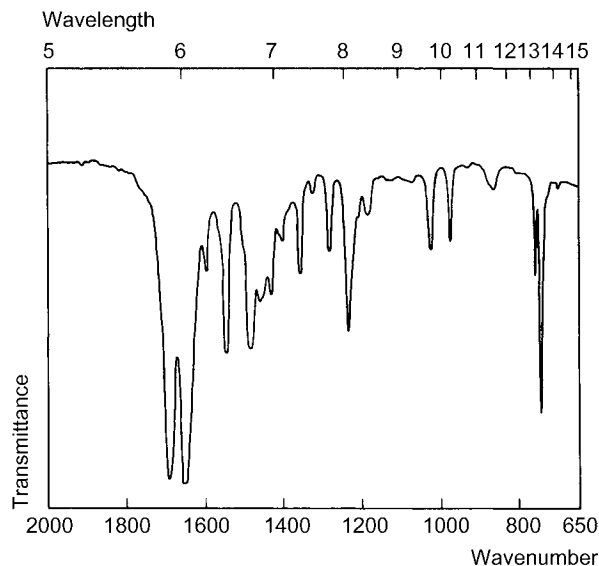
Gas Chromatography System GA—RI 1800, M (theobromine) RI 1807, M (theophylline) RI 2005; system GB—RI 1904, M (theobromine) RI 1920, M (theophylline) RI 1990; system GC—RI 2376; system GF—RI 2340.

High Performance Liquid Chromatography System HA— k 0.2; system HC— k 0.26; system HS— k 0.21; system HX—RI 305; system HY—RI 259; system HZ—RT 1.9 min; system HAA—RT 6.7 min; system HAM—RT 1.98 min; system HAX—RT 4.8 min; system HAY—RT 3.8 min; system HAZ— k 0.18.

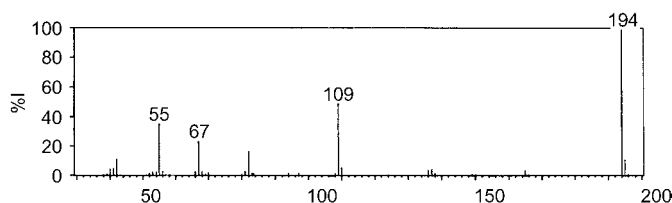
Ultraviolet Spectrum Aqueous acid—273 nm ($A_1^1 = 504a$). No alkaline shift.



Infrared Spectrum Principal peaks at wave numbers 1658, 1698, 747, 1548, 1242, 760 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at m/z 194, 109, 55, 67, 82, 195, 42, 110.



Quantification

Plasma GC Column: DB-17 (15 m $\times$ 530 μ m i.d., 1 μ m). Carrier gas: He, 21 mL/min. Temperature programme: 70° to 280°. NPD. Limit of detection, 50 μ g/L [Carregaro *et al.* 2001]. Column: OV-17. NSD. Limit of detection, 1 μ g/L [Teeuwen *et al.* 1991]. AFID. Limit of detection, 50 μ g/L [Bradbrook *et al.* 1979].

HPLC Column: Partisil-10 SCX. Mobile phase: 0.1 mol/L sodium phosphate buffer (pH 4.9):methanol (92:8), flow rate 1.5 mL/min. UV detection (λ =254 nm). Retention time: 5.1 min. Limit of detection, 10 μ g/L [Ghosheh *et al.* 2000]. Column: reversed phase. UV detection. Limit of detection, 0.1 mg/L for caffeine and theophylline [Schreiber-Deturmeny, Bruguerolle 1996]. Caffeine and its metabolites [Dobrocky *et al.* 1994]. Column: TSK gel ODS-80TM (150 $\times$ 4.6 mm i.d.). Mobile phase: methanol:0.1 mol/L sodium dihydrogen phosphate (30:70), flow rate 0.8 mL/min. UV detection (λ =274 nm). Limit of detection, 5 μ g/L for caffeine [Tanaka 1992]. UV detection. Limit of detection, 100 μ g/L [Haughey *et al.* 1982]. UV detection (λ =273 nm). Limit of detection, 100 μ g/L [O'Connell, Zurzola 1984].

LC-MS FAB. Limit of detection, ~5 μ g/L [Hieda *et al.* 1995].

Serum HPLC Caffeine and its metabolites. Column: MZ Kromasil C4 (250 $\times$ 4 mm i.d., 5 μ m). Mobile phase: acetate buffer (pH 3.5):methanol (97:3 to 80:20 in 20 min). UV detection (λ =275 nm). Limit of detection, 20 mg/L for caffeine [Georgia *et al.* 2001]. Caffeine and paraxanthine. Column: reversed phase. UV detection. Limit of detection, 0.3 mg/L for caffeine [Koch *et al.* 1999]. Caffeine and paraxanthine. UV detection. Limit of quantification, 50 μ g/L [Holland *et al.* 1998]. Caffeine and theophylline. Column: ODS reversed phase. Mobile phase: 0.2 mmol/L sodium hydrogen phosphate solution (pH 7.4). UV detection (λ =273 nm). Limit of detection, 0.5 mg/L for caffeine [Umamura *et al.* 1998]. See Plasma [Haughey *et al.* 1982].

Urine HPLC Caffeine, theobromine and theophylline. UV detection. Limit of detection, <1 ng/L [Bispo *et al.* 2002]. Caffeine metabolites. Column: Eclipse XDB-C₁₈. DAD [Caubet *et al.* 2002]. See Serum [Georgia *et al.* 2001]. Column: YMC-Pack Polyamine-II (250 $\times$ 4.6 mm i.d.). Mobile phase: acetonitrile:0.75% formic acid (91:9), flow rate 1.0 mL/min. UV detection (λ =260 and 280 nm) [Nyeki *et al.* 2001]. Caffeine and its metabolites. Column: ODS. Mobile phase: 0.05% acetic acid: methanol (92.5:7.5). UV detection (λ =280 nm) [Bendriess *et al.* 2000]. Caffeine metabolites. UV detection. Limit of detection, 1–2 μ mol [Rasmussen, Brosen 1996]. See Plasma [Dobrocky *et al.* 1994].

LC-MS See Plasma [Hieda *et al.* 1995].

CE Caffeine and its metabolites. Limit of detection, 2–5 μ mol/L [Rodopoulos, Norman 1994].

Oral Fluid GC See Blood [Teeuwen *et al.* 1991]. See Plasma [Haughey *et al.* 1982].

Sweat GC-MS [Tsuda *et al.* 2000].

Disposition in the Body Caffeine is rapidly absorbed after oral administration; bioavailability is almost 100%. It is widely distributed, crossing the placenta, entering the CNS and saliva; small amounts are excreted in breast milk. Metabolic reactions include *N*-demethylation, acetylation and oxidation to uric acid derivatives. Approximately 85% of a dose is excreted in the urine in 48 h with up to 40% of the dose as 1-methyluric acid, 10–15% as 1-methylxanthine and up to 35% as 5-acetylamino-6-formylamino-3-methyluracil and 5-acetylamino-6-amino-3-methyluracil; other metabolites excreted in the urine include theophylline, 1,7-dimethylxanthine (paraxanthine), 7-methylxanthine, and 1,3-dimethyluric acid. Approximately 1% is excreted in the urine as unchanged drug. The extent of *N*-acetylation is genetically determined. In neonates, caffeine is largely excreted unchanged in the urine, because the capacity to metabolise the drug is reduced until approx. 6 months of age. Caffeine, theophylline, theobromine and paraxanthine are found in plasma from dietary sources, especially coffee, tea and cocoa. An average cup of coffee or tea contains approximately 100 mg caffeine.

Therapeutic Concentration

Following a single dose of 130 mg caffeine to 36 subjects, peak plasma concentrations of 2.5–6.8 mg/L (mean, 4.0) were attained in 20–40 min [O'Connell, Zurzola 1984].

Following a single oral dose of 5 mg/kg to 10 subjects, peak plasma concentrations of 6.9–16.1 mg/L (mean, 10) were attained in ~0.5 h [Blanchard, Sawers 1983].

Toxicity Fatalities have occurred after the ingestion of 5–50 g caffeine, but recovery following the ingestion of 30 g has been reported. Toxic effects are associated with blood concentrations greater than 15 mg/L and fatalities with blood concentrations greater than 80 mg/L.

In a fatality resulting from ingestion of mail order diet pills containing caffeine as their only active ingredient, a 1560 mg/L postmortem serum concentration was reported in a 22-year-old woman [Mrvos *et al.* 1989].

The following postmortem tissue concentrations were reported in a fatality resulting from ingestion of 5.3 g caffeine by a 5-year-old child: blood 159 mg/L, kidney 230 μ g/g, liver 198 μ g/g and urine 114 mg/L [DiMaio, Garriott 1974].

In a death caused by the ingestion of 50 g caffeine, the following postmortem tissue concentrations were reported: blood 79 mg/L, brain 105 μ g/g, kidney 145 μ g/g, liver 214 μ g/g and urine 280 mg/L [Grusz-Harday 1973]. A 1-year-old girl survived after ingesting 2–3 g caffeine (200–300 mg/kg) despite having a maximum serum concentration of 385 mg/L 4 h after ingestion. Symptoms of toxicity included arrhythmias, seizures, metabolic disturbances and severe pulmonary oedema [Dietrich, Mortensen 1990].

Note For a review of caffeine toxicology, see Abbott [1986]. For a review of deaths involving caffeine, see Shum *et al.* [1997].

Half-life Plasma half-life, 2–10 h (mean, 4).

Volume of Distribution ~0.5 L/kg.

Clearance Plasma clearance, 1–2 mL/min/kg.

Distribution in Blood Plasma: whole blood ratio, 0.93;

Saliva Plasma: saliva ratio, ~1.3.

Protein Binding ~35%.

Dose As a mild stimulant 50 to 200 mg; maximum 1 g daily.

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Clomocycline

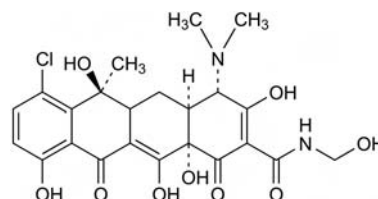
Antibacterial

C₂₃H₂₅ClN₂O₉ = 508.9

CAS—1181-54-0

IUPAC Name (2Z,4S,4aS,5aS,6S,12aS)-7-Chloro-4-(dimethylamino)-6,10,11,12a-tetrahydroxy-2-[hydroxy-(hydroxymethylamino)methylidene]-6-methyl-4,4a,5,5a-tetrahydrotetracene-1,3,12-trione

Synonyms 7-Chloro-4-(dimethylamino)-1,4,4a,5,5a,6,11,12a-octahydro-3,6,10,12,12a-pentahydroxy-N-(hydroxymethyl)-6-methyl-1,11-dioxo-2-naphthacene-carboxamide; chlormethylencycline; N²-(hydroxymethyl)chlortetracycline; methylolchlortetracycline.



Chemical Properties A yellow powder. It is sensitive to light and air, and undergoes decomposition between 145° and 170° without a definite melting point. Log *P* (octanol/water), –1.5.

Clomocycline Sodium

C₂₃H₂₄ClN₂NaO₉ = 530.9

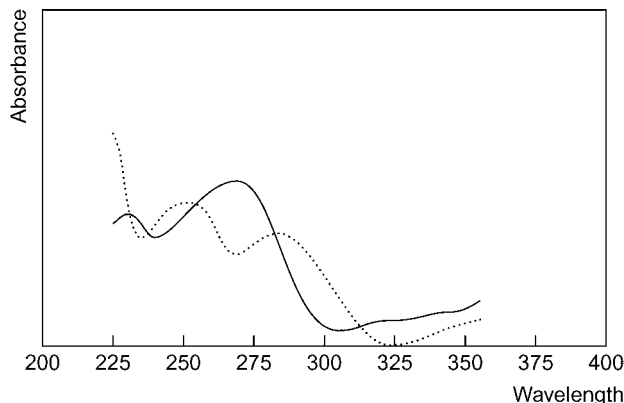
Proprietary Name *Megaclor*

Chemical Properties A yellow powder. It is sensitive to light and air, and undergoes decomposition without melting. Very soluble in water; soluble 1 in 250 of ethanol; practically insoluble in chloroform and ether.

Colour Tests Benedict's reagent—red; formaldehyde-sulfuric acid—orange-brown; Liebermann's reagent—black; Mandelin's test—violet→brown; Marquis test—orange-brown; sulfuric acid—blue-black.

Thin-layer Chromatography System TA—R_f 0.05, streaking (acidified potassium permanganate solution, strong reaction).

Ultraviolet Spectrum Aqueous acid—229, 268 nm (A₁ = 350b); aqueous alkali—251, 283 nm.



Infrared Spectrum Principal peaks at wavenumbers 1600, 1634, 1501, 1307, 1205, 1235 cm⁻¹ (clomocycline sodium, KBr disk).

Dose 0.68 to 1.36 g of clomocycline sodium daily.

Clonazepam

Anticonvulsant, Benzodiazepine

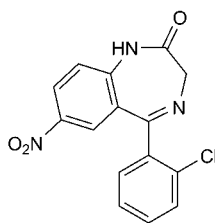
C₁₅H₁₀ClN₃O₃ = 315.7

CAS—1622-61-3

IUPAC Name 5-(2-Chlorophenyl)-7-nitro-1,3-dihydro-1,4-benzodiazepin-2-one

Synonyms 5-(2-Chlorophenyl)-1,3-dihydro-7-nitro-2H-1,4-benzodiazepin-2-one; clonazepamum; Ro-5-4023.

Proprietary Names *Antelapsin; Clonapam; Clonex; Clonopin; Iktorivil; Kenoket; Klonopin; Paxam; Rivotril.*



Chemical Properties White to light-yellow crystalline powder. Mp 236.5° to 238.5°. Protect from light. Solubility in mg/mL at 25°: acetone 31, chloroform 15, methanol 8.6, ether 0.7, benzene 0.5, water <0.1. pK_{a1} , 1.5; pK_{a2} , 10.5. Log *P* (octanol/water) 2.41. Extraction yield (chlorobutane), 1 [Demme *et al.* 2005].

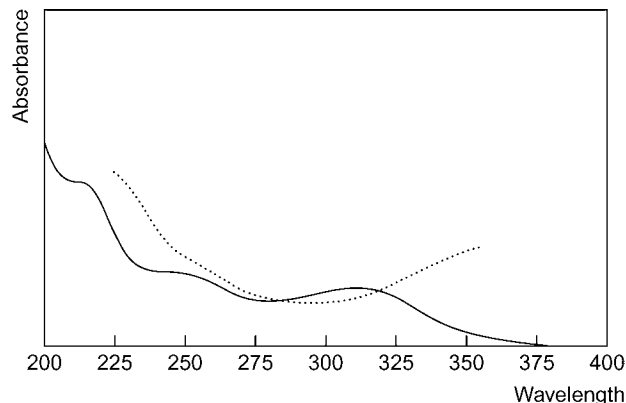
Colour Test Formaldehyde-sulfuric acid—orange.

Thin-layer Chromatography System TA— R_f 0.72; system TB— R_f 0.00; system TC— R_f 0.53; system TD— R_f 0.35; system TE— R_f 0.67; system TF— R_f 0.45; system TL— R_f 0.61; system TAD— R_f 0.56; system TAE— R_f 0.85; system TAF— R_f 0.87; system TAJ— R_f 0.50; system TAK— R_f 0.53; system TAL— R_f 0.91 (acidified iodoplatinate solution, positive; acidified potassium permanganate solution, positive).

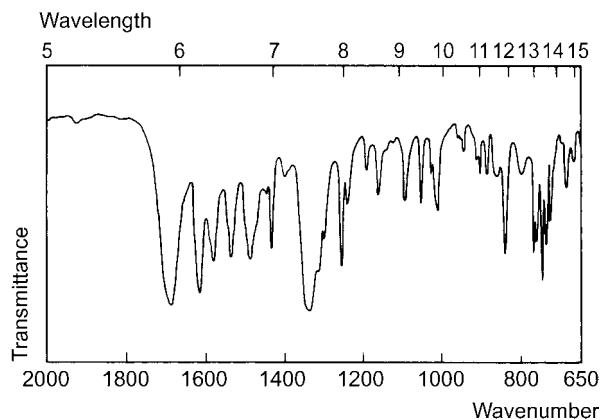
Gas Chromatography System GA—clonazepam RI 2823, M (7-amino-) RI 2890, M (amino-OH-) RI 2935; system GB—clonazepam RI 3000, clonazepam-TMS RI 2781, M (7-amino) RI 2996, M (7-amino)-TMS₂ RI 2742; system GG—RI 3600.

High Performance Liquid Chromatography System HPLC 27—clonazepam RT 3.43 min, 7-acetamidoclonazepam RT 2.37 min; system HI—*k* 2.85; system HK—*k* 0.35; system HX—RI 465; system HY—RI 403; system HZ—RT 4.6 min; system HAA—RT 17.4 min; system HAF—clonazepam RT 5.30 min; system HAK—clonazepam RT 6.2 min; system HAL—clonazepam RT 4.3 min, 7-acetamidoclonazepam RT 1.45 min, 7-aminoclonazepam RT 1.55 min; system HAM—clonazepam RT 4.3 min; system HAX—clonazepam RT 6.5 min; system HAY—clonazepam RT 6.0 min; system HAZ—clonazepam *k* 0.79; system HBH—clonazepam *k* 2.92; system HBI—clonazepam *k* 0.79.

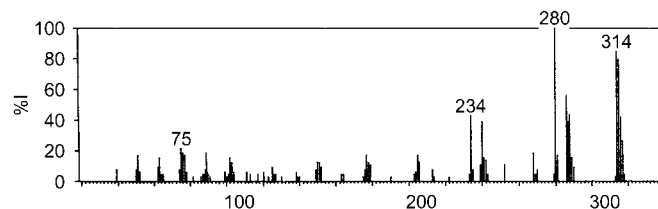
Ultraviolet Spectrum Aqueous acid—273 nm (A_1^1 = 645b); methanol—245 (A_1^1 = 460b), 309 nm (A_1^1 = 360b).



Infrared Spectrum Principal peaks at wavenumbers 1685, 1610, 748, 1255, 1578, 1532 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at *m/z* 280, 314, 315, 286, 234, 288, 316, 240; 7-acetamidoclonazepam 43, 327, 299, 298, 292, 329, 328, 256; 7-aminoclonazepam 285, 256, 257, 258, 44, 287, 110, 220.



Quantification

Blood HPLC Column: LiChrospher Select B C_{18} reversed phase (125 $\times$ 3.0 mm i.d., 5 μm). Mobile phase: 20 mmol/L potassium dihydrogen phosphate buffer (pH 2.1):acetonitrile (30:70 to 35:65 at 30 min), flow rate 0.5 mL/min to 0.3 mL/min at 30 min. UV detection (λ = 254 nm). Limit of quantification, 50 $\mu\text{g/L}$, limit of detection, 15 $\mu\text{g/L}$ [El Mahjoub, Staub 2001]. Column: Nova-Pak phenyl bonded (150 $\times$ 3.9 mm i.d., 4 μm). Mobile phase: acetonitrile:40 mmol/L potassium dihydrogen phosphate (pH 3.75; 28:72), flow rate 0.8 mL/min. UV-Vis detection (λ = 240 nm). Relative retention time: 1.99. Limit of quantification, 0.01 mg/L, limit of detection, 0.006 mg/L [Robertson, Drummer 1995].

Plasma GC Column: 3% OV-17 on Gas Chrom Q (1.2 m $\times$ 3 mm i.d.). Carrier gas: N_2 , 35 mL/min. Temperature: 270°. ECD. Limit of detection, 2.5 $\mu\text{g/L}$ [de Carvalho, Lanchote 1991]. Column: 3% SE-30 on 80/200 mesh Gas-Chrom Q (1.8 m $\times$ 2 mm i.d.). Carrier gas: He, 30 mL/min. Temperature: 215°. FID and NPD. Retention time: 8 min. Limit of detection, 1 $\mu\text{g/L}$ [Dhar, Kutt 1981]. Column: 3% OV-17 (10 m $\times$ 0.4 mm i.d.). Carrier gas: Ar: CH_3 (95:5), 10 mL/min. Temperature: 230°. ECD. Limit of detection, 1 $\mu\text{g/L}$ [de Boer *et al.* 1978]. Column: OV-17 Gas Chrom Q on 80/100 mesh. ECD. Retention time: 6.9 min. Limit of detection, 0.5 $\mu\text{g/L}$ [Berlin, Dahlström 1975].

GC-MS Column: HP-5 (12 m $\times$ 0.25 mm i.d., 0.33 μm). Carrier gas: He, 1 mL/min. Temperature programme: 180° to 300° at 30°/min. NCI, SIM acquisition mode. Limit of quantification, 0.25 $\mu\text{g/L}$, limit of detection, 0.1 $\mu\text{g/L}$ [Song *et al.* 1996]. Column: O V-17 on Celite JJ CQ 100/120 mesh (0.9 m $\times$ 4.0 mm i.d.). Carrier gas: Ar: CH_3 (90:10), 80 mL/min. Temperature: 240°. EI ionisation at 70 eV. Limit of detection, 3.0 $\mu\text{g/L}$ [Naestoft, Larsen 1974].

HPLC Column: Symmetry C_{18} (250 $\times$ 4.6 mm i.d.). Mobile phase: acetonitrile: phosphate buffer (1.09 g of potassium dihydrogen phosphate, pH 7.0; 50:50 to 70:30 at 15 min), flow rate 1.0 mL/min. UV detection (λ = 313 nm). Retention time: 6.20. Limit of detection, 2.0 $\mu\text{g/L}$ [Le Guellac *et al.* 1998]. Column: C_{18} Velosep (100 $\times$ 3.2 mm i.d., 3 μm). Mobile phase: acetonitrile: glacial acetic acid: water (32:0.5:67.5), flow rate 0.5 mL/min. UV detection (λ = 0.5 mL/min). Retention time: 7.0 min. Limit of quantification, 5 $\mu\text{g/L}$, limit of detection, 2 $\mu\text{g/L}$ [Sallustio *et al.* 1994]. Column: ODS C_{18} (150 $\times$ 4.0 mm i.d., 3 μm). Mobile phase: water: acetonitrile (1:1), flow rate 0.7 mL/min. UV detection (λ = 313 nm). Limit of detection, 4.0 $\mu\text{g/L}$ [de Carvalho, Lanchote 1991]. Column: μ Bondapak Phenyl (30 cm $\times$ 4 mm i.d.). Mobile phase: acetonitrile: 45 mmol/L potassium dihydrogen phosphate buffer (pH 3.0; 40:60), flow rate 1.5 mL/min. UV detection (λ = 313 nm). Retention time: 6.0 min. Limit of detection, 5 $\mu\text{g/L}$ [Dusci, Hackett 1987]. Column: Supelcosil LC-18 (150 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: acetonitrile: 0.1 mol/L sodium acetate (pH 7.7; 35:65), flow rate 3.0 mL/min. UV detection (λ = 254 nm). Limit of detection, 5 $\mu\text{g/L}$ [Petters *et al.* 1984].

Serum HPLC Column: C_{18} reversed phase TSK gel Super-ODS (1, 100 $\times$ 4.6 mm i.d., 2 μm) or Hypersil ODS (2, 100 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: acetonitrile: 5 mmol/L sodium dihydrogen phosphate (pH 6.0; 45:55), flow rate 0.65 mL/min. UV detection (λ = 254 nm). Retention time: 5.3 and 18.2 min for columns 1 and 2, respectively. Limit of quantification, 5 $\mu\text{g/L}$ [Tanaka *et al.* 1996; Tanaka *et al.* 1998]. Column: μ -Bondapak C_{18} (300 $\times$ 3.9 mm i.d., 10 μm). Mobile phase: acetonitrile: water (40:60), flow rate 1.5 mL/min. UV detection (λ = 254 nm). Retention time: 7.0 min. Limit of detection, 2.0 $\mu\text{g/L}$ [Furuno *et al.* 1991]. Column: Supelco C_8 reversed phase (150 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: acetonitrile: 1.75 mmol/L hydrochloric acid: 50 mmol/L sodium acetate (36:10:54), flow rate 1.5 mL/min. UV detection (λ = 220 nm). Limit of detection, 1.0 $\mu\text{g/L}$ [Zilli, Nisi 1986].

Urine GC-MS Column: DB-17 capillary (30 m $\times$ 0.32 mm i.d., 0.25 μm). Carrier gas: He, 80 kPa. Temperature programme: 150° for 1 min to 230° at 10°/min for 5 min to 300° at 10°/min for 9 min. ECD. Limit of detection, 80 $\mu\text{g/L}$ [Guan *et al.*]. Column: 5% phenyl methyl silicone (25 m $\times$ 0.2 mm i.d., 0.33 μm). Carrier gas: He, 0.5 mL/min. Temperature programme: 240° to 260° at 25°/min to 300° at 30°/min. EI ionisation at 70 eV, SIM acquisition mode. Retention time: 10.1 min. Limit of detection, 0.1 mg/L [Black *et al.* 1994].

HPLC Column: Supelcosil C_{18} (5.0 cm $\times$ 4.6 mm i.d., 5 μm). Mobile phase: water: methanol (52:48), flow rate 1.0 mL/min. UV detection (λ = 230 nm). Limit of detection, 600 $\mu\text{g/L}$ [Mullet, Pawliszyn 2002].

Hair GC-MS Column: HP-5MS capillary (30 m $\times$ 250 μm i.d., 0.25 μm). Carrier gas: He, 1.2 mL/min. Temperature programme: 60° for 1 min to 310° at 30°/min for 3 min. NCI, SIM acquisition mode. Retention time: 10.8 min. Limit of quantification, 10 ng/g, limit of detection, 5 ng/g [Negrusz *et al.* 2000].

Disposition in the Body Rapidly absorbed after oral administration with peak plasma levels found after about 4 h. The major metabolic reaction is catalysed by CYP3A4, is reduction of the nitro group to 7-aminoclonazepam, followed by acetylation to 7-acetamidoclonazepam; 7-aminoclonazepam is slightly active and is found in plasma at concentrations similar to those of clonazepam. 3-Hydroxylation of clonazepam and its 2 metabolites may also occur, followed by conjugation with glucuronic acid or sulfate. Clonazepam crosses the placental barrier and is excreted in breast milk. Less than 1% of a dose is excreted unchanged in the urine in 24 h. Urinary excretion accounts for up to 70% of a dose, as both free and conjugated metabolites, over 7 days, of which about 50% consists of 7-aminoclonazepam and 7-acetamidoclonazepam.

Therapeutic Concentration In plasma, usually in the range 0.02–0.07 mg/L.

A single oral dose of 2 mg given to 8 subjects produced peak plasma concentrations of 0.007–0.024 mg/L (mean 0.017) in 1–4 h [Berlin, Dahlström 1975].

After daily oral dosing of 25 subjects with 6 mg per day, steady-state plasma concentrations of 0.029–0.075 mg/L of clonazepam, 0.023–0.137 mg/L of 7-aminoclonazepam, and <0.003–0.013 mg/L of 7-acetamidoclonazepam were reported [Naestoft, Larsen 1974].

In 4 children whose epilepsy was unsatisfactorily controlled with valproate therapy, a single oral dose of clonazepam 0.05 mg/kg produced peak plasma concentrations of 0.02485–0.1138 mg/L at 1–1.5 h; valproate concentrations remained within the therapeutic range and were unaffected by clonazepam [Wang, Wang 2002].

Toxicity Toxicity is associated with plasma concentrations >0.1 mg/L.

Bioavailability ≈98%.

Half-life Plasma half-life, 20–40 h.

Volume of Distribution ≈3 L/kg.

Distribution in Blood Plasma : whole blood ratio, 1.54

Clearance Plasma clearance, ≈1.5 mL/min/kg.

Protein Binding ≈86%.

Note For reviews of the pharmacokinetics of clonazepam, see Berlin and Dahlström [1976] and Pinder [1976].

Dose 4 to 8 mg daily.

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Clonidine

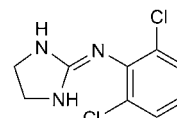
α₂ Adrenoceptor Agonist, Antihypertensive

C₉H₉Cl₂N₃ = 230.1

CAS—4205-90-7

IUPAC Name N-(2,6-Dichlorophenyl)-4,5-dihydro-1H-imidazol-2-amine

Synonyms Chlophazoline; ST-155-BS.



Chemical Properties White powder. Mp 130°. Soluble 1 in 8 of chloroform and 1 in 50 of ether. pK_a 8.3 [Baselt 2008], 8.2 [Nirogi *et al.* 2008]. Log P (octanol/water) 1.59. Stock solutions were stable at room temperature for 6 h and at 2 to 8° for 7 days. Clonidine was stable in plasma at room temperature for at least 4 h and for a minimum of 5 freeze-thaw cycles. Spiked plasma samples were stable for a minimum of 133 days when stored at –20°. Dry extract samples were stable for 19 h at 2 to 8° [Parekh *et al.* 2008]. No degradation was detected when plasma samples were stored at –20° for 12 days [Häring *et al.* 1988]. Stable in human plasma for 3 freeze-thaw cycles when stored below –50° and thawed at room temperature. Stable in plasma for 30 days at below –50°. Stock solutions are stable at room temperature for 6 and 22 h and at –4° for 30 days [Nirogi *et al.* 2008].

Clonidine Hydrochloride

C₉H₉Cl₂N₃ · HCl = 266.6

CAS—4205-91-8

Synonym ST-155

Proprietary Names Aruclonin; Atensina; Barclyd; Caprysin; Catanidin; Catapres (an); Clonistada; Clonnirit; Dispaclonidin; Dixarit; Duraclon; Edolglau; Haemiton; Isoglaucan; Mirfat; Normopresan; Paracefan; Tenso-Timelets. It is an ingredient of Clorpres; Combipres; Combipresan and Dilapres.

Chemical Properties White crystalline powder. Mp 305°. Soluble 1 in 13 of water, 1 in 25 of ethanol, 1 in 38 of dehydrated alcohol and 1 in 5000 of chloroform; practically insoluble in ether.

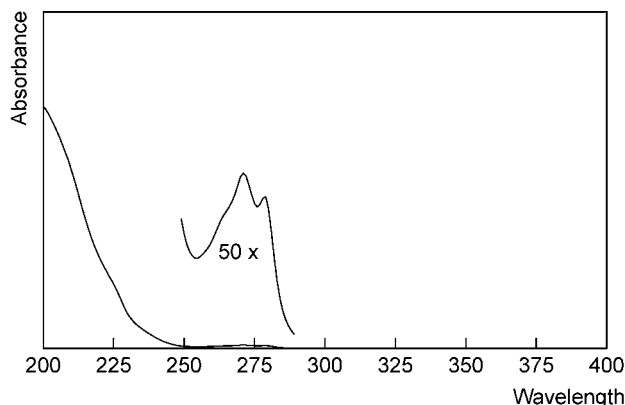
Colour Test Liebermann's reagent—yellow (→orange at 100°).

Thin-layer Chromatography System TA—R_f 0.62; system TB—R_f 0.08; system TC—R_f 0.31; system TE—R_f 0.70; system TL—R_f 0.53; system TAE—R_f 0.44; system TAF—R_f 0.76; system TAJ—R_f 0.09; system TAK—R_f 0.02; system TAL—R_f 0.51 (acidified iodoplatinate solution, positive).

Gas Chromatography System GA—clonidine RI 2090, art (dichlorophenylisocyanate) RI 1350, art (dichlorophenylmethylcarbamate) RI 1500; system GB—clonidine RI 2165.

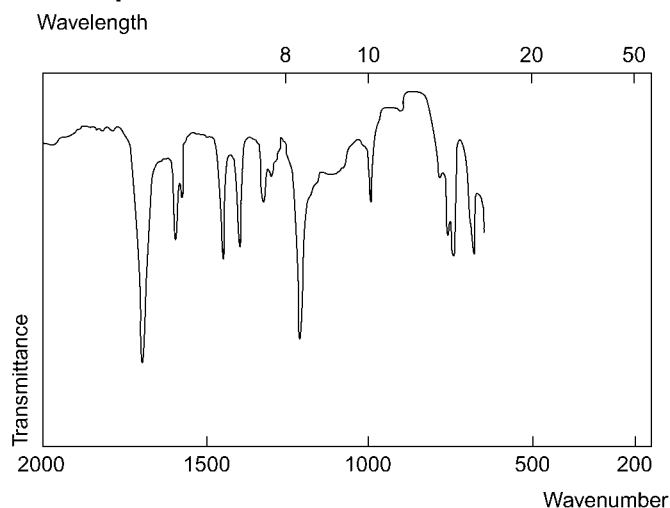
High Performance Liquid Chromatography System HA—k 1.2; system HX—RI 258; system HY—RI 194; system HZ—RT 2.5 min; system HAA—RT 6.1 min; system HAX—RT 7.8 min; system HAY—RT 4.3 min.

Ultraviolet Spectrum Aqueous acid—271 (A₁ = 21a), 278 nm.



Infrared Spectrum Principal peaks at wavenumbers 1653, 1610, 778, 795, 1295, 1568 cm^{–1} (clonidine hydrochloride) (KBr disk).

Infrared Spectrum



Mass Spectrum Principal ions at m/z 105, 77, 51.

Disposition in the Body Little is known on the uptake and distribution of CN following exposure. It is postulated that CN is converted to an electrophilic metabolite that reacts with sulfhydryl-containing enzymes, such as lactic dehydrogenase and nucleophilic sites of macromolecules.

Toxicity CN is the most toxic lachrymator (when compared with others in its class) used in tear gas formulations, causing burning in the eyes, nose and throat as well as a stinging or burning sensation on exposed skin. These symptoms occur almost instantaneously on exposure and generally resolve within ≈ 20 min. However, severe and permanent cornea injury has been demonstrated in laboratory animals. Odour threshold by humans is 0.035 ppm, the nose irritation threshold is 0.034 ppm, and the eye irritation threshold is 0.022 ppm [Amoore, Hautala 1983]. Human ocular irritancy thresholds and toxicity estimates are reported as follows: irritancy threshold 1 mg/m^3 , intolerable concentration 5 mg/m^3 , lethal concentration (10 min exposure) 850 mg/m^3 [Olajos, Salem 2001]. The inhalation toxicity of chemical warfare agents, military chemicals and riot control agents is, by convention, expressed by the notation Ct. It is defined as the product of the concentration in mg/m^3 multiplied by the exposure time (t) in minutes ($\text{mg}\cdot\text{min/m}^3$). The terms LC_{50} and IC_{50} describe the airborne dosages that are lethal (L) or incapacitating (I) to 50% of the exposed population. The minimal irritant concentration estimate in humans is 0.3 to $1 \text{ mg}\cdot\text{min/m}^3$ with LC_{50} and IC_{50} estimates reported as from 8500 to $25\,000 \text{ mg}\cdot\text{min/m}^3$ (also as $7000 \text{ mg}\cdot\text{min/m}^3$) and 20 to $50 \text{ mg}\cdot\text{min/m}^3$, respectively. Animal LC_{50} ($\text{mg}\cdot\text{min/m}^3$) values: rat 8750 (also reported as 3700), mouse 18 200 (also reported as 73 500), rabbit 11 480, guinea pig 13 140 (also reported as 3500). Animal LD_{50} (mg/kg) values: rat 127 (oral), 36 (IP), 41 (IV); mouse 81 (IV); rabbit 118 (oral), 31 (IV); guinea pig 158 (oral), 17 (IP). [Ballantyne, Swanston 1978; National Toxicology Program 1990; Olajos, Salem 2001; Punte *et al.* 1962]. Volunteers exposed to airborne concentrations of CN responded to its short-term irritant properties after 1 min of exposure at 210 mg/m^3 ; 2 min of exposure at 120 mg/m^3 , or 3 min of exposure at 90 mg/m^3 . The maximum safe inhaled dose of CN for humans is estimated at $500 \text{ mg}\cdot\text{min/m}^3$ [Punte *et al.* 1962]. CN has accounted for at least 5 deaths resulting from pulmonary injury and/or asphyxia [Chapman, White 1978; Stein, Kirwan 1964]. OSHA air monitoring/exposure levels for CN are 0.05 ppm for the permissible exposure limit (PEL) and 15 mg/m^3 for the immediately dangerous to life and health (IDLH) limit. Owing to its poor water solubility, CN formulations usually contain solvents as carriers as well as some kind of propellant gas. It is worth noting that some solvents may contribute to unwanted effects of CN use, such as corneal erosion, skin blistering, or even neurotoxicity [Olajos, Stopford 2004].

Note For a review and assessment of the toxicology and carcinogenesis of CN in rats and mice, see National Toxicology Program [1990]. For reviews of riot control agents, see Blain [2003]; Hu *et al.* [1989]; Olajos and Salem [2001]; Olajos and Stopford [2004].

Dose Used as riot control agent, having replaced adamsite (DM).

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Cocaine

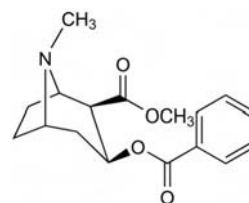
Anaesthetic (Local)

$\text{C}_{17}\text{H}_{21}\text{NO}_4 = 303.4$

CAS—50-36-2

IUPAC Name Methyl-(1S,3S,4R,5R)-3-benzoyloxy-8-methyl-8-azabicyclo-[3.2.1]octane-4-carboxylate

Synonyms Cocaina; ethyl-[1R-(exo,exo)]-3-(benzoyloxy)-8-methyl-8-azabicyclo-[3.2.1]octane-2-carboxylate; methyl benzoylcgonine. For names used to describe various forms of cocaine, see Sweetman [2007].



Chemical Properties An alkaloid obtained from coca, the dried leaves of *Erythroxylum coca* and other species of *Erythroxylum* (Erythroxylaceae) or by synthesis from ecgonine. A white crystalline, slightly volatile, powder. Mp 98° . Soluble 1 in 600 of water (1 in 270 of water at 80°), 1 in 6.5 of ethanol, 1 in 0.7 of chloroform and 1 in 3.5 of ether; also soluble in acetone, ethyl acetate and carbon disulfide. pK_a 8.7 [Needham *et al.* 2000], 8.6 (20°). Log P (octanol/water), 2.3. Extraction yield (chlorobutane), 1 [Demme *et al.* 2005]. Hydrolysed when stored at -16° [Javadi *et al.* 1978]. Brain tissue samples show degradation of 48% when stored at 4° for 1 month and 90% when stored for 2 months. Traces of cocaine were still detectable after 144 days storage at 4° [Hernandez *et al.* 1994].

Cocaine Hydrochloride

$\text{C}_{17}\text{H}_{21}\text{NO}_4 \cdot \text{HCl} = 339.8$

CAS—53-21-4

Synonyms Cocaininum chloratum; cocaini hydrochloridum.

Chemical Properties Hygroscopic colourless crystals or white crystalline powder. Mp $\sim 195^\circ$, with decomposition. Soluble 1 in 0.4 of water, 1 in 3.2 of cold alcohol, 1 in 2 of hot alcohol and 1 in 12.5 of chloroform; also soluble in glycerol and acetone; practically insoluble in ether.

Colour Tests Cobalt thiocyanate—blue; *p*-dimethylaminobenzaldehyde (100° for 3 min)—red.

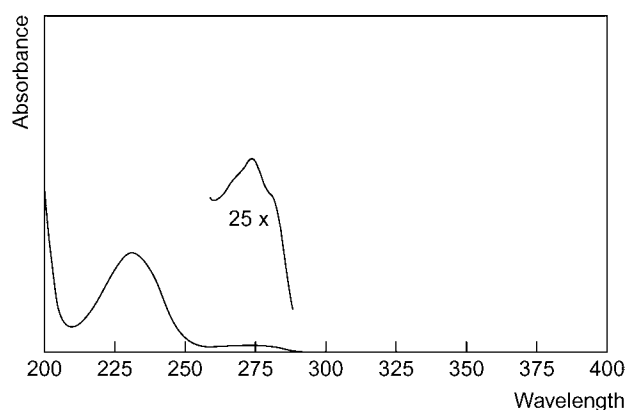
Thin-layer Chromatography System TA— R_f 0.65; system TB— R_f 0.45; system TC— R_f 0.47; system TE— R_f 0.77; system TL— R_f 0.54; system TAE— R_f 0.35; system TAF— R_f 0.30; system TAJ— R_f 0.13; system TAK— R_f 0.00; system TAL— R_f 0.02 (acidified iodoplatinate solution—positive; Dragendorff spray—positive).

Plate: silica 60 F₂₅₄ ($10 \times 10 \text{ cm}$). Mobile phase: hexane : toluene : diethylamine (65 : 20 : 5). UV detection. R_f 0.36 [Antonilli *et al.* 2001].

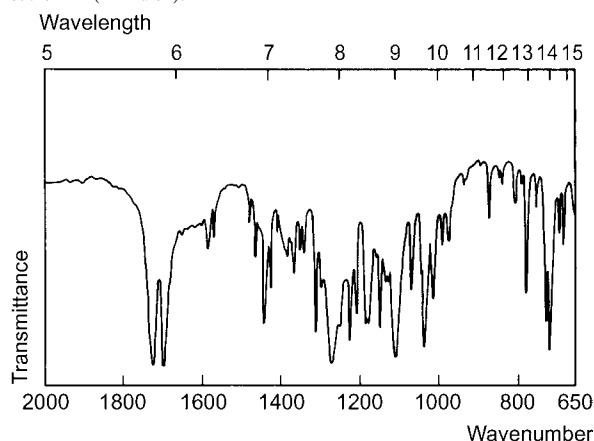
Gas Chromatography System GA—cocaine RI 2187, M (benzoylcgonine) RI 2570, M (ecgonine) not eluted, M (anhydroecgonine methylester) RI 1280, M (methylecgonine) RI 1472, M (nor-) RI 2162, M (*m*-OH-) RI 2460, M (OH-methoxy-) RI 2670, M (cocaethylene) RI 2250, M (nor-cocaethylene) RI 2115; system GB—cocaine RI 2289, M (ecgonine) not eluted, M (anhydroecgonine methylester) RI 1430, M (norecgonine) RI 1472, M (methylecgonine) RI 1530, M (ethylecgonine) RI 1602, M (nor-) RI 2259, M (benzoylcgonine) RI 2663, M (*m*-OH-) RI 2608, M (*p*-OH-) RI 2650, M (OH-methoxy-) RI 2729, M (cocaethylene) RI 2345, M (norcocaethylene) RI 2317, M (OH-cocaethylene) RI 2709, M (OH-methoxycocaethylene) RI 2779, M (methoxycocaethylene) RI 2663; system GF—RI 2550; system GQ—RT 7.85 min; system GAK—RT 12.7 min.

High Performance Liquid Chromatography System HA—cocaine k 2.8, M (benzoylcgonine) k 0.9 (tailing peak), M (ecgonine) k 1.1; system HQ—cocaine k 2.68, M (benzoylcgonine) k 5.68; system HX—RI 348; system HY—RI 289; system HZ—RT 3.3 min; system HAA—RT 11.9 min; system HAX—RT 10.0 min; system HAY—RT 5.0 min; system HBC—RT 5.2 min.

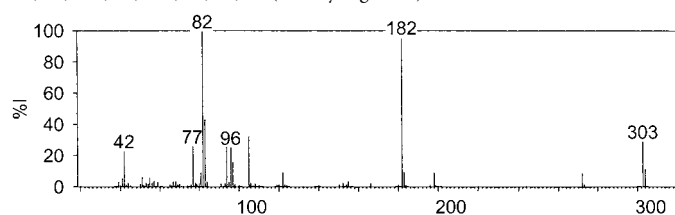
Ultraviolet Spectrum Aqueous acid—233 ($A_1 = 430a$), 275 nm.



Infrared Spectrum Principal peaks at wavenumbers 1710, 1738, 1275, 1110, 712, 1037 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at m/z 82, 182, 83, 105, 303, 77, 94, 96 (cocaine); 124, 82, 168, 77, 105, 42, 94, 83 (benzoylecgonine).



Quantification

Blood GC Column: 5% OV-225 on Gas-Chrom Q 100/120 mesh (3.0 m [10']). Temperature: 110°. ECD. Limit of detection not reported [Javaid *et al.* 1978]. AFID. Limit of detection, 0.02 mg/L [Dvorchik *et al.* 1977].

GC-MS SIM acquisition mode [Fleming *et al.* 2010]. SIM acquisition mode. Limit of detection, 25 $\mu\text{g/L}$ [Cardona *et al.* 2006]. Limit of detection, 0.5–4.0 $\mu\text{g/L}$ [Paul *et al.* 2005]. FID. Limit of detection, 71 $\mu\text{g/L}$ [Fernandez *et al.* 2004]. Column: DB-35 ms (30 m $\times$ 0.32 mm i.d., 0.25 μm). Carrier gas: He, 1.5 mL/min. Temperature programme: 120° for 1 min to 320° at 15°/min for 4 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of quantification, 10 $\mu\text{g/L}$, limit of detection, 2 $\mu\text{g/L}$ [Gunnar *et al.* 2004]. See also Abusada *et al.* [1993]; Aderjan *et al.* [1993]; Chasin, Midio [2000]; Corburt, Koves [1994]; and Smirnow, Logan [1996].

HPLC Column: Varian Pursuit C_{18} (100 $\times$ 3 mm i.d., 3 μm). Mobile phase: methanol:ammonium formate. Limit of quantification, 0.0005–0.01 mg/kg [Bjork *et al.* 2010]. Column: Zorbax C_{18} . Mobile phase: formic acid: water: acetonitrile. MRM acquisition mode. Limit of quantification, 0.008 mg/kg [Johansen, Bhatia 2007]. Column: Spherisorb 5 RP 8S (100 $\times$ 2.1 mm i.d., 5 μm). Mobile phase: water: acetonitrile containing 0.1% formic acid and 2 mmol/L ammonium formate (50:50), flow rate 400 $\mu\text{L/min}$. ESI, MRM acquisition mode. Limit of quantification, 5 $\mu\text{g/L}$ [Cailleux *et al.* 1999].

LC-MS Column: Gemini C6-Phenyl (50 $\times$ 3.0 mm i.d., 5 μm). Mobile phase: 0.1% formic acid in water:0.1% formic acid in acetonitrile (100:0 for 3.01 min to 98:2 at 3.05 min to 10:90 at 7.35 min until 8 min to 100:0 at 9.15 min until 10 min), flow rate 0.9 mL/min to 1.0 mL/min at 3.05 min until 10 min. TIS, positive ion mode, MRM acquisition mode. Limit of quantification, 8 $\mu\text{g/L}$, limit of detection, 3 $\mu\text{g/L}$ [Jagerdeo *et al.* 2008].

Plasma GC Column: HP Ultra-2 (12 m $\times$ 0.2 mm i.d., 0.33 μm). Temperature programme: 200° for 2 min to 280° at 30°/min until 11 min. NPD. Limit of

quantification, 4 $\mu\text{g/L}$ [Virag *et al.* 1994]. See Blood [Dvorchik *et al.* 1977; Javaid *et al.* 1978].

GC-MS Column: 5% phenylmethylsiloxane (12 m $\times$ 200 μm i.d., 0.33 μm). Temperature programme: 90° for 2 min to 215° at 30°/min for 5 min to 260° at 30°/min. SIM acquisition mode. Limit of quantification, 25 $\mu\text{g/L}$, limit of detection, 19 $\mu\text{g/L}$ [Alvarez *et al.* 2007]. See Blood [Gunnar *et al.* 2004]. Column: DB-1 (15 m $\times$ 0.53 mm i.d., 1.5 μm). Carrier gas: He, 15 mL/min. Temperature programme: 140° to 300° at 16°/min for 10 min. EI ionisation, positive ion mode, SIM acquisition mode (m/z 82). Limit of quantification, 100 $\mu\text{g/L}$. Limit of detection, 80 $\mu\text{g/L}$ [Ohshima, Takayasu 1999]. Column: HP-1 (12 m $\times$ 0.2 mm i.d., 0.33 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 70° for 1 min to 220° at 35°/min for 0.25 min to 250° at 10°/min for 3 min. SIM acquisition mode. Limit of detection, 1 $\mu\text{g/L}$ [Wang *et al.* 1994]. Column: HP-1 (12 m $\times$ 0.2 mm i.d., 0.33 μm). Carrier gas: He, 0.8 mL/min. Temperature programme: 70° to 220° at 35°/min for 15 s to 250° at 10°/min for 15 s to 260° at 3.5°/min for 15 s. Limit of detection, 1 $\mu\text{g/L}$ [Cone *et al.* 1994]. See also Abusada *et al.* [1993].

HPLC DAD. Limit of detection, 0.05 mg/L [Fernandez *et al.* 2007]. See Blood [Cailleux *et al.* 1999]. Column: Lichrospher RP18 (125 $\times$ 4 mm i.d.). Mobile phase: methanol (pH 7): phosphate buffer (70:30), flow rate 0.4 mL/min to 0.7 mL/min at 6 min to 1.0 mL/min at 8.0 min to 0.7 mL/min at 10 min to 0.7 mL/min at 11.0 min to 0.6 mL/min at 12 min to 0.4 mL/min. UV detection (λ =235 nm). Retention time: 8.8 min. Limit of detection, 12.5 $\mu\text{g/L}$ [Fernandez *et al.* 1996]. See also Evans, Morarity [1980] and Tagliaro *et al.* [1994].

LC-MS TIS, positive ion mode [Sergi *et al.* 2009]. Column: Zorbax Eclipse XDB-C₈ (150 $\times$ 2.1 mm i.d., 5 μm). Mobile phase: 20 mmol/L ammonium acetate buffer: acetonitrile-methanol (1:1, 99:1 for 3 min to 20:80 at 4 min for 11 min), flow rate 0.3 mL/min. TIS, MRM acquisition mode. Limit of quantification, 14.4 $\mu\text{g/L}$, limit of detection, 4.1 $\mu\text{g/L}$ [Klingmann *et al.* 2001]. Column: fused silica (500 $\times$ 0.1 mm i.d.) followed by C_{18} (30 $\times$ 4.6 mm i.d., 3 μm). Mobile phase: 25 mmol/L ammonium acetate in methanol water (50:50), flow rate 1.0 mL/min. APCI, MRM acquisition mode. Limit of detection, 2 $\mu\text{g/L}$ [Sosnoff *et al.* 1996].

Serum GC-MS Limit of quantification, 25 $\mu\text{g/L}$ [Maresova *et al.* 2008]. See Blood [Gunnar *et al.* 2004]. See Weinmann *et al.* [2000].

Urine GC Limit of quantification, 5 $\mu\text{g/L}$, limit of detection, 20 $\mu\text{g/L}$ [Farina *et al.* 2002]. See Blood [Javaid *et al.* 1978].

GC-MS Column: Oasis hydrophilic-lipophilic balance (20 $\times$ 3.9 mm i.d.) [Brunetto *et al.* 2010]. See Blood [Fleming *et al.* 2010]. Column: HP-5 (30 m $\times$ 250 μm i.d., 0.25 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 90° for 1 min to 190° for 1 min at 30°/min to 260° at 4°/min for 4 min to 290° for 10 min. EI ionisation at 70 eV, SIM acquisition mode. Retention time: 13.8 min. Limit of quantification, 20 ng/g, limit of detection, 5 ng/g [Lopez *et al.* 2009]. Column: HP-5MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Temperature programme: 100° for 3 min to 300° at 20°/min for 3 min. Limit of quantification, 100 $\mu\text{g/L}$, limit of detection, 50 $\mu\text{g/L}$ [Saito *et al.* 2007]. See Blood [Fernandez *et al.* 2004; Paul *et al.* 2005; Wang *et al.* 1994]. See also Aderjan *et al.* [1993], Cardenas *et al.* [1996], Cone *et al.* [1994], De Giovanni, Strano Rossi [1994], de la Torre *et al.* [1995], Elsohly *et al.* [1998], Gaillard, Pepin [1997], Garside *et al.* [1997], Gerlits [1993], Hall *et al.* [1999], Karacic, Skender [2000], Lombardero *et al.* [1993], Ohshima, Takayasu [1999], and Yonamine, Saviano [2006].

HPLC Limit of detection, 0.5 $\mu\text{g/L}$ [Jeanville *et al.* 2001]. Column: LiChroCART-LiChrospher 100 RP-18 (250 $\times$ 4 mm i.d., 5 μm). Mobile phase: 0.045 mol/L ammonium acetate solution in water-methanol-acetonitrile (80:10:10):0.045 mol/L ammonium acetate solution in water-methanol-acetonitrile (20:40:40, 100:0 to 47.2:52.8 in 20 min), flow rate 1 mL/min. UV detection, (λ =235 nm). Retention time: 14.8 min. Limit of quantification, 200 $\mu\text{g/L}$ [Antonilli *et al.* 2001]. See Blood [Cailleux *et al.* 1999]. See also Clauwaert *et al.* [1996], Nakashima *et al.* [1992], Nishikawa *et al.* [1994], and Roy *et al.* [1992].

LC-MS Limit of quantification, 25 $\mu\text{g/L}$ [Shakleya *et al.* 2010]. Column: XDB-C₈ (50 $\times$ 2.1 mm i.d., 1.8 μm). Mobile phase: 20 mmol/L ammonium formate (pH 2.7): methanol-acetonitrile (50:50). Flow rate 270 $\mu\text{L/min}$. Limit of detection, 1.0 $\mu\text{g/L}$ [Langman *et al.* 2009]. Column: Restek (30 $\times$ 2.1 mm i.d., 5 μm). Mobile phase: acetonitrile:5 mmol/L ammonium formate (pH 3.0). ESI, CID, MRM acquisition mode. Limit of quantification, 5.3 $\mu\text{g/L}$, limit of detection, 1.6 pg [Needham *et al.* 2000]. See Plasma. Limit of detection, 5 $\mu\text{g/L}$ [Fernandez *et al.* 1996].

Amniotic Fluid HPLC See Moore *et al.* [1992].

Breast Milk GC-MS Column: HP-5MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 90° for 1 min to 220° at 30°/min for 0.5 mL/min to 330° at 20°/min. SIM acquisition mode. Limit of quantification, 10 $\mu\text{g/L}$, limit of detection, 2.5 ng/mL [Winecker *et al.* 2001].

Sweat GC-MS Column: HP-1 (12 m $\times$ 0.2 mm i.d., 0.33 μm). Limit of quantification, 2.5 ng/patch, limit of detection, 1.25 ng/patch [Huestis *et al.* 1999].

Meconium GC-MS See Urine [Lopez *et al.* 2009]. Column: DB-1 (30 m $\times$ 0.32 mm i.d., 0.25 μm). Carrier gas: He, 6 psi. Temperature programme: 50° for 1 min to 280° at 33°/min. Limit of detection, 11 $\mu\text{g/L}$ [Lombardero *et al.* 1993]. See Blood [Abusada *et al.* 1993]. Limit of quantification, <0.25 $\mu\text{g/g}$ [Clark *et al.* 1992].

HPLC Column: C_{18} (100 $\times$ 4.6 mm i.d., 3 μm). Mobile phase: 0.01 mol/L sodium dihydrogen phosphate buffer (pH 2.0)-58 $\mu\text{mol/L}$ tetrabutylammonium hydroxide:acetonitrile (87:13), flow rate 1.0 mL/min. Limit of quantification, 50 ng/g [Murphy *et al.* 1993].

Oral Fluid LC-MS See Plasma [Sergi *et al.* 2009].

GC-MS See Plasma [Cone *et al.* 1994; Wang *et al.* 1994].

Bone GC-MS See McGrath, Jenkins [2009].

Brain GC-MS Column: DB 5 MS (15 m × 0.25 mm i.d., 0.25 μm). Temperature programme: 115° for 1.3 min to 290° at 15°/min for 2 min. EI ionisation, full scan mode. Limit of quantification, 50 ng/g, limit of detection, 25 ng/g [Hernandez *et al.* 1994].

HPLC See Browne *et al.* [1991].

Hair GC-MS See Gambelunghe *et al.* [2005]. Column: DB-5MS (15 m × 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 130° to 270° at 25°/min to 280° at 6°/min to 300° at 35°/min. PICI. Limit of quantification, 0.05 ng/mg, limit of detection, 0.01 ng/mg [Bourland *et al.* 2000]. Column: RTX 5 5% diphenylpolysiloxane: 95% dimethylpolysiloxane (30 m × 0.25 mm i.d.). Carrier gas: He, 1.0 mL/min. Temperature programme: 50° for 1 min to 300° at 50°/min for 6 min. SIM acquisition mode. Limit of detection, 0.2 ng/mg [Skender *et al.* 2002]. See Brewer *et al.* [2001], Girod, Staub [2000], Hold *et al.* [1998a], Hold *et al.* [1998b], Kintz, Mangin [1995], Pichini *et al.* [1999]. See Plasma [Moeller *et al.* 1993; Wang *et al.* 1994].

HPLC Column: Hypersil BDS-C₁₈ (150 × 2.1 mm i.d., 3 μm). Mobile phase: 45 mmol/L ammonium acetate-methanol: acetonitrile (84 : 8 : 8): methanol-acetonitrile-water (42 : 42 : 16, 100 : 0 for 7 min to 65 : 35 in 27 min for 3 min), flow rate 0.2 mL/min. Fluorescence detection (λ_{ex} = 242 nm, λ_{em} = 315 nm). Limit of detection, 0.2 ng/mg [Clauwaert *et al.* 2000]. Column: Hypersil BDS-C₁₈ (125 × 2.1 mm i.d., 3 μm). Mobile phase: ammonium acetate buffer: methanol: acetonitrile. Fluorescence detection (λ_{ex} = 242 nm, λ_{em} = 315 nm) [Clauwaert *et al.* 1998]. Column: Polystyrene divinylbenzene (5 μm). Mobile phase: 0.1 mol/L potassium phosphate (pH 3.0): methanol: tetrahydrofuran (70 : 25 : 5), flow rate 0.5 mL/min. Fluorescence detection (λ_{ex} = 230 nm, λ_{em} = 315 nm). Limit of detection, 1.5 μg/L [Tagliaro *et al.* 1993].

LC-MS Column: Zorbax SB-Phenyl (100 × 2.1 mm i.d., 3.5 μm). 25 mmol/L formic acid: 100% acetonitrile (98 : 2 to 80 : 20 for 3 min to 10 : 90 in 4 min for 2 min), flow rate 0.25 mL/min. ESI, positive ion mode, MRM acquisition mode. Limit of quantification, 0.0125 ng/mg [Hegstad *et al.* 2008].

Disposition in the Body Cocaine is normally only used as a surface anaesthetic in the eye, ear, nose and throat because of the possibility of systemic toxic effects when given by other routes. Addicts may inject it or use it as a snuff; it is, however, increasingly misused in its smokeable form ('crack cocaine') owing to the rapid absorption of the substance from the lungs. It is less toxic when taken orally owing to hydrolysis in the gastrointestinal tract. The main metabolites are benzoylecgonine, ecgonine and ecgonine methyl ester, all of which are inactive; some norcocaine, an active metabolite, may also be produced; other metabolites which have been reported are ethylecgonine, hydroxycocaine and methylecgonidine. Approximately 1–9% of a daily IV dose of 120 mg cocaine is excreted unchanged in the urine, together with 35–55% as benzoylecgonine. The excretion of unchanged drug is increased when the urine is acid; some ecgonine may also be present. After an intranasal dose of 1.5 mg/kg, up to 4% of the dose is excreted in the urine unchanged in 24 h, and 16–36% of the dose is excreted as benzoylecgonine. Approximately 30–60% of an oral, intranasal or intravenous dose is excreted in the urine as ecgonine methyl ester. No unchanged drug is eliminated in the faeces. Cocaine crosses the placenta and is found in breast milk.

Therapeutic Concentration

Six males with a history of cocaine abuse were administered with a single cocaine dose by the following routes: 25 mg cocaine hydrochloride IV, 32 mg cocaine hydrochloride intranasal and 42 mg cocaine base (smoked). Peak plasma concentrations of benzoylecgonine (BZE), ecgonine methyl ester (EME), norcocaine (NCOC), benzoynorecgonine (BNE), *meta*- and *para*-hydroxycocaine (*m*-HOCOC, *p*-HOCOC), *meta*- and *para*-hydroxybenzoylecgonine (*m*-HOBZE, *p*-HOBZE) and anhydroecgonine methyl ester (AEME) were determined by GC-MS and were as follows:

	IV (n = 6)		Intranasal (n = 6)		Smoked (n = 5)	
	Peak conc. (ng/mL)	Time (h)	Peak conc. (ng/mL)	Time (h)	Peak conc. (ng/mL)	Time (h)
Cocaine	775	3.9	412	5.1	707	2.6
BZE	15611	5.6	13681	7.8	9395	4.1
EME	4968	5	5831	5	3193	4.1
NCOC	19	5.1	11	4.3	4	3.3
BNE	255	0.6	409	8.5	160	5.1
<i>m</i> -HOCOC	43	0.08	47	5.3	48	3.7
<i>p</i> -HOCOC	6	0.01	7	4.6	1	4.8
<i>m</i> -HOBZE	131	0.45	208	10.2	131	6.9
<i>p</i> -HOBZE	247	0.43	353	7.5	258	4.1
AEME	0	n/a	0	n/a	23	2.3

Total percentage dose excreted in urine for the IV, intranasal and smoked routes were 57.1, 45.4 and 24.9%, respectively [Cone *et al.* 1998]. Ten volunteers were administered with single and multiple doses of cocaine; IV and by inhalation. On 4 separate study days, the volunteers smoked 2 doses of 25 mg cocaine with a 14-min interval (day 1), smoked 2 doses of 50 mg cocaine (day 2), were administered 2 doses of IV 16 mg cocaine with a 14-min

period between (day 3), and were again given 2 doses of IV 32 mg cocaine (day 4). Peak plasma concentrations of cocaine and benzoylecgonine were determined and were as follows:

		Cocaine	Benzoylecgonine
IV			
2 × 16 mg	peak (mg/L)	0.23	0.24
	time (min)	18	90
2 × 32 mg	peak (mg/L)	0.47	0.4
	time (min)	18	90
Smoked			
2 × 25 mg	peak (mg/L)	0.21	0.2
	time (min)	18	60
2 × 50 mg	peak (mg/L)	0.38	0.35
	time (min)	18	90

After multiple dosing in 3 volunteers (316-mg dose administered over 90 min), the peak plasma cocaine concentration was 1.2 mg/L [Isenschmid *et al.* 1992].

Toxicity The estimated minimum lethal dose is 1.2 g but susceptible persons have died from doses as small as 30 mg when applied to mucous membranes; addicts may be able to tolerate up to 5 g a day. Toxic effects have been noted with blood concentrations in the range 0.25–5 mg/L and fatalities have occurred with concentrations of 1 mg/L or more.

Postmortem concentrations of cocaine and benzoylecgonine in a 27-year-old man who died following the rupture in his gastrointestinal tract, of 4 cocaine-containing packages were reported as: blood 4.0 and 17.0 mg/L, urine 152.0 and 512 mg/L, bile 99.8 and 54.0 mg/L, vitreous humour 7.1 and 5.8 mg/L, brain 7.5 and 3.5 mg/g, and hair 55.5 and 27.7 μg/g [Furnari *et al.* 2002].

A 17-year-old male drug dealer died ~90 min after swallowing a package containing cocaine; the following postmortem tissue concentrations were reported for cocaine, benzoylecgonine and ecgonine methyl ester, respectively: blood 98.1, 86.1 and 51.4 mg/L, liver 128.1, 134.3 and 20.3 μg/g, kidney 92.5, 8.6 and 6.2 μg/g, brain 99.1, 32.1 and 17.6 μg/g, spleen 72.6, 4.7 and 6.9 μg/g, lung 121.3, 89.4 and 36.4 μg/g, and urine 10, 3.3 and 4.2 mg/L [Fineschi *et al.* 2002].

A teenage drug courier who died after a 12-h flight was found to have 24 packages of cocaine in her intestinal tract, several of which had burst or become damaged. Her postmortem blood concentration of cocaine was 104 mg/L [Patel 1996].

A blood cocaine concentration of 330 mg/L was found in a 26-year-old female who died as a result of cocaine intoxication; the concentrations of benzoylecgonine and ecgonine methyl ester were 50 and 18 mg/L, respectively. Such high concentrations suggested a massive administration but the manner of death was undetermined [Peretti *et al.* 1990].

Half-life Plasma half-life, 0.7–1.5 h (dose dependent).

Volume of Distribution ~1–3 L/kg.

Clearance Plasma clearance, 10–32 mL/min/kg (dose dependent).

Dose Cocaine hydrochloride is used in concentrations of 1 to 25%, for local anaesthesia.

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Aderjan RE *et al.* (1993). Determination of cocaine and benzoylecgonine by derivatization with iodomethane-D3 or PFPA/HFIP in human blood and urine using GC/MS (EI or PCI mode). *J Anal Toxicol* 17: 51–55.

Alvarez I *et al.* (2007). Determination of cocaine and cocaethylene in plasma by solid-phase micro-extraction and gas chromatography-mass spectrometry. *J Chromatogr B Analyt Technol Biomed Life Sci* 845: 90–94.

Antonilli L *et al.* (2001). Analysis of cocaethylene, benzoylecgonine and cocaine in human urine by high-performance thin-layer chromatography with ultraviolet detection: a comparison with high-performance liquid chromatography. *J Chromatogr B Biomed Sci Appl* 751: 19–27.

Bjork MK *et al.* (2010). Determination of 19 drugs of abuse and metabolites in whole blood by high-performance liquid chromatography-tandem mass spectrometry. *Anal Bioanal Chem* 396: 2393–2401.

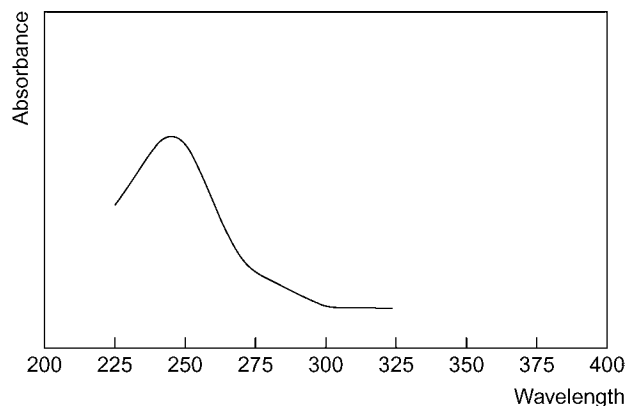
Bourland JA *et al.* (2000). Quantitation of cocaine, benzoylecgonine, cocaethylene, methylecgonine, and norcocaine in human hair by positive ion chemical ionization (PICI) gas chromatography-tandem mass spectrometry. *J Anal Toxicol* 24: 489–495.

Brewer WE *et al.* (2001). Analysis of cocaine, benzoylecgonine, codeine, and morphine in hair by supercritical fluid extraction with carbon dioxide modified with methanol. *Anal Chem* 73: 2371–2376.

Browne SP *et al.* (1991). A rapid method for the determination of cocaine in brain tissue. *J Forensic Sci* 36: 1662–1665.

Brunetto MD *et al.* (2010). Analysis of cocaine and benzoylecgonine in urine by using multi-syringe flow injection analysis-gas chromatography-mass spectrometry system. *J Sep Sci* 33: 1779–1786.

Cailleux A *et al.* (1999). Determination of opiates and cocaine and its metabolites in biological fluids by high-performance liquid chromatography with electrospray tandem mass spectrometry. *J Anal Toxicol* 23: 620–624.



Infrared Spectrum Principal peaks at wavenumbers 1653, 1508, 1529, 1238, 1136, 1597 cm^{-1} (KBr disk).

Diamorphine

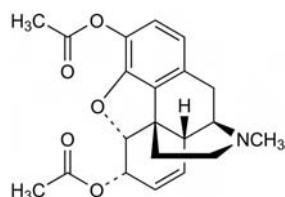
Narcotic Analgesic

$\text{C}_{21}\text{H}_{23}\text{NO}_5 = 369.4$

CAS—561-27-3

IUPAC Name (5 α ,6 α)-7,8-Didehydro-4,5-epoxy-17-methylmorphinan-3,6-diol diacetate (ester)

Synonyms Acetomorphine; diacetylmorphine; heroin.



Chemical Properties White crystals. Mp 173°. It is rapidly hydrolysed by alkalis. Soluble 1 in 1700 of water, 1 in 31 of ethanol, 1 in 1.5 of chloroform and 1 in 100 of ether. pK_a 7.6 (23°) Log *P* (ether/water pH 7.0), 0.2.

Diamorphine Hydrochloride

$\text{C}_{21}\text{H}_{23}\text{NO}_5 \cdot \text{HCl} \cdot \text{H}_2\text{O} = 423.9$

CAS—1502-95-0 (anhydrous)

Proprietary Names *Diagesil*; *Diamorf*.

Chemical Properties An almost white crystalline powder. Mp 243° to 244°. Soluble 1 in 2 of water, 1 in 11 of ethanol and 1 in 1.6 of chloroform; insoluble in ether. Diamorphine hydrolyses in aqueous solution to 3-*O*- and 6-*O*-monoacetylmorphine and morphine to a significant extent at room temperature.

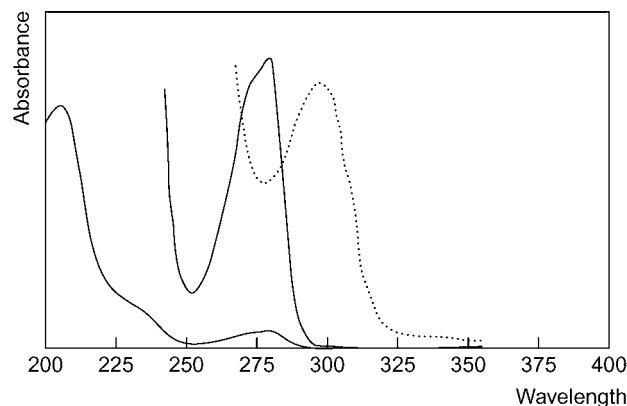
Colour Test Liebermann's reagent—black; Mandelin's test—blue-grey; Marquis test—violet.

Thin-layer Chromatography System TA— R_f 0.47; system TB— R_f 0.15; system TC— R_f 0.38; system TE— R_f 0.49; system TL— R_f 0.04; system TAE— R_f 0.26; system TAF— R_f 0.33; system TAJ— R_f 0.25; system TAK— R_f 0.05; system TAL— R_f 0.64 (acidified iodoplatinate solution—positive; Dragendorff spray—positive; Marquis test—violet).

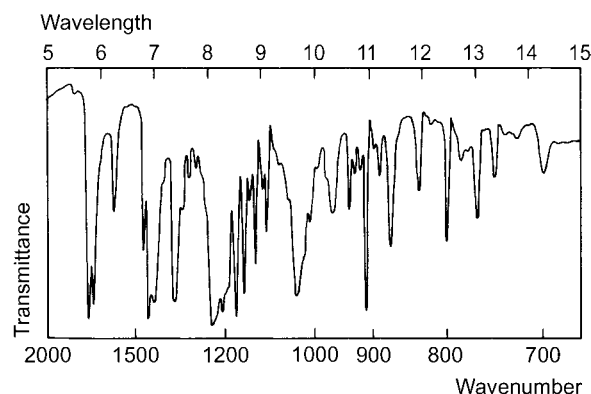
Gas Chromatography System GA—RI 2615, M (6-monoacetylmorphine) RI 2525, M (3-monoacetylmorphine) RI 2495, M (morphine) RI 2445; system GB—RI 2769, M (6-monoacetylmorphine) RI 2646, M (3-monoacetylmorphine) RI 2625, M (morphine) RI 2564; system GAK—RT 14.2 min.

High Performance Liquid Chromatography System HA— k 3.0 (tailing peak), M (6-monoacetylmorphine) k 3.6 (tailing peak), M (morphine) 3.8 (tailing peak); system HC— k 0.66, M (6-monoacetylmorphine) k 0.80, M (morphine) k 1.30; system HS— k 0.35, M (6-monoacetylmorphine) k 1.00, M (morphine) k 5.16; system HX—RI 340; system HY—RI 282; system HAX—RT 7.9 min; system HAY—RT 4.1 min; system HBC—RT 2.9 min.

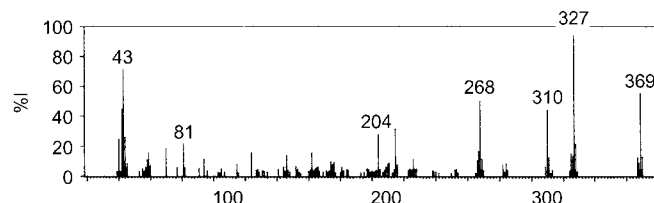
Ultraviolet Spectrum Aqueous acid—279 nm ($A_1^1 = 46a$); aqueous alkali—299 nm ($A_1^1 = 69a$)



Infrared Spectrum Principal peaks at wave numbers 1245, 1764, 1178, 1215, 911, 1736 cm^{-1} (diamorphine hydrochloride, Nujol mull). Two polymorphic forms may occur.



Mass Spectrum Principal ions at m/z (diamorphine) 327, 43, 369, 268, 310, 42, 215, 204; (6-monoacetylmorphine) 327, 268, 42, 43, 215, 44, 328, 269; (morphine) 285, 162, 42, 215, 286, 124, 44, 284.



Quantification

Blood GC Column: 2% OV-17 (2.13 m $\times$ 4.0 mm i.d.). Carrier gas: N_2 , 65 mL/min. Temperature: 250°. AFID. Limit of detection, 100 $\mu\text{g/L}$ [Smith, Cole 1975].

GC-MS Column: XTerra C_{18} . MRM acquisition mode, positive ion mode. Limit of quantification, 0.0007–0.02 mg/L [Karinen *et al.* 2009]. Column: 95% dimethylpolysiloxane, 5% diphenylpolysiloxane Rtx (15 m $\times$ 25 mm i.d., 0.1 μm). Carrier gas: He, 1.2 mL/min. Temperature programme: 150° for 1 min to 200° at 12.5°/min for 15 s to 290° at 30°/min for 4 min. SIM acquisition mode. Limit of quantification, 1 $\mu\text{g/L}$ [Goldberger *et al.* 1993].

HPLC Column: LiChrosorb Si-60 (300 $\times$ 4 mm i.d., 5 μm). Mobile phase: acetonitrile: methanol (pH 7, 75:25), flow rate 80 mL/h. UV detection ($\lambda = 218$ nm). Limit of detection, 12.5 $\mu\text{g/L}$ [Umans *et al.* 1982].

LC-MS Column: Gemini C_{18} reversed phase (100 $\times$ 2.0 mm i.d., 3 μm). Mobile phase: acetonitrile: ammonium acetate (pH 3.2, 15:85 for 9 min to 30:70 in 13 min to 80:20 in 10 min to 95:5 in 1 min), flow rate 150 $\mu\text{L/min}$. TIS, MRM acquisition mode. Limit of quantification, 0.008 mg/L, limit of detection, 0.002 mg/L [Gergov *et al.* 2009].

Plasma GC-MS Column: HP-1 (12 m $\times$ 0.2 mm i.d., 0.33 μm). Carrier gas: He, 1 mL/min. Temperature programme: 70° for 1 min to 220° at 35°/min for 0.25 min to 250° at 10°/min for 3 min. MSD, SIM acquisition mode. Limit of detection, 1 $\mu\text{g/L}$ [Wang *et al.* 1994]. See Blood [Goldberger *et al.* 1993].

HPLC Column: Spherisorb C_{18} ODS-2 (125 $\times$ 2 mm i.d., 3 μm). Mobile phase: water containing 8.5 g *o*-phosphoric acid (85%) and 0.39 g hexylamine/

1000 g; acetonitrile-water containing 8.5 g *o*-phosphoric acid and 1.1 g hexylamine/1000 g (98:2 to 97.5:2.5 at 2.5 min to 82.5:17.5 at 2.8 min to 97.5:2.5 in 3.5 min to 92.5:7.5 at 6 min to 85.5:14.5 at 7.5 min to 75:25 at 11 min), flow rate 0.2 mL/min. DAD (λ =210 nm). Limit of quantification, 25 µg/L [Bourquin *et al.* 1997].

LC-MS Column: Zorbax reversed phase (150 × 4.6 mm i.d., 5 µm). Mobile phase: 5 mmol/L ammonium formate (pH 4.0):acetonitrile (97:3 for 2 min to 87:13 at 2.6 min to 84.5:15.5 at 8 min to 20:80 at 8.1 min until 11 min to 97:3 at 11.1 min until 15 min), flow rate 1.0 mL/min. TIS, MRM acquisition mode. Retention time: 8.0 min. Limit of quantification, 5 µg/L [Rook *et al.* 2005].

Serum HPLC Column: Supelcosil LC-Si (250 × 2.1 mm i.d., 5 µm). Mobile phase: methanol:acetonitrile:water:formic acid (59.8:5.2:34.65:0.35), flow rate 230 µL/min. API. Limit of quantification, 0.5 µg/L [Zuccaro *et al.* 1997].

Urine GC-MS See Plasma [Wang *et al.* 1994]. Column: HP-1 (12 m × 0.2 mm i.d., 0.33 µm). Temperature programme: 120° for 1 min to 220° at 20°/min to 260° at 5°/min to 280° at 20°/min for 2 min. EI ionisation, SIM acquisition mode. Limit of detection, 50 µg/L [De Giovanni, Strano Rossi 1994]. See Blood [Goldberger *et al.* 1993].

HPLC Column: Silica (200 × 2 mm i.d., 3 µm). Mobile phase: dichloromethane: pentane: diethylamine: methanol. Limit of detection, 4–20 µg/L [Low, Taylor 1995]. Column: Supelcosil LC-18-DB (5 µm). Mobile phase: 0.1 mol/L sodium hydrogen phosphate: methanol (pH 7.3, 67.5:32.5), flow rate 1.0 mL/min Electrochemical detection. Retention time: 8.22 min. Limit of detection, 250 µg/L [Sawyer *et al.* 1988].

LC-MS See Blood [Gergov *et al.* 2009].

Meconium GC-MS See Salem *et al.* [2001].

Oral Fluid GC-MS See Plasma [Wang *et al.* 1994]. See Blood [Goldberger *et al.* 1993].

Braint GC-MS See Blood. Limit of quantification, 0.002–0.06 µg/g [Karinen *et al.* 2009].

Hair GC-MS Column: Restek Rtx-5 MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° for 1 min to 260° at 15°/min to 290° at 7°/min for 6 min. Limit of quantification, 0.02 ng/mg [Pichini *et al.* 1999]. See Plasma. Limit of detection, 0.1 ng/mg [Wang *et al.* 1994]. See Poletini *et al.* [1993].

LC-MS Column: Zorbax phenyl (50 × 2.1 mm i.d., 3 µm). Mobile phase: acetonitrile-methanol-20 µmol/L formate buffer (pH 3.0, 10:10:80): acetonitrile-methanol-20 µmol/L formate buffer (pH 3.0, 35:35:30; 100:0 to 35:65 from 0.5 to 0.7 min to 100:0 in 2 min), flow rate 0.25 mL/min. ESI. Limit of detection not reported [Kronstrand *et al.* 2004].

Other GC Illicit Heroin Samples. Column: 2.5% OV-1 or SE-52 on 80/100 mesh Chromosorb G (2 m × 2 mm i.d.). Carrier gas: N₂, 30 mL/min. Temperature programme: 240° for 2 min to 280° at 24°/min for 6 min. Limit of detection not reported [Machata, Vycudilik 1980].

HPLC Column: µBondapak C₁₈ (25 × 4.6 mm i.d.). Mobile phase: acetonitrile: water: phosphoric acid with 0.02 mol/L methanesulfonic acid (pH 2.2, 12:87:1), flow rate 3.0 mL/min. UV detection (λ =254 nm). Limit of detection, 0.5 g/L [Lurie *et al.* 1982].

Disposition in the Body Diamorphine is readily absorbed after oral administration or by injection. Following injection, it is rapidly hydrolysed to 6-monoacetylmorphine in blood and then more slowly metabolised to morphine, which is the major active metabolite; normorphine is also formed to a minor extent. Small quantities of codeine are occasionally seen in the urine of addicts, but this is thought to arise from the presence of acetylcodeine as an impurity in illicit heroin samples. The blood concentration of morphine depends on the route of administration of the drug, the dose, time since the last dose and the individual's metabolism. Orally administered diamorphine undergoes extensive first-pass metabolism to morphine. All metabolites may be conjugated with glucuronic acid. Up to 80% of a dose is excreted in the urine in 24 h, mainly as morphine 3-glucuronide, with ≈5–7% of the dose as free morphine, 1% as 6-monoacetylmorphine, 0.1% as unchanged drug and trace amounts of other metabolites. After inhalation, 14–20% of the dose appears in the urine; morphine metabolites are excreted in the bile. Diamorphine crosses the blood–brain barrier within 15–20 s and achieves relatively high brain levels; 68% of an IV dose is absorbed into the brain.

Therapeutic Concentration Because of its rapid hydrolysis, diamorphine is difficult to detect in plasma. Morphine plasma concentrations are usually in the range 0.01–0.07 mg/L.

Twenty-six newborn premature neonates, with a gestational age of 26–40 weeks (mean, 30.7), were administered with a loading dose of 50 µg/kg diamorphine over a 30 min period followed by a 15 µg/kg/h IV infusion for 14–149 h (mean, 60.2). The mean steady-state plasma morphine concentration after the 15 µg/kg/h dose was 62.5 µg/L (range, 20–98) [Barrett *et al.* 1991].

Nineteen newborn infants of 24–41 weeks of gestation (mean, 29.7) who were on ventilation were administered with a loading dose of 50 µg/kg or 200 µg/kg diamorphine followed by 15 µg/kg/h as an IV infusion. The mean steady-state plasma morphine concentration was 86 µg/L after the 50 µg/kg and 200 µg/kg doses. The steady-state morphine concentration was 86 µg/L (ranges, 20–215 for the 50 µg/kg loading dose; 14–146 for the 200 µg/kg dose). For morphine 3-glucuronide, steady state was 703 µg/L (ranges 135–1237 and 45 to 1401 for the 2 loading doses, respectively) and for morphine 6-glucuronide, 48 µg/L (ranges, 35–91 and 11–120 µg/L, respectively) [Barrett *et al.* 1996].

Toxicity Diamorphine is 2–3 times more potent than morphine. The estimated minimum lethal dose is 200 mg, but addicts may be able to tolerate up to 10 times as much; fatalities have occurred after doses of 10 mg.

A 17-year-old pregnant girl, a known drug abuser, was found dead in a public toilet with needle puncture marks. A postmortem carried out 48 h after death showed massive brain and lung oedema associated with an acute intoxication. In her femoral vein blood a 6-monoacetylmorphine concentration of 4 µg/L was detected as well as 280 µg/L morphine and 20 µg/L codeine. In her urine 1170 µg/L of 6-monoacetylmorphine was found. These were all consistent with a recent heroin intake. 6-Monoacetylmorphine, morphine, morphine 3-glucuronide and codeine were all detected in the fetus's amniotic fluid, brain and lung at high concentrations [Skopp *et al.* 1998].

A 40-year-old man with a long history of drug abuse was found dead after an overdose of heroin. Heroin was detected at a concentration of 0.109 mg/L in blood and 17 ng/g in gastrointestinal contents. Its metabolite, 6-monoacetylmorphine, was found at concentrations of 0.168 mg/L and 12 ng/g, respectively, and morphine at 1.14 mg/L and 425 ng/g, respectively. Morphine was also found in urine at a concentration of 3.65 mg/L [Rop *et al.* 1997].

Of 10 deaths resulting from heroin overdoses via non-injection routes, 4 involved inhalation ('chasing'), 4 involved nasal administration ('snorting') and 2 involved oral ingestion. In 1 case, the person was observed to smoke a heroin cigarette and then snort heroin immediately before losing consciousness. In 9 of the cases, venipuncture marks were detected at postmortem. The median blood morphine concentration was 0.31 mg/L (range, 0.06–0.99). Drugs other than morphine were also detected in 7 cases (alcohol (4), cocaine (2), dothiepin, methadone and paracetamol (1 each)) [Darke, Ross 2000].

A 46-year-old man died after injecting heroin into his penis. The concentration of morphine in body fluids at postmortem were blood 0.68 mg/L, urine 0.49 mg/L, bile 0.32 mg/L and vitreous humour 0.062 mg/L; ethanol was also detected in these fluids [Winek *et al.* 1999].

Of 10 deaths involving heroin body-packers, four had blood concentrations <1 mg/L. In five others, concentrations were 4.4, 6.7, 35.8, 39.4 and 52.6 mg/L. One victim, who died of peritonitis, had no detectable morphine in the blood [Wetli *et al.* 1997].

Half-life Plasma half-life, diamorphine ≈3 min, morphine ≈2–3 h.

Volume of Distribution Morphine ≈3–5 L/kg.

Clearance Plasma clearance, morphine ≈15–20 mL/min/kg; newborn infants (24–41 weeks of gestation), 4.6 mL/min/kg.

Protein Binding Morphine ≈20–35%.

Note For a review of the pharmacokinetics and pharmacodynamics of smoked heroin, see Jenkins *et al.* [1994]; for a review of the metabolism and excretion of diamorphine, see Boerner [1975]. For a review of fatal heroin overdose, see Darke, Zador [1996]; for reports of heroin use when driving, see Ceder, Jones [2002].

Dose For acute pain, 5 to 10 mg diamorphine hydrochloride every 4 h by SC or IM injection.

Barrett DA *et al.* (1991). Morphine kinetics after diamorphine infusion in premature neonates. *Br J Clin Pharmacol* 32: 31–37.

Barrett DA *et al.* (1996). Morphine, morphine-6-glucuronide and morphine-3-glucuronide pharmacokinetics in newborn infants receiving diamorphine infusions. *Br J Clin Pharmacol* 41: 531–537.

Boerner U (1975). The metabolism of morphine and heroin in man. *Drug Metab Rev* 4: 39–73.

Bourquin D *et al.* (1997). High-performance liquid chromatographic monitoring of intravenously administered diacetylmorphine and morphine and their metabolites in human plasma. *J Chromatogr B Biomed Sci Appl* 694: 233–238.

Ceder G, Jones AW (2002). Concentrations of unconjugated morphine, codeine and 6-acetylmorphine in urine specimens from suspected drug drivers. *J Forensic Sci* 47: 366–368.

Darke S, Ross J (2000). Fatal heroin overdoses resulting from non-injecting routes of administration, NSW, Australia, 1992–1996. *Addiction* 95: 569–573.

Darke S, Zador D (1996). Fatal heroin 'overdose': a review. *Addiction* 91: 1765–1772.

DeGiovanni N, Strano Rossi S (1994). Simultaneous detection of cocaine and heroin metabolites in urine by solid-phase extraction and gas chromatography–mass spectrometry. *J Chromatogr B Biomed Appl* 658: 69–73.

Gergov M *et al.* (2009). Simultaneous screening and quantification of 25 opioid drugs in post-mortem blood and urine by liquid chromatography–tandem mass spectrometry. *Forensic Sci Int* 186: 36–43.

Goldberger BA *et al.* (1993). Measurement of heroin and its metabolites by isotope-dilution electron-impact mass spectrometry. *Clin Chem* 39: 670–675.

Jenkins AJ *et al.* (1994). Pharmacokinetics and pharmacodynamics of smoked heroin. *J Anal Toxicol* 18: 317–330.

Karinen R *et al.* (2009). Determination of heroin and its main metabolites in small sample volumes of whole blood and brain tissue by reversed-phase liquid chromatography–tandem mass spectrometry. *J Anal Toxicol* 33: 345–350.

Kronstrand R *et al.* (2004). Screening for drugs of abuse in hair with ion spray LC-MS-MS. *Forensic Sci Int* 145: 183–190.

Low AS, Taylor RB (1995). Analysis of common opiates and heroin metabolites in urine by high-performance liquid chromatography. *J Chromatogr B Biomed Appl* 663: 225–233.

Lurie IS *et al.* (1982). High performance liquid chromatographic analysis of heroin by reverse phase ion-pair chromatography. *J Forensic Sci* 27: 519–526.

Machata G, Vycudilik W (1980). Gas chromatographic analysis of illicit heroin samples. *J Anal Toxicol* 4: 318–321.

Pichini S *et al.* (1999). Determination of opiates and cocaine in hair as trimethylsilyl derivatives using gas chromatography–tandem mass spectrometry. *J Anal Toxicol* 23: 343–348.

Poletini A *et al.* (1993). Rapid and highly selective GC/MS/MS detection of heroin and its metabolites in hair. *Forensic Sci Int* 63: 217–225.

Rook EJ *et al.* (2005). The quantitative analysis of heroin, methadone and their metabolites and the simultaneous detection of cocaine, acetylcodeine and their metabolites in human plasma by high-performance liquid chromatography coupled with tandem mass spectrometry. *J Chromatogr B Biomed Sci* 824: 213–221.

Rop PP *et al.* (1997). Concentrations of heroin, 6-monoacetylmorphine, and morphine in a lethal case following an oral heroin overdose. *J Anal Toxicol* 21: 232–235.

Salem MY *et al.* (2001). GC-MS determination of heroin metabolites in meconium: evaluation of four solid-phase extraction cartridges. *J Anal Toxicol* 25: 93–98.

Two healthy male volunteers were administered 2×25 mg Persantine tablets following a 12 h fast. Mean peak plasma concentrations reached $\approx 517 \mu\text{g/L}$ at ≈ 1 h [Williams *et al.* 1981].

A healthy volunteer was administered a single oral dose of 25 mg dipyridamole. The peak plasma concentration of $\approx 300 \mu\text{g/L}$ was reached after ≈ 1 h [Schmid *et al.* 1979].

After a single oral dose of 100 mg given to 4 subjects, peak serum concentrations of ≈ 2 mg/L were attained in 1 h [Nielsen-Kudsk, Pedersen 1979].

Toxicity

A 23-year-old female who suffered profound shock after deliberately ingesting 1750 mg of dipyridamole was successfully treated with aminophylline and dopamine [Chen *et al.* 1994].

Half-life Plasma half-life, ≈ 12 h.

Volume of Distribution ≈ 2.5 L/kg.

Clearance Plasma clearance, ≈ 2 mL/min/kg. Gastric pH has an impact on gastric emptying in elderly patients with a higher pH resulting in a slower emptying [Russell *et al.* 1994].

Protein Binding $>90\%$ [Szebeni, Weinstein 1991].

Dose 300 to 600 mg daily (antithrombotic); 150 mg daily (antianginal).

Barberi M *et al.* (1991). Sensitive determination of free and plasma protein-bound dipyridamole by high-performance liquid chromatography. *J Chromatogr* 565: 511–515.

Chen ZC *et al.* (1994). Profound shock resulting from a large dose of dipyridamole. *Int J Cardiol* 46: 75–78.

Dresse A *et al.* (1982). Pharmacokinetics of oral dipyridamole (Persantine) and its effect on platelet adenosine uptake in man. *Eur J Clin Pharmacol* 23: 229–234.

Mahony C *et al.* (1983). Plasma dipyridamole concentrations after two different dosage regimens in patients. *J Clin Pharmacol* 23: 123–126.

Nielsen-Kudsk F, Pedersen AK (1979). Pharmacokinetics of dipyridamole. *Acta Pharmacol Toxicol (Copenh)* 44: 391–399.

Pedersen AK (1979). Specific determination of dipyridamole in serum by high-performance liquid chromatography. *J Chromatogr* 162: 98–103.

Ricevuti G *et al.* (1991). Pharmacokinetics of dipyridamole-beta-cyclodextrin complex in healthy volunteers after single and multiple doses. *Eur J Drug Metab Pharmacokinet* 16: 197–201.

Rosenfeld J *et al.* (1982). High-performance liquid chromatographic determination of dipyridamole. *J Chromatogr* 231: 216–221.

Russell TL *et al.* (1994). pH-related changes in the absorption of dipyridamole in the elderly. *Pharm Res* 11: 136–143.

Schmid J *et al.* (1979). Rapid, sensitive determination of dipyridamole in human plasma by high-performance liquid chromatography. *J Chromatogr* 163: 239–243.

Steyn JM (1979). Spectrofluorimetric determination of dipyridamole in serum—a comparison of two methods. *J Chromatogr* 164: 487–494.

Szebeni J *et al.* (1991). Dipyridamole binding to proteins in human plasma and tissue culture media. *J Lab Clin Med* 117: 485–492.

Wang N *et al.* (2008). Simultaneous determination of dipyridamole and salicylic acid in human plasma by high performance liquid chromatography-mass spectrometry. *Biomed Chromatogr* 22: 149–156.

Williams C *et al.* (1981). High-performance liquid chromatographic assay for plasma dipyridamole monitoring. *J Chromatogr* 225: 225–230.

Wolfram KM, Björnsson TD (1980). High-performance liquid chromatographic analysis of dipyridamole in plasma and whole blood. *J Chromatogr* 183: 57–64.

Yadav M *et al.* (2008). Effect of collision-activated dissociation gas and collision energy on the fragmentation of dipyridamole and its rapid and sensitive liquid chromatography/electrospray ionization tandem mass spectrometric determination in human plasma. *Rapid Commun Mass Spectrom* 22: 511–518.

Dipyron

Analgesic, Antiinflammatory, Antipyretic

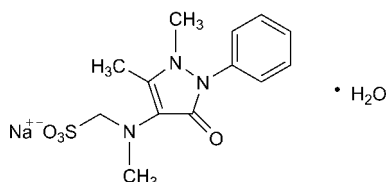
$\text{C}_{13}\text{H}_{16}\text{N}_3\text{NaO}_4\text{S} \cdot \text{H}_2\text{O} = 351.4$

CAS—68-89-3 (anhydrous); 5907-38-0 (monohydrate)

IUPAC Name Sodium[(1,5-dimethyl-3-oxo-2-phenylpyrazol-4-yl)-methylamino]methanesulfonate

Synonyms Aminopyrine-sulfonate sodium; analginum; metamizole; metamizole sodium; metamizolum natrium; methampyrone; methylmelubrin; natrium novaminsulfonicum; noramidazophenum; noraminophenazonum; novamidazofen; NSC-73205; sodium-[(2,3-dihydro-1,5-dimethyl-3-oxo-2-phenyl-1H-pyrazol-4-yl)methylamino]methanesulphonate monohydrate; sodium noramidopyrine methanesulphonate; sulpyrine.

Proprietary Names Algi; Analgin; Baralgin; Berlosin; Conmel; Dolemicin; Dolocalma; Dolalgol; Inalgol; Lasain; Metilon; Neo Melubrina; Nevralgina; Nolotil; Novalgin(e); Novaminsulfon; Optalgin; Pyril; Trisalgin; Unagen. It is an ingredient of many proprietary preparations—see Sweetman [2007].



Chemical Properties White or yellowish-white crystalline powder. Mp 172° . Soluble 1 in 1.5 of water and 1 in 30 of ethanol; practically insoluble in ether, acetone, benzene and chloroform.

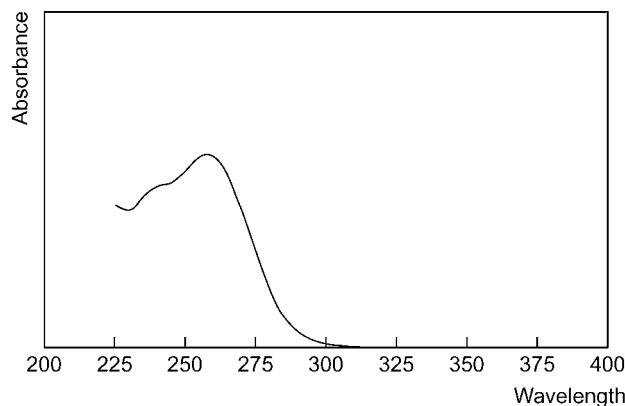
Colour Tests Ferric chloride—violet; Folin-Ciocalteu reagent—blue; Liebermann's reagent (100°)—blue; Mandelin's test—brown; nitrous acid—transient blue.

Thin-layer Chromatography System TA— R_f 0.84; system TB— R_f 0.00; system TC— R_f 0.01; system TD— R_f 0.00; system TE— R_f 0.02; system TF— R_f 0.00; system TL— R_f 0.02; system TAD— R_f 0.02; system TAE— R_f 0.85; system TAF— R_f 0.59 (acidified iodoplatinate solution, white).

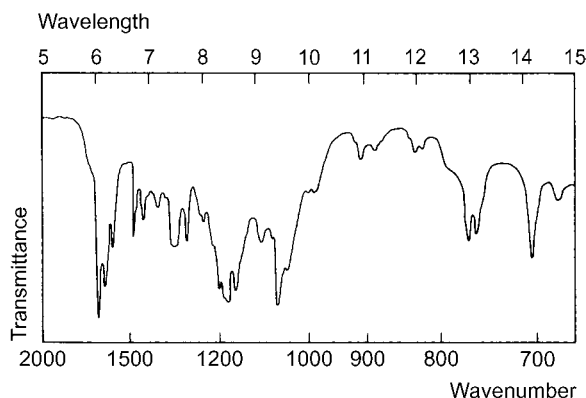
Gas Chromatography System GA—M (bis-desalkyl-) RI 1955, M (desalkyl-)—AC RI 2395; system GB—dipyron RI 2069.

High Performance Liquid Chromatography System HD— k 0.1; system HW— k 0.45; system HX—RI 316; system HY—RI 194; system HZ—RT 1.4 min.

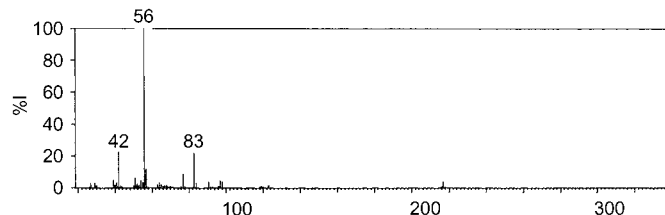
Ultraviolet Spectrum Aqueous acid—258 nm ($A_1^1 = 266a$).



Infrared Spectrum Principal peaks at wavenumbers 1672, 1064, 1179, 1163, 1208, 1639 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at m/z 56, 42, 83, 57, 77, 51, 97, 54.



Quantification

Plasma HPLC Column: OD-MP Spheri-5 RP-18 (10 cm $\times$ 4.6 mm i.d.). Mobile phase: 0.1 mol/L SDS with 2.5% pentanol, flow rate 1.0 mL/min. UV detection ($\lambda = 262$ nm). Limit of detection, 14.4 and 18 mg/L for FAA and AA, respectively [Carretero *et al.* 1995]. Column: Spherisorb ODS (125 $\times$ 4.5 mm i.d., 5 μm). Mobile phase: 0.05 mol/L TEAP buffer (pH 2.8): acetonitrile: methanol (100:10:8), flow rate 1.1 mL/min. UV detection ($\lambda = 265$ nm). Limit of detection, 0.1 mg/L [Damm 1989]. Column: μ Bondapak C₁₈ (300 $\times$ 3.9 mm i.d., 10 μm). Mobile phase: methanol: 0.01 mol/L sodium acetate (pH 3.0; 8:92), flow rate 1.6 mL/min. UV detection ($\lambda = 257$ nm). Retention time: 14, 17, 21, and 25 min for MAA, AA, FAA and AAA, respectively. Limit of detection, 0.1 mg/L [Katz *et al.* 1984]. Column: μ Bondapak C₁₈ (30 cm $\times$ 3.9 mm i.d., 10 μm). Mobile phase: water: methanol: acetic acid (75:20:5), flow rate 1.0 mL/min. UV detection ($\lambda = 254$ and 280 nm). Limit of quantification, 1 mg/L for dipyron and 250 ng/mL for 4-methylaminophenazone Asmardi, Jamali 1983].

Urine HPLC See Plasma. Limit of detection, 2 mg/L [Damm 1989]. Column: Spherisorb ODS (250 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: water: methanol: triethylamine: acetic acid (70.9:27.7:0.9:0.5), flow rate 1.0 mL/min. UV detection

($\lambda = 254$ nm). Retention time: 22.2 min for MMA. Limit of detection, 30 μ g/L [Agúndez *et al.* 1994]. See Plasma [Asmardi, Jamali 1983].

Other HPLC Human Liver Microsomes. Column: Lichrospher 60 RP (125 $\times$ 3.0 mm i.d., 5 μ m). Mobile phase: methanol:0.05 mol/L sodium acetate buffer (pH 2.2; 1:99 to 4:96 at 6 min to 7:93 at 13 min to 9:91 at 18 min for 8 min to 1:99 at 27 min), flow rate 0.9 mL/min. UV detection ($\lambda = 257$ nm). Limit of quantification, 50 μ g/L for all metabolites [Geisslinger *et al.* 1996].

Disposition in the Body Rapidly hydrolysed by desmethane-sulfonation in the gastrointestinal tract after oral administration to the active metabolite 4-methylamino-antipyrine (MMA). MMA undergoes incomplete oxidative demethylation to 4-formylamino-antipyrine (FAA) and demethylation to 4-amino-antipyrine (AA), which in turn is acetylated to form 4-acetylamino-antipyrine (AAA). About 70% of a dose is excreted in the urine in 24 h as metabolites. Dipyrone metabolites are excreted in breast milk.

Therapeutic Concentration

In 12 subjects given single oral doses of a test and reference preparation of dipyrone 1 g, the mean peak plasma concentrations of 4-methylamino-antipyrine, 4-formylamino-antipyrine, 4-amino-antipyrine, and 4-acetylamino-antipyrine were 17.19 and 16.92 mg/L (test and reference), 1.6 and 1.84 mg/L, 2.94 and 2.89 mg/L, and 1.93 and 1.86 mg/L at 1.42 and 1.46 h, 6.2 and 7.0 h, 4.4 and 4.4 h, and 11.3 and 11.6 h, respectively. Four of the subjects were rapid and 8 were slow acetylators; the peak plasma concentration (or the test preparation) differed significantly between the 2 acetylator types only for 4-amino-antipyrine (16.98 vs 18.52 for slow vs rapid acetylators) and 4-acetyl-amino-antipyrine (1.38 vs 4.38 mg/L) [Bacacheva *et al.* 1995].

The formation and urinary excretion of dipyrone metabolites was measured in 12 healthy volunteers following a single oral 1.0 g dose of dipyrone. The group consisted of 3 slow and 9 rapid acetylators. Renal clearance of all metabolites was similar for both phenotypes. Formation of the metabolites AA and AAA were significantly different between the 2 phenotypes: AA, 0.25 (slow) vs 0.1 mL/min/kg (rapid) AAA, 0.75 vs 7.53 mL/min/kg (rapid) [Zylber-Katz *et al.* 1992].

A single oral dose of 1 g dipyrone sodium was administered to 12 healthy volunteers. Pharmacokinetic values for the main metabolites are shown below.

	MAA	AA	FAA	AAA
C _{max} (mg/L)	12.2 $\pm$ 1.4	1.5 $\pm$ 0.8	2.4 $\pm$ 0.5	1.8 $\pm$ 0.6
t _{max} (h)	1.3 $\pm$ 0.4	4.4 $\pm$ 0.9	5.8 $\pm$ 1.3	15.0 $\pm$ 6.4
t _{1/2} (h)	2.7 $\pm$ 0.5	3.7 $\pm$ 1.3	11.2 $\pm$ 1.5	9.5 $\pm$ 1.5
CL _{ren} (mL/min)	5.0 $\pm$ 1.8	37.8 $\pm$ 12.8	48.5 $\pm$ 4.5	60.6 $\pm$ 8.3

[Damm 1989].

After a single oral dose of 1 g given to 12 slow acetylators and 11 rapid acetylators (determined with reference to dapson), the mean peak plasma concentrations (mg/L, time in hours) were as shown below.

	Slow acetylators	Rapid acetylators
4-Acetylamino-phenazone	1.6 (16.1)	4.4 (10.0)
4-Amino-phenazone	2.7 (6.7)	1.6 (3.2)
4-Formylamino-phenazone	1.8 (7.7)	2.5 (6.6)
4-Methylamino-phenazone	10.0 (1.6)	11.0 (1.2)

[Levy *et al.* 1984]

Toxicity Dipyrone is rarely used in the US due to an association with agranulocytosis. Other adverse reactions include skin rashes, anaphylaxis, thrombocytopenia, hepatocellular damage, and CNS symptoms. The estimated minimum lethal dose is 5 g, but fatalities from acute poisoning are rare.

A 56-year-old male died after ingesting up to 20 g dipyrone, 2 g baclofen and 0.15 g oxybutynin; terazosin, aspirin, defibrotide, and laxatives also may have been ingested. Postmortem, carried out a few hours after death, revealed dipyrone, baclofen and oxybutynin concentrations, respectively, of 106, 669 and 0.1 mg/L in the blood and 774 mg/L, 215 mg/L, and trace amounts in the urine [De Giovanni, d'Aloja 2001].

Note For a case of a death after the IV administration of Avafortan (24 mg/mL avaprazone and 240 mg/mL dipyrone) see Fosseus and Straughan [1983].

Half-life Plasma half-life, 4-acetylamino-antipyrine about 10.6 h, 4-amino-antipyrine $\approx$ 5 h, 4-formylamino-antipyrine $\approx$ 10 h, and 4-methylamino-antipyrine $\approx$ 3.3 h.

Dose 0.5 to 4 g daily in divided doses.

Agúndez JA *et al.* (1994). Determination of aminopyrine, dipyrone and its metabolites in urine by high-performance liquid chromatography. *Ther Drug Monit* 16: 316–322.

Asmardi G, Jamali F (1983). High-performance liquid chromatography of dipyrone and its active metabolite in biological fluids. *J Chromatogr* 277: 183–189.

Bacacheva N *et al.* (1995). Comparative bioavailability of two oral metamizole formulations. Influence of the acetylation phenotype. *Arzneimittelforschung* 45: 282–285.

Carreiro I *et al.* (1995). Determination of antipyrine metabolites in human plasma by solid-phase extraction and micellar liquid chromatography. *Analyst* 120: 1729–1732.

Damm D (1989). Simultaneous determination of the main metabolites of dipyrone by high-pressure liquid chromatography. *Arzneimittelforschung* 39: 1415–1417.

De Giovanni N, d'Aloja E (2001). Death due to baclofen and dipyrone ingestion. *Forensic Sci Int* 123: 26–32.

Fosseus CG, Straughan JL (1983). Sudden death due to intravenous avaprazone and dipyrone. *S Afr Med J* 64: 81.

Geisslinger G *et al.* (1996). High-performance liquid chromatographic analysis of dipyrone metabolites to study their formation in human liver microsomes. *Pharm Res* 13: 1272–1275.

Katz EZ *et al.* (1984). Simultaneous determination of dipyrone metabolites in plasma by high-performance liquid chromatography. *J Chromatogr* 305: 477–484.

Levy M *et al.* (1984). Plasma kinetics of dipyrone metabolites in rapid and slow acetylators. *Eur J Clin Pharmacol* 27: 453–458.

Sweetman S, ed. (2007). *Martindale: The Complete Drug Reference*, 35th edn. London: Pharmaceutical Press.

Zylber-Katz E *et al.* (1992). Formation and excretion of dipyrone metabolites in man. *Eur J Clin Pharmacol* 42: 187–191.

Diquat

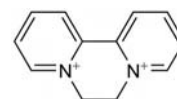
Herbicide

C₁₂H₁₂N₂ = 184.2

CAS = 2764-72-9

IUPAC Name 6,7-Dihydrodipyrido[1,2-a:2',1'-c]pyrazinediium ion

Synonym Deiquat



Note Unless otherwise stated, the analytical information given below refers to diquat dibromide. For information on the monoene and diene reduction products for GC-MS analysis, see under Quaternary ammonium herbicides in Chapter 16.

Diquat Dibromide

C₁₂H₁₂Br₂N₂ = 344.0

CAS = 85-00-7

Proprietary Names Aquacide; Reglone. It is an ingredient of *Pathclear* and *Weedol*.

Chemical Properties Yellow crystals. Mp $\sim$ 335°. Soluble in water. Log *P* (octanol/water), 2.4.

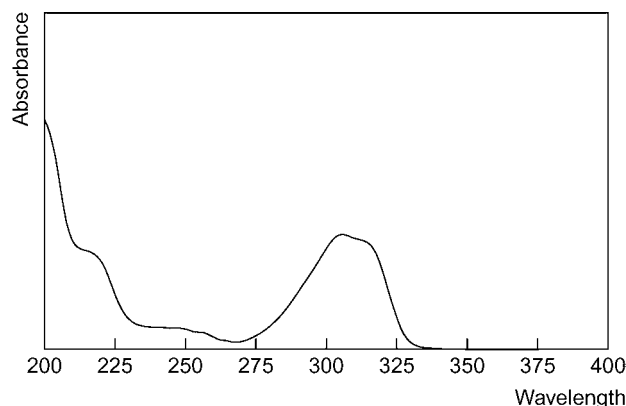
Colour Test Sodium dithionite—green.

Thin-layer Chromatography System TA—R_f 0.00; system TAB—R_f 0.00; system TAC—R_f 0.00; system TAE—R_f 0.00; system TAF—R_f 0.00; system TL—R_f 0.00 (acidified iodoplatinate solution, positive).

Gas Chromatography System GA—diquat dibromide not eluted; system GK—monoene reduction product RT 0.40 min; diene reduction product RT 0.49 min (both relative to caffeine).

High Performance Liquid Chromatography System HX—RI 00; system HAO—diquat dibromidek 99.9; system HAP—diquat dibromide k 0.00.

Ultraviolet Spectrum Aqueous acid—310 nm (A₁ = 518b)



Infrared Spectrum Principal peaks at wavenumbers 1522, 1498, 1140, 1166, 1097, 1632 cm⁻¹ (KBr disk).

Toxicity No special hazard to humans, anaemia observed in repeated dose toxicity studies.

Bioavailability Approximately 35% (oral).

Half-life Elimination half-life, $t_{1/2}$ (α -phase) 0.27 to 0.37 h and (β -phase) 1.59 to 3.10 h.

Volume of Distribution 181 L.

Clearance Total clearance, 800 mL/min.

Protein Binding Extensively binds to plasma proteins and serum albumin ~98%.

Dose A usual dose of 200 mg entacapone, up to ten times a day, with a maximum of 2000 mg.

Demme U *et al.* (2005). Systematic evaluation of 1-chlorobutane for liquid-liquid extraction of drugs. *Proceedings of the 12th TIAFT*, Seoul: 481–486.

Karlsson M, Wikberg T (1992). Liquid chromatographic determination of a new catechol-O-methyltransferase inhibitor, entacapone, and its Z-isomer in human plasma and urine. *J Pharm Biomed Anal* 10(8): 593–600.

Keranen T *et al.* (1994). Inhibition of soluble catechol-O-methyltransferase and single-dose pharmacokinetics after oral and intravenous administration of entacapone. *Eur J Clin Pharmacol* 46(2): 151–157.

Wikberg T *et al.* (1993). Identification of major metabolites of the catechol-O-methyltransferase inhibitor entacapone in rats and humans. *Drug Metab Dispos* 21: 81–92.

Epalrestat

Antidiabetic, Thiazolidine

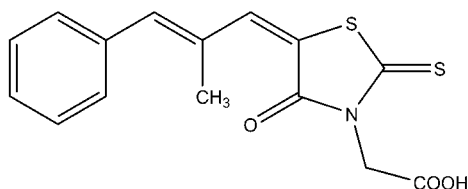
$C_{15}H_{13}NO_3S_2 = 319.4$

CAS—82159-09-9

IUPAC Name 2-[(5Z)-5-[(E)-2-Methyl-3-phenylprop-2-enylidene]-4-oxo-2-sulfanylidene-1,3-thiazolidin-3-yl]acetic acid

Synonyms 3-Carboxymethyl-5-(2-methylcinnamylidene)rhodanine; 5-[(E,E)- β -methylcinnamylidene]-4-oxo-2-thioxo-3-thiazolidine-acetic acid; (5Z)-5-[(2E)-2-methyl-3-phenyl-2-propenylidene]-4-oxo-2-thioxo-3-thiazolidineacetic acid; ONO-2235.

Proprietary Names Kinedak; Sorbistat.



Chemical Properties Crystals from ethanol–water. Mp 210° to 217°.

Epalrestat N-Methyl-D-Glucamine

$C_{22}H_{30}N_2O_8S_2 = 514.6$

Chemical Properties Crystals from methanol. Mp 163° to 165°.

Disposition in the Body Metabolised to mono- and dihydroxyphenylepalrestat as sulfate conjugates and, together with unchanged drug, these are present in both serum and urine. There is some evidence that epalrestat binds to serum albumin.

Therapeutic Concentration

Single oral doses of 50, 100 and 200 mg epalrestat were administered to 7 healthy male volunteers. Mean peak plasma concentrations of the drug and its metabolites were as follows:

Dose (mg)	Compound	C_{max} (mg/L)	Time (h)
50	Epalrestat	3.9	1
	4-OH Metabolite	0.5	2
	3,4-(OH) ₂ Metabolite	–	–
100	Epalrestat	7.4	2
	4-OH Metabolite	1.2	2
	3,4-(OH) ₂ Metabolite	0.4	2
200	Epalrestat	13.6	2
	4-OH Metabolite	2.10	2
	3,4-(OH) ₂ Metabolite	0.70	2

The half-life was in the range 0.8 to 1.1 h [Steele *et al.* 1993].

Half-life Approximately 1 h.

Dose Recommended, 50 mg three times daily before meals.

Steele JW *et al.* (1993). Epalrestat. A review of its pharmacology, and therapeutic potential in late-onset complications of diabetes mellitus. *Drugs Aging* 3: 532–555.

Ephedrine

Sympathomimetic

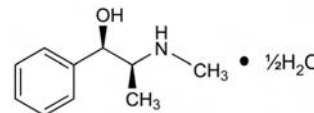
$C_{10}H_{15}NO, \frac{1}{2}H_2O = 174.2$

CAS—50906-05-3

IUPAC Name (1R,2S)-2-(Methylamino)-1-phenylpropan-1-ol hydrate

Synonyms Hydrated ephedrine; (α R)- α -[(1S)-1-(methylamino)ethyl]benzene-methanol hemihydrate.

Proprietary Names It is an ingredient of *Franolyn Expectorant* and *Letigen*.



Chemical Properties An alkaloid obtained from species of *Ephedra*, or prepared synthetically. Colourless crystals or white crystalline powder or granules that decompose on exposure to light. Mp 40° (hemihydrate); in warm weather it slowly volatilises. Soluble 1 in 20 of water and 1 in <1 of ethanol; soluble in chloroform with turbidity owing to separation of water; soluble in ether. Extraction yield (chlorobutane), 0.2 [Demme *et al.* 2005].

Anhydrous Ephedrine

$C_{10}H_{15}NO = 165.2$

CAS—299-42-3

Synonym Ephedrine

Proprietary Names *Ephed 20th*; *Kondon's Nasal*; *Tendrin*.

Chemical Properties Unctuous deliquescent colourless crystals, or white crystalline powder. It rapidly absorbs carbon dioxide and decomposes on exposure to light. Mp 38.1°. Soluble 1 in 20 of water; very soluble in ethanol; soluble in chloroform; freely soluble in ether.

Ephedrine Hydrochloride

$C_{10}H_{15}NO, HCl = 201.7$

CAS—50-98-6

Synonyms Ephedrine chloride; ephedrinium chloratum; l-ephedrinum hydrochloricum.

Proprietary Names CAM; it is an ingredient of many proprietary preparations [Sweetman 2007].

Chemical Properties Colourless crystals or white crystalline powder. Mp 217° to 220°. Soluble 1 in 3 to 1 in 4 of water and 1 in 14 to 1 in 17 of ethanol; very slightly soluble in chloroform; practically insoluble in ether.

Ephedrine Sulfate

$(C_{10}H_{15}NO)_2, H_2SO_4 = 428.5$

CAS—134-72-5

Proprietary Names It is an ingredient of *Franol Plus*.

Chemical Properties Fine white crystals or powder that darken on exposure to light. Mp 245° with decomposition [O'Neil *et al.* 2006]. Soluble 1 in 1.3 of water and 1 in 90 of ethanol. pK_a 9.6 (25°). Log *P* (octanol/water), 1.1.

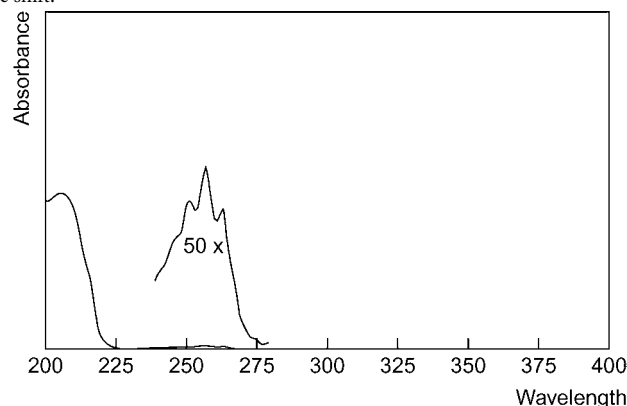
Colour Test Liebermann's reagent—red-orange.

Thin-layer Chromatography System TA— R_f 0.30; system TB— R_f 0.05; system TC— R_f 0.05; system TE— R_f 0.25; system TL— R_f 0.01; system TAE— R_f 0.10; system TAF— R_f 0.64; system TAJ— R_f 0.00; system TAK— R_f 0.01; system TAL— R_f 0.29 (acidified iodoplatinate solution, positive; acidified potassium permanganate solution, positive; Dragendorff spray, positive; Marquis test, brown; ninhydrin spray, positive).

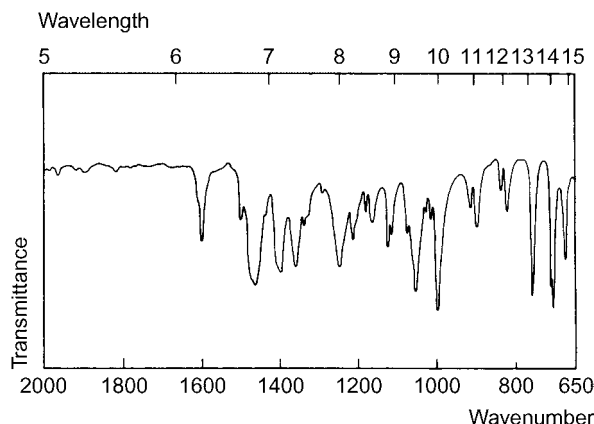
Gas Chromatography System GA—ephedrine RI 1365, M (-TFA₂) RI 1345, M (-PFP₂) RI 1370, M (-TMS₂) RI 1620, M (-AC₂) RI 1795, M (nor-) RI 1360, M (nor-) TFA₂ RI 1355, M (nor-) PFP₂ RI 1380, M (nor-) TMS₂ RI 1555, M (nor-) AC₂ RI 1805, M (OH-) RI 1875, M (OH-) AC₃ RI 2145; system GB—ephedrine RI 1410, M (nor-) RI 1356; system GC—ephedrine RI 1467, M (nor-) RI 1383.

High Performance Liquid Chromatography System HA—ephedrine *k* 1.0, M (nor-) *k* 0.9; system HB—ephedrine *k* 5.68, M (nor-) *k* 3.87; system HC—ephedrine *k* 1.79, M (nor-) *k* 0.70; system HX—RI 221; system HY—RI 227; system HZ—RT 2.1 min; system HAA—RT 5.7 min; system HAM—ephedrine not detected.

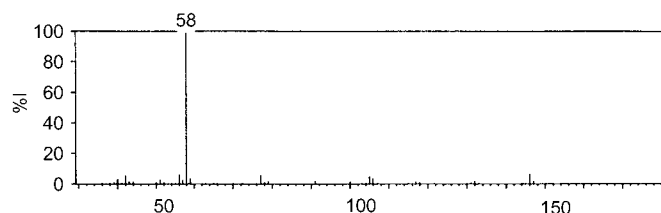
Ultraviolet Spectrum Aqueous acid—251, 257 ($A_1^1 = 12a$), 263 nm. No alkaline shift.



Infrared Spectrum Principal peaks at wavenumbers 994, 699, 754, 1049, 1242, 670 cm^{-1} (ephedrine hydrochloride, KBr disk).



Mass Spectrum Principal ions at m/z 58, 146, 56, 105, 77, 42, 106, 40; norephedrine 44, 77, 79, 51, 45, 42, 107, 105.



Quantification

Blood GC-MS Column: 5% phenyl silicone capillary (12.5 m $\times$ 0.2 mm i.d., 0.33 μm). Carrier gas: He, 20 mL/min. Temperature programme: 70° to 175° at 25°/min to 280° at 50°/min. SIM acquisition mode [Backer *et al.* 1997].

LC-MS Column: XBridge Shield RP-18 (250 $\times$ 3.0 mm i.d., 5 μm). Mobile phase: acetonitrile: 10 mmol/L ammonium bicarbonate (pH 10.5, 100:0 to 40:60 at 3.5 min to 80:20 at 7.5 min), flow rate 0.5 mL/min. ESI, MRM acquisition mode. Limit of quantification, 0.2 $\mu\text{g/L}$ [Qiu *et al.* 2008].

Plasma GC Column: 3% phenyl cyanopropyl methyl silicone on 100/120 mesh (0.9 m $\times$ 2.0 mm i.d.). Carrier gas: Ar: CH₄ (95:5), 93 mL/min. Temperature: 235°. ECD. Limit of detection, 2 $\mu\text{g/L}$ [Midha *et al.* 1979].

GC-MS Column: HP-5MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1 mL/min. Temperature programme: 50° for 0.5 min to 200° at 300°/min to 300° at 10°/min for 5 min. EI ionisation, SIM acquisition mode. Limit of detection, 2–5 $\mu\text{g/L}$ [Frison *et al.* 2005]. Limit of detection, 0.82 $\mu\text{g/L}$ [Nakano *et al.* 2000].

HPLC Column: C₁₈ (150 $\times$ 4.6 mm). Mobile phase: acetonitrile: water (52:48). Fluorescence detection (λ =264 and 313 nm). Limit of detection, 2 $\mu\text{g/L}$ for ephedrine and 5 $\mu\text{g/L}$ for norephedrine [Aymard *et al.* 2000]. Column: reversed phase. Mobile phase: 0.6% phosphate buffer (pH 6.5): methanol (3:8). Fluorescence detection. Limit of detection, 0.5 ng [Shao *et al.* 1995].

LC-MS Column: Zorbax SB-C₁₈ (150 $\times$ 4.6 mm). Mobile phase: methanol: water: formic acid (80:20:0.5), flow rate 0.5 mL/min. ESI, positive ion mode, SRM acquisition mode. Limit of quantification, 0.5 $\mu\text{g/L}$ [Ren *et al.* 2006]. Column: Hypersil Phenyl BDS (50 $\times$ 4.6 mm i.d.). Mobile phase: 10 mmol/L ammonium formate with 0.1% formic acid: 10 mmol/L ammonium formate with 0.1% formic acid (100:0 to 80:20 over 7 min to 0:100 over 0.5 min for 1 min to 100:0 for 3.5 min). APCI, positive ion mode, SRM acquisition mode. Limit of quantification, 1 $\mu\text{g/L}$ [Jacob *et al.* 2004]. Column: YMC phenyl LC (250 $\times$ 2 mm, 5 μm). Mobile phase: 3.8 g ammonium acetate: 20 mL glacial acetic acid: 30 mL acetonitrile: to 1000 mL with water. Limit of quantification, 100 $\mu\text{g/L}$ for ephedrine alkaloids [Trujillo, Sorenson 2003].

CE Buffer: 0.1 mol/L phosphate electrolyte (pH 2.5): acetonitrile (90:10). Limit of detection, 5.3 $\mu\text{g/L}$ [Wei *et al.* 2007].

Serum CE UV detection (λ =192 nm). Limit of detection, 0.15–0.25 $\mu\text{g/L}$ [Fang *et al.* 2006a].

Urine GC Column: Rtx-5 5% diphenyl 95% dimethylpolysiloxane (15 m $\times$ 0.25 mm i.d., 1.0 μm). Carrier gas: He. Temperature programme: 100° to 105° at 0.5°/min to 118° at 2°/min to 280° at 50°/min for 2 min. NPD. Retention time: 14.8 min. Limit of detection, 4 $\mu\text{g/L}$ [Van Eenoo *et al.* 2001]. See Plasma [Midha *et al.* 1979].

GC-MS Column: 5% phenylmethyl silicone (17 m $\times$ 0.2 mm i.d., 0.33 μm). Temperature programme: 85° for 1.5 min to 270° at 15°/min to 290° at 50°/min for 2.5 min. EI ionisation at 70 eV, scan mode. Limit of quantification, 10 $\mu\text{g/L}$, limit of detection, 5 $\mu\text{g/L}$ [Strano-Rossi *et al.* 2008]. Column: HP-5MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Temperature programme: 100° for 3 min to 300° at 20°/min for 3 min. MSD. Retention time: 7.67 min. Limit of quantification, 12.5 $\mu\text{g/L}$, limit of detection, 5 $\mu\text{g/L}$ [Saito *et al.* 2007]. Column: HP Ultra 2 5% diphenyl 95% dimethylsiloxane (12.5 m $\times$ 0.2 mm i.d., 0.33 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 80° for 2 min to 150° at 10°/min to 310° at 10°/min for

1 min. EI ionisation at 70 eV, MSD, scan mode. Limit of detection, 5.0 mg/L [Spyridaki *et al.* 2001]. See Plasma [Frison *et al.* 2005].

HPLC Column: Nucleosil-100 C₁₈ (125 $\times$ 4 mm i.d., 5 μm). Mobile phase: acetonitrile: 0.03 mol/L sodium phosphate buffer (pH 3.0, 7:93 to 30:70 in 10 min), flow rate 1.0 mL/min. UV detection (λ =210 nm). Limit of quantification, 10 $\mu\text{g/L}$, limit of detection, 5 $\mu\text{g/L}$ [Bagheri *et al.* 2008]. Column: Hypersil C₁₈ (250 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: 10 mmol/L TEA and 20 mmol/L potassium dihydrogen phosphate (pH 3): acetonitrile (88:12 for 6 min to 45:55 at 25 min), flow rate 1.0 mL/min. UV detection (λ =210 nm). Limit of detection, 0.05 mg/L [Zhang *et al.* 2008]. Column: Spherisorb C₁₈. Mobile phase: tetraethylammonium phosphate: methanol. UV detection (λ =206 nm). Limit of detection, 0.1–0.3 mg/L [Chan *et al.* 2005]. Column: Hypersil BDS C₁₈ (150 $\times$ 3 mm i.d., 3 μm). Mobile phase: 3% acetonitrile in 0.1% sulfuric acid. Retention time: 11.9 min. Limit of quantification, 5.5 mg/L [Gmeiner *et al.* 2002]. Column: Symmetry Shield (compared with other columns). Mobile phase: 50 mmol/L phosphate buffer: 25 mmol/L TEA, flow rate 1.5 mL/min to 2.0 mL/min in 10 min. UV detection (λ =215 nm). Limit of detection, 0.3 mg/L [Imaz *et al.* 2000]. Column: Sep Spherisorb ODS 1. Mobile phase: tetraethylammonium phosphate: methanol. UV detection (λ =214 nm). Limit of detection, 0.5 mg/L [van der Merwe *et al.* 1994]. See also Imaz *et al.* [1993].

LC-MS Column: C₈. Mobile phase: water: acetonitrile (98:2) containing 0.1% HAc and 0.01% trifluoroacetic acid. Limit of detection, 2.5–5 mg/L [Deventer *et al.* 2009]. Column: Hypersil GOLD (100 $\times$ 2.1 mm i.d., 5 μm). Mobile phase: 1% acetonitrile in 10 mmol/L formic acid: 60% acetonitrile in 10 mmol/L formic acid (100:0 to 0:100 at 10 min to 100:0 for 4 min), flow rate 200 $\mu\text{L/min}$. ESI, positive ion mode. Limit of quantification, 5 $\mu\text{g/L}$ [Björnstad *et al.* 2009]. Limit of detection, 1 mg/L [Trujillo, Sorenson 2003]. See Blood [Qiu *et al.* 2008]. See Plasma [Jacob *et al.* 2004].

CE Buffer: 0.1 mol/L phosphate electrolyte (pH 2.5): acetonitrile (90:10). Limit of detection, 8 $\mu\text{g/L}$ [Wei *et al.* 2007]. Limit of detection, 3 $\mu\text{g/L}$ for ephedrine and 5 $\mu\text{g/L}$ for pseudoephedrine [Fang *et al.* 2006b]. See Serum [Fang *et al.* 2006a].

Note For an ion mobility spectroscopy method for the analysis of ephedrine, see Lokhnauth, Snow [2005].

Oral Fluid GC-MS See Urine. SIM acquisition mode [Strano-Rossi *et al.* 2008].

Brain GC-MS See Blood [Backer *et al.* 1997].

Hair GC-MS See Plasma. Limit of detection, 0.1–0.2 $\mu\text{g/g}$ [Frison *et al.* 2005]. Column: V-1 capillary (equivalent). Temperature programme: 60° to 280° at 20°/min. Limit of detection, <50 pg derivatised ephedrine [Nakahara, Kikura 1997].

Kidney GC-MS See Blood [Backer *et al.* 1997].

Liver GC-MS See Blood [Backer *et al.* 1997].

Disposition in the Body Ephedrine is readily absorbed after oral or percutaneous administration. It is metabolised by *N*-demethylation to norephedrine (phenylpropanolamine) and by oxidative deamination followed by conjugation. It accumulates in the liver, lungs, kidneys, spleen and brain. Approximately 90% of a dose is excreted in the urine in 24 h, with ~55–75% of the dose as unchanged drug, 8–20% as norephedrine and 4–13% as deaminated metabolites such as benzoic acid, hippuric acid and 1-phenylpropane-1,2-diol. In acidic urine, excretion of unchanged drug is increased slightly, whereas in alkaline urine, ~20–35% of the dose is excreted unchanged and the proportion of norephedrine is increased.

Ephedrine is a metabolite of methylephedrine.

Therapeutic Concentration In plasma, usually in the range 0.035–0.08 mg/L.

After a single oral dose of 22 mg of ephedrine hydrochloride to 10 subjects, peak plasma concentrations of 0.04–0.14 mg/L (mean 0.08) were attained; after daily oral doses of 33 mg of ephedrine hydrochloride to 10 subjects, peak plasma concentrations of 0.07–0.12 mg/L (mean 0.08) were obtained [Costello *et al.* 1975].

Ephedrine toxicity has been linked to nutritional supplements containing *Ephedra sinica* (ma huang) and a number of deaths have occurred. The pharmacokinetics of 3 commercially available ma huang products (mean ephedrine content per dose: 27, 25.6 and 23.6 mg, respectively) were compared with those of ephedrine hydrochloride capsules (25 mg dose) in 10 subjects. The mean peak plasma concentrations of ephedrine (and the time at which the peak occurred) were ephedrine hydrochloride 86 $\mu\text{g/L}$ (at 2.81 h) and the ma huang products 100.1 $\mu\text{g/L}$ (at 2.68 h), 86.2 $\mu\text{g/L}$ (at 2.61 h) and 73.4 $\mu\text{g/L}$ (at 3.05 h), respectively [Gurley *et al.* 1998].

A single 25-mg dose of ephedrine may result in urine levels above 10 mg/L if tested within ~8 h after administration [Tseng *et al.* 2006].

For the pharmacodynamics and pharmacokinetics of single oral and nasal doses of ephedrine, see Berlin *et al.* [2001].

Toxicity The estimated minimum lethal dose in children up to 2 years of age is 200 mg and for adults 2 g, but fatalities are rare. Single doses of up to 400 mg have been given without causing serious toxic effects.

In a 28-year-old woman whose death was attributed to the ingestion of ephedrine with suicidal intent, the following postmortem tissue concentrations were reported: blood 11 mg/L, liver 24 $\mu\text{g/g}$, kidney 14 $\mu\text{g/g}$ and brain 8.9 $\mu\text{g/g}$. Amitriptyline was also detected [Backer *et al.* 1997].

In a review of 127 autopsies where ephedrine was identified postmortem, blood ephedrine levels were found to be <0.49 mg/L in 50%, with a range of 0.07–11.73 mg/L in trauma victims and 0.02–12.35 mg/L in non-trauma cases. Norephedrine and pseudoephedrine were present in the blood of 22.8% and 6.3%, respectively. More than 88% of decedents tested positive for other drugs [Blechman *et al.* 2004].

For reports of sudden deaths related to ephedrine toxicity from a ma huang-containing drink, see Theoharides [1997] and Karch [1999]. For reports of 3 deaths after overdoses with Letigen (caffeine 200 mg/ephedrine 20 mg), see Hedetoft *et al.* [1999] and Kanstrup, Petersen [2003].

Half-life Plasma half-life, 3–6 h but may be increased when the urine is alkaline and decreased when it is acid.

Clearance Renal clearance 0.34 L/min.

Volume of Distribution 181 L.

Dose Ephedrine hydrochloride or sulfate 90 to 180 mg daily orally in divided doses; in concentrations of 0.5 to 1% in nasal drops.

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Wei F *et al.* (2007). Combining poly(methacrylic acid-co-ethylene glycol dimethacrylate) monolith microextraction and on-line pre-concentration-capillary electrophoresis for analysis of ephedrine and pseudoephedrine in human plasma and urine. *J Chromatogr B Analyt Technol Biomed Life Sci* 850: 38–44.

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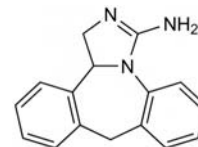
Epinastine

Antihistamine

C₁₆H₁₅N₃ = 249.3

CAS—80012-43-7

Synonyms 9,13b-Dihydro-1H-dibenzo[*c,f*]imidazo[1,5-*a*]azepin-3-amine; WAL-801.



Chemical Properties A crystalline powder. Mp 205° to 208°. pK_a 11.2. Log P (octanol/water), 3.51; log P (octanol/aqueous buffer, pH 7.4 (20°)), –0.70.

Epinastine Hydrobromide

C₁₆H₁₅N₃·HBr = 330.2

Chemical Properties A crystalline powder. Mp 284° to 286°.

Epinastine Hydrochloride

C₁₆H₁₆ClN₃ = 285.8

CAS—108929-04-0

Synonym WAL-801-Cl

Proprietary Names Alesion, Flurinol.

Chemical Properties A crystalline powder. Mp 273° to 275°. It is freely soluble in water.

Quantification

Plasma HPLC Column: 5C18-MS Cosmosil (150 × 4.6 mm i.d.). Mobile phase: triethylamine (0.3%, pH 4.5): methanol (64:36), flow rate 1.3 mL/min. UV detection (λ=220 nm). Internal standard (IS): diphenidol. Retention time: epinastine, 8.1 min; IS, 14.8 min. Limit of detection, 4 μg/L for epinastine hydrochloride [Ohtani *et al.* 1996].

Disposition in the Body Epinastine is excreted mainly via urine and faeces unchanged.

Dose A usual dose of 20 mg daily is administered.

Ohtani H *et al.* (1996). Quantitative determination of epinastine in plasma by high-performance liquid chromatography. *J Chromatogr B Biomed Appl* 683(2): 281–284.

Epirubicin

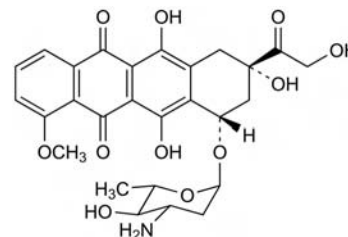
Antineoplastic

C₂₇H₂₉NO₁₁ = 543.5

CAS—56420-45-2

IUPAC Name (7S,9S)-7-[(2R,4S,5R,6S)-4-Amino-5-hydroxy-6-methyloxan-2-yl]oxy-6,9, 11-trihydroxy-9-(2-hydroxyacetyl)-4-methoxy-8,10-dihydro-7H-tetra-cene-5,12-dione

Synonyms (8S,10S)-10-[(3-Amino-2,3,6-trideoxy-α-L-arabino-hexopyranosyl)oxy]-7,8,9,10-tetrahydro-6,8,11-trihydroxy-8-(hydroxyacetyl)-1-methoxy-5,12-naphthacenedione; 4'-epiadriamycin; 4'-epi-DX; 4'-Epidoxorubicin; IMI-28; pldorubicin.



Epirubicin Hydrochloride

C₂₇H₂₉NO₁₁·HCl = 580.0

CAS—56390-09-1

Proprietary Names Farmorubicin; Farmorubicina; Farmorubicine; Pharmorubicin.

Fenproporex

Anorectic

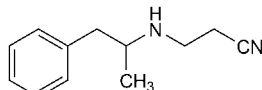
$C_{12}H_{16}N_2 = 188.3$

CAS—15686-61-0; 18305-29-8 (hydrochloride)

IUPAC Name 3-(1-Phenylpropan-2-ylamino)propanenitrile

Synonyms *N*-2-Cyanoethylamphetamine; 3-[(1-methyl-2-phenylethyl)amino]propanenitrile.

Proprietary Names *Antioibes*; *Delgafen*; *Desobesi-M*; *Dicel*; *Drenur*; *Feniseq*; *Gacilin*; *Igrasmin*; *Ifa Dex*; *Ifa Diety*; *Inobesin*; *Lipomax*; *Nobese*; *Obisin*; *Pesex-R*; *Solvolip*; *Tegisec* (includes proprietary names of the hydrochloride salt).



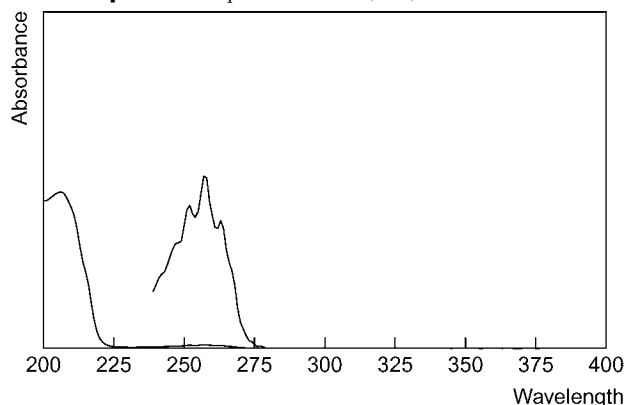
Chemical Properties Log *P* (octanol/water), 1.7.

Thin-layer Chromatography System TE—*R_f* 0.77.

Gas Chromatography System GA—fenproporex RI 1585, fenproporex-TFA RI 1705, fenproporex-PFP RI 1685, fenproporex-AC RI 1915; system GB—fenproporex RI 1648.

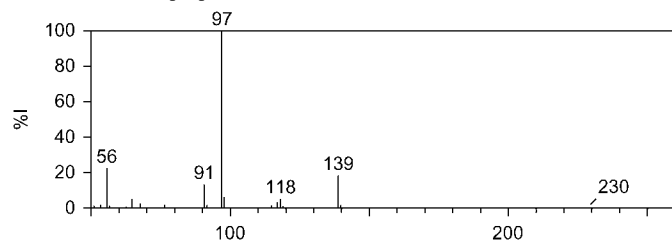
High Performance Liquid Chromatography System HY—RI 226; system HAA—RT 19.3 min.

Ultraviolet Spectrum Aqueous acid—252, 257, 263 nm.



Infrared Spectrum Principal peaks at wavenumbers 745, 1583, 697, 1019, 707, 1139 cm^{-1} (Fenproporex hydrochloride), (KBr disk).

Mass Spectrum Principal ions at *m/z* 97, 56, 91, 68, 132, 173; fenproporex-AC *m/z* 97, 56, 139, 91, 118, 65 [Kraemer *et al* 2000]; fenproporex-PFP *m/z* 243, 118, 190, 56, 91, 202; fenproporex-TFA *m/z* 193, 118, 140, 91, 56, 152.



Quantification

Blood GC Column: DB-17 (15 m × 0.32 mm i.d., 0.15 μm). Carrier gas: He, 17.5 psi. Temperature programme: 130° for 1 min to 300° at 15°/min for 1 min. NPD. Limit of detection, 1 $\mu g/L$ [Bell *et al* 2001].

GC-MS Column: HP cross-linked methylsilicone (12 m × 0.2 mm i.d., 330 nm). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° for 3 min to 310° at 30°/min for 8 min. EI ionisation at 70 eV, full scan mode. Limit of detection, 50 $\mu g/L$ [Kraemer *et al* 2000].

Urine GC See Blood [Bell *et al* 2001].

Gastric Contents GC See Blood [Bell *et al* 2001].

Liver GC See Blood [Bell *et al* 2001].

Disposition in the Body

Toxicity

In a fatality attributed to fenproporex in a young adult, the following tissue concentrations were found for fenproporex and amphetamine, respectively: inferior vena cava blood, 0.90 and 0.084 mg/L; urine, 1.2 and 0.94 mg/L; and total gastric content, 120 and 0.14 mg; other drugs detected in the blood included diazepam (0.54 mg/L), nordiazepam (0.46 mg/L), diphenhydramine (0.12 mg/L) and gamma hydroxybutyric acid (1100 mg/L) [Bell *et al* 2001].

Dose Fenproporex has been given as the hydrochloride in usual doses equivalent to 20 mg of the base daily.

Bell RR *et al*. (2001). A contemporaneous finding of fenproporex in a polydrug suicide. *J Anal Toxicol* 25: 652–656.

Kraemer T *et al*. (2000). Studies on the metabolism and toxicological detection of the amphetamine-like anorectic fenproporex in human urine by gas chromatography-mass

pectrometry and fluorescence polarization immunoassay. *J Chromatogr B Biomed Sci Appl* 738: 107–118.

Fentanyl

Narcotic Analgesic

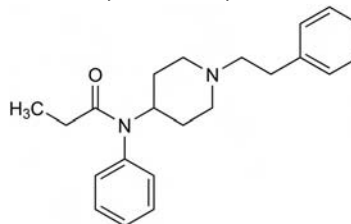
$C_{22}H_{28}N_2O = 336.5$

CAS—437-38-7

IUPAC Name *N*-[1-Phenethylpiperidin-4-yl]-*N*-phenylpropanamide

Synonym *N*-Phenyl-*N*-[1-(2-phenylethyl)-4-piperidinyl]propanamide

Proprietary Names *Fentanylum*; *Phentanyl*.



Chemical Properties Crystals. Mp 83° to 84°. Sparingly soluble in water. Extraction yield (chlorobutane), 1 [Demme *et al* 2005].

Fentanyl Citrate

$C_{22}H_{28}N_2O \cdot C_6H_8O_7 = 528.6$

CAS—990-73-8

Synonyms Fentanyl citras; McN-JR-4263-49; R-4263

Proprietary Names *Actiq*; *Durogesic*; *Fentanest*; *Haldid*; *Leptanal*; *Sinteny*; *Sublimaze*; *Tanyl*. It is an ingredient of *Fentazin* (vet.); *Hypnorm* (vet.); *Inoval*; *Innovar*; *Leptofen*; *Marcan with Fentanyl*; *Thalamonal*.

Chemical Properties White granules or a white glistening crystalline powder. Mp 149° to 151°. Soluble 1 in 40 of water, 1 in 140 ethanol, 1 in 350 of chloroform, and 1 in 10 of methanol; slightly soluble in ether. Log *P* (octanol/water pH 7.4), 2.3.

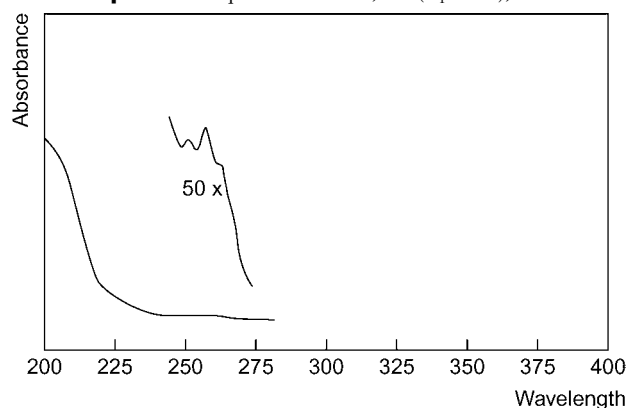
Colour Test Marquis test—orange.

Thin-layer Chromatography System TA—*R_f* 0.70; system TB—*R_f* 0.43; system TC—*R_f* 0.74; system TE—*R_f* 0.78; system TL—*R_f* 0.58; system TAE—*R_f* 0.70; system TAF—*R_f* 0.77 (acidified iodoplatinate solution, positive).

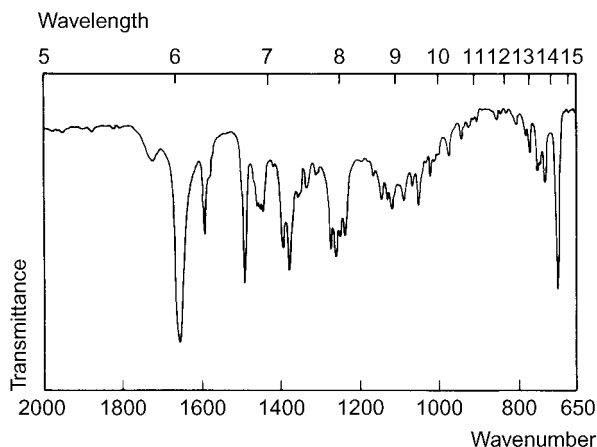
Gas Chromatography System GA—RI 2720; system GB—RI 2899.

High Performance Liquid Chromatography System HA—*k* 0.8; system HC—*k* 1.11; system HX—RI 373; system HY—RI 299; system HAA—RT 14.2 min; system HAX—RT 11.4 min; system HAY—RT 6.0 min.

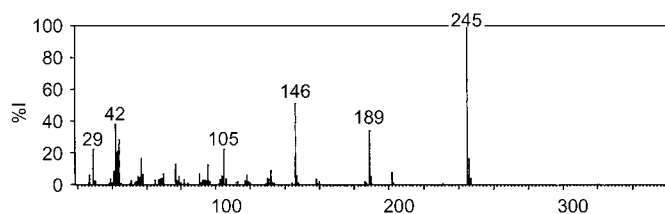
Ultraviolet Spectrum Aqueous acid—251, 257 (*A*₁ = 13a), 263 nm.



Infrared Spectrum Principal peaks at wavenumbers 1660, 701, 1493, 1263, 1273, 1236 cm^{-1} (KBr disk, see below).



Mass Spectrum Principal ions at m/z 245, 146, 42, 189, 44, 105, 29, 43.



Quantification

Blood GC NPD. Limit of detection not reported [Kowalski *et al.* 1987].

GC-MS Column: HP-5MS. SIM acquisition mode. Limit of detection, 2.5 µg/L [Wong *et al.* 2008]. Column: DB-5 (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 6 mL/min. Temperature: 270°. SIM acquisition mode. Limit of detection not reported [Martin *et al.* 2006]. Column: HP-5 (15 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 3.0 mL/min. SIM acquisition mode. Limit of detection, 1.67 µg/L [Anderson, Muto 2000].

LC-MS Column: Genesis C₁₈ reversed phase (100 × 2.0 mm i.d., 3.0 µm). Mobile phase: acetonitrile:0.1% formic acid (15:85 for 9 min to 30:70 over 13 min to 80:20 over 10 min to 95:5 for 1 min), flow rate 150 µL/min. Quadrupole MS detection, positive ion mode, MRM acquisition mode. Retention time: 13.3 min. Limit of detection, 0.08 µg/L [Gergov *et al.* 2009]. Column: XterraMS C₁₈ (100 × 2.1 mm i.d., 3.5 µm). Mobile phase: 0.15% formic acid: 0.15% acetonitrile (90:10 for 2 min to 1:100 over 6 min for 5 min to 90:10 in 1 min), flow rate 0.2 mL/min. MS/MS detection, positive ion mode, MRM acquisition mode. Limit of detection not reported [Coopman *et al.* 2007]. Column: LiChroCART LiChrospher 60 reversed phase select B. Mobile phase: 0.1% formic acid in water: acetonitrile. Limit of quantification, 0.25 µg/L, limit of detection, 0.05 µg/L [Skulska *et al.* 2007].

Plasma GC Column: HP-1 (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1 mL/min. Temperature: 220°. Quadrupole MS detection, EI ionisation, SIM acquisition mode. Limit of detection, 0.03 µg/L [Bagheri *et al.* 2007]. Column: HP Ultra (12 m × 0.2 mm i.d., 0.33 µm). Carrier gas: He, 120 kPa. SIM acquisition mode. Limit of quantification, 0.01 µg/L [Moisés *et al.* 2005]. AFID detection. Column: HP-5MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.55 mL/min. MS detection, SIM acquisition mode. Limit of detection, 20 ng/L [Fryirs *et al.* 1997]. Limit of detection, 3.3 µg/L [Van Rooy *et al.* 1981].

HPLC Column: Shim-pack CLC-C₈ (250 × 4.6 mm i.d., 5 µm). Mobile phase: 0.02% perchloric acid: methanol (48:52), flow rate 1 mL/min. Limit of detection, 0.1 µg/L [Ebrahimzadeh *et al.* 2008]. MS-MS fluorescence detection [Nitsun *et al.* 2006]. Column: Spherisorb silica (250 × 4.5 mm i.d., 5 µm). Mobile phase: methanol:0.02% perchloric acid (30:70). UV detection (λ =200 nm). Limit of detection, 200 ng/L [Portier *et al.* 1999]. Column: Econosphere reverse-phase (250 × 4.6 mm i.d., 5 µm). Mobile phase: potassium dihydrogen phosphate adjusted with 85% orthophosphoric acid: acetonitrile (65:35), flow rate 1.4 mL/min. UV detection (λ =195 nm). Limit of quantification, 2.0 µg/L, limit of detection, 0.25 µg/L [Kumar *et al.* 1996].

LC-MS Column: YMC Pro C₁₈ (50 × 2 mm i.d.). Mobile phase: acetonitrile:5 mmol/L formic acid (18:82). Triple quadrupole detection, positive ion mode. Limit of detection, 0.02 µg/L [Huynh *et al.* 2005]. Column: Betasil silica (50 × 3 mm i.d., 2 µm). Mobile phase: acetonitrile:water containing formic acid (90:10 to 50:50 over 2 min to 90:10 over 2.1 min), flow rate 0.5 mL/min. Limit of quantification, 40 ng/L [Naidong *et al.* 2002].

Serum GC Column: Chrompack fused silica (10 m × 0.32 mm i.d.). Carrier gas: He, 5 mL/min. TSD detection. Limit of detection, 0.25 µg/L [Woestenborghs *et al.* 1987].

Urine GC-MS See Blood [Wong *et al.* 2008]. Column: HP-5 (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 20 mL/min. SIM acquisition mode. Limit of detection, 3 µg/L [Poklis, Backer 2004]. Column: DB35-MS (30 m × 0.25 mm i.d., 0.15 µm). Carrier gas: He, 2.5 mL/min. EI ionisation. Limit of detection, 2.5 ng/L [Van Nimmen *et al.* 2004]. See Blood [Anderson, Muto 2000]. Limit of detection, 0.3 µg/L [Valaer *et al.* 1997].

HPLC See Plasma [Ebrahimzadeh *et al.* 2008]. DAD [Klinke, Linnet 2007].

LC-MS See Blood. Limit of detection, 0.10 µg/L [Gergov *et al.* 2009]. See Blood [Coopman *et al.* 2007]. See Blood. Limit of quantification, 2.1 µg/L, limit of detection, 1.9 µg/L [Skulska *et al.* 2007]. Column: YMC Pro C₁₈ (50 × 2 mm i.d.). Mobile phase: acetonitrile:5 mmol/L formic acid (2:98 to 30:70 over 4 min for 1 min). Triple quadrupole detection, positive ion mode. Limit of detection, 0.1 µg/L [Huynh *et al.* 2005].

Bile GC-MS See Blood [Anderson, Muto 2000].

LC-MS See Blood [Coopman *et al.* 2007].

Gastric Contents GC-MS See Blood [Anderson, Muto 2000].

LC-MS See Blood [Coopman *et al.* 2007].

Milk HPLC See Plasma [Nitsun *et al.* 2006].

Oral Fluid GC-MS Column: DB-35ms (30 m × 0.32 mm i.d., 0.25 µm). Carrier gas: He, 2.0 mL/min. Temperature: 90°. MS detection, EI ionisation, SIM acquisition mode. Retention time: 7.33 min. Limit of quantification, 5.42 µg/L [Gunnar *et al.* 2005].

Sweat LC-MS See Hair. Limit of quantification, 0.09 ng per half patch, limit of detection, 0.04 ng per half patch [Schneider *et al.* 2008].

Vitreous Humour GC-MS See Blood [Anderson, Muto 2000].

LC-MS See Blood [Coopman *et al.* 2007].

Hair GC-MS Column: 5% phenyl 95% methyl silicone DB-5 (15 m × 0.25 mm i.d., 0.25 µm). MSD, EI ionisation. Limit of quantification, 5 ng/g [Moore *et al.* 2008]. Column: HP5-MS 5% phenyl 95% methylsiloxane (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas, He, 1.0 mL/min. Temperature programme: 100° for 1 min to 295° at 30°/min for 5 min. MS-MS detection, EI ionisation. Retention time: 11.9 min. Limit of quantification, 1 ng/g [Kintz *et al.* 2005].

LC-MS Column: Xterra MS C₁₈ (150 × 3.9 mm i.d., 5 µm). Mobile phase: 20 mmol/L ammonium acetate (pH 3.8):acetonitrile (62:38), flow rate 1.0 mL/min. Retention time: 5.59 min. MS-MS detection, EI ionisation. Limit of quantification, 0.06 µg/g, limit of detection, 0.03 µg/g [Schneider *et al.* 2008]. MS-MS detection [Muschhoff *et al.* 2007].

Kidney GC-MS See Blood [Anderson, Muto 2000].

LC-MS See Blood [Coopman *et al.* 2007].

Liver GC-MS See Blood [Anderson, Muto 2000].

LC-MS See Blood [Coopman *et al.* 2007].

Disposition in the Body Fentanyl is rapidly metabolised in the liver. Two metabolites, norfentanyl and despropionylfentanyl, have been detected in plasma at concentrations similar to those of fentanyl. Approximately 70% of a dose is excreted in the urine in 72 h, mostly as metabolites, with ~10–20% of a dose being excreted as unchanged drug in 48 h. Approximately 9% of a dose is eliminated in the faeces. Fentanyl crosses the placenta and small amounts may be found in breast milk.

Therapeutic Concentration

Following IV injection of 200 µg to 6 subjects, plasma concentrations of ~2 µg/L were reported after 2 min [Michiels *et al.* 1977].

Following a single IV dose of 60 µg/kg to 5 subjects, plasma concentrations of 30–200 µg/L (mean 0.1) were reported at 1 min, decreasing to 10 µg/L at 1 h [Bovill, Sebel 1980].

After 10 subjects received 200 µg fentanyl by IV injection or 2000 µg of 50% free and 50% liposome-encapsulated fentanyl by aerosol inhalation, the mean peak plasma fentanyl level was significantly greater for the IV route (4.67 ± 1.87 at 3.6 min vs 1.15 ± 0.36 µg/L at 22.7 min); but at 8 and 24 h after administration the plasma levels were greater following inhalation (0.16 ± 0.10 vs 0.25 ± 0.14 µg/L at 8 h, and 0.05 ± 0.06 vs 0.12 ± 0.16 µg/L at 24 h, for IV and inhalation, respectively) [Hung *et al.* 1995].

In 10 children receiving treatment with fentanyl transdermal patches, the time to reach peak plasma concentration was 18 to >66 h for patches releasing 25 µg/h fentanyl. Peak plasma concentrations of 0.91 µg/L were obtained with 25 µg/h patches and 39.3 µg/L with 200 µg/h patches [Collins *et al.* 1999].

Administration of fentanyl (10 to 15 µg/kg) via an oral transmucosal delivery system to 17 children resulted in peak plasma concentrations of 1.03 ± 0.31 µg/L. The children were thought to have swallowed a large fraction of the dose, leading to a relatively late and variable peak concentration time of 53 ± 40 min [Wheeler *et al.* 2002].

Toxicity The estimated minimum lethal dose is 2 mg.

The following postmortem tissue concentrations were reported in a fatality resulting from the self-administration of fentanyl (in µg/L or µg/g): serum 17.7, blood 27.5, urine 92.7, bile 58.2, liver 0.0775, kidney 0.0415, brain 0.0302, lung 0.0834, stomach 31.6. A partly filled syringe containing 2800 µg/L fentanyl was found near the scene of death [Chaturvedi *et al.* 1990].

In a review of 25 fatalities involving fentanyl transdermal patches, the following post-mortem tissue concentrations were reported (in µg/L or µg/g): heart blood 1.89–139 (23 cases), femoral blood 3.1–43 (13 cases), vitreous humour 2–20 (4 cases), liver 0.0058–0.613 (22 cases), bile 3.5–262 (15 cases), urine 2.9–895 (19 cases), gastric 0–122 µg total (17 cases), spleen 0.0078–0.079 (3 cases), kidney 0.011 (1 case), lung 0.031 (1 case) [Anderson, Muto 2000].

An 83-year-old female who was found dead with three 100 µg/h fentanyl patches on her chest had the following tissue concentrations at postmortem: blood 25 µg/L, brain 0.054 µg/g, heart 0.094 µg/g, kidney 0.069 µg/g, liver 0.104 µg/g [Edinboro *et al.* 1997].

A postmortem serum fentanyl concentration of 2 µg/L was reported in a 35-year-old woman who had intravenously injected the contents of a transdermal fentanyl patch (5 mg, shared with another person who survived); 0.16 µg/L ethanol was also found in the serum [Reeves, Ginfier 2002].

A 63-year-old man who was found dead with 20 fentanyl patches of different strengths with a total dose of 1350 µg/h had the following tissue concentrations at postmortem: femoral blood 0.0949 µg/g, heart blood (left) 0.0459 µg/g, heart blood (right) 0.0748 µg/g, urine 101 µg/L, bile 468 µg/L, gastric contents 745 µg/L, CSF 78.4 µg/L, vitreous humour 133 µg/L [Wiesbrock *et al.* 2008].

Half-life Plasma half-life, ~3.7 h (dose dependent; also increased in the elderly and premature infants and during cardiopulmonary by-pass surgery).

Volume of Distribution Approximately 4 L/kg.

Clearance Plasma clearance, ~13 mL/min/kg.

Protein Binding ~80%.

Dose For analgesia in surgery, initially, the equivalent of 50 to 200 µg fentanyl. IV; supplementary doses of 50 µg. With assisted ventilation, an initial dose of 300 to 3500 µg may be given.

Anderson DT, Muto JJ (2000). Duragesic transdermal patch: postmortem tissue distribution of fentanyl in 25 cases. *J Anal Toxicol* 24: 627–634.

1 min to 300° at 15°/min. Elionisation at 70 eV, SIM acquisition mode. Limit of quantification, 10 µg/L [Tsujikawa *et al.* 2006].

LC-MS Dried Mushrooms. Column: Symmetry C₁₈ (150 × 2.1 mm i.d., 3.5 µm). Mobile phase: 10 mmol/L ammonium acetate:acetonitrile (70:30 for 1 min to 10:90 at 25 min for 5 min to 70:30 at 31 min for 15 min), flow rate 0.2 mL/min. DAD (λ = 256 nm). Limit of quantification, 25.9 µg/L; limit of detection, 7.8 µg/L [Tsujikawa *et al.* 2007].

Note For a GLC-mass spectral analysis of fungal metabolites, see Repke *et al.* [1978].

Disposition in the Body After ingestion, ibotenic acid hydrolysed to muscimol in the low pH of the gastric fluid; muscimol proceeds to the brain or is eliminated via the systemic circulation [Michelot, Melendez-Howell 2003]. Both ibotenic acid and muscimol are found in urine 1 h after consumption [Satora *et al.* 2006].

Toxicity The LD₅₀ in mice is 15 mg/kg (IV) and 38 mg/kg (oral); in rats it is 42 mg/kg (IV) and 129 mg/kg (oral) [O'Neil *et al.* 2006]. For an overview of the features of various types of mushroom poisoning, see DiPalma [1981].

A 47-year-old mother and 27-year-old daughter ingested 5 fried mushroom caps thinking that they were parasol mushrooms *Macrolepiota procera*. Approximately 2 h later, the women suffered from nausea, stomach ache, and diarrhoea and had vomited several times. After several days in hospital, both patients were discharged [Satora *et al.* 2006].

A 48-year-old man consumed a plate of what he thought were *A. caesarea* mushrooms. Half an hour later he became nauseous, vomited and fell asleep. His wife found him 3 h later comatose and having a seizure-like episode. Examination of the mushrooms he had picked revealed *A. muscaria* among them. Ten hours after ingestion he awoke and was completely oriented. Eight hours later, however, his condition deteriorated and he became confused and uncooperative. He developed visual and auditory hallucinations as well as paranoia. He refused treatment and was diagnosed with paranoid psychosis. He was transferred to a psychiatric hospital 3 days later. On the fifth day he started to drink and take tablets and on the sixth day all symptoms of psychosis disappeared and he was discharged [Brvar *et al.* 2006].

Note For a study investigating the ibotenic acid-muscimol content of various *Amanita* species from several continents, see Chilton and Ott [1976].

Dose Used as a neurobiological tool.

ACD (2007) *ACD/LogP DB*. Toronto ON: Advanced Chemistry Development www.acdlabs.com (accessed November 2007).

Brvar M *et al.* (2006). Prolonged psychosis after *Amanita muscaria* ingestion. *Wien Klin Wochenschr* 118: 294–297.

Chilton W, Ott SJ (1976). Toxic metabolites of *Amanita pantherina*, *A. cothurnata*, *A. muscaria* and other *Amanita* species. *Lloydia* 39: 150–157.

DiPalma JR (1981). Mushroom poisoning. *Am Fam Physician* 23: 169–172.

Michelot D, Melendez-Howell LM (2003). *Amanita muscaria*: chemistry, biology, toxicology, and ethnomycology. *Mycol Res* 107: 131–146.

O'Neil MJ *et al.* (2006). *The Merck Index, An Encyclopedia of Chemicals, Drugs and Biologicals*, 14 edn. Whitehouse Station, NJ: Merck Research Laboratories.

Repke DB *et al.* (1978). GLC-mass spectral analysis of fungal metabolites. *J Pharm Sci* 67: 485–487.

Satora L *et al.* (2006). Panther cap *Amanita pantherina* poisoning case report and review. *Toxicol* 47: 605–607.

Tsujikawa K *et al.* (2006). Analysis of hallucinogenic constituents in *Amanita* mushrooms circulated in Japan. *Forensic Sci Int* 164: 172–178.

Tsujikawa K *et al.* (2007). Determination of muscimol and ibotenic acid in *Amanita* mushrooms by high-performance liquid chromatography and liquid chromatography–tandem mass spectrometry. *J Chromatogr B Analyt Technol Biomed Life Sci* 852: 430–435.

Ibuprofen

Analgesic, COX Inhibitor, NSAID

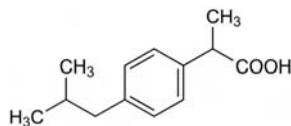
C₁₃H₁₈O₂ = 206.3

CAS—15687-27-1

IUPAC Name 2-[4-(2-Methylpropyl)phenyl]propanoic acid

Synonyms Ibuprofenum; RD-13621; U-18573.

Proprietary Names Advil; Apsifen; Arthrofen; Brufen; Cuprofen; Ebufac; Fenbid; Galprofen; Genpril; Haltran; Ibrufthalal; Ibu; Ibufac; Ibufen; Ibugel; Ibular; Ibuleve; Ibumousse; Ibuspray; Inovon; Isisfen; Junifen; Librofen; Lidifen; Manorfen; Menadol; Migrafen; Motrin; Novaprin; Nuprin; Nurofen; Obifen; Orbifen; Pacifene; PhorPain; Proflex; Relcofen; Rimafen; Saletol-200; Seclofin; Uniprofen. Ibuprofen is an ingredient of many proprietary preparations [Sweetman 2007].



Chemical Properties A white powder or crystals. Mp 75° to 77°. Practically insoluble in water; soluble 1 in 1.5 of ethanol, 1 in 1 of chloroform and 1 in 2 of ether. pK_{a1} 4.4, pK_{a2} 5.2. Log P (octanol/water), 4.0. Extraction yield (chlorobutane), 0 [Demme *et al.* 2005].

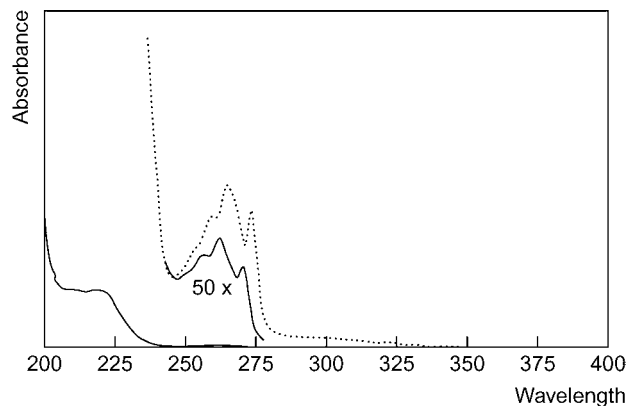
Colour Tests Liebermann's reagent—brown-orange; Marquis test—brown (→orange at 100°).

Thin-layer Chromatography System TD—R_f 0.46; system TE—R_f 0.06; system TF—R_f 0.57; system TG—R_f 0.18; system TAD—R_f 0.54; system TAE—R_f 0.75; system TAJ—R_f 0.59; system TAK—R_f 0.76; system TAL—R_f 0.93.

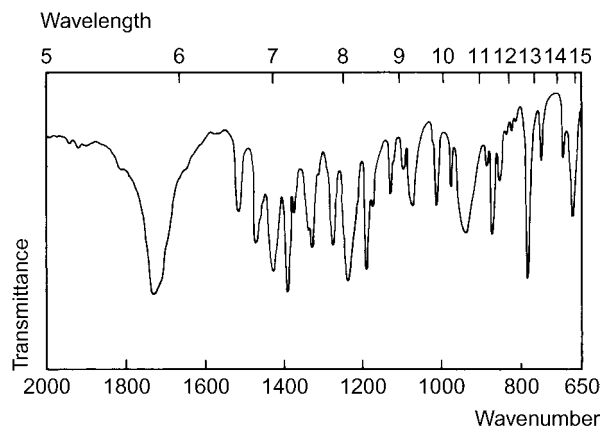
Gas Chromatography System GA—ibuprofen RI 1615, ibuprofen-Me RI 1510, M (3-OH-)-Me RI 1630, M (OH-)-Me RI 1750, M (COOH-)-Me₂ RI 1765; system GB—ibuprofen RI 1637, M (2-OH-)- RI 2096; system GD—ibuprofen-Me RRT 0.89 (relative to *n*-hexadecane); system GL—ibuprofen-Me RI 1505, M (3-OH-)-Me RI 1680.

High Performance Liquid Chromatography System HD—k 15.1; system HX—RI 616; system HY—RI 598; system HZ—RT 16.5 min; system HAA—RT 23.8 min; system HAX—RT 8.1 min; system HAY—RT 10.5 min.

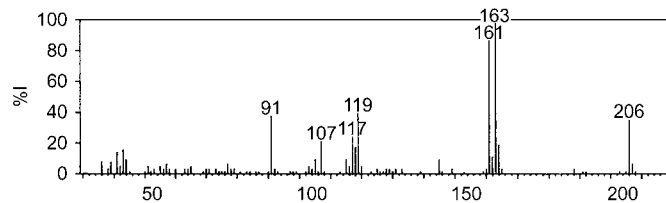
Ultraviolet Spectrum Aqueous alkali—265 (A₁ = 18.5a), 273 nm.



Infrared Spectrum Principal peaks at wavenumbers 1721, 1232, 779, 1185, 1273, 870 cm⁻¹ (KBr disk).



Mass Spectrum Principal ions at m/z 163, 161, 119, 91, 206, 117, 107, 164.



Quantification

Blood GC-MS Quantification of ibuprofen [Huber, Garg 2010]. Column: 3% OV-17 on 100/120 mesh Gas Chrom Q (2.0 m × 4 mm i.d.). Carrier gas: N₂, 70 mL/min. Temperature: 275°. ECD. Limit of detection, ~20 pg [Montagna, Groppi 1980].

Plasma HPLC Column: Waters Symmetry C₁₈ (150 × 4.6 mm i.d., 5 µm). Mobile phase: acetonitrile: water (pH 2.6, 60:40), flow rate 2 mL/min. UV detection (λ = 220 nm). Limit of quantification, 1.56 mg/L [Farrar *et al.* 2002]. Column: Waters Symmetry C₁₈ (150 × 4.6 mm i.d., 5 µm). Mobile phase: acetonitrile: water: acetic acid: triethylamine (pH 5.0, 60:40:0.1:0.02 to 100:0:0:0 at 20 min for 5 min), flow rate 1.6 mL/min. Fluorescence detection (λ_{ex} = 280 nm, λ_{em} = 320 nm). Limit of quantification, 0.1 mg/L [Canaparo *et al.* 2000]. UV detection (λ = 254 nm). Limit of detection, 50 ng [Save *et al.* 1997]. Column: C₁₈ (33 × 4.6 mm i.d., 3 µm). Mobile phase: concentrated phosphoric acid: water: acetonitrile (4:600:400), flow rate 1.5 mL/min. UV detection (λ = 220 nm). Retention time: 3.9 min. Limit of detection, 25 mg/L [Rifai *et al.* 1996]. Column: C₁₈ Econosil ODS (250 × 10 mm i.d., 10 µm). Mobile phase: methanol: 10 mmol/L acetic acid (pH 3.0, 65:35), flow rate 3.5 mL/min, UV detection (λ = 225 or 214 nm). Retention time: 24.8 min. Limit of detection, 0.5 mg/L [Castillo, Smith 1993]. C₁₈

reversed phase (100 × 4.6 mm i.d.). Mobile phase: acetonitrile: water: acetic acid: triethylamine (60:40:0.1:0.02), flow rate 1.2 mL/min. Fluorescence detection (λ_{ex} = 280 nm, λ_{em} = 320 nm) [Lemko *et al.* 1993]. Spherisorb ODS (125 × 4.6 mm i.d., 5 μ m). Mobile phase: methanol: water (pH 3.3, 65:35), flow rate 1.0 mL/min. UV detection (λ = 280 nm). Retention time: 8 min. Limit of detection, 5 mg/L [Blagbrough *et al.* 1992]. Column: C₁₈ reversed phase. Mobile phase: acetonitrile: phosphoric acid (pH 2.2, 1:1). UV detection (λ = 220 nm). Limit of detection, 25 μ g/L [Rustum 1991].

GC-MS Column: DB-1 (15 m × 0.22 mm i.d.). Temperature programme: 100° to 280° at 12°/min. Retention time: 17 min. Limit of detection, 5 mg/L [Way *et al.* 1997]. Column: 3% phenylmethylsiloxane on Chromosorb WHP 100/120 mesh (2.0 m × 2 mm i.d.). Carrier gas: He, 20 mL/min. Temperature: 320°. CI at 100 eV, SIM acquisition mode. Retention time: 2.15 min for (S)-(+)-ibuprofen and 2.55 min for (R)-(-)-ibuprofen. Limit of detection, 5 μ g/L [Zhao *et al.* 1994]. Column: cross-linked methyl silicone HP-1 (25 m × 0.32 mm i.d., 0.17 μ m). Carrier gas: He, 55 cm/s. Temperature programme: 60° to 200° at 30°/min to 240° at 8°/min to 290° at 30°/min. EI ionisation, SIM acquisition mode. Retention time: 8.86 min for (S)-ibuprofen and 8.96 min for (R)-ibuprofen. Limit of quantification, 3 μ g/L, limit of detection, <1 μ g/L [Jack *et al.* 1992].

Serum HPLC Column: YMC ODS (70 × 4.6 mm i.d., 5 μ m) or TSK ODS 80 TM (150 × 4.6 mm i.d., 5 μ m). Mobile phase: 20 mmol/L phosphate buffer (pH 3.5): acetonitrile (60:40), containing 5 mmol/L tetra-*n*-butylammonium bromide or 20 mmol/L phosphate buffer (pH 7.0): acetonitrile (70:30) containing 5 mmol/L tetra-*n*-butylammonium bromide or 20 mmol/L phosphate buffer (pH 7.0): acetonitrile (70:30), flow rate 1.0 mL/min. UV detection (λ = 221 nm). Limit of detection, 0.5 μ g/L [Yamashita *et al.* 1991].

GC Column: 10% 3-cyanopropyl silicone on 100/120 mesh Gas Chrom Q (1.83 m × 3 mm i.d.). Carrier gas: CH₄:Ar (5:95), 10 mL/min. Temperature: 190°. ECD. Retention time: 9.1 min. Limit of detection, 100 μ g/L [Kaiser, Martin 1978]. Column: 5% FFAP on Gas Chrom W (HP) 80/100 mesh (1.83 m × 2 mm i.d.). Carrier gas: N₂, 35 mL/min. Temperature: 220°. FID. Retention time: 4.3 min. Limit of detection, 500 μ g/L [Hoffman 1977].

Urine HPLC Column: Inertsil ODS-2 (150 × 4.6 mm, i.d., 5 μ m). Mobile phase: 50 mmol/L phosphate buffer (58:42, pH 5.0), flow rate 0.9 mL/min. Limit of detection, 50 μ g/L [Hirai *et al.* 1997]. Column: Polygosi 60-5 (120 × 4.6 mm, i.d., 5 μ m). Mobile phase: acetonitrile: 0.2 mol/L potassium phosphate (pH 7.4): water (18:15:67), flow rate 1.0 mL/min. UV detection (λ = 220 nm). Limit of quantification, 4–23.3 mg/L, limit of detection, 1–12 mg/L for phase I and II metabolites [Kepp *et al.* 1997]. Column: Partisil silica column (250 × 4.6 mm i.d., 5 μ m). Mobile phase: hexane: ethanol (98.2:1.8) containing 0.05% trifluoroacetic acid, flow rate 2 mL/min. UV detection (λ = 220 nm). Limit of quantification, 10 mg/L [Tan *et al.* 1997].

GC-MS See Blood [Montagna, Groppi 1980].

Biological Fluids, Plasma and Urine HPLC Column: C₁₈ (100 × 4.6 mm i.d., 5 μ m). Mobile phase: acetonitrile: water: acetic acid: TEA (46.5:53.5:0.1:0.03), flow rate 1.6 mL/min. UV detection (λ = 225 nm). Retention time: 17.7 min for (S)-ibuprofen and 15.7 min for (R)-ibuprofen. Limit of detection, 0.25 mg/L [Wright *et al.* 1992]. Column: Partisil 10 ODS-3 (250 × 4.5 mm i.d.). Mobile phase: acetonitrile: water (28:72): acetonitrile-0.05 mol/L monobasic potassium phosphate (50:50), flow rate 2.0 mL/min. UV detection (λ = 220 nm). Limit of detection, 1 mg/L for ibuprofen in plasma, 5 mg/L for ibuprofen or the hydroxy or carboxy metabolites in urine [Lockwood, Wagner 1982].

GC-MS Column: 3% OV-17 on Chromosorb WAW DMCS 120/140 mesh (0.91 m × 2 mm i.d.). Carrier gas: CH₄, 20 mL/min. Temperature programme: 150°. CI at 110 eV. Limit of detection, 50 pg [Whitlam, Vine 1980].

Synovial Fluid GC-MS See Plasma [Jack *et al.* 1992].

HPLC See Plasma [Blagbrough *et al.* 1992].

Disposition in the Body Ibuprofen is readily and almost completely absorbed after oral administration. >60% of a dose is excreted in the urine in 24 h, including ~9% of the dose as the 2-hydroxy metabolite, 2-[4-(2-hydroxy-2-methylpropyl)phenyl]propionic acid; ~17% as the conjugated hydroxy metabolite; ~16% as the 2-carboxy metabolite, 2-[4-(carboxypropyl)phenyl]propionic acid; and ~19% as the conjugated carboxy metabolite (both metabolites are inactive); <10% of a dose is excreted unchanged. The remainder of the dose is probably eliminated in the faeces after excretion in the bile; excretion is virtually complete within 24 h. It is found in breast milk in very low concentrations. Ibuprofen's disposition is stereoselective and there is some metabolic conversion of the inactive (R)-(-)-enantiomer to the active (S)-(+)-form (dexibuprofen).

Therapeutic Concentration In plasma, usually in the range 20–30 mg/L.

In 98 patients receiving high doses of ibuprofen (20–30 mg/kg IV as a single dose), peak plasma concentrations of 21–150 mg/L (mean, 83) were attained in 1–3 h (mean, 1.77) [Murry *et al.* 1999].

Ibuprofen lysine (10 mg/kg) given as an IV bolus within the first 3 h after birth to 21 premature neonates produced an ibuprofen plasma concentration of 180.6 mg/L ($\pm$ 11.1) after 1 h. In 5 of the neonates, maintenance doses of 5 mg/kg once daily on days 2 and 3 resulted in mean plasma concentrations of 116.6 mg/L ($\pm$ 54.5) and 113.6 mg/L ($\pm$ 58.2), respectively [Aranda *et al.* 1997].

Percutaneous application of ibuprofen 500 mg (10 g of a 5% gel under occlusion for 2 h, on the back 20 × 20 cm) produced a peak plasma concentration of 7.1 mg/L ($\pm$ 4.4) at 2.4 h ($\pm$ 0.8) in 18 subjects. In the same subjects, a single oral dose of 400 mg produced a peak plasma concentration of 36.7 mg/L ($\pm$ 7.5) at 1.1 h ($\pm$ 0.8) [Kleinbloesem *et al.* 1995].

After a single oral dose of 200 mg to 2 subjects, peak plasma concentrations of 18 and 24 mg/L ibuprofen were attained in 1.5 h; peak plasma

concentrations of 0.6 and 1 mg/L for the hydroxy metabolite and 1.7 and 2.1 mg/L for the carboxy metabolite were attained in 3 h; plasma concentrations after oral doses of 200 mg three times daily for 14 days were of the same order [Mills *et al.* 1973].

Toxicity

A 26-year-old male with a history of ibuprofen overdose was found dead after having recently been issued with a prescription for ibuprofen. The following postmortem tissue concentrations were reported: heart blood 518.0 mg/L, femoral blood 348.3 mg/L, liver 942 μ g/g, brain 283.9 μ g/g, gastric contents 131 mg total [Kunsmann, Rohrig 1993].

In 2 overdoses caused by ibuprofen, plasma concentrations of 400 and 711 mg/L were reported ~1.5 h after ingestion. In the first case, the estimated dose was 14–16 g; both subjects recovered [Court *et al.* 1981].

In an attempted suicide involving the ingestion of 12 g ibuprofen, an initial serum concentration of 840 mg/L was reported; this declined to 220 mg/L at 3 h; chlorpheniramine (chlorphenamine) was also detected in the urine. The subject was comatose but recovered within 24 h [Hunt, Leigh 1980].

Half-life Plasma half-life, ~2 h.

Volume of Distribution ~0.1 L/kg.

Protein Binding ~99%.

Note For a review of the clinical pharmacokinetics of ibuprofen, see Davies [1998].

Dose 0.6 to 2.4 g daily.

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Way BA *et al.* (1997). Measurement of plasma ibuprofen by gas chromatography–mass spectrometry. *J Clin Lab Anal* 11: 336–339.

Whitlam JB, Vine JH (1980). Quantitation of ibuprofen in biological fluids by gas chromatography–mass spectrometry. *J Chromatogr* 181: 463–468.

Wright MR *et al.* (1992). Improved high-performance liquid chromatographic assay method for the enantiomers of ibuprofen. *J Chromatogr* 583: 259–265.

Yamashita K *et al.* (1991). Column-switching techniques for high-performance liquid chromatography of ibuprofen and mefenamic acid in human serum with short-wavelength ultraviolet detection. *J Chromatogr* 570: 329–338.

Zhao MJ *et al.* (1994). Gas chromatographic-mass spectrometric determination of ibuprofen enantiomers in human plasma using R(-)-2,2,2-trifluoro-1-(9-anthryl)ethanol as derivatizing reagent. *J Chromatogr B Biomed Sci Appl* 656: 441–446.

Kanamycin

Antibiotic

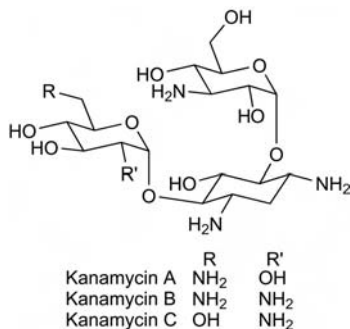
$C_{18}H_{36}N_4O_{11}$ = 484.5

CAS—59-01-8

IUPAC Name O-3-Amino-3-deoxy- α -D-glucopyranosyl-(1 $\rightarrow$ 6)-O-[6-amino-6-deoxy- α -D-glucopyranosyl-(1 $\rightarrow$ 4)]-2-deoxy-D-streptamine

Synonym Kanamycin A

Proprietary Names Kanapant; Kanibel.



Chemical Properties An antimicrobial substance produced by the growth of *Streptomyces kanamyceticus*. A whitish powder. pK_a 7.2. Log *P* (octanol/water), -6.7.

Kanamycin Acid Sulfate

$C_{18}H_{36}N_4O_{11}$, $1.7H_2SO_4$ = 651.2

Proprietary Names Kangen; Kannasyn.

Chemical Properties A white hygroscopic powder. Soluble 1 in 1 of water; practically insoluble in ethanol, acetone, chloroform and ether.

Kanamycin Sulfate

$C_{18}H_{36}N_4O_{11}$, H_2SO_4 = 582.6

CAS—25389-94-0

Synonym Kanamycin A sulfate

Proprietary Names Kamycine; Kanacolibrio; Kanamytrex; Kanasis; Kana-Stulln; Kancin; Kanescin; Kan-Ophtal; Kantrex.

Chemical Properties A white crystalline powder. Soluble 1 in 8 of water; practically insoluble in ethanol, acetone, chloroform and ether.

Thin-layer Chromatography System TA— R_f 0.01; system TAE— R_f 0.00 (acidified potassium permanganate solution, positive).

Ultraviolet Spectrum Aqueous acid—249 (A_1^1 = about 3b), 306 nm.

Infrared Spectrum Principal peaks at wavenumbers 1033, 1068, 1143, 1123, 980, 1516 cm^{-1} (kanamycin sulfate, KBr disk).

Quantification

Serum HPLC Fluorescence detection. For method for quantification of kanamycin and dibekacin, see Kubo *et al.* [1985].

Immunoassay Fluorescence detection DeCastro *et al.* [1986].

Spectrofluorimetry Limit of detection, 50 $\mu g/L$ [Csiba 1979].

Bioassay For method of quantification of aminoglycoside antibiotics in biological fluids, see Broughall [1978].

Urine Spectrofluorimetry See Serum [Csiba 1979].

Disposition in the Body Kanamycin is poorly absorbed after oral administration, but rapidly and completely absorbed after IM injection. Up to 95% of a dose is excreted unchanged in the urine in 24 h, most being excreted within the first 6 h. A small amount is excreted in the bile.

Therapeutic Concentration During treatment, the serum concentration should be in the range 20–25 mg/L and should be monitored regularly, especially in patients who have renal insufficiency. During multiple dosing, the trough concentration immediately preceding a dose should not exceed 10 mg/L.

Toxicity Toxic effects may be produced at serum concentrations of 30 mg/L or more, or during chronic treatment if the trough serum concentration exceeds 10 mg/L.

Half-life Plasma half-life, about 2 to 4 h; increased to 4–5 days in renal failure.

Volume of Distribution About 0.3 L/kg.

Clearance Plasma clearance, about 1.4 mL/min/kg.

Protein Binding Not significantly bound.

Dose The equivalent of up to 1.5 g of kanamycin daily, by IM injection.

Broughall JM (1978). Aminoglycosides. In: Reeves DS *et al.*, eds. *Laboratory Methods in Antimicrobial Chemotherapy*. Edinburgh: Churchill Livingstone, 194–207.

Csiba A (1979). Spectrofluorimetric method for aminoglycoside antibiotics. *J Pharm Pharmacol* 31: 115–116.

DeCastro AF *et al.* (1986). Determination of kanamycin concentration in serum by substrate-labeled fluorescent immunoassay. *Antimicrob Agents Chemother* 29: 961–964.

Kubo H *et al.* (1985). Rapid method for determination of kanamycin and dibekacin in serum by use of high-pressure liquid chromatography. *Antimicrob Agents Chemother* 28: 521–523.

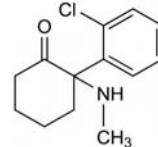
Ketamine

Anaesthetic (General)

$C_{13}H_{16}ClNO$ = 237.7

CAS—6740-88-1

IUPAC Name 2-(2-Chlorophenyl)-2-(methylamino)cyclohexanone



Chemical Properties Crystals Mp 92° to 93°. pK_a 7.5. Log *P* (octanol/water), 3.1. Extraction yield (chlorobutane), 1 [Demme *et al.* 2005].

Ketamine Hydrochloride

$C_{13}H_{16}ClNO$, HCl = 274.2

CAS—1867-66-9

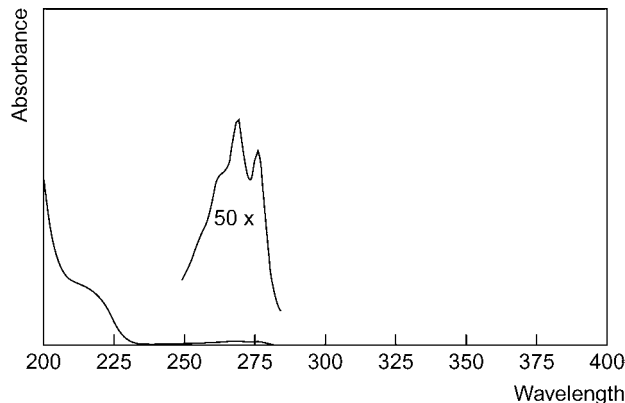
Chemical Properties A white crystalline powder. Mp 262° to 263°. Soluble 1 in 4 of water, 1 in 14 of ethanol and 1 in 6 of methanol; sparingly soluble in chloroform; practically insoluble in ether.

Thin-layer Chromatography System TA— R_f 0.63; system TB— R_f 0.37; system TC— R_f 0.63; system TE— R_f 0.79; system TL— R_f 0.64; system TAE— R_f 0.68; system TAF— R_f 0.72; system TAJ— R_f 0.47; system TAK— R_f 0.04; system TAL— R_f 0.43 (acidified iodoplatinate solution, positive).

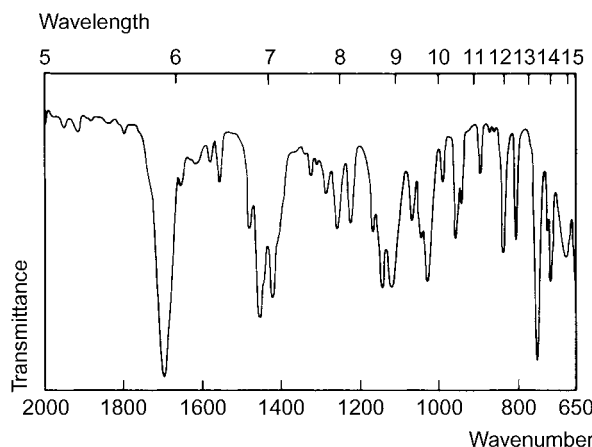
Gas Chromatography System GA—RI 1840, M (nor-) RI 1810, M (nor-OH-) -H₂O RI 1960, M (nor-OH-) -NH₃ RI 1740, M (nor-di-OH-) -2H₂O RI 1920; system GB—RI 1939, M (nor-) RI 1907, M (nor-OH-) -H₂O RI 2058, M (nor-OH-) -NH₃ RI 1840, M (nor-di-OH-) -2H₂O RI 2009, M (nor-) -H₂O RI 1931; system GM—RRT 0.427, M (nor-) RRT 0.423 (both relative to iprindole); system GV—RRT 10.4 min.

High Performance Liquid Chromatography System HX—RI 311; system HY—RI 262; system HZ—RT 2.4 min; system HAA—RT 9.6 min.

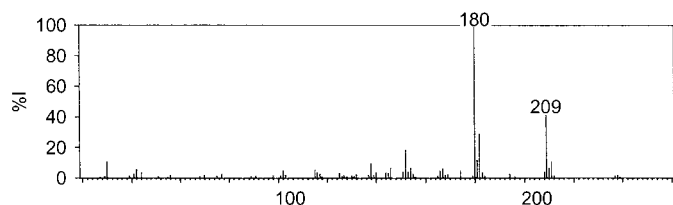
Ultraviolet Spectrum Aqueous acid—269 (A_1^1 = 25a), 276 nm.



Infrared Spectrum Principal peaks at wavenumbers 1696, 747, 1142, 1120, 712, 1027 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at m/z 180, 209, 182, 152, 181, 30, 211, 138.



Quantification

Plasma GC Column: SE-30 or OV-17 (1.8 m $\times$ 2 mm i.d.). Carrier gas: N₂, 30 mL/min. Temperature programme: 180° for 8 min to 225° at 15°/min for 3 min. NSD. Limit of detection, 80 μ g/L [Stiller *et al.* 1982].

GC-MS Column: DB-5 capillary (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He. Temperature programme: 90° to 260° at 25°/min for 10 min. EI ionisation, SIM acquisition mode. Limit of detection, 10 μ g/L [Feng *et al.* 1995].

HPLC Column: Chiralcel OD. Mobile phase: *n*-hexane:propan-2-ol (98:2). UV detection (λ = 215 nm). Limit of detection, 5 μ g/L [Yanagihara *et al.* 2000]. Column: Cyano CN (250 $\times$ 4.6 mm, 5 μ m). Mobile phase: methanol:acetonitrile:orthophosphoric acid:0.01 mol/L sodium dihydrogen phosphate (200:80:2:718). UV detection (λ = 215 nm) [Gross *et al.* 1999]. Column: Purospher RP-18. Mobile phase: acetonitrile:0.03 mol/L phosphate buffer (23:77, pH 7.2). Limit of quantification, 5 μ g/L [Bolze, Bouliou 1998]. Column: Chiral AGP. Mobile phase: 16% methanol and a 10 μ mol/L phosphate buffer (pH 7.0). UV detection (λ = 220 nm). Limit of detection, 5 μ g/L [Svensson, Gustafsson 1996]. Column: C₁₈. Mobile phase: acetonitrile:0.05 mol/L sodium phosphate buffer (pH 4.45, 30:70), flow rate 1 mL/min. UV detection (λ = 210 nm). Limit of detection, 5 μ g/L [Adams *et al.* 1992]. See also Geisslinger *et al.* [1991].

Serum GC FID. Limit of detection, 25 μ g/L [Wieber *et al.* 1975].

Urine GC See Plasma [Stiller *et al.* 1982]. See Serum [Wieber *et al.* 1975].

GC-MS Column: DB 5MS (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1.5 mL/min. Temperature programme: 90° for 2 min to 160° at 30°/min to 275° at 10°/min for 15 min. MSD, EI ionisation, SIM acquisition mode. Retention time: 10.4 min. Limit of detection, 5 μ g/L [Olmos-Carmona, Hernandez-Carrasquilla 1999].

Disposition in the Body Ketamine rapidly distributes into tissues including the brain following parenteral administration. It crosses the placenta. It has a bi- or triexponential elimination pattern. In the first phase, ketamine exerts its anaesthetic effects in the CNS, and it is then redistributed into peripheral tissues and metabolised in the liver. Approximately 90% of a dose is excreted in the urine in 72 h, with ~2% of the dose as unchanged drug, 2% as norketamine, 16% as dehydronorketamine and 80% as conjugates of hydroxylated metabolites. Norketamine (which has ~17% of the potency of ketamine) and dehydronorketamine are found in the serum in concentrations similar to those of ketamine. It has been suggested that dehydronorketamine may be an analytical artefact rather than a metabolite.

Therapeutic Concentration

Following single IV injections of 2.5 mg/kg to 5 subjects, serum concentrations of ~1 mg/L were reported after 15 min [Wieber *et al.* 1975].

During continuous IV infusion at an average rate of 41 μ g/kg/min to 31 subjects, a mean steady-state plasma concentration of 2.2 mg/L was reported; norketamine and dehydronorketamine had mean peak plasma concentrations of 1.1 mg/L and 0.7 mg/L, respectively, attained at ~3 h. The subjects awoke at an average plasma concentration of 0.64 mg/L [Idvall *et al.* 1979].

Following an IM injection of 0.5 mg/kg in 6 subjects, peak plasma concentrations of 0.10–0.43 mg/L (mean 0.24) were attained in ~0.3 h; norketamine attained a mean peak plasma concentration of 0.09 mg/L at ~1 h [Clements *et al.* 1982].

Toxicity Ketamine produces hallucinogenic effects similar to those of phencyclidine and may be subject to abuse.

The following postmortem tissue concentrations were reported in a fatality involving ketamine: blood 1.8 mg/L, urine 2.0 mg/L, brain 4.3 μ g/g, spleen 6.1 mg/g, liver 4.9 μ g/g, kidney 3.6 μ g/g; ethanol was also detected at a blood concentration of 1700 mg/L [Moore *et al.* 1997].

In a fatality caused by homicidal overdose of ketamine, tissue concentrations at postmortem were as follows: blood 27.4 mg/L, urine 8.51 mg/L, bile 15.2 mg/L, brain 3.24 mg/L, liver 6.6 mg/L, kidney 3.38 mg/L; norketamine was also detected in all samples but not quantified [Licata *et al.* 1994].

Half-life Plasma half-life, ~2–3 h.

Volume of Distribution ~4 L/kg.

Clearance Plasma clearance, ~17 mL/min/kg.

Distribution in Blood Plasma: whole blood ratio, ~0.6.

Protein Binding ~20–50%.

Note For a study of the stability of ketamine and its metabolites in human biological samples, see Hijazi *et al.* [2001]. For a review of the recent trend of ketamine misuse in the USA, see Moore *et al.* [2001].

Dose The equivalent of 1 to 4.5 mg/kg ketamine, by slow IV injection or 6.5 to 13 mg/kg, IM.

Adams HA *et al.* (1992). [The simultaneous determination of ketamine and midazolam using high pressure liquid chromatography and UV detection (HPLC/UV)]. *Anaesthesiology* 41: 619–624. Bolze S, Bouliou R (1998). HPLC determination of ketamine, norketamine, and dehydronorketamine in plasma with a high-purity reversed-phase sorbent. *Clin Chem* 44: 560–564. Clements JA *et al.* (1982). Bioavailability, pharmacokinetics, and analgesic activity of ketamine in humans. *J Pharm Sci* 71: 539–542.

Demme U *et al.* (2005). Systematic evaluation of 1-chlorobutane for liquid–liquid extraction of drugs. *Proceedings of the 12th TIAFT*, Seoul: 481–486.

Feng N *et al.* (1995). Development of a gas chromatography–mass spectrometry method for determination of ketamine in plasma and its application to human samples. *Ther Drug Monit* 17: 95–100. Geisslinger G *et al.* (1991). Stereoselective high-performance liquid chromatographic determination of the enantiomers of ketamine and norketamine in plasma. *J Chromatogr* 568: 165–176.

Gross AS *et al.* (1999). Simultaneous analysis of ketamine and bupivacaine in plasma by high-performance liquid chromatography. *J Chromatogr B Biomed Sci Appl* 728: 107–115.

Hijazi Y *et al.* (2001). Stability of ketamine and its metabolites norketamine and dehydronorketamine in human biological samples. *Clin Chem* 47: 1713–1715.

Idvall J *et al.* (1979). Ketamine infusions: pharmacokinetics and clinical effects. *Br J Anaesth* 51: 1167–1173.

Licata M *et al.* (1994). A fatal ketamine poisoning. *J Forensic Sci* 39: 1314–1320.

Moore KA *et al.* (1997). Tissue distribution of ketamine in a mixed drug fatality. *J Forensic Sci* 42: 1183–1185.

Moore KA *et al.* (2001). Urine concentrations of ketamine and norketamine following illegal consumption. *J Anal Toxicol* 25: 583–588.

Olmos-Carmona ML, Hernandez-Carrasquilla M (1999). Gas chromatographic–mass spectrometric analysis of veterinary tranquilizers in urine: evaluation of method performance. *J Chromatogr B Biomed Sci Appl* 734: 113–120.

Stiller RL *et al.* (1982). Gas chromatographic analysis of ketamine and norketamine in plasma and urine: nitrogen-sensitive detection. *J Chromatogr* 232: 305–314.

Svensson JO, Gustafsson LL (1996). Determination of ketamine and norketamine enantiomers in plasma by solid-phase extraction and high-performance liquid chromatography. *J Chromatogr B Biomed Sci Appl* 678: 373–376.

Wieber J *et al.* (1975). Pharmacokinetics of ketamine in man. *Anaesthesist* 24: 260–263.

Yanagihara Y *et al.* (2000). Stereoselective high-performance liquid chromatographic determination of ketamine and its active metabolite, norketamine, in human plasma. *J Chromatogr B Biomed Sci Appl* 746: 227–231.

Ketanserin

Antihypertensive, 5-HT_{2A} Receptor Antagonist

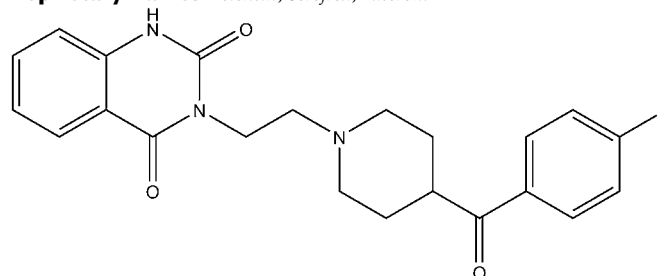
C₂₂H₂₂FN₃O₃ = 395.4

CAS—74050-98-9

IUPAC Name 3-[2-[4-(4-Fluorobenzoyl)-1-piperidinyl]ethyl]-2,4[1H,3H]-quinazolin-2-one

Synonym R-41468

Proprietary Names Ketensin; Serefrex; Taseron.



Chemical Properties Crystals. Mp 227° to 235°. Solubility 0.001 g/100 mL water, 0.038 g/100 mL ethanol, and 2.34 g/100 mL DMF, pK_a 7.5 [O'Neil MJ *et al.* 2006]. Log P (octanol/water), 3.21 [ACD 2007]. Stability of standard solutions was verified at 6° for 3 months [Kurowski 1985]. Ketanserin was stable in plasma at –20° for 3 months and there was no effect of freeze-thaw [Okonkwo *et al.* 1983; Yassen *et al.* 2003].

Ketanserin Tartrate

C₂₂H₂₂FN₃O₃ · C₄H₆O₆ = 545.5

CAS—83846-83-7

Synonyms R-49945; Ket.

Proprietary Names Perketan; Serepress; Sufrexal.

High Performance Liquid Chromatography Column: Waters μ Bondapak phenyl reversed phase (300 $\times$ 3.9 mm i.d., 10 μ m). Mobile phase: 5.44 g potassium dihydrogen phosphate in approx. 1200 mL water with 608 mL acetonitrile, pH 2.3 made up to 2 L, flow rate 1.0 mL/min. UV detection (λ = 254 nm). Limit of detection, 50 μ g/L [Simon, Somani 1982].

Quantification

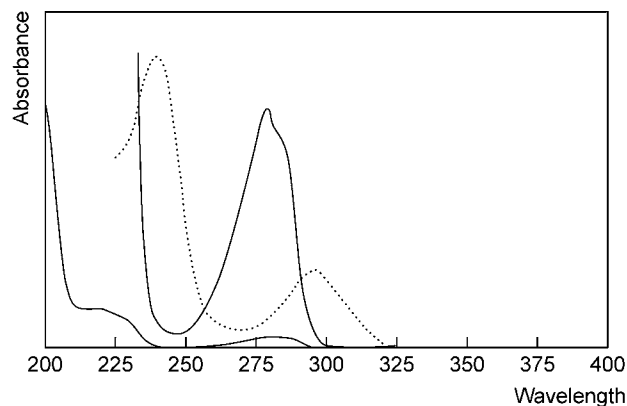
Blood HPLC Column: Macherey Nagel Nucleosil reversed phase (250 $\times$ 2 mm i.d., 10 μ m). Mobile phase: water:acetonitrile (70:30) with 0.2% diethylamine, flow rate 1.0 mL/min. Fluorometric detection (λ_{ex} = 270 nm, λ_{em} = 410 nm). Limit of detection, 0.5 μ g/L [Okonkwo *et al.* 1983].

Plasma HPLC Column: Hypersil BDS (100 $\times$ 4.5 mm i.d., 3 μ m). Mobile phase: 0.01 mol/L acetate buffer (pH 4.9):methanol:acetonitrile (52:40:8), flow rate 0.9 mL/min. Fluorescence detection (λ_{ex} = 332 nm, λ_{em} = 410 nm). Limit of quantification, 2 μ g/L for ketanserin, 1.7 μ g/L for ketanserinol [Yassen *et al.* 2003]. Column: RP-18 Chromosorb LC-7 (30 cm $\times$ 4.6 mm i.d., 5 μ m). Mobile phase: 0.1 mol/L ammonium acetate (pH 6.8):1.0 mol/L ammonium acetate (pH 6.8):methanol:acetonitrile (10:45:45; 95:5 to 30:70 at 50 min), flow rate 1.5 mL/min. UV detection (λ = 230 or 240 nm). Limit of detection, 1 μ g/L [Meuldermans *et al.* 1988]. Column: Macherey Nagel Nucleosil (250 $\times$ 4 mm i.d., 5 μ m). Mobile phase: 2% acetic acid:0.17 mol/L ammonium acetate:acetonitrile (50:10:40), flow rate 1.2 mL/min. Fluorescence detection (λ_{ex} = 310 nm, λ_{em} = 370 nm). Limit of detection, 200 ng/L for ketanserin, 100 ng/L for ketanserinol [Kurowski 1985]. Column: Chrompack CP Spher C₈ (250 $\times$ 4.6 mm i.d., 8 μ m).

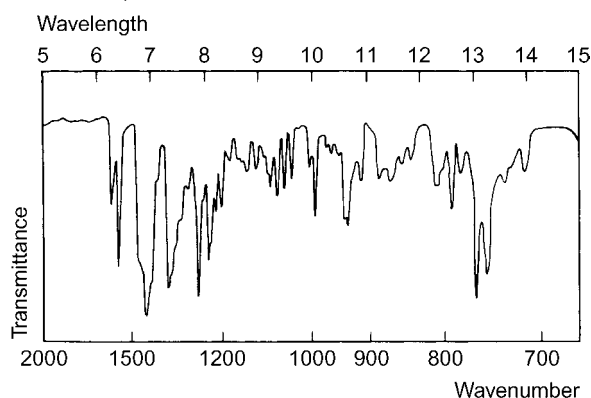
Gas Chromatography System GA—levallorphan RI 2355, levallorphan-AC RI 2390; system GB—levallorphan RI 2460.

High Performance Liquid Chromatography System HA— k 1.9 (tailing peak); system HC— k 1.46; system HX—RI 356; system HY—RI 291.

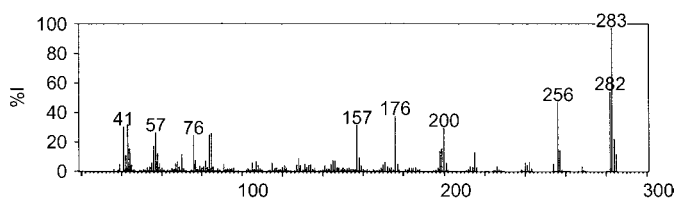
Ultraviolet Spectrum Aqueous acid—279 nm ($A_1^1=71a$); aqueous alkali—240 ($A_1^1=312a$), 299 nm.



Infrared Spectrum Principal peaks at wavenumbers 759, 1271, 747, 1575, 1242, 932 cm^{-1} (Nujol mull).



Mass Spectrum Principal ions at m/z 283, 282, 256, 176, 157, 43, 41, 57.



Disposition in the Body Levallorphan is effective within 1 min of IV injection and the effects may last up to 4 h. It is metabolised by *N*-dealkylation and glucuronic acid conjugation.

Toxicity The estimated minimum lethal dose is 0.2 g or 2 g for an addict.

Dose 0.5 to 1 mg of levallorphan tartrate intravenously, repeated if necessary.

Levamisole

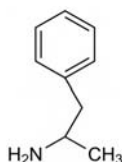
Anorectic, CNS Stimulant

$\text{C}_9\text{H}_{13}\text{N}$ = 135.2

CAS—156-34-3

IUPAC Name (–)- α -Methylbenzeneethanamine

Synonyms (–)-Amfetamine; laevo-amfetamine; levamphetamine.



Chemical Properties pK_a 10.1 (20°). Log *P* (octanol/water), 1.8.

Note For analytical data see under Amfetamine.

Levamisole

Anthelmintic

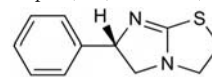
$\text{C}_{11}\text{H}_{12}\text{N}_2\text{S}$ = 204.3

CAS—14769-73-4

IUPAC Name (6*S*)-2,3,5,6-Tetrahydro-6-phenylimidazo[2,1-*b*]thiazole

Synonym 1-Tetramisole

Proprietary Names Anthelpor (vet.); Levadin (vet.).



Chemical Properties A white crystalline powder. Mp 60° to 61.5°. Soluble in dilute acetic acid and in chloroform. pK_a 8.0. Log *P* (octanol/water), 1.8.

Levamisole Hydrochloride

$\text{C}_{11}\text{H}_{12}\text{N}_2\text{S}\cdot\text{HCl}$ = 240.8

CAS—16595-80-5

Proprietary Names Armadose (vet.); Ascaridil; Chanaverm (vet.); Decaris; Decazole (vet.); Ergamisol; Ketrax; Levacide (vet.); Levacur (vet.); Levasole (vet.); Nemicide (vet.); Nilverm (vet.); Ripercol (vet.); Solaskil; Spartakon (vet.); Sure (vet.); Tramisol (vet.); Vermisole (vet.); Wormaway (vet.).

Chemical Properties A white to pale cream-coloured crystalline powder. Mp 227° to 229°. Soluble 1 in 2 of water and 1 in 5 of methanol; practically insoluble in ether.

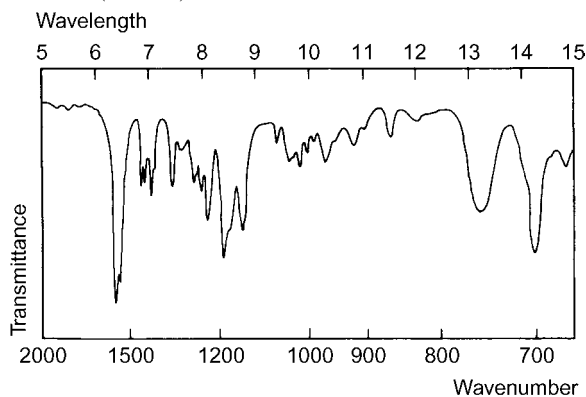
Colour Tests Liebermann's reagent—red-orange; Mandelin's test—orange→grey-green.

Thin-layer Chromatography System TA— R_f 0.62; system TB— R_f 0.18; system TC— R_f 0.48; system TE— R_f 0.65; system TL— R_f 0.42; system TAE— R_f 0.53; system TAF— R_f 0.52 (acidified iodoplatinate solution, positive).

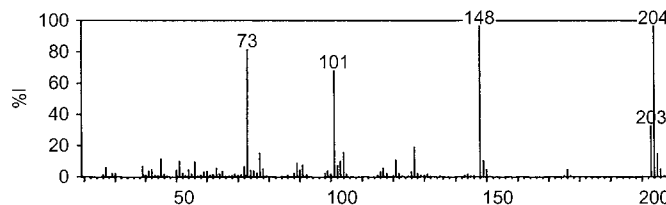
Gas Chromatography System GA—RI 1928.

High Performance Liquid Chromatography System HY—RI 238; system HAA—retention time 7.0 min.

Infrared Spectrum Principal peaks at wavenumbers 1587, 1575, 1197, 699, 1150, 1248 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at m/z 204, 148, 73, 101, 203, 127, 104, 205.



Quantification

Plasma GC FID. Limit of detection, 2 $\mu\text{g/L}$ [Kouassi *et al.* 1986]. AFID. Limit of detection, 4 $\mu\text{g/L}$ [Rousseau *et al.* 1981].

HPLC UV detection. Limit of detection, 21 $\mu\text{g/L}$ [Vandamme *et al.* 1995].

Urine GC FID. For method, see Kouassi *et al.* [1986].

HPLC UV detection. Limit of detection, 0.5 mg/L for *p*-hydroxylevamisole [Kouassi *et al.* 1986].

Other UV detection (in sheep or cattle plasma). Limit of detection 20 $\mu\text{g/L}$, UV detection [Marriner *et al.* 1980].

Disposition in the Body Levamisole is readily absorbed after oral administration, but extensively metabolised. It is almost completely excreted in the urine and faeces within 48 h.

Therapeutic Concentration

Following single oral doses of 2.5 mg/kg and 5 mg/kg to 11 subjects, mean peak plasma concentrations of 0.7 and 1.5 mg/L, respectively, were attained within 2 h [Luyckx *et al.* 1982].

Half-life Plasma half-life, about 4 h.

Dose The equivalent of 2.5 to 5 mg/kg of levamisole daily, for 2 or 3 days.

- Kouassi E *et al.* (1986). Novel assay and pharmacokinetics of levamisole and p-hydroxylevamisole in human plasma and urine. *Biopharm Drug Dispos* 7: 71–89.
- Luyckx M *et al.* (1982). Pharmacokinetics of levamisole in healthy subjects and cancer patients. *Eur J Drug Metab Pharmacokinet* 7: 247–254.
- Marriner S *et al.* (1980). Determination of the anthelmintic levamisole in plasma and gastrointestinal fluids by high-performance liquid chromatography. *Analyst, Lond* 105: 993–996.
- Rousseau F *et al.* (1981). Gas-chromatographic determination of levamisole in human plasma—normalization and reliability of the method. *Eur J Drug Metab Pharmacokinet* 6: 281–288.
- Vandamme TF *et al.* (1995). Quantitation of levamisole in plasma using high performance liquid chromatography. *Eur J Drug Metab Pharmacokinet* 20(2): 145–149.

Levetiracetam

Antiepileptic

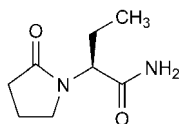
C₈H₁₄N₂O₂ = 170.2

CAS—102767-28-2

IUPAC Name (2*R*)-2-(2-Oxopyrrolidin-1-yl)butanamide

Synonyms (α*S*)-α-Ethyl-2-oxo-1-pyrrolidineacetamide; (S)-α-Ethyl-2-oxo-1-pyrrolidine acetamide; SIB-S1; UCB-L059.

Proprietary Name Keppra



Chemical Properties White to off-white crystalline powder with a faint odour and bitter taste. Crystals from ethyl acetate. Mp 117°. Extraction yield (chlorobutane), 0 [Demme *et al.* 2005]. It is very soluble in water (104 g/100 mL); freely soluble in chloroform (65.3 g/100 mL) and methanol (53.6 g/100 mL); soluble in ethanol (16.5 g/100 mL); sparingly soluble in acetonitrile (5.7 g/100 mL); practically insoluble in *n*-hexane. Stable in plasma in the autosampler at 25° for 24 h, after 5 freeze–thaw cycles and for 30 days when kept frozen at –80° [Matar 2008]. Stock solutions were stable at room temperature for 24 h and at 2 to 8° for 17 days. Levetiracetam in control human plasma was stable at room temperature for at least 6 h. Levetiracetam was stable for at least 3 freeze–thaw cycles. Levetiracetam-spiked plasma samples stored at –70° were stable for 66 days [Jain *et al.* 2006].

Gas Chromatography System GB—RI 1629.

Column: (analytical) DB-FFAP megabore (30 m × 0.53 mm i.d., 1 μm). Carrier gas: He, 30 mL/min. Temperature programme: 70° to 190° at 40°/min to 250° at 5°/min for 2 min. IS: UCB–G025. NPD. Retention time: levetiracetam, 14.0 min; IS, 11.5 min [Vermeij, Edelbroek 1994].

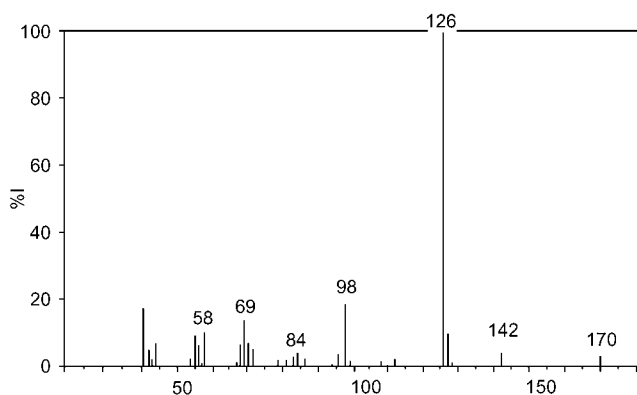
Gas Chromatography–Mass Spectrometry Column: 6-TBDMS-2,3-perme-β-cyclodextrin (20% w/w) in SE52 fused-silica capillary (10 m × 0.25 mm i.d., 0.5 μm). Carrier gas: He, 40 cm/s. Temperature programme: 110° for 2 min to 170° at 10°/min for 5 min. IS: *N*-dimethylvalproyl glycineamide. SIM acquisition mode (*m/z*: 69, 98, 126 for levetiracetam; *m/z*: 129, 157, 186 for IS). Retention time: (S)-levetiracetam, 8.6 min; (R)-levetiracetam, 8.8 min; IS, 10.3 min. [Isoherranen *et al.* 2000].

High Performance Liquid Chromatography Column: LiChrospher, 60 RP-select B (250 × 4 mm i.d., 5 μm). Mobile phase: acetonitrile: 50 mmol/L phosphate buffer (pH 5.6; 15:85), flow rate 0.8 mL/min. IS: UCB-17025. UV detection (λ = 220 nm). Retention time: levetiracetam, 5.4 min; IS, 6.8 min [Ratnaraj *et al.* 1996].

Column: Spherisorb, RP 3ODS2 (150 × 4.6 mm i.d.). Mobile phase: acetonitrile: water (6:94 to 20:80 in 5 min to 40:60 in 1 min for 10 min to 6:94 in 1 min), flow rate 1.0 mL/min. IS: UCB-G025. DAD (λ = 205 nm). Retention time: levetiracetam 6.4 min; IS 7.8 min [Vermeij, Edelbroek 1994].

Capillary Electrophoresis Column: uncoated fused silica (total/effective length 50/41 cm, 50 μm i.d.). Rinse: 1.0 mol/L sodium hydroxide for 10 min, 0.1 mol/L sodium hydroxide for 10 min, distilled water for 10 min and background electrolyte (BGE) for 10 min. Microemulsions: with 1-propanol and 1-butanol as cosurfactants: 10 mmol/L aqueous borate buffer (pH 9.2): *n*-octane: SDS: alcohol (93.76:0.48:1.80:3.96); with 1-pentanol as cosurfactant: aqueous borate buffer: *n*-octane: SDS: alcohol (96.41:0.50:2.07:1.02); with 1-hexanol as cosurfactant: aqueous borate buffer: *n*-octane: SDS: alcohol (96.87:0.51:2.10:0.52). Limit of quantification, ≈10 ppm [Ivanova *et al.* 2003].

Mass Spectrum Principal ions at *m/z* 126, 98, 41, 69, 58, 55, 127, 44.



Quantification

Plasma GC Column: HP-FFAP fused silica capillary (25 m × 0.32 mm i.d.). NPD. Limit of quantification, 4.0 mg/L, limit of detection, 0.5 mg/L [Fay *et al.* 2005]. Column: fused silica capillary FFAP (15 m × 0.32 mm i.d.). Carrier gas: 20 mL/min. Temperature: 230°. NPD. Retention time: 7 min. Limit of quantification, 0.5 mg/L [Perucca *et al.* 2003].

GC-MS Column: 6-TBDMS-2,3-perme-β-cyclodextrin (10%) in SE52 (10 m × 0.25 mm i.d., 0.5 μm). Carrier gas: He, 40 cm/s. Temperature programme: 110° for 2 min to 170° at 10°/min for 5 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of quantification, 2.35 pg/L, limit of detection, 0.94 pg/L [Isoherranen *et al.* 2000].

HPLC Column: Synergi Hydro-RP (150 × 4.6 mm, 4 μm). Mobile phase: 50 mmol/L potassium dihydrogen phosphate buffer (pH 4.5): acetonitrile (94:6), flow rate 1.5 mL/min. UV detection (λ = 205 nm). Retention time: 7.2 min. Limit of quantification, 2.0 mg/L, limit of detection, 1.0 mg/L [Contin *et al.* 2008]. Column: XTerra MS C₁₈ (250 × 4.6 mm i.d., 5 μm). Mobile phase: methanol: 10 mmol/L ammonium buffer (pH 10; 30:70), flow rate 0.8 mL/min. DAD (λ = 200 nm). Retention time: 6 min. Limit of quantification, 0.5 mg/L [Lancelin *et al.* 2007]. Column: Luna phenyl hexyl (150 × 2.0 mm i.d., 5 μm). Mobile phase: water: dibutylammonium phosphate: methanol (100:1 vial: 100), flow rate 0.5 mL/min. UV detection (λ = 220 nm). Retention time: 1.33 min. Limit of quantification, 5 mg/L [Juenke *et al.* 2006]. Column: ResElut reversed phase C₈ (150 × 4.6 mm i.d., 5 μm). Mobile phase: methanol: acetonitrile: 3 mmol/L phosphate buffer containing 0.5 mL triethylamine (6:5:89), flow rate 1.0 mL/min. DAD (λ = 205 nm). Retention time: 7.3 min. Limit of quantification, 500 μg/L, limit of detection, 150 μg/L [Pucci *et al.* 2004]. Column: LiChrospher 60 RP-select B (250 × 4 mm i.d., 5 μm). Mobile phase: acetonitrile: 50 mmol/L phosphate buffer (pH 5.6; 15:85), flow rate 0.8 mL/min. UV detection (λ = 220 nm). Retention time: 5.4 min. Limit of quantification, 5 μmol/L, limit of detection, 1 μmol/L [Ratnaraj *et al.* 1996].

LC-MS Column: Symmetry C₁₈ (50 × 3.9 mm i.d., 5 μm). Mobile phase: methanol: water: formic acid (97:0.3:0.25), flow rate 0.2 mL/min. ESI, positive ion mode. Limit of quantification, 1.0 mg/L [Matar 2008]. Column: Supelco LC-18-DB (3.3 cm × 3.0 mm i.d., 3 μm). Mobile phase: 15 mmol/L ammonium acetate: methanol (98:2): 16 mmol/L ammonium acetate: methanol (3:97) both containing 0.1% acetic acid (100:0 for 2.5 min to 0:100 at 2.51 min for 2.5 min), flow rate 1.0 mL/min. ESI, positive ion mode, MRM acquisition mode. Retention time: 2.9 min. Limit of detection, 0.1 mg/L [Guo *et al.* 2007]. Column: Alltech C₁₈ (100 × 4.6 mm i.d., 3 μm). Mobile phase: 2 mmol/L ammonium acetate: acetonitrile: 0.1% formic acid (60:30:30). Limit of detection, 0.1 mg/L [Zhao *et al.* 2007]. Column: Betasil C₁₈ (100 × 3.0 mm i.d., 3 μm). Mobile phase: buffer (pH 3.2): acetonitrile (20:80), flow rate 500 μL/min. TIS, positive ion mode, SRM acquisition mode. Retention time: 0.87 min. Limit of quantification, 0.5 mg/L [Jain *et al.* 2006].

CE Capillary: fused silica (total/effective length 32/23.6 cm, 50 μm i.d.). Running buffer: 100 mmol/L phosphate buffer (pH 8.0). UV detection (λ = 215 nm). Retention time: <8 min [Mertzman, Foley 2004]. Capillary: uncoated fused silica (total/effective length 50/41 cm, 50 μm i.d., 360 μm o.d.). Running buffer: 30 mmol/L borate buffer (pH 9.2) with 50 mmol/L SDS. UV detection (λ = 214 nm). Limit of quantification, ≈10 ppm [Ivanova *et al.* 2003].

Serum GC Column: DB-FFAP (30 m × 0.53 mm i.d., 1 μm). Carrier gas: He, 30 mL/min. Temperature programme: 70° to 190° at 40°/min to 250° at 5°/min for 2 min. NPD. Retention time: 14 min. Limit of quantification, 0.29 mg/L [Vermeij, Edelbroek 1994].

HPLC Column: Thermo Hypercarb (150 × 4.6 mm i.d., 5 μm). Mobile phase: 0.423% hydrogen phosphate: acetonitrile (95:5 to 70:30 at 8 min to 0:100 for 4 min), flow rate 1 mL/min. UV detection (λ = 205 nm). Retention time: 6.8 min. Limit of detection, 0.1 mg/L [Martens-Lobenhoffer, Bode-Böger 2005]. See Plasma [Ratnaraj *et al.* 1996]. Column: Spherisorb 3ODS2 (15 × 0.46 cm i.d.). Mobile phase: acetonitrile: water (6:94 to 20:80 at 5 min to 40:60 at 6 min for 4 min to 6:94 at 11 min for 4 min. DAD (λ = 205 nm). Retention time: 6.4 min. Limit of quantification, 0.36 mg/L [Vermeij, Edelbroek 1994].

LC-MS See Plasma [Guo *et al.* 2007].

CE Column: untreated capillary (35 cm × 50 μm). Running buffer: 140 mmol/L boric acid (pH 8.9) containing 40 g (13.8 mmol/L) SDS and 250 mL methanol/L. UV detection (λ = 214 nm). Retention time: 6.5 min. Limit of detection, ≈3 mg/L [Shihabi *et al.* 2003].

Urine GC-MS See Plasma. Limit of quantification 15 pg/L [Isoherranen *et al.* 2000].

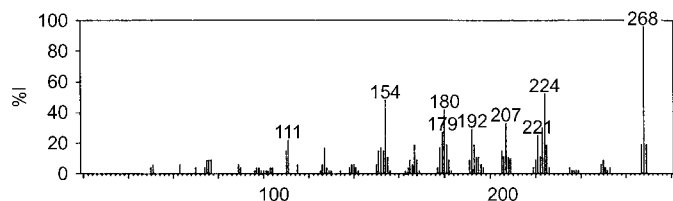
Oral Fluid LC-MS See Plasma [Guo *et al.* 2007].

Disposition in the Body Levetiracetam is rapidly and almost completely absorbed after oral administration. The extent of absorption is not affected by food [Fay *et al.* 2005] but peak plasma concentrations may be reduced by ≈20%, and time to reach these concentrations may be prolonged by ≈1.5 h. Peak plasma concentrations are reached, generally, within 1 h and steady state achieved after two days of multiple twice-daily dosing. The major metabolic pathway is enzymatic hydrolysis, which is not extensive and occurs in several tissues (including blood cells). A carboxylic acid metabolite, ucb LO57, is produced. Two other minor metabolites have been detected; they are products of hydroxylation. Levetiracetam is not metabolised by the hepatic CYP450 system and thus avoids interactions with other antiepileptic drugs. The drug and its metabolites are excreted renally, with 66% of a dose excreted unchanged in urine, whereas 27% as an inactive hydrolysis product (on the acetamido group) [Neels *et al.* 2004; Radtke 2001].

Therapeutic Concentration In serum, 10–37 mg/L [Neels *et al.* 2004].

Sixty-nine patients (39 males and 30 females) aged 18–78 years were treated with 500–3000 mg/day levetiracetam associated with one or more

Mass Spectrum Principal ions at m/z 268, 224, 154, 180, 207, 223, 192, 179.



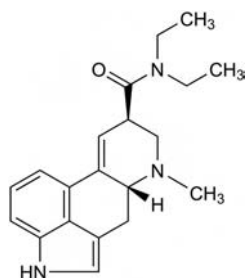
Lysergide

Hallucinogen

$C_{20}H_{25}N_3O = 323.4$

CAS—50-37-3

Synonyms 9,10-Didehydro-*N,N*-diethyl-6-methylergoline-8 β -carboxamide; *N*, *N*-diethyl-*d*-lysergamide; LSD; LSD-25; lysergic acid diethylamide.



Chemical Properties A colourless crystalline substance. Mp 80° to 85°. Soluble in water. pK_a 7.5. Log *P* (octanol/water), 2.9. Extraction yield (chlorobutane), 0.95 [Demme *et al.* 2005]. There is evidence that LSD degrades readily in biological specimens when exposed to light or elevated temperatures. It may also bind to glass containers in acidic solutions.

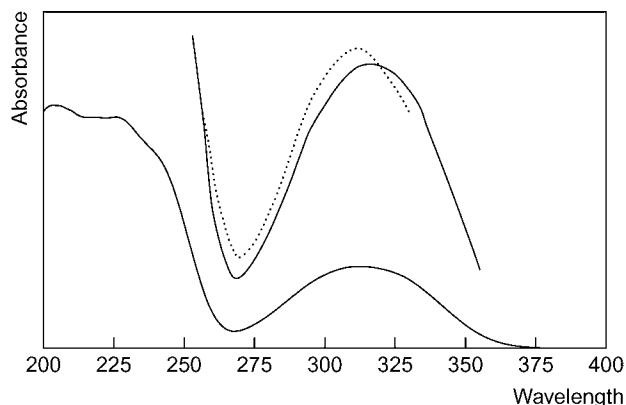
Colour Tests *p*-Dimethylaminobenzaldehyde—violet; Mandelin's test—grey; Marquis test—grey.

Thin-layer Chromatography System TA— R_f 0.60; system TB— R_f 0.03; system TC— R_f 0.39; system TD— R_f 0.30; system TE— R_f 0.56; system TL— R_f 0.24; system TM— R_f 0.70; system TAE— R_f 0.60; system TAF— R_f 0.59; system TAJ— R_f 0.33; system TAK— R_f 0.02; system TAL— R_f 0.59 (acidified iodoplatinate solution—positive; Dragendorff spray—positive; Marquis test—grey; van Urk reagent—blue).

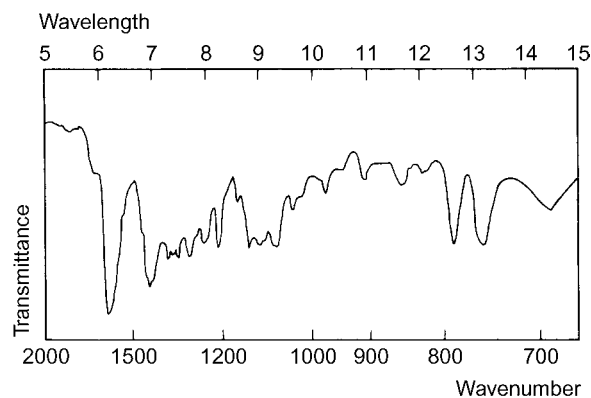
Gas Chromatography System GA—LSD RI 3445, LSD-TMS RI 3595, iso-LSD-TMS RI 3515, M (nor-)-TMS RI 3705, M (nor-)-TMS₂ RI 3515, M (2-oxo-3-OH-)-TMS₂ RI 3430; system GB—RI 3332; system GAL—RT 7.3 min.

High Performance Liquid Chromatography System HA— k 0.7; system HP—LSD k 1.83, M (2-oxy-)- k 0.92; system HX—RI 362; system HAA—RT 12.0; system HBE—RT 5.3 min.

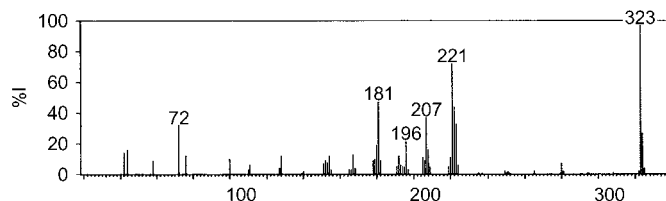
Ultraviolet Spectrum Aqueous acid—315 nm ($A_1^1 = 225a$); aqueous alkali—310 nm.



Infrared Spectrum Principal peaks at wavenumbers 1626, 1307, 1136, 1066, 1212, 749 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at m/z 323, 221, 181, 222, 207, 72, 223, 324.



Quantification

Blood GC-MS Column: CP-Sil 8CB-MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.4 mL/min. Temperature programme: 200° for 1.5 min to 220° at 30°/min to 280° at 20°/min to 300° at 2°/min for 4.83 min. CI and EI modes. Limit of detection, 0.02 $\mu g/L$ [Libong *et al.* 2003]. Column: Ultra-2 (12 m $\times$ 0.2 mm i.d., 0.33 μm). Carrier gas: H₂, 67 cm/s. Temperature programme: 175° for 0.8 min to 298° at 19°/min for 3–5 min. Desorption CI. Limit of quantification, 20 ng/L [Nelson, Foltz 1992].

HPLC Column: LiChrosphere 60 RP-Select B (5 μm). Fluorescence detection (λ_{ex} = 320 nm, λ_{em} = 420 nm). Limit of detection, 0.05 $\mu g/L$ [Bergemann *et al.* 1999].

LC-MS Column: BEH C₁₈ (50 $\times$ 2.5 mm i.d., 1.7 μm). Mobile phase: 20 mmol/L ammonium acetate buffer (pH 4.0):acetonitrile (90:10 for 1 min to 75:25 in 10 min for 1 min), flow rate 0.2 mL/min. API, ESI, MRM acquisition mode. Limit of quantification, 20 ng/L for LSD and iso-LSD, and 50 ng/L for nor-LSD and 2-oxo-3-OH-LSD, limit of detection, 5 ng/L for LSD and iso-LSD, and 10 ng/L for nor-LSD and 2-oxo-3-OH-LSD [Chung *et al.* 2009]. Column: Luna CN (150 $\times$ 2 mm i.d., 5 μm). Mobile phase: water-0.1% formic acid-2 mmol/L ammonium formate (pH 3.0): acetonitrile-0.1% formic acid-2 mmol/L ammonium formate (80:20 to 20:80 in 8 min for 4 min), flow rate 0.3 mL/min. ESI. Limit of quantification, 20 ng/L for LSD and nor-LSD, 10 ng/L for iso-LSD, limit of detection, 10 ng/L for LSD and nor-LSD [Favretto *et al.* 2007]. Column: Zorbax SB-C₁₈ (30 $\times$ 2.1 mm i.d., 3.5 μm). Mobile phase: 5% acetonitrile in 0.05% formic acid: 100% acetonitrile with 0.05% formic acid (5:95 for 2 min to 40:60 over 10 min), flow rate 0.2 mL/min. ESI, MRM acquisition mode. Limit of quantification, 0.01 $\mu g/kg$ for LD and iso-LSD [Johansen, Jensen 2005]. Column: Spherisorb 5 RP 8S (100 $\times$ 2.1 mm i.d., 5 μm). Mobile phase: water:acetonitrile (40:60) containing 0.1% formic acid and 2 mmol/L ammonium formate, flow rate 400 $\mu L/min$. ESI, MRM acquisition mode. Retention time: 4.3 min. Limit of quantification, 0.02 $\mu g/L$ for LSD and iso-LSD [Canezin *et al.* 2001]. Column: Eclipse XDB-C₁₈ (75 $\times$ 4.6 mm i.d., 3.5 μm). Mobile phase: 20 mmol/L ammonium formate (pH 4.3):methanol (70:30 for 5 min to 66:44 over 5.5 min for 11 min in total) flow rate 0.7 mL/min. ESI. SIM acquisition mode. Limit of detection, 100 ng/L for LSD and 400 ng/L for OH-LSD [Sklerov *et al.* 2000]. See also de Kanel *et al.* [1998] and Bogusz *et al.* [1998].

Plasma GC-MS Column: Ultra-1 dimethylsilicone (12.5 m $\times$ 0.2 mm i.d., 0.33 μm) or DB-5 (10 m $\times$ 0.25 mm i.d., 1.0 μm). Carrier gas: H₂, 90 cm/s. Temperature programme: 180° for 1 min to 300° at 20°/min for 1 min. ECD. Limit of detection, 0.1 $\mu g/L$ [Papac, Foltz 1990].

LC-MS Column: Zorbax SB-phenyl (75 $\times$ 4.6 mm i.d., 3.5 μm). Mobile phase: ammonium acetate buffer (pH 4.0):acetonitrile-propan-1-ol (35:65), flow rate 250 $\mu L/min$. ESI, positive ion mode, MRM acquisition mode. Limit of quantification, 0.05 $\mu g/L$ for LSD, limit of detection, 0.025 $\mu g/L$ for LSD and *N*-demethyl-LSD [de Kanel *et al.* 1998].

Serum GC-MS Column: HP-5MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Temperature programme: 150° for 2 min to 290° at 30°/min for 25 min. EI ionisation, SIM acquisition mode. Limit of detection, 0.1 $\mu g/L$ [Musshoff, Daldrup 1997].

HPLC Column: LiChrospher 60 RP Select B HP-5 MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Mobile phase: 0.05 mol/L phosphate buffer (pH 2.3):acetonitrile (75:25), flow rate 0.5 mL/min. Fluorescence detection (λ_{ex} = 231 nm, λ_{em} = 420 nm). Retention time: 14.4 min for LSD, 15.3 min for LAMPA. Limit of detection, 50 pg [Röhrich *et al.* 2000]. Comparison with radioimmunoassay, limit of detection, 0.5 $\mu g/L$ [McCarron *et al.* 1990].

LC-MS Column: Superspher RP-18 (125 $\times$ 3 mm i.d., 4 μm). Mobile phase: acetonitrile:50 mmol/L ammonium formate (pH 3.0, 25:75), flow rate 0.5 mL/min.

APCI, SIM acquisition mode. Limit of detection, 0.5 µg/L [Bogusz *et al.* 1998]. See Plasma [de Kanel *et al.* 1998].

Urine GC-MS Column: HP-5MS (30 m × 0.25 mm i.d., 0.25 µm). Temperature programme: 200° for 1 min to 260° at 50°/min to 290° at 10°/min to 320° at 50°/min for 5 min. SIM acquisition mode. Limit of quantification, 1.0 µg/L for 2-oxo-3-OH-LSD; limit of detection, 0.5 µg/L for 2-oxo-3-OH-LSD [Burnley, George 2003]. Column: Eclipse XDB-C₁₈ (150 × 4.6 mm i.d., 3.5 µm). Mobile phase: 0.01 mol/L ammonium acetate buffer (pH 8.0): acetonitrile (80:20) with 0.02% TEA. APCI, positive ion mode. Limit of quantification, 400 ng/L for OH-LSD, limit of detection, 200 ng/L for OH-LSD [Poch *et al.* 1999]. Column: ZB-5 (15 m × 0.25 mm i.d., 0.25 µm). Carrier gas: H₂, 20 psi (137.9 kPa). Temperature programme: 175° for 0.8 min to 298° at 20°/min. PICI, SRM acquisition mode. Limit of detection, 10 ng/L for LSD and 2-oxo-3-OH-LSD [Reuschel *et al.* 1999a]. Column: ZB-5 (15 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.4 mL/min. Temperature programme: 150° for 1.5 min to 220° at 30°/min to 280° at 20°/min to 300° at 10°/min for 1.5 min. EI ionisation, CID. Limit of quantification, 80 ng/L, limit of detection, 20 ng/L, [Sklerov *et al.* 1999]. See Blood [Nelson, Foltz 1992]. See also Clarkson *et al.* [1998], Francom *et al.* [1988], Lim *et al.* [1988], Paul *et al.* [1990].

HPLC See Blood [Bergemann *et al.* 1999]. Limit of detection, <1 µg/L [Blum *et al.* 1990]. See Serum [McCarron *et al.* 1990].

LC-MS See Blood. Limit of quantification, 20 ng/L for LSD and iso-LSD, and 50 ng/L for nor-LSD, 2-oxo-3-OH-LSD; limit of detection, 10 ng/L for LSD, iso-LSD, nor-LSD and 2-oxo-3-OH-LSD [Chung *et al.* 2009]. Column: RP C₁₈ (150 × 2.1 mm i.d., 3.5 µm). Mobile phase: water:acetonitrile (80:20) with 0.1% formic acid, flow rate 200 µL/min. ESI. Limit of quantification, 0.2 ng/L [Chapuis-Hugon *et al.* 2009]. See Blood [Favretto *et al.* 2007]. Column: Atlantis dC₁₈ (100 × 2.1 mm i.d., 3 µm). Mobile phase: acetonitrile:ammonium formate buffer (pH 3.0, 0:100 for 1 min to 50:50 in 10 min for 2.5 min to 0:100 in 0.5 min), flow rate 0.2 mL/min. ESI, positive ion mode, MRM acquisition mode. Retention time: 10.4 min. Limit of quantification, 0.2 µg/L, limit of detection, 0.1 µg/L, [Concheiro *et al.* 2007]. Column: Zorbax Eclipse XDB C₈ (150 × 4.6 mm i.d., 5 µm). Mobile phase: 2 mmol/L ammonium acetate in acetonitrile:water (70:30), flow rate 0.8 mL/min. ESI, SRM acquisition mode. Limit of quantification, 1 µg/L [Cui *et al.* 2004]. Mobile phase: buffer:acetonitrile (79:21 for 2 min to 63:37 over 9 min), flow rate 0.8 mL/min. SIM acquisition mode. Limit of quantification, 250 ng/L for LSD and 2-oxo-3-OH-LSD, limit of detection, 250 ng/L for LSD and 2-oxo-3-OH-LSD [Horn *et al.* 2003]. Column: Luna C₁₈ (100 × 2.0 mm i.d., 3 µm). Mobile phase: 20 mmol/L acetic acid-5 mmol/L ammonia:acetonitrile (pH 4.5, 80:20), flow rate 0.3 mL/min. API, ESI. Limit of detection, 0.02 µg/L [Bodin, Svensson 2001]. See Blood [Canezin *et al.* 2001; Johansen, Jensen 2005; Sklerov *et al.* 2000]. See Plasma [de Kanel *et al.* 1998]. See Serum [Bogusz *et al.* 1998]. See also Hoja *et al.* [1997], Klette *et al.* [2002], Poch *et al.* [2000], Webb *et al.* [1996], White *et al.* [1997] and White *et al.* [1999].

CE Capillary: fused silica (65/60 cm total/effective length, 50 µm i.d.). Running buffer: acetonitrile:methanol:water (5:35:60) containing 100 mmol/L SDS, 3 mmol/L Brij-30 and 50 mmol/L hydrogen phosphate. LTFS detection. Limit of detection, 20 ppt (with sweeping-MEKC) and 60 ppt (with CSEI-sweep-MEKC) [Fang *et al.* 2002].

Note For a comparison of EMIT and CEDIA photometric immunoassays with DPC radioimmunoassay, see Wiegand *et al.* [2002].

Body Fluids HPLC-radioimmunoassay Fluorescence detection. Limit of detection, 0.5 µg/L [Twitchett *et al.* 1978].

Fluoroimmunoassay Limit of detection, 4 µg/L [Hubbard *et al.* 1983].

Vitreous Humour LC-MS See Blood [Favretto *et al.* 2007].

Hair HPLC See Serum [Röhrich *et al.* 2000].

LC-MS See Urine. Limit of detection, 0.1 µg/g [Chapuis-Hugon *et al.* 2009].

Other HPTLC Illicit dosage forms. Limit of detection, 2 ng [Kraus *et al.* 1980].

Note For reviews of methods for determination of LSD and other drugs of abuse for investigation of drug-related driving, see Moeller, Kraemer [2002]; for methods for the determination of LSD in body fluids, see Reuschel *et al.* [1999b], Schneider *et al.* [1998] and Smith, Robinson [1985].

Note Known interferences to LSD detection include ergometrine, ergonovine, methylegometrine, dihydroergotamine, ergocornine, ergocristine, methysergide and ergotamine. Several of these agents are used clinically for various medical conditions.

Disposition in the Body LSD is readily absorbed from all mucous membranes. It can be administered through inhalation and topical application, although oral ingestion is more common. LSD-paper can be placed on skin or in the conjunctival sac for absorption of the drug. It is believed to undergo extensive biotransformation in the liver (hydroxylation and glucuronide conjugation) to inactive metabolites. A very small percentage crosses the blood-brain barrier and it is extensively bound to plasma proteins. Approximately 1% of a dose is excreted unchanged in the urine in 24 h. 2-Oxy-LSD is a major metabolite although 13-OH-LSD and 14-OH-LSD also appear.

As little as 0.5–1.5 µg/kg bodyweight is known to produce hallucinations. LSD is thought to produce its effects by binding to and activating the 5-hydroxytryptamine receptor subtype 2 (5-HT₂).

Single doses of 200–400 µg were administered to 8 subjects; the urinary concentration of LSD or a closely related metabolite ranged from 0.001–0.055 mg/L in the 24 h after ingestion [Taunton-Rigby *et al.* 1973].

Blood Concentration

A single oral dose of 160 µg was administered to 13 subjects; plasma concentrations measured at intervals over a period of 2.5 h varied considerably but were in the range 0–0.009 mg/L [Upshall, Wailing 1972].

A peak plasma concentration of 1.9 µg/L LSD was found 3 h after the administration of an oral dose of 1 µg/kg to a healthy subject. Apparent plasma half-life was 5.1 h [Papac, Foltz 1990].

Toxicity Hallucinatory effects are produced with doses of 200–400 µg but recovery has occurred after much larger doses (up to 10 mg). Plasma concentrations >0.001 mg/L have been associated with toxic effects.

Following a dinner party, 4 men and 4 women (age range 19–39 years) were admitted to hospital after 'snorting' 2 lines of white powder believed to be cocaine. Analysis of the white powder identified it as almost pure (80–90%) LSD tartrate. Emesis and collapse occurred along with sign of sympathetic overactivity, hyperthermia, coma and respiratory arrest. Concentrations of LSD tartrate ranged from 2.1–26 µg/L in serum and 1000–7000 µg/100 mL in gastric fluid. All patients recovered with supportive care [Klock *et al.* 1975].

A 25-year-old man died 16 h after being admitted to hospital. Postmortem toxicological analysis by radioimmunoassay (RIA) and HPLC revealed the following LSD concentrations (µg/L):

Sample	RIA	HPLC
Antemortem serum	14.4	–
Antemortem plasma	14.8	8
Postmortem blood	4.8	–
Stomach contents	55.2	60
Liver blood	7.2	–

[Fysh *et al.* 1985].

An 8-month-old baby was brought to the emergency department by his father after the baby had ingested 5 blotters of LSD. Charcoal and magnesium sulfate were administered. Both urine and gastric acid tested positive for LSD. The baby was admitted to the paediatric intensive care unit for observation but no further treatment was necessary [Maslanka, Scott 1992].

For a report of the clinical findings in 5 patients with suspected LSD intoxication, see [Blaho *et al.* 1997].

Half-life Plasma half-life, ~2.5 h.

Volume of Distribution ~0.3 L/kg.

Protein Binding Approximately 90%, decreasing to ~65% in overdose subjects.

Dose In psychotherapy, doses of 100 to 750 µg have been given.

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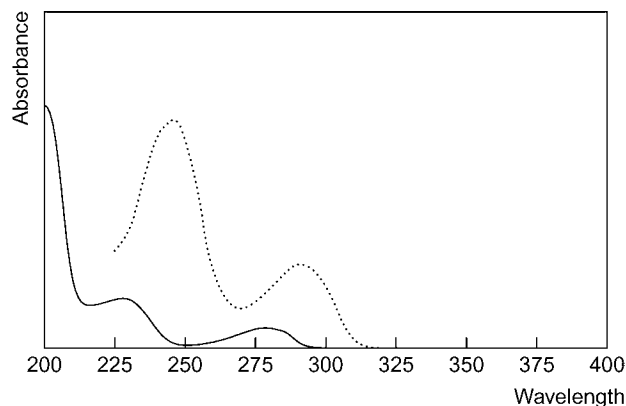
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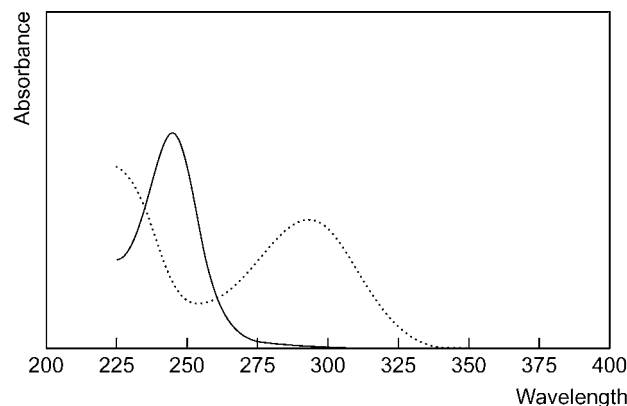
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Quantification See Normetadrenaline.

Disposition in the Body Metadrenaline is a metabolite of adrenaline.



Infrared Spectrum Principal peaks at wavenumbers 1524, 1577, 1209, 1025, 1147, 1258 cm^{-1} (KBr disk).

Metaldehyde

Molluscicide

$(\text{C}_2\text{H}_4\text{O})_x$

CAS—9002-91-9

IUPAC Name 2,4,6,8-Tetramethyl-1,3,5,7-tetraoxocane

Proprietary Names It is an ingredient of many proprietary preparations used for the destruction of slugs. As a compressed fuel, it is known as *Meta*.

Chemical Properties A polymer of acetaldehyde. A white crystalline solid which burns readily with a non-luminous carbon-free flame and sublimes at 112° . In the presence of acids it decomposes slowly to acetaldehyde. Practically insoluble in water; soluble in benzene and chloroform; sparingly soluble in alcohol and ether.

Colour Test Add 1 drop of sulfuric acid to a small amount of solid, then add a trace of pyrocatechol—violet-red.

Gas Chromatography System GA—RI 1020.

Infrared Spectrum Principal peaks at wavenumbers 1094, 1113, 1163, 1074, 975, 1205 cm^{-1} (KBr disk).

Quantification

Plasma GC [Booze, Oehme 1985].

Urine GC [Booze, Oehme 1985].

Disposition in the Body

Toxicity Metaldehyde probably decomposes slowly in the body to acetaldehyde which is further oxidised. Fatalities have occurred after the ingestion of about 3 g.

Booze TF, Oehme FW (1985). Gas chromatographic analysis of metaldehyde in urine and plasma. *J Anal Toxicol* 9: 172–173.

Metallibure

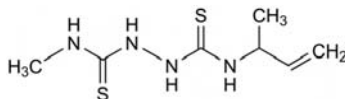
Hypothalamus and Pituitary Suppressant

$\text{C}_7\text{H}_{14}\text{N}_4\text{S}_2 = 218.3$

CAS—926-93-2

IUPAC Name 1-But-3-en-2-yl-3-(methylcarbamothioylamino)thiourea

Synonym *N*-Methyl-*N'*-(1-methyl-2-propenyl)-1,2-hydrazinedicarbothioamide



Chemical Properties A white powder. Mp 198° to 200° , with decomposition. Practically insoluble in water and chloroform; soluble 1 in 100 of acetone and 1 in 200 of methanol; soluble in pyridine. Log *P* (octanol/water), 0.3.

Caution Protective gloves should be worn when handling metallibure.

Colour Tests Nessler's reagent (100°)—black; palladium chloride—orange.

Thin-layer Chromatography System TA— R_f 0.79 (acidified iodoplatinate solution, positive).

Gas Chromatography System GA—not eluted.

Ultraviolet Spectrum Aqueous acid—243 nm ($A_1^1=1340b$); aqueous alkali—293 nm.

Metamfetamine

Central Stimulant, Phenethylamine

$\text{C}_{10}\text{H}_{15}\text{N} = 149.2$

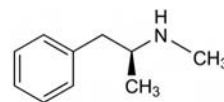
CAS—537-46-2

IUPAC Name (2*S*)-*N*-Methyl-1-phenylpropan-2-amine

Synonyms *d*-Deoxyephedrine; desoxyephedrine; (αS)-*N*, α -dimethylbenzene-neethanamine; methamphetamine; methylamfetamine; methylamphetamine; phenylmethylaminopropane.

Note Metamfetamine in a smokeable and snortable form: crank; crystal; crystal meth; ice; meth; speed.

Proprietary Name *Norodin*



Chemical Properties A clear, colourless, slowly volatile, mobile liquid. Bp $\sim 214^\circ$. Slightly soluble in water; miscible with ethanol, chloroform and ether. pK_{a1} 10.1 [Huestis, Cone 2007], pK_{a2} 9.87 [Fan *et al.* 2005]. Log *P* (octanol/water), 2.07 [Fan *et al.* 2005]. Extraction yield (chlorobutane), 0.7 [Demme *et al.* 2005].

Metamfetamine Hydrochloride

$\text{C}_{10}\text{H}_{15}\text{N}\cdot\text{HCl} = 185.7$

CAS—51-57-0

IUPAC Name (2*S*)-*N*-methyl-1-phenylpropan-2-amine hydrochloride

Proprietary Names *Amphedroxyn*; *Desfedrin*; *Desoxyfed*; *Desoxyn*; *Destim*; *Doxephlin*; *Drinalfa*; *Gerobit*; *Hiropon*; *Isophen*; *Madrine*; *Methampex*; *Methedrine*; *Methylisomyn*; *Pervitin*; *Soxysympamine*; *Syndrox*; *Tonedron*.

Chemical Properties White crystals or crystalline powder. Mp 170° to 175° . Soluble 1 in 2 of water, 1 in 4 of ethanol and 1 in 5 of chloroform; practically insoluble in ether. Log *P* (octanol/water), -0.66 [Meylan, Howard 1995].

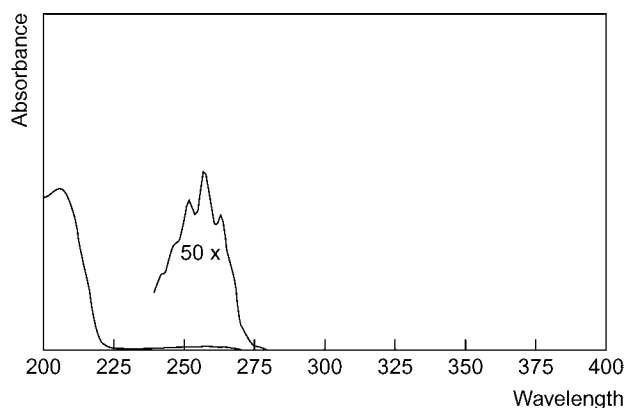
Colour Test Marquis test—orange.

Thin-layer Chromatography System TA— R_f 0.31; system TB— R_f 0.28; system TC— R_f 0.13; system TE— R_f 0.42; system TL— R_f 0.05; system TAE— R_f 0.09; system TAF— R_f 0.63; system TAJ— R_f 0.00; system TAK— R_f 0.03; system TAL— R_f 0.45 (acidified iodoplatinate solution—positive; acidified potassium permanganate—positive; Dragendorff spray—positive; Marquis test—brown; ninhydrin (spray)—positive).

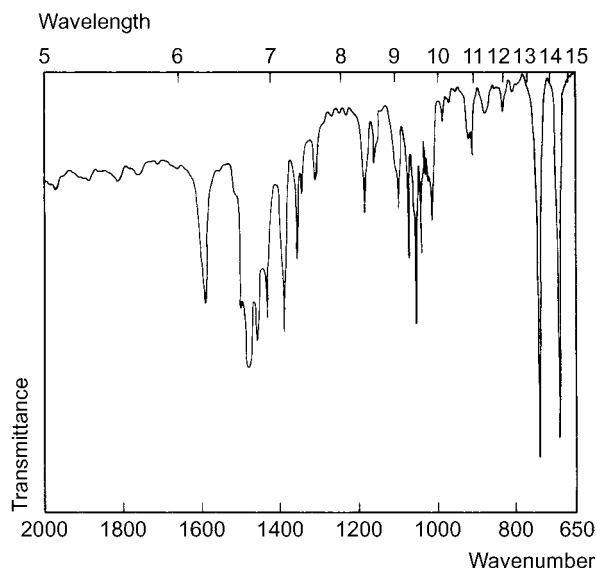
Gas Chromatography System GA—metamfetamine RI 1175, amfetamine RI 1125, M (4-OH-)(pholedrine) RI 1885, M (OH-methoxy-) RI 1810; system GB—metamfetamine RI 1200, amfetamine RI 1150; system GC—metamfetamine RI 1722, amfetamine RI 1536; system GF—metamfetamine RI 1335, amfetamine RI 1315; system GAD—metamfetamine RT 8.5 min, amfetamine RT 7.5 min.

High Performance Liquid Chromatography System HA—metamfetamine *k* 2.0, amfetamine *k* 0.9; system HB—metamfetamine *k* 10.52, amfetamine *k* 8.48; system HC—metamfetamine *k* 2.07, amfetamine *k* 0.98; system HX—RI 262; system HY—RI 216; system HZ—RT 2.4 min; system HAA—RT 8.4 min.

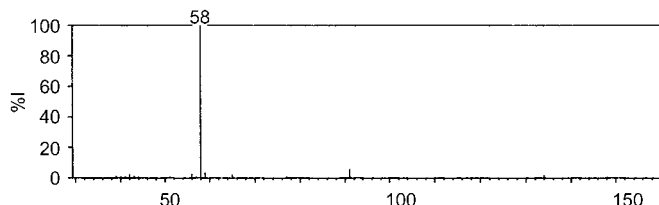
Ultraviolet Spectrum Aqueous acid—252, 257 ($A_1^1=12.1a$), 263 nm.



Infrared Spectrum Principal peaks at wavenumbers 747, 698, 1060, 1491, 1590, 1085 cm^{-1} (metamfetamine hydrochloride, KBr disk).



Mass Spectrum Principal ions at m/z 58, 91, 59, 134, 65, 56, 42, 57 (metamfetamine); 44, 91, 40, 42, 65, 45, 39, 43 (amfetamine).



Quantification

Blood GC Limit of detection, 220 $\mu\text{g/L}$ [Sun *et al.* 2007].

GC-MS Column: HP-5MS (15 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He. Temperature programme: 60° for 1 min to 220° at 20°/min. EI ionisation at 70 eV, SIM acquisition mode. Retention time: 5.3 min for metamfetamine, 4.66 min for amfetamine, 7.14 min for 3,4-methylenedioxymethamfetamine (MDMA) [Gunn *et al.* 2010]. Column: DB-5HT (15 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 2.5 to 3.5 mL/min. Temperature programme: 130° for 0.1 min to 230° at 70°/min for 0.17 min. EI ionisation, SIM acquisition mode. Limit of quantification, 25 $\mu\text{g/L}$ for metamfetamine and amfetamine [Gunnar *et al.* 2007]. EI ionisation, positive ion mode, SIM acquisition mode. Limit of detection, 1.5 $\mu\text{g/L}$ for metamfetamine and 1.1 $\mu\text{g/L}$ for amfetamine [Hasegawa *et al.* 2007]. Column: HP-5MS. SIM acquisition mode [Rasmussen *et al.* 2006]. Column: HP-5MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 0.8 mL/min. Temperature programme: 100° for 1 min to 300° at 10°/min for 3 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of detection, 5.0 ng/g for metamfetamine and amfetamine [Nishida *et al.* 2002; Nishida *et al.* 2003]. Column: PTE-5 (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 0.4 mL/min. Temperature programme: 60° for 3 min to 280° at 20°/min for 3 min. EI ionisation, SIM acquisition mode. Limit of detection, 0.5 ng/g for metamfetamine and amfetamine [Okajima *et al.* 2001]. See also Nagasawa *et al.* [1996], Namera *et al.* [2000] and Sato and Mitsui [1997].

LC-MS Column: Agilent HPLC 1100 with a Varian Pursuit 3 C₁₈ column (3 $\times$ 100 mm, 3 μm). Mobile phase: methanol-ammonium formate. Limit of

quantification, 0.0005–0.01 mg/kg for metamfetamine and other drugs of abuse [Bjork *et al.* 2010]. Column: Nucleodur Sphinx RP. Mobile phase: 10 mmol/L ammonium formate buffer and acetonitrile. Limit of quantification, 2.5 $\mu\text{g/L}$ for metamfetamine and related drugs, limit of detection, 0.05–0.5 $\mu\text{g/L}$ for metamfetamine and related drugs, [Fernández Mdel *et al.* 2009]. Column: BEH C₁₈ (50 $\times$ 2.1 mm i.d.). Mobile phase: aqueous pyrrolidine: methanol (52:48), flow rate 0.4 mL/min. ESI, positive ion mode, SIR acquisition mode. Retention time: 1.81 min for metamfetamine, 1.2 min for amfetamine [Apollonio *et al.* 2006]. Column: Uptispher ODB C₁₈ (150 $\times$ 2.1 mm, 5 μm). Mobile phase: acetonitrile and 2 mmol/L formate buffer (pH 3.0). Limit of quantification, 0.1 $\mu\text{g/L}$ for metamfetamine and other amfetamines [Chèze *et al.* 2007].

Plasma GC Column: HP-5 5% phenylmethyl (25 $\times$ 0.32 m, 0.52 μm). Carrier gas: He, 2.5 mL/min. Temperature programme: 70° to 120° at 30°/min to 185° at 5°/min to 300° at 70°/min for 1.5 min. NPD. Limit of detection, 1 $\mu\text{g/L}$ [Cheung *et al.* 1997]. Column: HP Ultra-1 (25 m $\times$ 0.2 mm i.d., 0.33 μm). Carrier gas: He, 2 mL/min. Temperature programme: 110° for 0.5 min to 150° at 5°/min for 7.25 min to 300° at 50°/min for 3 min. Retention time: 11.72 min for metamfetamine, 9.76 min for amfetamine. Limit of detection, <10 $\mu\text{g/L}$ for metamfetamine and amfetamine [Jacob *et al.* 1995].

GC-MS See Blood [Gunn *et al.* 2010]. Limit of quantification, 2.5 $\mu\text{g/L}$ for metamfetamine and metabolites [Huestis, Cone 2007]. Column: 5% phenylmethylsiloxane HP-5MS (30 m $\times$ 0.25 mm i.d., 250 nm). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° to 180° at 30°/min to 230° at 5°/min to 310° at 30°/min. NCI, SIM acquisition mode. Limit of detection, <1 $\mu\text{g/L}$ [Peters *et al.* 2003a]. Column: HP-5MS capillary 5% phenylmethylsiloxane (30 m $\times$ 0.25 mm i.d., 250 nm). Carrier gas: He, 0.6 mL/min. Temperature programme: 100° to 250° at 10°/min to 310° at 30°/min for 1 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of quantification, 5 $\mu\text{g/L}$ for metamfetamine and related drugs [Peters *et al.* 2003b]. Column: HP-5MS (30 m $\times$ 0.25 mm i.d., 250 nm). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° to 180° at 30°/min to 230° at 5°/min to 310° at 30°/min. NCI, SIM acquisition mode. Limit of quantification, 5 $\mu\text{g/L}$ for metamfetamine and amfetamine, limit of detection, 1 $\mu\text{g/L}$ for metamfetamine and amfetamine [Peters *et al.* 2002].

HPLC Column: Daisopak SP-120-5-ODS-BP (250 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: acetonitrile: 0.01 mol/L citrate buffer (pH 4.0, 55:45), flow rate 1.0 mL/min. Fluorescence detection (λ_{ex} = 325 nm, λ_{em} = 430 nm). Limit of detection, 0.87 $\mu\text{g/L}$ for metamfetamine and 0.46 $\mu\text{g/L}$ for amfetamine [Nakashima *et al.* 2003].

LC-MS Column: Alltima C₁₈ (250 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: acetonitrile: 5 mmol/L formic acid (10:90 for 2 min to 25:75 in 16 min to 35:65 in 0.1 min to 100:0 in 12 min. TIS, positive ion mode, MRM acquisition mode. Retention time: 14 min. Limit of quantification, 1.2 $\mu\text{g/L}$; limit of detection, 0.2 $\mu\text{g/L}$ [Sergi *et al.* 2009]. Column: CAPCELL PAK SCX UG 80 (250 $\times$ 1.5 mm i.d.). Mobile phase: 25 mmol/L ammonium acetate (pH 4.0):acetonitrile (3:7), flow rate 0.2 mL/min. ESI, SRM acquisition mode. Limit of detection, 50 $\mu\text{g/L}$ for metamfetamine and related drugs and metabolites [Kuwayama *et al.* 2008]. Column: Hypersil BDS-C₁₈ (100 $\times$ 2.1 mm i.d., 3.5 μm). Mobile phase: 10 mmol/L ammonium acetate:acetonitrile (75:25), flow rate 0.3 mL/min. ESI, positive ion mode. Limit of quantification, 0.5 $\mu\text{g/L}$, limit of detection, 0.2 $\mu\text{g/L}$ [Wood *et al.* 2003].

Serum GC-MS See Blood [Gunn *et al.* 2010]. See Plasma [Peters *et al.* 2002]. Column: DB-5 MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 60° to 120° at 30°/min to 180° at 20°/min to 250° at 30°/min for 0.5 min. Limit of detection, 50 ng/L for metamfetamine and 80 ng/L for amfetamine [Lee *et al.* 2000]. Column: DB5-MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Temperature programme: 80° for 1 min to 200° at 20°/min to 240° at 5°/min to 310° at 30°/min for 6 min. EI ionisation at 70 eV, SIM acquisition mode. Retention time: 6.1 min for metamfetamine, 5.3 min for amfetamine. Limit of quantification, 1.8 $\mu\text{g/L}$ for metamfetamine, 4.9 $\mu\text{g/L}$ for amfetamine, limit of detection, 1.4 $\mu\text{g/L}$ for metamfetamine, 1.5 $\mu\text{g/L}$ for amfetamine [Weinmann *et al.* 2000].

Note For a fluoroimmunoassay, see Kimura *et al.* [2005].

Urine TLC Limit of detection, for *p*-hydroxymetamfetamine, 10 ng [Kato *et al.* 2005].

GC Limit of detection, 10 $\mu\text{g/L}$ [Wang *et al.* 2008]. Column: HP-5 (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 120° for 1.5 min to 300° at 12.5 min for 7 min. FID. Limit of quantification, 0.1 mg/L, limit of detection, 0.05 mg/L [Shakleya *et al.* 2006]. Column: AT-5 (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 40° for 1 min to 280° at 20°/min for 5 min. FID. Limit of detection, 30 $\mu\text{g/L}$ for metamfetamine and amfetamine [Raikos *et al.* 2003]. Column: CB-5 methylphenylpolysiloxane (15 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 110° to 140° at 10°/min to 20° at 30°/min for 3 min. NPD. Limit of detection, ~35 $\mu\text{g/L}$ for metamfetamine and amfetamine [Taylor *et al.* 1989]. Column: 2% Thermo-3000 on Chromosorb W (AW-DMCS) 80/100 mesh (2 m $\times$ 2 mm i.d.). Carrier gas: N₂, 50 mL/min. Temperature: 200°. ECD. Retention time: 3.3 min for metamfetamine and 6.9 min for amfetamine. Limit of detection, 10 $\mu\text{g/L}$ for metamfetamine and amfetamine [Terada *et al.* 1982]. See Plasma [Cheung *et al.* 1997; Jacob *et al.* 1995].

GC-MS Column: HP-5MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 127 kPa. Temperature programme: 70° for 1 min to 300° at 20°/min for 2 min. EI ionisation at 70 eV, full scan mode. Limit of detection, 5 $\mu\text{g/L}$ for metamfetamine and MDMA, 10 $\mu\text{g/L}$ for amfetamine and methylenedioxiamfetamine (MDA) [Nakamoto *et al.* 2010]. Column: DB-5MS (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° for 1 min to 160° at 60°/min for 0.5 min to 210° at 50°/min for 0.5 min to 250° at 40°/min for 0.5 min to

280° at 30°/min for 2 min. NCI. Limit of quantification, 43.5 ng/L for metamfetamine and 4 ng/L for amfetamine, limit of detection, 13.04 ng/L for metamfetamine and 1.20 ng/L for amfetamine [Chung *et al.* 2009]. Column: DB-1MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.0 mL/min. Temperature programme: 60° for 2 min to 140° at 20°/min to 190° at 10°/min to 250° at 20°/min for 1 min. SIM acquisition mode. Limit of detection, 0.1 µg/L for metamfetamine and 0.5 µg/L for amfetamine [He *et al.* 2009]. Column: HP-5MS (20 m × 250 µm i.d., 0.25 µm). Carrier gas: He, 1.0 mL/min. Temperature programme: 50° for 1 min to 190° at 20°/min for 2 min to 250° at 5°/min for 1 min. EI ionisation, SIM acquisition mode. Limit of quantification, <7.7 µg/L, limit of detection, <2.3 µg/L [Aasim *et al.* 2008]. See Plasma [Huestis, Cone 2007]. Column: DB-5MS (30 m × 0.32 mm i.d., 0.25 µm). Carrier gas: He, 2 mL/min. Temperature programme: 60° for 1 min to 200° at 20°/min to 300° at 40°/min for 5 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of detection, 0.08 µg/L for metamfetamine and 0.1 µg/L for amfetamine [Kumazawa *et al.* 2007]. See also Miranda *et al.* [2007], Saito *et al.* [2007], Fuh *et al.* [2006], Fujii *et al.* [2006], Lua *et al.* [2006], Maresová *et al.* [2006], Holler *et al.* [2005], Klette *et al.* [2005], Wang *et al.* [2005], Dasgupta, Spies [1998], Paul *et al.* [2004], Huang *et al.* [2002], Namera *et al.* [2002], Stout *et al.* [2002], Yamada *et al.* [2002], Wu *et al.* [2001], Jurado *et al.* [2000], Hensley, Cody [1999], Myung *et al.* [1998], Dallakian *et al.* [1996], Meatherall [1995], Platoff *et al.* [1992], Gan *et al.* [1991], Taylor *et al.* [1989]. See Blood. Limit of detection, 5.0 µg/L for metamfetamine and amfetamine [Nishida *et al.* 2003; Nishida *et al.* 2002].

HPLC Column: CAPCELL PAK SXC. Mobile phase: 0.1 mol/L borate buffer (pH 9.4). UV detection ($\lambda = 210$ nm). Limit of quantification, 0.005 mg/L [Kumihashi *et al.* 2007]. Column: Symmetryshield RP8 (150 × 4.6 mm i.d., 3.5 µm). Mobile phase: water:acetonitrile (90:10) with 25 mmol/L phosphate (pH 3.0), flow rate 1.0 mL/min. UV detection ($\lambda = 210$ nm). Limit of detection, 0.3 µg/L for metamfetamine and amfetamine [He *et al.* 2007]. Column: C₈ Genesis (150 × 4.6 mm, 5 µm). Mobile phase: acetonitrile:phosphate buffer (pH 2.5) containing 0.3% TEA (12:88). Metamfetamine and related drugs [Bugamelli *et al.* 2006]. Column: Kromasil ODS (250 × 4.6 mm i.d., 5 µm). Mobile phase: acetonitrile:0.02 mol/L disodium hydrogen phosphate-0.02 mol/L methanesulfonic acid (pH 2.9, 12.5:87.5), flow rate 1.0 mL/min. UV detection ($\lambda = 210$ nm). Limit of detection, 1.4–4.0 µg/L for metamfetamine and related drugs [Fan *et al.* 2005]. Column: CAPCELL PAK C₁₈ UG 120 S5 (250 × 1.5 mm i.d., 5 µm). Mobile phase: 0.1 mol/L Tris-HCl buffer (pH = 7.0):acetonitrile (45:55), flow rate 0.2 mL/min. Fluorescence detection ($\lambda_{ex} = 330$ nm, $\lambda_{em} = 440$ nm). Limit of detection, 0.67 fmol/injection for metamfetamine and 0.14 fmol/injection for amfetamine [Al Dirbashi *et al.* 2000a]. See also al Dirbashi *et al.* [1999a]; al Dirbashi *et al.* [1998a]; al Dirbashi *et al.* [1998b]; al Dirbashi *et al.* [1997]; Makino *et al.* [1999], Foster *et al.* [1998], Campins-Falcó *et al.* [1996], Campins Falcó [1995], Molins Legua *et al.* [1995], Hayakawa *et al.* [1993].

LC-MS Column: Nucleodur Sphinx RP. Mobile phase: 10 mmol/L ammonium formate buffer:acetonitrile. Limit of quantification, 25 µg/L, limit of detection, 0.25–2.5 µg/L, for metamfetamine and related drugs [Fernández Mdel *et al.* 2009]. Column: C₁₈ (100 × 2 mm i.d., 3 µm). Mobile phase: 25 mmol/L formic acid-1% acetonitrile:25 mmol/L formic acid-90% acetonitrile (91:9 to 80:20 in 4 min to 0:100 in 3 min to 91:9 for 5 min), flow rate 0.3 mL/min. ESI, SRM acquisition mode. k' 3.1 for metamfetamine, 2.8 for amfetamine. Limit of quantification, 28 µg/L for metamfetamine, 143 µg/L for amfetamine, limit of detection, 8 µg/L for metamfetamine, 43 µg/L for amfetamine [Andersson *et al.* 2008]. Column CAPCELL PAK C₁₈ MG II (150 × 2 mm i.d., 5 µm). Mobile phase: 5 mmol/L ammonium formate (pH 4) acetonitrile, flow rate 230 µL/min. Limit of detection, <1.95 µg/L for metamfetamine and related drugs [Kim *et al.* 2008]. See Plasma [Kuwayama *et al.* 2008]. Column: Atlantis dC₁₈ (100 × 2.1 mm i.d., 3 µm). Mobile phase: acetonitrile: ammonium formate buffer (pH 3.0, 0:100 for 1 min to 50:50 in 10 min for 2.5 min to 0:100 in 0.5 min for 5 min), flow rate 0.2 mL/min. ESI, positive ion mode, SRM acquisition mode. Retention time: 8.5 min. Limit of quantification, 0.5 µg/L for metamfetamine and 1–2 µg/L for amfetamine, limit of detection, 0.5 µg/L for metamfetamine and amfetamine [Concheiro *et al.* 2007a; Concheiro *et al.* 2007b]. Column: Alltech Platinum EPS C₁₈ (30 × 2.1 mm i.d., 1.5 µm). Mobile phase: 0.01 mol/L ammonium formate (pH 3):acetonitrile (77:23), flow rate 0.2 mL/min. ESI, positive ion mode. Limit of quantification, 0.05 mg/L for metamfetamine and amfetamine [Cheng *et al.* 2007]. See Blood [Chèze *et al.* 2007]. See also Katagi *et al.* [1996], Katagi *et al.* [2000] and Wu, Fuh [2005].

CE Column: fused silica capillary. Mobile phase: 50 mmol/L phosphate buffer (pH 3):acetonitrile (80:20) and 100 mmol/L SDS. Limit of detection, 20 µg/L for metamfetamine and 15 µg/L for amfetamine and *p*-hydroxymetamfetamine [Cheng *et al.* 2006]. Column: fused silica capillary. Mobile phase: 25 mmol/L phosphate buffer (pH 2.5):methanol (80:20) and 100 mmol/L SDS. Limit of detection, 15 µg/L [Lin *et al.* 2006].

Note For a comparison of 6 immunoassays for detection of metamfetamine and other amfetamines, see Verstraete, Heyden [2005]; for a fluoroimmunoassay in serum, see Kimura *et al.* [2005]; for a radioimmunoassay see Inayama *et al.* [1980].

Bile LC-MS See Plasma [Kuwayama *et al.* 2008].

Oral Fluid GC-MS Comparison with immunoassay [Crooks, Brown 2010]. See Plasma [Huestis, Cone 2007]. SIM acquisition mode. Amfetamine and related drugs [Scheidweiler, Huestis 2006].

LC-MS See Plasma. Limit of quantification, 0.4 µg/L, limit of detection, 0.2 µg/L [Sergi *et al.* 2009]. See Plasma [Wood *et al.* 2003].

Sweat GC-MS Column: DB-1 (10 m × 0.18 mm i.d.). Temperature programme: 100° to 240° at 20°/min to 300° at 50°/min. SIM acquisition mode. Retention time:

5.2 min for metamfetamine and 4.8 min for amfetamine. Limit of detection, 3.69 µg/L [Fay *et al.* 1996].

Hair GC Column: CBJ-17. NPD. Limit of detection, 0.4 µg/g for metamfetamine and 0.1 µg/g for amfetamine [Koide *et al.* 1998].

GC-MS Limit of detection, <0.028 µg/g for metamfetamine and other amfetamines [Kim *et al.* 2010]. Limit of quantification, 0.24–0.46 µg/g for metamfetamine and other amfetamines, limit of detection, 0.07–0.14 µg/g for metamfetamine and other amfetamines [Johansen, Jørnild 2009]. SIM acquisition mode. Limit of detection, ~50 ng/g for metamfetamine and other amfetamines [Meng *et al.* 2009]. PTV. Limit of quantification, 0.20 µg/g for metamfetamine and amfetamine [Miyaguchi *et al.* 2009]. Column: Uptispher ODB C₁₈ (150 × 2.1 mm, 5 µm). Mobile phase: acetonitrile:2 mmol/L formate buffer (pH 3.0). Limit of quantification, 5 ng/g for metamfetamine, *N*-methyl-*p*-methoxyamfetamine (MMA), methylenedioxyethylamfetamine (MDEA) and methylbenzodioxylbutanamine (MBDB), and 14.7 ng/g for amfetamine [Chèze *et al.* 2007]. See also Ando *et al.* [2004], Gottardo *et al.* [2007], Kim *et al.* [2005], Kintz *et al.* [1995], Martins *et al.* [2006], Miki *et al.* [2008], Pujadas *et al.* [2003], Skender *et al.* [2002], Villamor *et al.* [2005], Wu *et al.* [2001] Wu *et al.* [2008] and Zhu *et al.* [2007]. See Blood. Limit of detection, 0.25 ng/mg for metamfetamine and amfetamine [Nishida *et al.* 2003].

HPLC See Plasma. Mobile phase: methanol:acetonitrile: 0.01 mol/L citrate buffer (pH 4.0), flow rate 1.1 mL/min. Limit of detection, 0.12 µg/g for metamfetamine and 0.08 µg/g for amfetamine [Nakashima *et al.* 2003]. Fluorescence detection [al Dirbashi *et al.* 1999b; Al Dirbashi *et al.* 2000b].

LC-MS Limit of quantification, 0.5 µg/g for metamfetamine and other amfetamines, limit of detection, 0.1 µg/g for metamfetamine and other amfetamines [Tabernero *et al.* 2009]. ESI. Metamfetamine and amfetamine enantiomers and selegiline and metabolites [Nishida *et al.* 2006].

Note For a review of methods for the analysis of metamfetamine and its metabolites in hair, see Takayama *et al.* [2003].

Other HPLC Clothing. UV detection ($\lambda = 330$ nm) or fluorescence detection ($\lambda_{ex} = 440$ nm). Limit of detection, 37.3 pg (fluorescence) and 0.4 pg (UV) for metamfetamine and amfetamine [al Dirbashi *et al.* 2001].

Note For a review of the determination of metamfetamine enantiomers, see Jirovský *et al.* [1998].

Disposition in the Body Metamfetamine is readily absorbed after oral administration. Approximately 70% of a dose is excreted in the urine in 24 h. Under normal conditions, up to 43% of a dose is excreted as unchanged drug, up to 15% as 4-hydroxymetamfetamine, and ~5% as amfetamine, the major active metabolite. A number of other metabolites have been identified. Excretion of unchanged drug is dependent on the urinary pH, being increased in acidic urine and greatly reduced (to ~2% of a dose) if the urine is alkaline.

Metamfetamine is a metabolite of benzfetamine and selegiline.

Therapeutic Concentration In plasma, usually in the range 0.01–0.05 mg/L.

Following a single oral dose of 12.5 mg metamfetamine hydrochloride to 10 subjects, a mean peak blood concentration of ~0.02 mg/L was attained in ~2 h [Driscoll *et al.* 1971].

Average concentrations of 111 and 281 µg/L (metamfetamine) and 4 and 15 µg/L (amfetamine) were found in breast milk 24 h after IV administration of amfetamines to 2 breastfeeding mothers [Bartu *et al.* 2009].

Toxicity The estimated minimum lethal dose is 1 g, but fatalities attributed to metamfetamine are rare.

The following postmortem tissue concentrations, mg/L or µg/g, were reported in a 25-year-old female found dead after the IV abuse of metamfetamine followed by ingestion of ~1.5 g:

	Metamfetamine	Amfetamine
Blood	43	0.35
Brain	101.8	0.86
Kidney	75.5	0.72
Liver	174.6	1.30
Urine	277.5	10.0

[Kojima *et al.* 1984].

A 37-year-old man who died following mixed drug abuse had the following postmortem metamfetamine concentrations: blood 0.7 mg/L, bile 21.7 mg/L, urine 32.0 mg/L, gastric contents 2.9 mg/L, liver 2.2 µg/g and brain 2.7 µg/g [Moore *et al.* 1996].

Of 146 deaths in which metamfetamine was detected in the blood, 52 were directly attributed to the drug and 92 were classified as drug-related; a large proportion of the deaths resulted from homicidal (27%) or suicidal (15%) violence. The metamfetamine blood concentration in the 92 drug-related deaths was reported to be in the range 0.05–9.30 mg/L (median, 0.42), with 90% of the population having concentrations <2.20 mg/L. In 17 metamfetamine-related traffic deaths, the blood concentration was reported to be 0.05–2.60 mg/L, with most deaths occurring at concentrations >0.5 mg/L [Logan *et al.* 1998].

In 413 deaths where metamfetamine was detected, blood levels of metamfetamine and amfetamine were 2.08 mg/L and 0.217 mg/L, respectively, in cases where metamfetamine was related to the cause of

death. These levels were not significantly different from those where metamfetamine use was not related to the cause of death (1.78 and 0.19 mg/L, respectively) [Karch *et al.* 1999].

A pilot who died in a civil aircraft accident had postmortem blood levels of metamfetamine (1.13 mg/L) and amfetamine (0.022 mg/L). Urine levels of metamfetamine and amfetamine were 59.2 and 1.50 mg/L, respectively. Metamfetamine was distributed throughout the body, including the brain. Metamfetamine levels in gastric contents were 575-fold higher than amfetamine levels [Chaturvedi *et al.* 2004].

A 20-year-old woman who had drugs enclosed in plastic bags in her vagina, developed metamfetamine toxicity. Serum levels of metamfetamine and amfetamine were 3.1 and 0.11 mg/L, respectively [Kashani, Ruha 2004].

In another case, a man who died within 1 h of an IV injection of metamfetamine had a femoral blood level of 1.5 mg/L. Levels of 132 µg/g were found in the first three 2 cm hair samples and overall the levels in hair provided evidence of chronic metamfetamine use for >8 months [Beránková *et al.* 2005].

A man who collapsed shortly after an IV injection of metamfetamine was in a coma for 9 days before he died. More than 9 days after administration, metamfetamine was not detected in postmortem blood, urine or liver, but levels in hair were 1.1 µg/g [Ago *et al.* 2006].

Another case of 'body stuffing' in which a 25-year-old man had delayed onset toxicity after ingesting metamfetamine wrapped in plastic has been reported [Hendrickson *et al.* 2006].

There is a further report of 3 men in their thirties who attempted to smuggle metamfetamine 498, 292, and 73 g by 'body packing' the drug as packages each containing ~14 g wrapped in plastic film and Scotch tape. One man died from acute metamfetamine poisoning from ~20 g metamfetamine that had leaked from the packages into his stomach. The plasma level of metamfetamine was 8.6 mg/L ~17 h before he died. Postmortem levels of metamfetamine and amfetamine in cardiac blood were 63.5 and 1.2 mg/L, respectively. Urine levels were 4518 and 72.4 mg/L, respectively; gastric contents were 8490 and 16.9 mg/L, respectively [Takekawa *et al.* 2007].

In a retrospective review of metamfetamine-related deaths, the median blood level was 0.2 mg/L (range, 0.02–15); other drugs, mainly bezodiazepines and morphine, were identified in 89% of cases [Kaye *et al.* 2008].

A woman in her late thirties who was hospitalised for 13 days following a road traffic accident suffered a cardiopulmonary arrest and died. At postmortem it was concluded that her injuries were not the cause of death. Postmortem cardiac blood levels of metamfetamine and amfetamine were 0.178 and 0.051 mg/L, respectively. Metamfetamine was also found in hair samples indicating metamfetamine use for at least 3 months before death [Ago *et al.* 2009].

A 49-year-old man who believed he was being followed by the police swallowed an '8-ball of meth' (one-eighth of an ounce of metamfetamine, equivalent to ~3 g). Levels of 3 mg/L metamfetamine were found in blood samples taken in hospital; postmortem femoral blood levels were 30 mg/L [Kiely *et al.* 2009].

There is another report of a fatality in a 39-year-old man who at postmortem was found to have 158 columned-shaped packages wrapped in tinfoil and plastic containing a total of 390 g metamfetamine in his alimentary tract. One package in his stomach had ruptured and was empty. Postmortem levels of metamfetamine were cardiac blood 24.5 mg/L, urine 191 mg/L, liver 116 mg/L and gastric contents 1045 mg/L [Li *et al.* 2009].

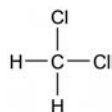
For other reports of fatalities associated with metamfetamine, see Logan *et al.* [1996], Molina, Jejurikar [1999], Sribanditmongkol *et al.* [2000] and Yamamoto *et al.* [1991]. For a report of recovery after overdose with ~2.3 g metamfetamine administered IV, see Buffum, Shulgin [2001].

Half-life Plasma half-life, ~9 h.

Dose 2.5 to 25 mg of metamfetamine hydrochloride daily, orally; 15 to 20 mg IM, or 10 to 15 mg IV.

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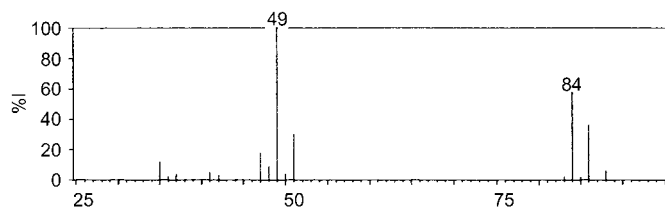
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Chemical Properties Dichloromethane A clear, colourless, volatile liquid. Mass per mL about 1.32 g. Bp 39° to 41°. Soluble 1 in 50 of water; miscible with ethanol and ether. Log *P* (octanol/water), 1.2.

Gas Chromatography System GA—RI 515; system GI—retention time 1.9 min.

Mass Spectrum Principal ions at *m/z* 49, 84, 86, 51, 47, 35, 88, 41.



Quantification

Blood GC FID. Methylene chloride and chloroform. Limit of detection, 0.3 mg/L in blood [Seno *et al.* 1999].

Urine GC Limit of detection, 0.2 mg/L in urine, see Blood [Seno *et al.* 1999].

Disposition in the Body

Toxicity Methylene chloride is widely used in paint strippers and several non-fatal and fatal cases of accidental inhalation have been reported. The maximum permissible atmospheric concentration is 200 ppm and the temporary estimated acceptable daily intake is up to 500 µg/kg.

The following tissue concentrations were reported in a fatality due to accidental methylene chloride poisoning: blood 252 mg/L, brain 75 µg/g, heart 30 µg/g [Kim *et al.* 1996].

Blood concentrations collected at autopsy in 2 cases of fatal methylene chloride poisoning were 572 and 601 mg/L; air concentrations a few hours after discovery of the bodies were up to 168 000 ppm [Manno *et al.* 1992].

The following postmortem tissue concentrations were reported in a 47-year-old factory worker who died while making an inventory of the annual stock of methylene chloride contained in several tanks, one of which overflowed: blood 150 mg/L, urine 2.0 mg/L, gastric content 5.6 mg/L, brain 122 µg/g, fat 99 µg/g, liver 44 µg/g, lung 20 µg/g, kidney 15 µg/g [Goulle *et al.* 1999].

Goulle JP *et al.* (1999). Fatal case of dichloromethane poisoning. *J Anal Toxicol* 23: 380–383.

Kim NY *et al.* (1996). Two fatal cases of dichloromethane or chloroform poisoning. *J Forensic Sci* 41 (3): 527–529.

Manno M *et al.* (1992). Double fatal inhalation of dichloromethane. *Hum Exp Toxicol* 11: 540–545.

Seno H *et al.* (1999). Extraction of chloroform and methylene chloride in human whole blood and urine by headspace solid phase microextraction (SPME). *Med Sci Law* 39: 332–336.

Methylenedioxyamfetamine

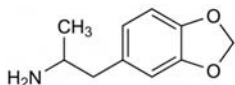
Hallucinogen

C₁₀H₁₃NO₂ = 179.2

CAS—4764-17-4

IUPAC Name 1-(1,3-Benzodioxol-5-yl)propan-2-amine

Synonyms MDA; α-methyl-1,3-benzodioxole-5-ethanamine; methylenedioxy-amphetamine; N-methyl-3,4-methylenedioxyamfetamine; tenamfetamine; SKF-5.



Chemical Properties Almost colourless oil. p*K_a* 9.67 (25°). Log *P* (octanol/water), 1.64.

Methylenedioxyamfetamine Hydrochloride

C₁₀H₁₃NO₂·HCl = 215.7

CAS—6292-91-7

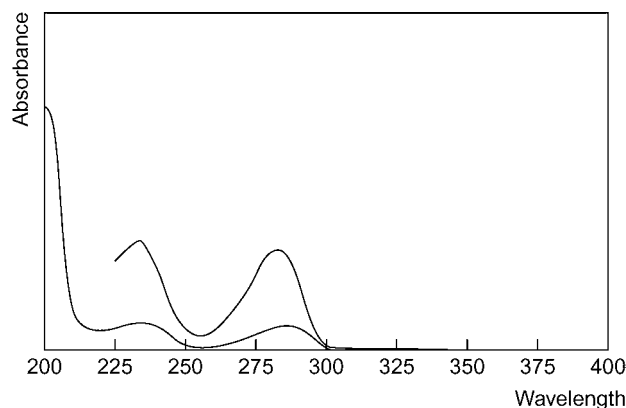
Colour Tests Liebermann's reagent—black; Mandelin's test—green→blue; Marquis test—blue-black; sulfuric acid—violet.

Thin-layer Chromatography System TA—R_f 0.39; system TB—R_f 0.18; system TC—R_f 0.12; system TE—R_f 0.42; system TL—R_f 0.17; system TAE—R_f 0.10; system TAF—R_f 0.76 (acidified potassium permanganate solution, positive).

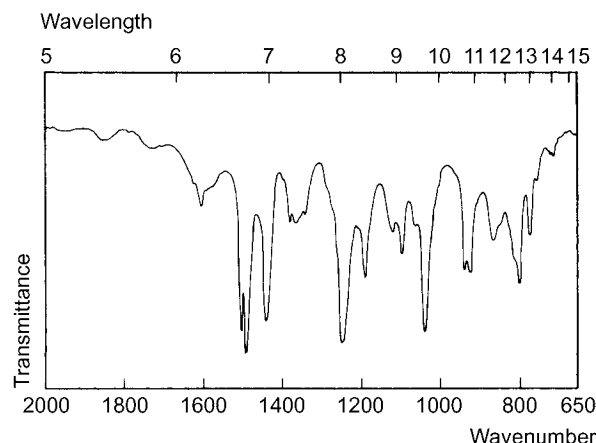
Gas Chromatography System GA—MDA RI 1480, MDA-TFA RI 1615, MDA-PFP RI 1605, MDA-AC RI 1860, art (formyl) RI 1520; system GB—MDA RI 1512, art (formyl) RI 1689.

High Performance Liquid Chromatography System HC—*k* 0.98; system HX—RI 266; system HY—RI 248; system HZ—retention time 2.1 min; system HAA—retention time 8.1 min; system HAX—retention time 6.9 min; system HAY—retention time 3.8 min.

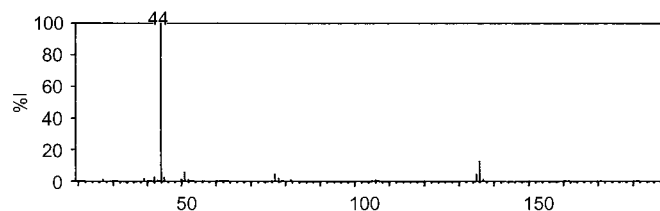
Ultraviolet Spectrum Aqueous acid—233 (A₁ = 216b), 285 nm.



Infrared Spectrum Principal peaks at wavenumbers 1490, 1257, 1038, 1504, 799, 1188 cm⁻¹ (KBr disk).



Mass Spectrum Principal ions at *m/z* 44, 136, 51, 135, 77, 42, 78, 45.



Quantification

Blood HPLC Limit of quantification, 2 µg/L [Clauwaert *et al.* 2000]. Electrochemical detection. Limit of detection, 1 µg/L [Michel *et al.* 1993].

Plasma GC Limit of detection, <1.6 µg/L [Ortuno *et al.* 1999].

HPLC Limit of detection, 10 µg/L [Herraez-Hernandez *et al.* 2001].

Serum GC-MS Methylenedioxyamfetamine and other drugs of abuse [Weinmann *et al.* 2000].

HPLC See Blood [Clauwaert *et al.* 2000].

Urine GC See Plasma. Limit of detection, 47 µg/L [Ortuno *et al.* 1999]. NP. Limit of detection, 15 µg/L [Ugland *et al.* 1999].

GC-MS Methylenedioxyamfetamine and related drugs. Limit of detection for MDA, ~31 µg/L [Stout *et al.* 2002]. Limit of quantification, 20 µg/L [Jurado *et al.* 2000]. Methylenedioxyamfetamine and other amphetamines. Limit of detection, ~10 µg/L [Hensley, Cody 1999]. Limit of detection, 15 µg/L [Ugland *et al.* 1999].

HPLC See Plasma [Herraez-Hernandez *et al.* 2001]. See Blood. Limit of quantification, 100 µg/L [Clauwaert *et al.* 2000].

Vitreous Humour HPLC See Blood [Clauwaert *et al.* 2000].

Hair GC-MS Methylenedioxyamfetamine and other drugs of abuse. Limits of detection, 0.05–0.3 µg/g [Skender *et al.* 2002]. Limit of detection, 0.01 µg/g [Rohrich, Kauert 1997]. Limit of detection, 0.05 µg/g [Kintz *et al.* 1995].

HPLC Limit of detection, <1 µg/L (0.1 µg/g) [Tagliaro *et al.* 1999].

Other HPLC Biological Tissues. See Blood [Michel *et al.* 1993].

Disposition in the Body

Toxicity The estimated lethal dose is 0.5 g.

A 21-year-old man was found dead in bed 7 h after being last seen in a 'normal' state. *Temazepam* tablets as well as 2 types of white tablets were found by his bedside. Toxicological analysis revealed the presence of the following drugs:

	Blood (mg/L)	Stomach contents (mg/L)	Urine
Amfetamine	0.256	<0.1	Present
MDA	8.5	299	Present
MDMA	2.1	96	Present
MDEA	3.5	324	Present

MDMA, methylenedioxyamfetamine; MDEA, methylenedioxyethylamfetamine. [Forrest *et al.* 1994].

In 5 fatalities due to the ingestion of methylenedioxyamfetamine, the following postmortem tissue concentrations were reported: blood 6–26 mg/L (mean 14, 5 cases), bile 5–9 mg/L (mean 6, 3 cases), liver 8–17 µg/g (mean 11, 3 cases), urine 46–160 mg/L (mean 102, 3 cases) [Cimbura 1972].

In a fatality in which death occurred shortly after the ingestion of methylenedioxyamfetamine, the following postmortem concentrations were reported: blood 2.3 mg/L, bile 7 mg/L, liver 11 µg/g, urine 175 mg/L [Lukasewski 1979].

Note For a review of the pharmacology of MDA, see Simpson and Rumack [1981]; for a distribution study of MDMA and MDA in a fatal overdose, see De Letter *et al.* [2002].

Cimbura G (1972). 3,4-methylenedioxyamfetamine (MDA): analytical and forensic aspects of fatal poisoning. *J Forensic Sci* 17: 329–333.

Clauwaert KM *et al.* (2000). Determination of the designer drugs 3,4-methylenedioxyamfetamine, 3,4-methylenedioxyethylamfetamine, and 3,4-methylenedioxyamfetamine with HPLC and fluorescence detection in whole blood, serum, vitreous humor, and urine. *Clin Chem* 46: 1968–1977.

De Letter EA *et al.* (2002). Distribution study of 3,4-methylenedioxyamfetamine and 3,4-methylenedioxyethylamfetamine in a fatal overdose. *J Anal Toxicol* 26: 113–118.

Forrest AR *et al.* (1994). A fatal overdose with 3,4-methylenedioxyamfetamine derivatives. *Forensic Sci Int* 64: 57–59.

Hensley D, Cody JT (1999). Simultaneous determination of amphetamine, methamphetamine, methylenedioxyamfetamine (MDA), methylenedioxyethylamfetamine (MDMA), and methylenedioxyethylamfetamine (MDEA) enantiomers by GC-MS. *J Anal Toxicol* 23: 518–523.

Herraez-Hernandez R *et al.* (2001). Sensitive determination of methylenedioxyethylamphetamines by liquid chromatography. *Analyst* 126(5): 581–586.

Jurado C *et al.* (2000). Rapid analysis of amphetamine, methamphetamine, MDA, and MDMA in urine using solid-phase microextraction, direct on-fiber derivatization, and analysis by GC-MS. *J Anal Toxicol* 24: 11–16.

Kintz P *et al.* (1995). Simultaneous determination of amphetamine, methamphetamine, 3,4-methylenedioxyamfetamine and 3,4-methylenedioxyethylamfetamine in human hair by gas chromatography-mass spectrometry. *J Chromatogr B Biomed Sci Appl* 670: 162–166.

Lukasewski T (1979). 3,4-Methylenedioxyamfetamine overdose. *Clin Toxicol* 15: 405–409.

Michel RE *et al.* (1993). High-pressure liquid chromatography/electrochemical detection method for monitoring MDA and MDMA in whole blood and other biological tissues. *J Neurosci Methods* 50: 61–66.

Ortuno J *et al.* (1999). Quantification of 3,4-methylenedioxyamfetamine and its metabolites in plasma and urine by gas chromatography with nitrogen-phosphorus detection. *J Chromatogr B Biomed Sci Appl* 723(1–2): 221–232.

Rohrich J, Kauert G (1997). Determination of amphetamine and methylenedioxy-amphetamine derivatives in hair. *Forensic Sci Int* 84: 179–188.

Simpson DL, Rumack BH (1981). Methylenedioxyamfetamine. Clinical description of overdose, death, and review of pharmacology. *Arch Intern Med* 141: 1507–1509.

Skender L *et al.* (2002). Quantitative determination of amphetamines, cocaine, and opiates in human hair by gas chromatography/mass spectrometry. *Forensic Sci Int* 125: 120–126.

Stout PR *et al.* (2002). Rapid simultaneous determination of amphetamine, methamphetamine, 3,4-methylenedioxyamfetamine, 3,4-methylenedioxyethylamfetamine, and 3,4-methylenedioxyethylamfetamine in urine by solid-phase extraction and GC-MS: a method optimized for high-volume laboratories. *J Anal Toxicol* 26: 253–261.

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Ugland HG *et al.* (1999). Automated determination of 'Ecstasy' and amphetamines in urine by SPME and capillary gas chromatography after propylchloroformate derivatisation. *J Pharm Biomed Anal* 19(3–4): 463–475.

Weinmann W *et al.* (2000). Automated solid-phase extraction and two-step derivatisation for simultaneous analysis of basic illicit drugs in serum by GC/MS. *Int J Legal Med* 113: 229–235.

Methylenedioxyamfetamine

Hallucinogen, Phenethylamine, Stimulant

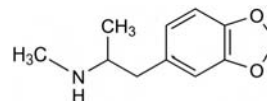
C₁₁H₁₅NO₂ = 193.2

CAS—42542-10-9

IUPAC Name 1-(1,3-Benzodioxol-5-yl)-N-methylpropan-2-amine

Synonyms (2-Benzol[1,3] dioxol-5-yl-1-methyl-ethyl)-methyl-amine; N,α-dimethyl-1,3-benzodioxole-5-ethanamine; MDMA; 3,4-methylenedioxyamfetamine; 3,4-methylenedioxyethylamfetamine; N-methyl-3,4-methylene-dioxyphenylisopropylamine.

Street Names A bean; Adam; Apples; Baby slits; Brownies; Burgers; Chocolate chips; Clarity; Crowns; Dennis the Menace; Diamond Whites; Disco biscuits; Doctor; Doves; E; Ecstasy; Elaine; Essence; Euphoria; Love doves; Lovers Speed; M, M; MDM; Mitsubishi; Mitsies; New Yorkers; Roll; Rhubarb, Custards; Stacy; Tangos; X; XTC.



Chemical Properties A viscous, colourless oil. Bp 100° to 110°. pK_a (pH 9.0) (benzene) 9.41, (hexane) 8.69, (ethyl acetate) 8.84. Extraction yield (chlorobutane), 0.7 [Demme *et al.* 2005].

Methylenedioxyamfetamine Hydrochloride

C₁₁H₁₅NO₂·HCl = 229.75

CAS—64057-70-1

IUPAC Name 1-(1,3-Benzodioxol-5-yl)-N-methylpropan-2-amine hydrochloride

Chemical Properties A white to off-white crystalline powder with a bitter taste. Mp 147° to 148° for crystals from propan-2-ol/n-hexane, 152° to 153° for crystals from propan-2-ol/ether.

Colour Test Marquis reagent—black with dark purple halo.

Thin-layer Chromatography System TA—R_f 0.33; system TB—R_f 0.24; system TE—R_f 0.39; system TF—R_f 0.20; system TAE—R_f 0.08; system TAJ—R_f 0.03; system TAK—R_f 0.17; system TAL—R_f 0.57.

Gas Chromatography System GA—RI 1585, methylenedioxyamfetamine (MDA) RI 1480, MDMA-TFA RI 1720, MDMA-PFP RI 1830, MDMA-TMS RI 1710, MDMA-AC RI 2140; system GB—RI 1572, MDA RI 1512; system GAK—RT 8.7 min; system GAD—RT 10.9 min.

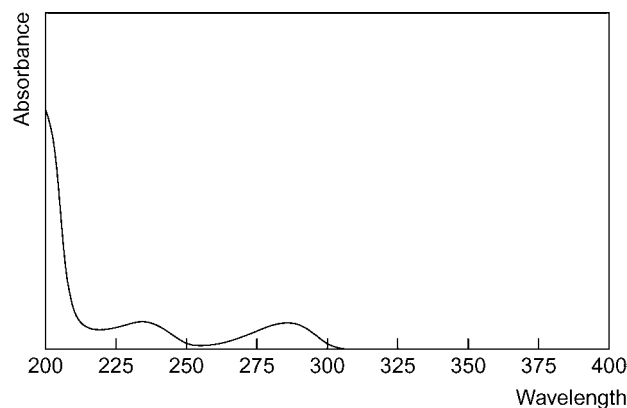
Column: 3% OV-1 Chromosorb WHP 80/100 mesh (1.2 m × 100 mm (4 ft × 4 in) i.d.). Column temperature: 200°. Carrier gas: N₂, 32 mL/min. FID. RI: 1559 [Mills, Roberson 1993].

High Performance Liquid Chromatography System HX—RI 278; system HY—RI 252; system HZ—RT 2.2 min; system HAA—RT 9.1 min; system HBC—RT 2.42 min; system HBD—RT 5.31 min.

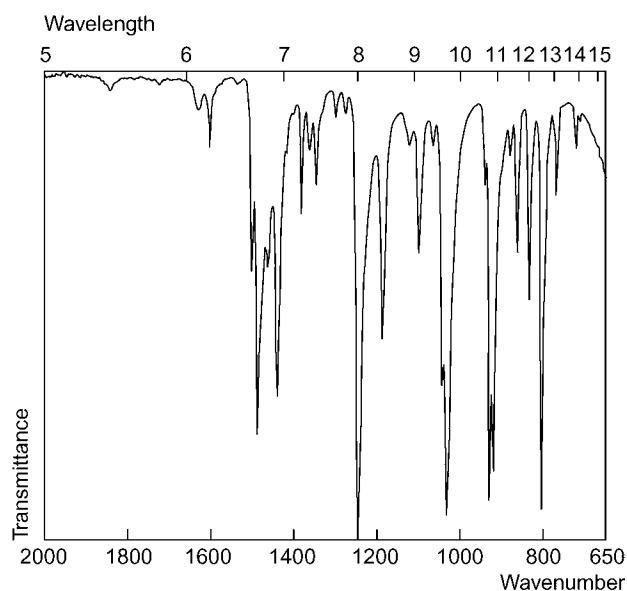
Column: ODS Hypersil Si-10 (300 × 4.0 mm i.d., 10 µm). Mobile phase: methanol with 1% ammonium hydroxide:methylene chloride (10:90), flow rate 2 mL/min. UV detection (λ = 254 nm). Retention time: 4.4 min [Mills, Roberson 1993].

Column: Zorbax CN (150 × 4.6 mm i.d., 5 µm). Mobile phase: methanol:0.1 mol/L acetate buffer (pH 4.56, 50:50). UV detection (λ = 280 and 284 nm). Retention time: 6.69 min for MDMA and 5.83 min for MDA [Garrett *et al.* 1991].

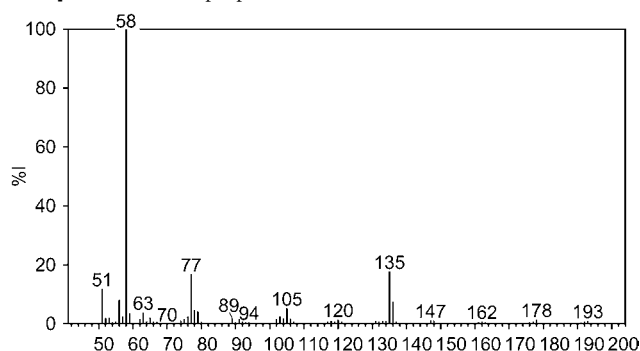
Ultraviolet Spectrum Aqueous acid (0.2 mol/L sulfuric acid)—234, 285 nm; (0.025 mol/L sulfuric acid)—198, 234.5, 282.5 nm; (0.1 mol/L ammonium formate, pH 3)—285; basic—232, 285 nm.



Infrared Spectrum Principal peaks at wavenumber 1033, 930, 805, 1489, 1442, 1189.2 cm⁻¹ (hydrochloride)



Mass Spectrum Principal peaks at m/z 58, 77, 135, 51, 105, 63, 89, 178.



Quantification

Blood GC-MS Column: HP-5MS (15 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He. Temperature programme: 60° for 1 min to 220° at 20°/min. EI ionisation, SIM acquisition mode. Retention time: 7.14 min. Limit of detection not reported [Gunn *et al.* 2010]. SIM acquisition mode. MDMA, MDA and other amfetamines [Rasmussen *et al.* 2006]. Column: HP-5 5% phenylmethylsiloxane (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 0.7 mL/min. Temperature programme: 60° for 2 min to 250° at 20°/min for 5 min. SIM acquisition mode. Limit of quantification, 3.59 μ g/g, limit of detection, 1.19 μ g/g [Gentili *et al.* 2002]. Column: HP 5MS capillary (30 m $\times$ 0.32 mm i.d., 0.25 μ m). Carrier gas: He, 50 kPa. Temperature programme: 60° for 1 min to 140° at 15°/min to 212° at 30°/min for 3 min to 285°. EI ionisation at 70 eV, SIM acquisition mode. Retention time: 10.48 min. Limit of quantification, 10 μ g/L for MDMA and 3,4-methylenedioxy-*N*-ethylamphetamine (MDEA) and 50 μ g/L for MDA, limit of detection, 0.5–0.8 μ g/L for MDMA, MDA and other amfetamines [Marquet *et al.* 1997].

HPLC Column: ODS. Mobile phase: 10 mmol/L citric acid: 20 mmol/L sodium hydrogen phosphate buffer (pH 4.0): acetoxime (50:45:5). Fluorescence detection. Limit of detection, 0.36–0.83 μ g/L for MDMA and MDA [Tomita *et al.* 2006]. Column: C₁₈ BDS Hypersil (100 $\times$ 2.1 mm i.d., 3 μ m). Mobile phase: 0.1 mol/L ammonium acetate-methanol-acetonitrile (90:10:10): methanol-acetonitrile-water (45:45:10, 100:0 for 6 min to 30:70 in 14 min), flow rate 1 mL/min. I.S. 3,4-methylenedioxyphenyl-*N*-methyl propylamine (MDMPA). Fluorescence detection (λ_{ex} = 288 nm, λ_{em} = 324 nm). Retention time: 14.1 min for MDMA, 13.1 min for MDA, 15.1 min for MDEA and 17.1 min for IS Limit of quantification, 2 μ g/L, limit of detection, 0.8 μ g/L for MDMA, MDA and MDEA [Clauwaert *et al.* 2000]. Column: Whatman (250 $\times$ 4.6 mm i.d., 5 μ m). Mobile phase: 0.1 mol/L sodium acetate (pH 4.25): methanol (10:3), flow rate 0.8 mL/min. Electrochemical detection. Limit of detection, 1 μ g/L for MDMA, MDA and MDE [Michel *et al.* 1993].

LC-MS Column: Agilent HPLC 1100 with Varian Pursuit 3 C₁₈ (100 $\times$ 3 mm i.d., 3 μ m). Mobile phase: methanol: ammonium formate. MRM acquisition mode, positive ion mode. Limit of quantification, 0.0005–0.01 mg/kg for MDMA, MDA and other drugs of abuse [Bjork *et al.* 2010]. Column: Nucleodur Sphinx RP. Mobile phase: 10 mmol/L ammonium formate buffer: acetonitrile. Limit of quantification, 2.5 μ g/L, limit of detection, 0.05–0.5 μ g/L for MDMA, MDA and other amfetamines [Fernández Mdel *et al.* 2009]. Column: BEH C₁₈ (50 $\times$ 2.1 mm i.d.). Mobile phase: aqueous pyrrolidine: methanol (52:48), flow rate 0.4 mL/min. ESI, positive ion mode. Retention time: 2.04 min. Limit of detection not reported [Apollonio *et al.* 2006].

Note For a review of methods published between 2002 and 2007 for the determination of MDMA, MDA, other amfetamines and drugs of abuse in blood, see Kraemer, Paul [2007].

Plasma GC Column: Ultra-2 (12 m $\times$ 0.2 mm i.d., 0.33 μ m). Carrier gas: He, 0.8 mL/min. Temperature programme: 70° for 2 min to 100° at 30°/min to 200° at 20°/min to 280° at 25°/min. NPD. Limit of quantification, 5.3 μ g/L, limit of detection, 1.6 μ g/L [Ortuño *et al.* 1999].

GC-MS Column: VF-5MS (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° for 1 min to 300° at 15°/min for 10 min. EI ionisation, SIM acquisition mode. Retention time: 8.7 min. Limit of quantification, 5 μ g/L, limit of detection, 2 μ g/L for MDMA, MDA and related drugs [da Silva *et al.* 2010]. See Blood [Gunn *et al.* 2010]. Column: Zorbax 300-SCX (150 $\times$ 2.1 mm i.d., 5 μ m). Mobile phase: 5 mmol/L aqueous ammonium formate (pH 3.0): acetonitrile (70:30). ESI, positive ion mode. Limit of detection not reported [Mueller *et al.* 2009]. Column: ZB-50 (30 m $\times$ 0.32 mm i.d., 0.25 μ m). Temperature programme: 70° to 150° at 35°/min for 0.5 min to 195° at 10°/min for 0.11 min to 100° at 50°/min to 275° at 800°/min to 190° at 10°/min. EI ionisation, SIM acquisition mode. Limit of quantification, 2.5 μ g/L, limit of detection, 0.5 μ g/L [Kolbrich *et al.* 2008a]. Column: HP-5MS (30 m $\times$ 0.25 mm i.d., 250 nm). Carrier gas: He, 0.6 mL/min. Temperature programme: 100° to 250° at 10°/min to 310° at 30°/min for 1 min. EI ionisation, SIM acquisition mode. Limit of quantification, 5 μ g/L [Peters *et al.* 2003a; Peters *et al.* 2003b; Peters *et al.* 2005]. Column: 5% phenylmethylsiloxane (12 m $\times$ 0.22 mm i.d., 0.33 μ m). Carrier gas: He, 1.2 mL/min. Temperature programme: 150° for 1 min to 290° at 20°/min for 7 min. EI ionisation, SIM acquisition mode. Limit of quantification, 7.3 μ g/L for (R)-MDMA and 9.3 μ g/L for (S)-MDMA, limit of detection, 2.4 μ g/L for (R)-MDMA and 3.1 μ g/L for (S)-MDMA [Pizarro *et al.* 2002; Pizarro *et al.* 2003].

HPLC See Blood [Tomita *et al.* 2006]. UV and fluorescence detection. Limit of detection, 25 μ g/L for MDMA and MDE and 10 μ g/L for MDA [Herráez-Hernández *et al.* 2001]. Column: ODS Hypersil C₁₈ (130 mm). Mobile phase: methanol: 0.1 mol/L acetate buffer (pH 4.54 50:50), flow rate 0.7 mL/min. UV detection (λ = 280 nm). Limit of detection, 3.68 μ g/L for MDMA and 2.0 μ g/L for MDA [Garrett *et al.* 1991].

LC-MS Column: C₁₈ Alltima (250 $\times$ 4.6 mm i.d., 5 μ m). Mobile phase: acetonitrile: 5 mmol/L formic acid (10:90 for 2 min to 25:75 in 16 min to 35:65 in 0.1 min to 100:0 in 12 min), flow rate 1.0 mL/min. MRM acquisition mode, positive ion mode. Limit of quantification, 1.8 μ g/L, limit of detection, 0.9 μ g/L [Sergi *et al.* 2009]. Column: CAPCELL PAK SCX UG 80 (250 $\times$ 4.6 mm i.d.). Mobile phase: 25 mmol/L ammonium acetate (pH 4.0): acetonitrile (3:7), flow rate 0.2 mL/min. ESI, SRM acquisition mode. Limit of detection, 50 μ g/L [Kuwayama *et al.* 2008]. Column: Hypersil BDS-C₁₈ (100 $\times$ 2.1 mm i.d., 3.5 μ m). Mobile phase: 10 mmol/L ammonium acetate: acetonitrile (75:25), flow rate 0.3 mL/min. ESI, positive ion mode, MRM acquisition mode. Limit of detection, <2 μ g/L for MDMA, MDA and other amfetamines [Wood *et al.* 2003].

Serum GC-MS See Blood [Gunn *et al.* 2010]. Column: DB5-MS (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Temperature programme: 80° for 1 min to 200° at 20°/min to 240° at 5°/min to 310° at 30°/min for 6 min. EI ionisation, SIM acquisition mode. Limit of quantification, 10.4 μ g/L [Weinmann *et al.* 2000].

HPLC See Blood [Clauwaert *et al.* 2000].

Urine TLC Fluorescence detection (250 to 400 nm). Limit of detection, 50 ng for MDMA, MDA and other amfetamines [Kato *et al.* 2008].

GC Limit of detection, 20 μ g/L for MDMA and MDA [Wang *et al.* 2008]. Column: AT-5 (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1.0 mL/min. Temperature programme: 40° for 1 min to 280° at 20°/min for 5 min. FID. Limit of detection, 35 μ g/L [Raikos *et al.* 2003]. NPD. Limit of detection, 47 μ g/L for MDMA, MDA and other metabolites [Ortuño *et al.* 1999]. Column: SPB-1 polymethylsiloxane (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1.0 mL/min. Temperature: 180°. Limit of detection, 5 μ g/L for MDMA and MDEA and 15 μ g/L for MDA [Ugland *et al.* 1999].

GC-MS SIM acquisition mode. Limit of quantification, 3.5 μ g/L for MDMA, MDA and related drugs [da Silva *et al.* 2010]. Column: HP-5MS (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 127 kPa. Temperature programme: 70° for 1 min to 300° at 20°/min for 2 min. EI ionisation at 70 eV. Limit of detection, 5 μ g/L for MDMA and 10 μ g/L for MDA [Nakamoto *et al.* 2010]. Column: Varian VF-5ht (30 m $\times$ 0.25 mm i.d., 0.10 μ m). Temperature programme: 150° to 270° at 10°/min. EI ionisation, full scan mode. Limit of quantification, 0.1 mg/L, limit of detection, 0.1 mg/L [Brown *et al.* 2007]. Column: DB-35MS (15 m $\times$ 0.32 mm i.d., 0.25 μ m). Carrier gas: He, 2.0 psi. Temperature programme: 70° for 2 min to 160° at 20°/min for 2 min to 200° at 15°/min to 300° for 1 min. EI ionisation, SIM acquisition mode. Limit of quantification, 25 μ g/L, limit of detection, 10 μ g/L for MDMA, MDA and other metabolites [Pirnay *et al.* 2006]. Limit of quantification, 10 μ g/L, limit of detection, 5 μ g/L for MDMA, MDA and other amfetamines [Maresová *et al.* 2006]. See Blood [Gentili *et al.* 2002]. See Plasma [Pizarro *et al.* 2002; Pizarro *et al.* 2003]. See also Centini *et al.* [1996], Dallakian *et al.* [1996], de Boer *et al.* [1997], Gan *et al.* [1991], Hensley, Cody [1999], Jurado *et al.* [2000], Klette *et al.* [2005], Namera *et al.* [2002], Paul *et al.* [2004], Pellegrini *et al.* [2002] and Stout *et al.* [2002].

HPLC Column: Genesis C8 (150 $\times$ 4.6 mm i.d., 5 μ m). Mobile phase: acetonitrile: phosphate buffer (pH 2.5) containing 0.3% TEA (12:88). Fluorometric detection [Bugamelli *et al.* 2006]. UV detection. Limit of detection, 5.3–84 ng for MDMA, MDA and other amfetamines [Soares *et al.* 2004]. See Plasma [Herráez-Hernández *et al.* 2001]. See Blood. Limit of quantification, 0.1 mg/L, limit of detection, 2.5 μ g/L for MDMA, MDA and MDEA [Clauwaert *et al.* 2000].

LC-MS See Blood. Limit of quantification, 25 μ g/L, limit of detection, 0.25–2.5 μ g/L for MDMA, MDA and other amfetamines [Fernández Mdel *et al.* 2009].

Column: C₁₈ (100 × 2 mm i.d., 3 µm). Mobile phase: acetonitrile:25 mmol/L formic acid (91:9 for 5 min to 80:20 in 4 min to 0:100 in 3 min to 91:9 for 5 min), flow rate 0.3 mL/min. ESI, SRM acquisition mode. Limit of quantification, 26 µg/L, limit of detection, 8 µg/L [Andersson *et al.* 2008]. Column: CAPCELL PAK C₁₈ MG-II (150 × 2 mm i.d., 5 µm). Mobile phase: 5 mmol/L ammonium formate (pH 4.0):acetonitrile. MRM acquisition mode. Limit of detection, <1.95 µg/L for MDMA, MDA and other amfetamines and metabolites [Kim *et al.* 2008]. Column: Atlantis dC₁₈ (20 × 2.1 mm i.d., 3 µm). Mobile phase: acetonitrile:ammonium formate buffer (pH 3.0, 0:100 to 10:90 in 2 min for 1.5 min to 100:0 in 1 min to 0:100 in 0.5 min). ESI, SRM acquisition mode. Limit of quantification, 1 µg/L for MDMA and 2 µg/L for MDA, limit of detection, 0.2 µg/L for MDMA and 1 µg/L for MDA [Concheiro *et al.* 2007a; Concheiro *et al.* 2007b]. Column: ODS (150 × 1.5 mm i.d., 5 µm). Mobile phase: methanol:10 mmol/L ammonium formate buffer (pH 3.5, 15:85 to 35:65 over 25 min). ESI, CID. Limit of detection, 5 nmol/L [Shima *et al.* 2007]. See Plasma [Kuwayama *et al.* 2008]. See Hair [Wu *et al.* 2001]. See also Chèze *et al.* [2007], Jenkins *et al.* [2004] and Wu, Fuh [2005].

CE Column: fused silica (245/230 mm effective/paced length, 75 µm). Mobile phase: 20 mmol/L sodium phosphate buffer:acetonitrile (80:20). Limit of quantification, 12 µg/L, limit of detection, 5–15 µg/L for MDMA, MDA and other drugs of abuse [Aturki *et al.* 2009]. Column: monolithic capillary (245/336 mm effective/total length, 75 µm i.d.). Limit of quantification, 7.5 mg/L, limit of detection, 1 mg/L for MDMA and metabolites [Choodum *et al.* 2009].

Note For a comparison of a range of immunoassays for MDMA, MDA and other amfetamines, see Verstraete, Heyden [2005] or Hsu *et al.* [2003].

Bile LC-MS See Plasma [Kuwayama *et al.* 2008].

Oral Fluid GC-MS Column: Agilent HP-5MS (30 m × 0.32 mm i.d., 0.25 µm). Carrier gas: He, 1.5 mL/min. Temperature programme: 70° to 170° at 25°/min for 0.2 min to 195° at 5°/min for 0.5 min to 300° at 30°/min for 1 min. SIM acquisition mode. Limit of quantification, 5 µg/L, limit of detection, 2.5 µg/L [Scheidweiler, Huestis 2006]. See Blood [Gentili *et al.* 2002].

LC-MS See Sergi *et al.* [2009] and Wood *et al.* [2003]

Note For a review of the determination of MDMA, MDA and other 'Ecstasy' components in oral fluid, sweat and hair, see Kintz, Samyn [1999]. For the analysis of MDMA and its metabolites in oral fluid and correlation with plasma levels, see Navarro *et al.* [2001].

Vitreous Humour HPLC See Blood [Clauwaert *et al.* 2000]

Sweat See Oral Fluid [Kintz, Samyn 1999].

Hair GC-MS Column: DB-5MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.1 mL/min. Temperature programme: 90° for 0.5 min to 150° at 25°/min to 230° at 15°/min for 1.0 min to 300° at 40°/min for 1.3 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of quantification, 0.05 ng/mg, limit of detection, 0.011 µg/g [Kim *et al.* 2010]. Limit of quantification, 0.24–0.46 µg/g, limit of detection, 0.07–0.14 µg/g for MDMA, MDA and other amfetamines [Johansen, Jørnild 2009]. Column: DB-5 (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° for 1 min to 210° at 10°/min for 3 min. EI ionisation, SIM acquisition mode. Retention time: 11.7 min. Limit of quantification, 50 ng/g, limit of detection, 20 ng/g for MDMA, MDA and other amfetamines [Meng *et al.* 2009]. Column: HP-5MS 5% phenylmethylsiloxane (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.0 mL/min. Temperature programme: 150° for 0.2 min to 280° at 20°/min for 3 min. EI ionisation, SIM acquisition mode. Retention time: 4.98 min. Limit of quantification, 0.05 ng/mg, limit of detection, 0.03 µg/g [Wu *et al.* 2008a; Wu *et al.* 2008b]. Column: HP-5MS (30 m × 0.25 mm i.d., 0.25 µm). Carrier gas: He, 1.0 mL/min. Temperature programme: 150° for 2 min to 220° at 20°/min to 260° at 5°/min to 305° at 30°/min. NCI. Limit of detection, 0.002 ng/mg [Martins *et al.* 2007]. See Blood [Gentili *et al.* 2002]. See also Kim *et al.* [2007], Kintz *et al.* [1995], Martins *et al.* [2006], Pujadas *et al.* [2003], Röhrich, Kauert [1997], Skender *et al.* [2002], Villamor *et al.* [2005] and Zhu *et al.* [2007].

HPLC Column: CAPCELL PAK C₁₈ UG120 (250 × 1.5 mm i.d., 5 µm). Mobile phase: 20 mmol/L imidazole-nitric acid buffer (pH 7.0):acetonitrile:tetrahydrofuran (51.5:45:3.5), flow rate 0.1 mL/min. Electrochemical detection. Limit of quantification, 10 ng/mg, limit of detection, 3 ng/mg [Nakamura *et al.* 2007]. Column: Daisopak SP-120-5-ODS-BP (250 × 4.6 mm i.d., 5 µm). Mobile phase: acetonitrile:methanol:water (30:40:30), flow rate 1.0 mL/min. Fluorescence detection (λ_{ex} = 325 nm, λ_{em} = 430 nm). Limit of detection, 0.2 ng/mg [Kaddoumi *et al.* 2004]. Column: reversed phase (250 × 4.6 mm i.d., 5 µm). Mobile phase: 0.1 mol/L potassium phosphate (pH 3):acetonitrile (82:18). Fluorescence detection (λ_{ex} = 285 nm, λ_{em} = 320 nm). Limit of detection, <1 µg/L for MDMA, MDA and MDEA [Tagliaro *et al.* 1999].

LC-MS Column: Synergi Polar (150 × 2 mm i.d., 4 µm). Mobile phase: 0.01% acetic acid: methanol (50:50), flow rate 0.2 mL/min. ESI, positive ion mode. Limit of quantification, 0.5 µg/g, limit of detection, 0.1 µg/g for MDMA, MDA and other amfetamines [Taberner *et al.* 2009]. Column: Supelcosil LC-CN (33 × 4.6 mm i.d., 3 µm). Mobile phase: acetonitrile:50 mmol/L ammonium acetate (15:85), flow rate 0.4 mL/min. ESI, SIM acquisition mode. Limit of detection, 8–56 ng/L for MDMA and related drugs [Wu *et al.* 2001].

CE-MS Column: uncoated fused silica capillary (100 cm total length, 75 µm). ESI, SIM acquisition mode. Limit of quantification, 0.04 ng/mg, limit of detection, 0.02 ng/mg [Gottardo *et al.* 2007].

Note For a comparison of CE and GC-MS for the analysis of MDMA, see Fang *et al.* [2002].

Disposition in the Body MDMA is absorbed into the bloodstream after ingestion and excreted in urine, the majority of the dose being unchanged (65% within 3 days). Metabolism occurs by a number of routes: *N*-demethylation of the parent compound to MDA (7%) with further *O*-demethylation to 3,4-

dihydroxymetamphetamine (HHMA) and 3,4-dihydroxymetamphetamine (HHA). Both HHMA and HHA are subsequently *O*-methylated by catechol-*O*-methyltransferase (COMT) mainly to 4-hydroxy-3-methoxymetamphetamine (HMMA) and 4-hydroxy-3-methoxyamfetamine (HMA). These 4 metabolites are excreted in the urine as the conjugated glucuronide or sulfate metabolites.

Therapeutic Concentration

After the administration of a single oral dose of 1.5 mg/kg body weight MDMA to 2 patients, plasma and urine samples were collected over periods of 9 and 22 h, respectively. Peak plasma concentrations of MDMA and MDA were 331 µg/L after 2 h and 15 µg/L after 6.3 h, respectively. Peak concentrations of 28.1 µg/L MDMA in urine appeared after 21.5 h. Up to 2.3 µg/L MDA, 35.1 µg/L HMMA and 2.1 µg/L HMA were measured within 16 to 21.5 h, also in urine [Helmlin *et al.* 1996].

Eight healthy male volunteers, aged between 21 and 31 years, were given a 75 mg dose of MDMA. Mean peak plasma concentration was 0.13 mg/L after 1.8 h. Mean peak plasma concentrations of MDA were 7.8 µg/L approximately 5 h after administration [de la Torre *et al.* 2000].

In a double-blind, randomised study, 17 healthy subjects were given single oral doses of 1.0 mg/kg MDMA, 1.6 mg/kg MDMA or placebo. Peak plasma levels of MDMA were ≈163 and 292 µg/L after the low and high doses, respectively. Peak plasma levels of MDA were 8.4 and 13.8 µg/L following the low and high doses of MDMA, respectively. Mean half-lives of MDMA and MDA were 7–8 h and 10.5–12.5 h, respectively [Kolbrich *et al.* 2008b].

In a further study, 30.2–34.3% of urinary excretion was found to occur in the first 24 h following single 1.0 or 1.6 mg/kg doses of MDMA [Abraham *et al.* 2009].

Toxicity Fatalities with doses of 300 mg have been reported. Risk of adverse reaction is high but the pattern is varied. Severe acute toxic effects and morbidity are noted. MDMA is capable of causing severe toxicity and the pattern of acute toxicity reflects the circumstances in which it is misused. Circumstances associated with ingestion and direct effects of the drug are thought to influence toxicity. The amount of MDMA ingested is difficult to determine because the concentration in tablets is unknown and it may be mixed with other substances. A lethal toxic reference concentration of 0.4–0.8 mg/L has been noted, although some overdose cases report concentrations ten times this amount without fatality.

An 18-year-old man ingested 3 tablets and collapsed. He was unconscious on the floor for 60 min and died 2.5 h after admission to hospital. A plasma concentration of 0.36 mg/L MDMA was detected [Henry *et al.* 1992].

A 16-year-old woman ingested 1 tablet and suffered from seizures and experienced hallucinations. The MDMA plasma concentration was 0.424 mg/L. She died 36 h after admission to hospital [Henry *et al.* 1992].

A 21-year-old woman ingested an unknown number of tablets and suffered from hyperactivity leading to convulsions and repeated seizures. Plasma concentration of MDMA was 0.11 mg/L. She underwent a liver transplant after 4 days but died after 18 days [Henry *et al.* 1992].

An 18-year-old man ingested 3 tablets and suffered from seizures and sweating. The MDMA concentration was 1.26 mg/L in plasma. He died 6.5 h after admission [Henry *et al.* 1992].

A 23-year-old man ingested 3 tablets and suffered from convulsions and agitation. The MDMA plasma concentration was 0.2 mg/L. He was hospitalised for 33 days [Henry *et al.* 1992].

A 19-year-old woman ingested 2 tablets and suffered from vomiting, watery diarrhoea and was in a coma. Plasma concentrations were MDMA 0.24 mg/L, MDA 0.02 mg/L and amfetamine 0.02 mg/L. She recovered [Henry *et al.* 1992].

A report of a traffic fatality in a 29-year-old man attributed the cause of death to the use of MDMA. Concentrations of MDMA were: blood 2.32 mg/L (clotted blood) and 2.14 mg/L (sodium fluoride–potassium oxalate anticoagulated blood), vitreous humour 1.11 mg/L and urine 118.8 mg/L. Levels of MDA were <0.25 mg/L in blood and vitreous humour and 3.86 mg/L in urine [Crifasi, Long 1996].

Following a fatal poisoning with MDMA, postmortem blood levels of (S)-MDMA and (R)-MDMA were 1.3 and 1.6 mg/L, respectively. Levels of (S)- and (R)-MDA were 0.8 mg/L [Moore *et al.* 1996].

A 30-year-old man was admitted to hospital after taking 50 ecstasy tablets, 10 oxazepam tablets (10 mg) and 5 units of alcohol over a period of 4 to 5 h. Presenting features were unconsciousness, apnoea and convulsions. Toxicological analysis revealed the following MDMA and MDA concentrations:

Time after ingestion (h)	Sample	MDMA (mg/L)		MDA (mg/L)	
		R-(–)	S-(+)	R-(–)	S-(+)
4	Urine	44	42	nd	0.4
	Gastric lavage	70	72	nd	nd
36	Urine	36	17	3	7
	Urine	18	10	5	4
60	Urine	18	10	5	4
	Serum	0.8	0.2	nd	nd

The individual recovered after a few days without sequelae [Ramcharan *et al.* 1998].

Twelve hours after ingesting a maximum of 40 ecstasy tablets, a 19-year-old man was admitted to an intensive care unit. Toxicological analysis revealed the following MDMA concentrations:

Time after ingestion (h)	Serum MDMA (mg/L)	Urine MDMA (mg/L)
13	4.3	630
24	1.4	180
30	0.75	105

The individual recovered after a few days without sequelae [Regenthal *et al.* 1999].

A 28-year-old man was found unconscious in a bar. A postmortem examination was carried out roughly 28 h later, which revealed the following MDMA and MDA concentrations:

Sample	MDMA	MDA
Subclavian blood (mg/L)	3.5	0.09
Femoral blood (mg/L)	3.1	0.093
Serum (subclavian blood) (mg/L)	4.2	0.1
Vitreous humour (mg/L)	3.4	0.06
Urine (mg/L)	170.9	4
Bile (mg/L)	14.2	0.32
Liver (µg/g)	26.2	1.203
Stomach content (µg/g)	118.1	0.448
Spleen (µg/g)	10	0.264
Brain (frontal lobe) (µg/g)	17.4	0.296
Brain (parietal lobe) (µg/g)	17.1	0.362
Brainstem (µg/g)	13.2	0.22
Cerebellum (µg/g)	11.7	0.225

Cotinine, caffeine and trace amounts of benzoylcegonine were also detected [De Letter *et al.* 2002].

A 19-year-old woman died after taking ecstasy tablets. Blood levels of MDMA were 3.8 mg/L and traces of MDA were also found [Libiseller *et al.* 2005].

A 17-year-old man died after ingestion of ecstasy. Postmortem levels were highest in the stomach (835.97 µg/g) and kidney (801.14 µg/g). Lowest levels were found in the liver (22.26 µg/g) [Dordevic, Tomasevic 2007].

MDMA and MDA blood levels of 1.42 mg/L and 0.17 mg/L, respectively, were found at postmortem following fatal ecstasy poisoning in a 22-year-old man. Hair levels of 8.74–15.51 µg/g in 2 cm strands of hair confirmed chronic drug use [Klys *et al.* 2007].

A 14-month-old infant who swallowed a portion of an ecstasy tablet had a generalised convulsion 40 min after ingestion. He presented with hyperthermia, hypertension, ventricular extrasystoles, tachypnoea and mydriasis. Urine amphetamine levels were greater than 16 mg/L 5 h later and the serum level of MDMA was 0.591 mg/L at 8 h. He recovered after general supportive measures and IV benzodiazepines [Melian *et al.* 2004].

In another case of MDMA intoxication in an infant after accidental ingestion, MDMA and HMMA levels in the infant's hydrolysed urine were 11.7 and 34.4 mg/L, respectively [Garcia-Algar *et al.* 2005].

For other reports of ecstasy poisoning in young children, see Boucher *et al.* [2009] and Cassidy *et al.* [2009]. For other studies of MDMA and MDA distribution following fatalities associated with MDMA, see Dams *et al.* [2003], De Letter *et al.* [2004], [2007], Liu *et al.* [2006] and Valtier *et al.* [2007]. For an overview of ecstasy-related fatalities, see Schifano [2004]; for a review of MDMA deaths in New York City, see Gill *et al.* [2002]; for reviews of deaths in the UK associated with ecstasy use, see Schifano *et al.* [2003]; Schifano [2004]; and Schifano *et al.* [2006]; for a review of MDMA-related deaths in Taiwan, see Lin *et al.* [2009]; for a report on MDMA and driving impairment, see Logan, Couper [2001], see also Brookhuis *et al.* [2004], Kuypers *et al.* [2006], Lamers *et al.* [2003] and Moeller, Hartung [1997].

Half-life ≈6–7 h.

Clearance The mean total clearance of MDMA for a 75 mg dose is 86.9 L/h.

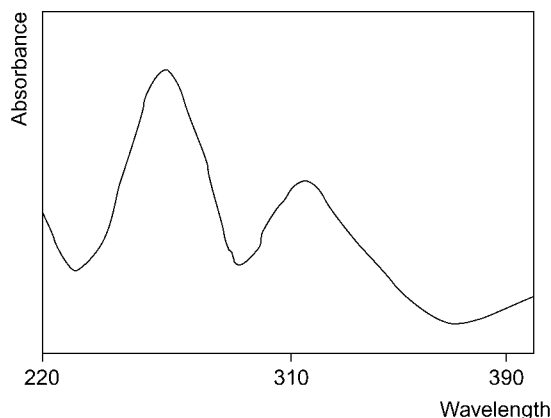
Protein Binding ≈65%

Note For methods of analysis and ecstasy tablet profiling, see Bonadio *et al.* [2009], Cheng *et al.* [2003], Choodum *et al.* [2009], Gimeno *et al.* [2002], Inoue *et al.* [2009], Marquis *et al.* [2008], Milliet *et al.* [2009], Parrott [2004], Rashid *et al.* [2000], Sherlock *et al.* [1999], Shetab Boushehri *et al.* [2009], Tanner-Smith [2006] and Vogels *et al.* [2009]. For a review of the history, neurochemistry and toxicology of ecstasy, see Rochester, Kirchner [1999]; see also Maurer [1996], Ricaurte *et al.* [2000] and Rogers *et al.* [2009].

Dose The usual dose is between 80 and 200 mg (more often 100 to 150 mg).

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Mass Spectrum Principal ions at m/z 369, 394, 315, 370, 333, 395 (glucuronide) [Herber *et al.* 1995]

Disposition in the Body Up to 99.5% of an IV dose is excreted in the urine as sulfo- and glucuroconjugates. The oxo group is reduced first in the presence of NADPH or NADH, followed readily by glucuronidation of the phenolic moiety, leading to a 1 β -O-glucuronide [Herber *et al.* 1995]

Toxicity IV administration of 2.5 g/kg naftazone to mice did not produce toxic symptoms.

Dose Up to 300 μ g daily.

Herber R *et al.* (1995). Reduction and glucuronidation of naftazone by human and rat liver microsomes. *Drug Metab Dispos* 23: 1305–1314.

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Naproxen

Analgesic

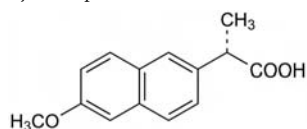
$C_{14}H_{14}O_3 = 230.3$

CAS—22204-53-1

IUPAC Name (2S)-2-(6-Methoxynaphthalen-2-yl)propanoic acid

Synonyms (α S)-6-Methoxy- α -methyl-2-naphthaleneacetic acid; naproxenum; RS-3540.

Proprietary Names Arthrosin; Arthroten; Condrotex; Laraflex; Laser; Napratec; Naprel; Naprex; Naprius; Naprobene; Naprocoat; Naprodil; Naprometin; Napromex; Napronet; Naproscrip; Naprosyn(e); Naproval; Naproxii; Napxen; Nycopren; Proxen; Rimoxyn; Timpron; Xenar.



Chemical Properties A white crystalline powder. Mp 152° to 154° (crystals from acetone-hexane). Practically insoluble in water; soluble 1 in 25 of ethanol, 1 in 15 of chloroform, and 1 in 40 of ether. pK_a 4.2 (25°). Log P (octanol/water), 3.2.

Naproxen Sodium

$C_{14}H_{13}NaO_3 = 252.2$

CAS—26159-34-2

Synonym RS-3650

Proprietary Names Aleve; Anaprox; Naprelan; Naprodil; Naprogesic; Naproxen; Naproso; Naprosyn; Naprovite; Synflex.

Chemical Properties Mp 244° to 246° (crystals from acetone).

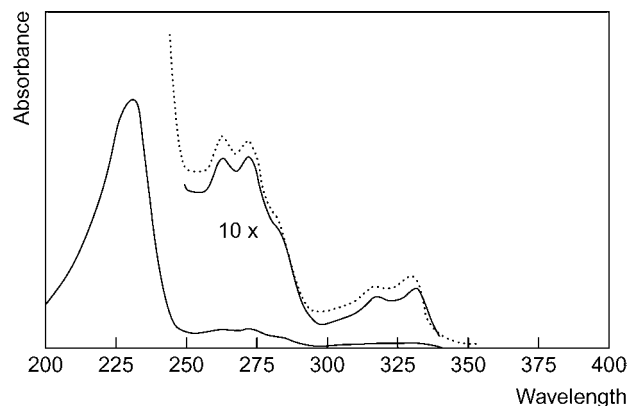
Colour Tests Liebermann's reagent—black-green; Marquis test—brown; sulfuric acid—orange.

Thin-layer Chromatography System TD—R_f 0.33; system TE—R_f 0.06; system TF—R_f 0.38; system TG—R_f 0.14; system TAD—R_f 0.44; system TAE—R_f 0.82; system TAJ—R_f 0.60; system TAK—R_f 0.75; system TAL—R_f 0.93 (Ludy Tenger reagent, orange).

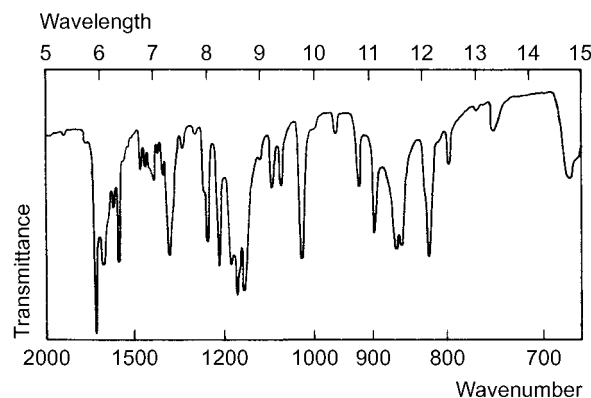
Gas Chromatography System GA—naproxen RI 2045, naproxen-Me RI 1980, M (O-desmethyl-)—Me₂ RI 1980, M (OH-)—Me₂ RI 1800; system GB—naproxen RI 2337, M (O-desmethyl-) RI 2396; system GD—naproxen-Me RRT 1.37 and RRT 1.18 (relative to *n*-C₁₆H₃₄); system GL—naproxen-Me RI 2120, M (O-desmethyl-)—Me₂ RI 2120, M (OH-)—Me₂ RI 1800.

High Performance Liquid Chromatography System HD—*k* 3.3; system HF—RRT 0.67 (relative to meclofenamic acid); system HX—RI 501; system HY—RI 468; system HZ—RT 6.8 min; system HAX—RT 7.0 min; system HAY—RT 7.2 min.

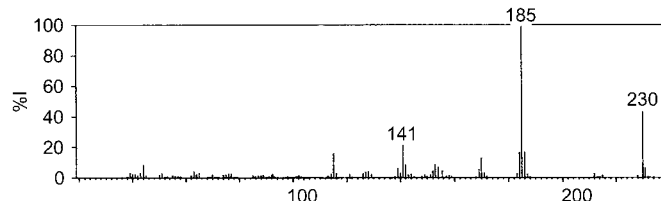
Ultraviolet Spectrum Aqueous acid—262 ($A_1^1=208b$), 272 ($A_1^1=215b$), 315 ($A_1^1=52b$), 328 nm ($A_1^1=63b$); aqueous alkali—261 ($A_1^1=218b$), 271 ($A_1^1=218b$), 316, 330 nm.



Infrared Spectrum Principal peaks at wavenumbers 1724, 1174, 1155, 1223, 1190, 1681 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at m/z 185, 230, 141, 186, 184, 115, 170, 153.



Quantification

Plasma HPLC For method of quantification for naproxen and 6-O-desmethylnaproxen and conjugates, see Andersen and Hansen [1992]. For method of quantification for naproxen, ibuprofen and diclofenac, see Blagbrough *et al.* [1992]. Limit of quantification, 1.5 mg/L for naproxen and 6-O-desmethylnaproxen and conjugates [Vree *et al.* 1992]. Fluorescence detection. Limit of detection, 100 μ g/L for naproxen and 6-O-desmethylnaproxen [van Loenhout *et al.* 1982].

Serum Spectrofluorimetry Limit of detection, 500 μ g/L [Anttila 1977].

UV spectrophotometry See Holzbecher *et al.* [1979].

GC FID. Limit of detection, 2 mg/L [Weber *et al.* 1981].

Urine GC FID. For method of quantification for naproxen and metabolites, see Wan and Matin [1979].

HPLC UV detection. Limit of detection, 3 ng for naproxen naproxen and nabumetone and its metabolite [Mikami *et al.* 2000]. UV detection. Limit of detection, 5 μ g/L for naproxen and other NSAIDs [Hirai *et al.* 1997]. See Plasma [Andersen, Hansen 1992]. See Plasma [Vree *et al.* 1992]. See Plasma [van Loenhout *et al.* 1982].

Synovial Fluid HPLC For method of quantification for naproxen, ibuprofen and diclofenac, see Blagbrough *et al.* [1992].

Disposition in the Body Readily and almost completely absorbed after oral or rectal administration. Naproxen crosses the placenta and small amounts are found in breast milk; it also diffuses into synovial fluid. About 50% of a dose is excreted in the urine in 24 h and ~94% in 5 days together with ~1–2% in the faeces. Of the material excreted in the urine, ~60% is conjugated naproxen, 5% is 6-O-desmethylnaproxen, and 20% is conjugated desmethylnaproxen. <10% of the excreted material is unchanged drug.

Hydrolysis of conjugated naproxen has been reported during storage of urine samples, and assays carried out immediately after collection suggested that only very small amounts of unchanged drug were excreted [Upton *et al.* 1980.]

Therapeutic Concentration Plasma concentrations are dose-dependent at doses of up to 500 mg twice daily; at larger doses the increase in concentration is non-linear and renal clearance is accelerated.

Following single oral doses of 250 mg to 6 subjects, peak plasma concentrations of 26–51 mg/L (mean, 38) were attained in 2–4 h; peak plasma concentrations of 49–69 mg/L (mean, 60) were attained in 3 subjects, 2 h after a single oral dose of 500 mg; peak concentrations of the desmethyl metabolite were about 0.1–0.2 mg/L [Tomson *et al.* 1981.]

Toxicity

A serum concentration of 414 mg/L was reported in 1 subject 15 h after the ingestion of 25 g of naproxen; the only toxic effect was mild gastrointestinal distress [Fredell, Strand 1977].

A 15-year-old girl who had rapidly developed severe metabolic acidosis and seizures after ingesting ~50 × 275 mg naproxen sodium tablets (each containing 250 mg naproxen and 25 mg sodium) had a serum naproxen concentration of 12.9 g/L 14.5 h after the ingestion; serum bicarbonate levels returned to normal after 12.5 h [Martinez *et al.* 1989.]

Half-life Plasma half-life, 10–20 h (mean, 14).

Volume of Distribution About 0.16 L/kg.

Clearance Plasma clearance, 0.13 mL/min/kg.

Protein Binding >99% at normal therapeutic concentrations; decreased at higher plasma concentrations and in subjects with liver disease.

Note For a review of naproxen, see Brogden *et al.* [1979] and for a review of its pharmacokinetics, see Davies and Anderson [1997].

Dose Usually 0.5 to 1 g daily.

Andersen JV, Hansen SH (1992). Simultaneous quantitative determination of naproxen, its metabolite 6-O-desmethylnaproxen and their five conjugates in plasma and urine samples by high-performance liquid chromatography on dynamically modified silica. *J Chromatogr* 577: 325–333.

Anttila M (1977). Fluorometric determination of naproxen in serum. *J Pharm Sci* 66: 433–434.

Blagbrough IS *et al.* (1992). High-performance liquid chromatographic determination of naproxen, ibuprofen and diclofenac in plasma and synovial fluid in man. *J Chromatogr* 578: 251–257.

Brogden RN *et al.* (1979). Naproxen up to date: a review of its pharmacological properties and therapeutic efficacy and use in rheumatic diseases and pain states. *Drugs* 18: 241–277.

Davies NM, Anderson KE (1997). Clinical pharmacokinetics of naproxen. *Clin Pharmacokinet* 32: 268–293.

Fredell EW, Strand LJ (1977). Naproxen overdose. *JAMA* 238: 938.

Hirai T *et al.* (1997). Simultaneous analysis of several non-steroidal anti-inflammatory drugs in human urine by high-performance liquid chromatography with normal solid-phase extraction. *J Chromatogr B Biomed Sci Appl* 692: 375–388.

Holzbecher M *et al.* (1979). An ultraviolet spectrophotometric procedure for the routine determination of naproxen. *Clin Biochem* 12: 66–67.

Martinez R *et al.* (1989). Severe metabolic acidosis after acute naproxen sodium ingestion. *Ann Emerg Med* 18: 1102–1104.

Mikami E *et al.* (2000). Simultaneous analysis of naproxen, nabumetone and its major metabolite 6-methoxy-2-naphthylacetic acid in pharmaceuticals and human urine by high-performance liquid chromatography. *J Pharm Biomed Anal* 23(5): 917–925.

Tomson G *et al.* (1981). Relation of naproxen kinetics to effect on platelet prostaglandin release in men and dysmenorrheic women. *Clin Pharmacol Ther* 29: 168–173.

Upton RA *et al.* (1980). Negligible excretion of unchanged ketoprofen, naproxen, and probenecid in urine. *J Pharm Sci* 69: 1254–1257.

van Loenhout JWA *et al.* (1982). *J Liq Chromatogr* 5: 549–561.

Vree TB *et al.* (1992). Determination of naproxen and its metabolite O-desmethylnaproxen with their acyl glucuronides in human plasma and urine by means of direct gradient high-performance liquid chromatography. *J Chromatogr* 578: 239–249.

Wan SH, Matin SB (1979). Quantitative gas-liquid chromatographic analysis of naproxen, 6-O-desmethylnaproxen and their conjugates in urine. *J Chromatogr* 170: 473–478.

Weber SS *et al.* (1981). Effect of Mylanta on naproxen bioavailability. *Ther Drug Monit* 3: 75–83.

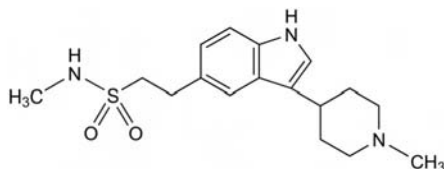
Naratriptan

Antimigraine

C₁₇H₂₅N₃O₂S = 335.4

CAS—121679-13-8

IUPAC Name N-Methyl-3-(1-methyl-4-piperidinyl)-1H-indole-5-ethanesulfonamide
Synonym GR-85548X



Chemical Properties Crystals from ethyl acetate. Mp 170° to 171°.

Naratriptan Hydrochloride

C₁₇H₂₅N₃O₂S·HCl = 371.9

CAS—121679-19-4; 143388-64-1

Synonym GR-85548A

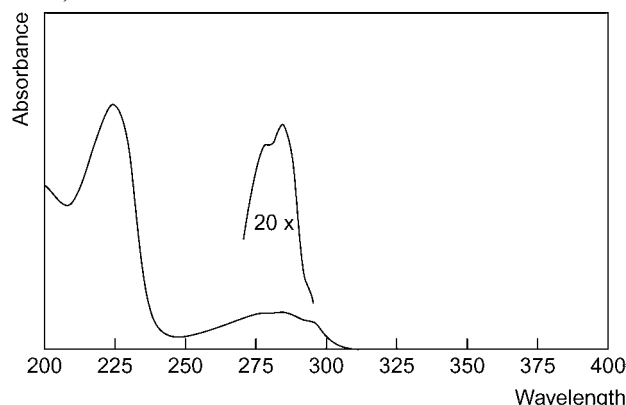
Proprietary Names *Amerge*; *Naramig*.

Chemical Properties Microcrystals. Mp 237° to 239°.

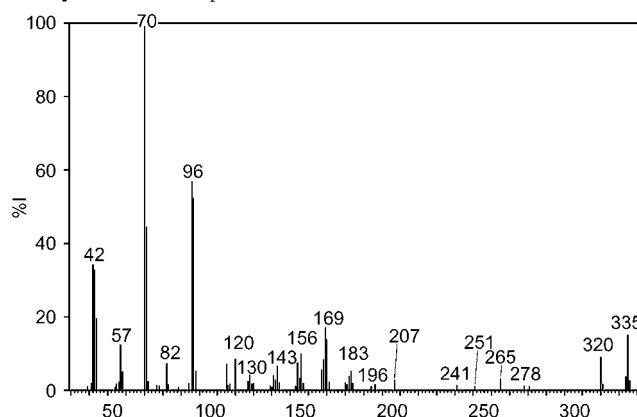
High Performance Liquid Chromatography System HZ—retention time 2.1 min.

Column: silica Alltech (150 × 2.1 mm i.d., 5 μm). Mobile phase: 20 mmol/L ammonium acetate (pH 4 to pH 2.7) : methanol : acetonitrile (80 : 10 : 10), flow rate 300 μL/min. IS: bufotamine. MS—MS detection (ESI, MRM with m/z: 336 to 98 for naratriptan and 205 to 160 for IS). Retention time: naratriptan, 2.71 min; IS, 2.35 min [Vishwanathan *et al.* 2000].

Ultraviolet Spectrum Aqueous acid (pH 2.38)—224, 278, 284, 295 nm; ethanol—285, 296 nm.



Mass Spectrum Principal ions at m/z 70, 96, 97, 71, 42, 43, 44, 169.



Quantification

Serum HPLC—MS MS—MS detection (ESI). Limit of detection, 0.1 μg/L [Vishwanathan *et al.* 2000].

Disposition in the Body Naratriptan is rapidly and well absorbed after administration, and undergoes some hepatic metabolism by a wide range of cytochrome P450 isoenzymes. Renal excretion is the main route of elimination and the drug is predominately excreted in urine, with 50% of the dose being recovered as the unchanged drug and 30% as the inactive metabolite. It displays linear pharmacokinetics over the therapeutic range and repeat administration does not lead to accumulation of the drug. Food does not alter the absorption.

Therapeutic Concentration

Seven adolescent migraine patients, aged between 12 and 16 years, were administered a single dose of 2.5 mg naratriptan outside a migraine attack.

The mean naratriptan plasma concentration was 8.0 μg/L (range, 5.9 to 10.7) observed between 1.5 and 4 h after dosing [Christensen *et al.* 2001].

Bioavailability Men, 63%; women, 74%.

Half-life Approximately 4.5 to 5.5 h; prolonged in patients with renal and hepatic impairment.

Volume of Distribution 170 L.

Clearance Systemic, 6.6 mL/min/kg; reduced by 26% in the elderly and by 50% in renal impaired individuals.

Protein Binding 28 to 31% (concentration range, 50 to 1000 μg/L).

Note For a review of naratriptan use in migraine treatment, see Dulli [1999].

Dose Oral administration: initial dose is 2.5 mg in the UK; 1 to 2.5 mg in the US and a second dose if required. The maximum dose is 5 mg daily.

Christensen ML *et al.* (2001). Pharmacokinetics of naratriptan in adolescent subjects with a history of migraine. *J Clin Pharmacol* 41(2): 170–175.

Dulli DA (1999). Naratriptan: an alternative for migraine. *Ann Pharmacother* 33: 704–711.

Vishwanathan K *et al.* (2000). Determination of antimigraine compounds rizatriptan, zolmitriptan, naratriptan and sumatriptan in human serum by liquid chromatography/electrospray tandem mass spectrometry. *Rapid Commun Mass Spectrom* 14(3): 168–172.

Narceine

Alkaloid

C₂₃H₂₇NO₈ = 445.5

CAS—131-28-2 (anhydrous)

IUPAC Name 6-[[6-[2-(Dimethylamino)ethyl]-4-methoxy-1,3-benzodioxol-5-yl]-acetyl]-2,3-dimethoxybenzoic acid

Uses Phenacaine hydrochloride has been used as a 1% aqueous solution or as a 1 or 2% ointment.

Phenacemide

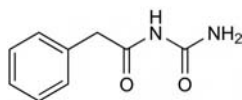
Anticonvulsant

CAS—63-98-9

IUPAC Name *N*-carbamoyl-2-phenylacetamide

Synonyms *N*-(aminocarbonyl)benzeneacetamide; carbamidum phenylacetici; phenacetylurea.

Proprietary Name *Phenurone*



Chemical Properties $C_9H_{10}N_2O_2$ = 178.2. A white crystalline powder. Mp about 212° to 216°. Very slightly soluble in water; slightly soluble in acetone, ethanol, chloroform, ether and methanol. Log *P* (octanol/water), 0.9.

Colour Tests Liebermann's reagent—red-orange; Nessler's reagent—brown-orange (slow).

Thin-layer Chromatography System TD— R_f 0.22; system TE— R_f 0.65; system TF— R_f 0.40; system TAD— R_f 0.50.

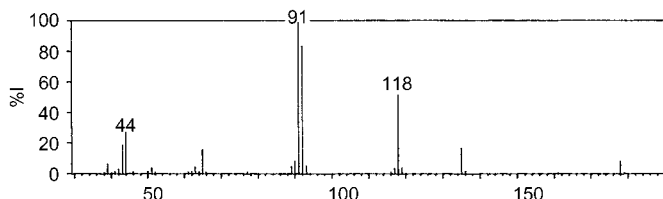
Gas Chromatography System GA—RI 1473.

High Performance Liquid Chromatography System HX—RI 339; system HY—RI 266.

Ultraviolet Spectrum Methanol—258 (A_1^1 =18a), 265 nm.

Infrared Spectrum Principal peaks at wavenumbers 1660, 1090, 716, 1618, 1170, 970 cm^{-1} (KBr disk).

Mass Spectrum Principal ions at *m/z* 91, 92, 118, 44, 43, 135, 65, 178.



Disposition in the Body Well absorbed after oral administration. It is extensively metabolised to inactive metabolites, principally by *p*-hydroxylation of the phenyl ring; the metabolites are excreted in the urine.

Toxicity The estimated minimum lethal dose is 5 g.

Dose 1.5 to 3 g daily.

Phenacetin

Analgesic

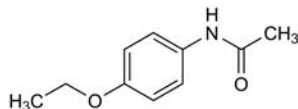
$C_{10}H_{13}NO_2$ = 179.2

CAS—62-44-2

IUPAC Name *N*-(4-Ethoxyphenyl)acetamide

Synonyms *p*-Acetiphenetidine; aceto-*p*-phenetidine; acetophenetidin; acetylphenetidin; fenacetina; paracetophenetidin.

Proprietary Names It is an ingredient of *APC tablets* and was included in *Gripponyl*, *Hemagene Tailleux*, *Polypirine* and *Cratodin Rectal*.



Chemical Properties White glistening crystalline scales or fine white crystalline powder. Mp 134° to 135°. Soluble 1 in 1300 of water, 1 in 15 of ethanol, 1 in 14 of chloroform and 1 in 90 of ether. pK_a 2.2. Log *P* (octanol/water), 1.6.

Colour Tests Liebermann's reagent—violet; boil 0.1 g with 1 mL of hydrochloric acid for 3 min, dilute with 10 mL of water, cool, filter, and add one drop of 0.02 mol/L potassium dichromate to the filtrate—violet→red.

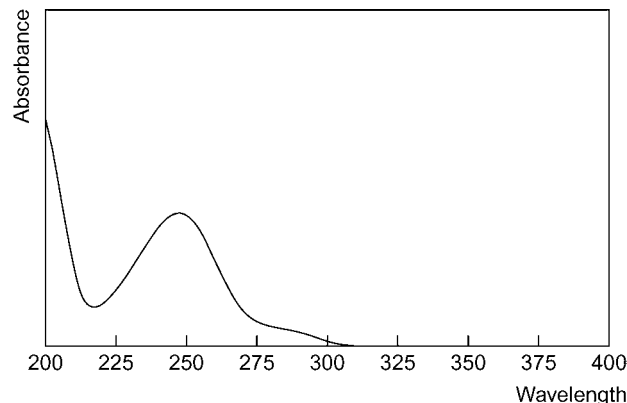
Thin-layer Chromatography System TD— R_f 0.38; system TE— R_f 0.68; system TF— R_f 0.37; system TAD— R_f 0.52; system TAE— R_f 0.83; system TAJ— R_f 0.58; system TAK— R_f 0.41; system TAL— R_f 0.89 (acidified potassium permanganate solution, positive). See also under Paracetamol.

Gas Chromatography System GA—phenacetin RI 1675, M (hydroquinone) RI 1240, M (*p*-phenetidine) RI 1275, paracetamol RI 1687; system GB—RI 1730; system GF—RI 2325.

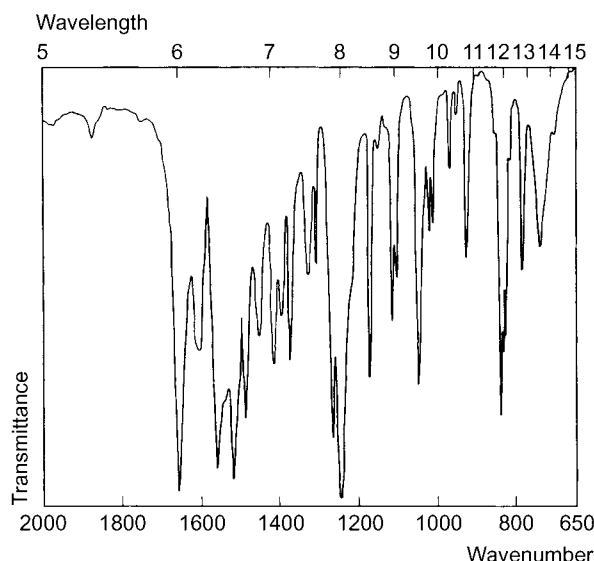
High Performance Liquid Chromatography System HD—phenacetin *k* 0.6, paracetamol *k* 0.1; system HW—phenacetin *k* 4.40, paracetamol *k* 0.30; system HX—phenacetin RI 377, paracetamol RI 264; system HY—phenacetin RI 335,

paracetamol RI 241; system HZ—phenacetin retention time 3.0 min, paracetamol retention time 1.9 min.

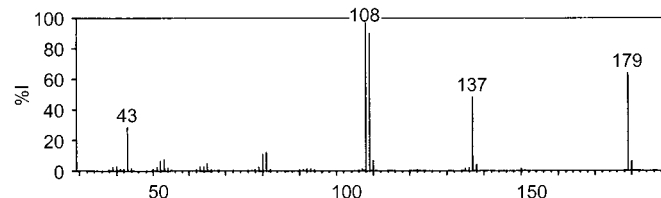
Ultraviolet Spectrum Aqueous acid—244 nm (A_1^1 =649a). No alkaline shift.



Infrared Spectrum Principal peaks at wavenumbers 1244, 1655, 1513, 1555, 1265, 836 cm^{-1} (KBr disk).



Mass Spectrum Principal ions at *m/z* 108, 109, 179, 137, 43, 81, 80, 110; paracetamol 109, 151, 43, 80, 108, 81, 53, 52.



Quantification

Plasma GC FID. Phenacetin and paracetamol. Limit of detection, <100 µg/L [Evans, Harbison 1977].

GC-MS Limit of detection, 1 pg [Murray, Boobis 1991]. Phenacetin and paracetamol. Limit of detection, 1 µg/L for phenacetin and 100 µg/L for paracetamol [Garland *et al.* 1977].

HPLC UV detection. Phenacetin and paracetamol. Limit of detection, 500 µg/L [Gotelli *et al.* 1977].

See also under Paracetamol.

Disposition in the Body Readily absorbed after oral administration, but subject to extensive first-pass metabolism. It is metabolised mainly in the liver by *O*-dealkylation to paracetamol and acetaldehyde, followed by conjugation of the paracetamol with sulfate or glucuronic acid (see under Paracetamol); other reactions are deacetylation to phenetidine (*p*-ethoxyaniline), *N*-, 2-, and α -hydroxylation forming mainly 2-hydroxyphenetidine and also 2-hydroxyphenacetin, sulfate conjugation of deacetylated metabolites, and glutathione conjugation to form *S*-(1-acetamido-4-hydroxyphenyl)cysteine and its corresponding mercapturic acid.

2-Hydroxyphenetidine appears to be the nephrotoxic metabolite and is possibly involved in the formation of methaemoglobinaemia. As the dose is increased so the percentage of the dose which is deacetylated is increased, producing proportionately

more phenetidine and 2-hydroxyphenetidine; production of these metabolites is also increased by concomitant administration of aspirin, caffeine, and codeine. Phenacetin is not hepatotoxic despite being converted to paracetamol; this is because the enzyme systems which convert paracetamol to its toxic metabolite are also involved in the conversion of phenacetin to paracetamol and the two conversions therefore compete, resulting in reduced paracetamol oxidation. Phenacetin metabolism is stimulated by cigarette smoking. The extent of *O*-dealkylation appears to be genetically determined. About 80 to 90% of a dose is excreted in the urine in 24 h with 50 to 80% of the dose as the sulfate and glucuronide conjugates of paracetamol, 6 to 8% as 2-hydroxyphenetidine sulfate, about 2% as *S*-(1-acetamido-4-hydroxyphenyl)cysteine, about 0.3% as phenetidine and 2-hydroxyphenacetin, 0.2% as unchanged drug, and 2 to 3% as unconjugated paracetamol.

Therapeutic Concentration

After a single oral dose of 0.9 g to 9 subjects, peak plasma-phenacetin concentrations of 0.2 to 7.4 mg/L (mean, 2.3) were attained in 1 to 2 h, and peak plasma-paracetamol concentrations of 3.1 to 11.9 mg/L (mean, 7.9) were attained in 2 to 3.5 h; plasma-phenacetin concentrations were reported to be lower in 9 smokers [Pantuck *et al.* 1974].

Toxicity The estimated minimum lethal dose is 5 g. Prolonged use of phenacetin is associated with analgesic nephropathy; it may also produce haemolytic anaemia and methaemoglobinemia.

In a fatality involving the intentional overdosage of a preparation containing codeine, aspirin, phenacetin and caffeine, a blood-phenacetin concentration of 136 mg/L was reported; the codeine and salicylate blood concentrations were 5.3 and 265 mg/L respectively [Wright *et al.* 1975].

Half-life Plasma half-life, phenacetin 0.7 to 1.5 h, paracetamol 1.5 to 3 h.

Volume of Distribution About 1 to 2 L/kg.

Clearance Plasma clearance, about 20 mL/min/kg.

Distribution in Blood Plasma: whole blood ratio, 0.93.

Protein Binding About 30%.

Note For a review of the general properties of phenacetin, see Clissold [1986].

Dose Phenacetin has been given as a single dose of 300 to 600 mg; maximum daily dose of 3 g.

Clissold SP (1986). Paracetamol and phenacetin. *Drugs* 32: 446–59.

Evans MA, Harbison RD (1977). GLC microanalyses of phenacetin and acetaminophen plasma levels. *J Pharm Sci* 66: 1628–1629.

Garland WA *et al.* (1977). Quantitative determination of phenacetin and its metabolite acetaminophen by GLC-chemical ionization mass spectrometry. *J Pharm Sci* 66: 340–344.

Gotelli GR *et al.* (1977). Determination of acetaminophen and phenacetin in plasma by high-pressure liquid chromatography. *Clin Chem* 23: 957–959.

Murray S, Boobis AR (1991). Combined assay for phenacetin and paracetamol in plasma using capillary column gas chromatography-negative-ion mass spectrometry. *J Chromatogr* 568: 341–350.

Pantuck EJ *et al.* (1974). Effect of cigarette smoking on phenacetin metabolism. *Clin Pharmacol Ther* 15: 9–17.

Wright JA *et al.* (1975). Blood codeine concentrations in fatalities associated with codeine. *Clin Toxicol* 8: 457–463.

Phenadoxone

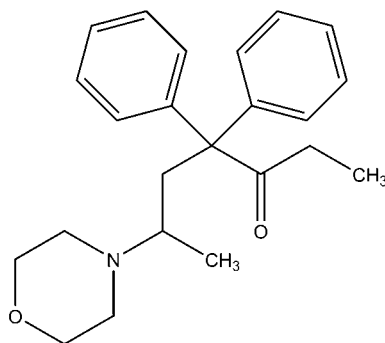
Narcotic Analgesic

$C_{23}H_{29}NO_2 = 351.5$

CAS—467-84-5

IUPAC Name 6-Morpholin-4-yl-4,4-di(phenyl)heptan-3-one

Synonyms Heptaxone; morphodone; 6-morpholino-4,4-diphenylheptan-3-one; 6-(4-morpholinyl)-4,4-diphenyl-3-heptanone; 6-tetrahydrooxazine-4,4-diphenyl-3-heptanone.



Chemical Properties Mp 75° to 76°. Log *P* (octanol/water) 4.2 [National Institutes of Health 2008], 1.56 [Meylan, Howard, 1995]. Phenadoxone is extracted by organic solvents from aqueous alkaline solutions; hydrolysis of biological material is required to improve yield.

Phenadoxone Hydrochloride

$C_{23}H_{29}NO_2 \cdot HCl = 387.9$

CAS—545-91-5

Synonyms CB-11; hepagin; heptalin; heptazone; Hoechst 10600.

Proprietary Name Heptalgin

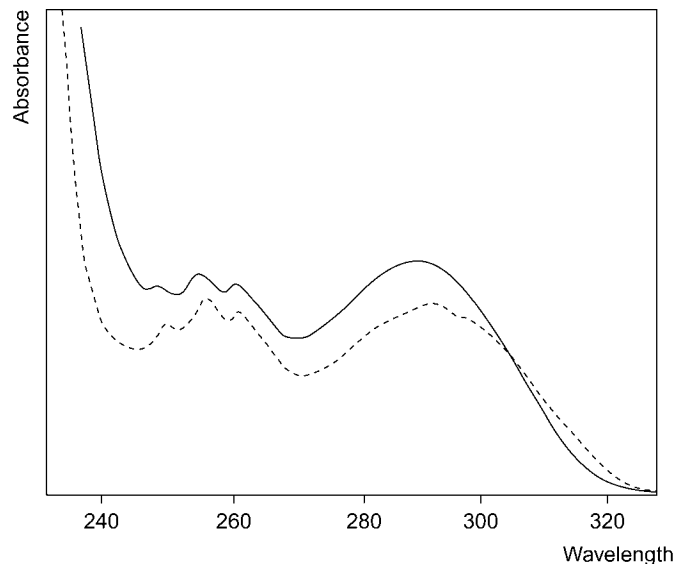
Chemical Properties A colourless crystalline powder. Mp 224° to 225° (dec). Soluble 1 in 25 of water and 1 in 10 of ethanol; very soluble in chloroform; almost insoluble in benzene and ethyl acetate. pKa (25°) 6.7 in 40% alcohol [O'Neil *et al.* 2006].

Colour Test Ammonium vandate test—deep green→blue (limit of detection, 0.5 µg).

Thin-layer Chromatography System T10—*R_f* 0.77 (location reagent acidified iodoplatinate spray, positive reaction).

Gas Chromatography System G2/225—retention time 1.60 (relative to codeine); system G4—retention time 0.96 (relative to codeine).

Ultraviolet Spectrum Ethanol—259, 296 nm; aqueous acid (0.1 N sulfuric acid)—253, 259, 266, 292 nm.



Note For a review of the pharmacology of phenadoxone, see Basil *et al.* [1950].

Dose Up to 50 mg orally, up to 15 mg SC or IM.

References

- Basil B *et al.* (1950). The pharmacology of phenadoxone or dl-6-morpholino-4:4-diphenylheptan-3-one hydrochloride. *Br J Pharmacol Chemother* 5: 125–141.
Meylan WM, Howard PH (1995). Atom/fragment contribution method for estimating octanol-water partition coefficients. *J Pharm Sci* 84: 83–92.
National Institutes of Health (2008). Phenadoxone. http://pubchem.ncbi.nlm.nih.gov/summary/summary.cgi?cid=10089&loc=ec_rcs. (accessed 1 July 2008).
O'Neil MJ *et al.* (2006). *The Merck Index: An Encyclopedia of Chemicals, Drugs and Biologicals*, 14 edn. Whitehouse Station, NJ: Merck & Co., Inc.

Phenaglycodol

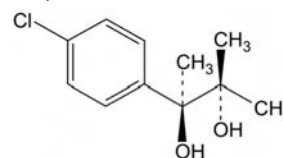
Tranquilliser

$C_{11}H_{15}ClO_2 = 214.7$

CAS—79-93-6

IUPAC Name 2-(4-Chlorophenyl)-3-methylbutane-2,3-diol

Proprietary Name *Felixyn*



Chemical Properties A crystalline solid. Mp 77° to 78°. Practically insoluble in water; soluble in ethanol.

Colour Tests Liebermann's reagent—brown; Mandelin's test—blue.

Thin-layer Chromatography System TE—*R_f* 0.71; system TAE—*R_f* 0.84.

Ultraviolet Spectrum Ethanol—259, 266 (*A*₁—13.5a), 275 nm.

Infrared Spectrum Principal peaks at wavenumbers 1095, 1085, 1015, 1180, 807, 830 cm⁻¹ (KBr disk).

Mass Spectrum Principal ions at *m/z* 43, 59, 155, 121, 156, 31, 157, 158.

Quantification

Blood UV spectrophotometry [Wallace 1968].

Urine UV spectrophotometry See Blood [Wallace 1968].

Dose 0.9 to 1.2 g daily.

Wallace JE (1968). Determination of phenaglycodol in biologic specimens by ultraviolet spectrophotometry. *J Pharm Sci* 57: 426–429.

Demme U *et al.* (2005). Systematic evaluation of 1-chlorobutane for liquid-liquid extraction of drugs. *Proceedings of the 12th TIAFT*, Seoul: 481–486.

Tzeng TB *et al.* (1994). Sensitive method for the assay of sertindole in plasma by high-performance liquid chromatography and fluorimetric detection. *J Chromatogr B Biomed Appl* 661: 299–306.

Wong SL, Granneman GR (1998). Modeling of sertindole pharmacokinetic disposition in healthy volunteers in short term dose-escalation studies. *J Pharm Sci* 87: 1629–1631.

Wong SL *et al.* (1997). Pharmacokinetics of sertindole in healthy young and elderly male and female subjects. *Clin Pharmacol Ther* 62: 157–164.

Sertraline

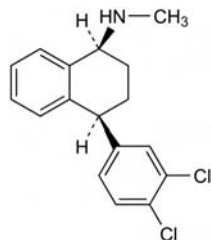
Antidepressant

$C_{17}H_{17}Cl_2N$ = 306.2

CAS—79617-96-2

IUPAC Name (1*S*,4*S*)-4-(3,4-Dichlorophenyl)-1,2,3,4-tetrahydro-*N*-methyl-1-naphthalenamine

Synonym CP-51974



Chemical Properties Crystals. Mp $\sim 207^\circ$. Blood, plasma and cardiac tissue samples were stable for at least 6 months at -80° [Läer *et al.* 1996]. Extraction yield (chlorobutane), 1 [Demme *et al.* 2005].

Sertraline Hydrochloride

$C_{17}H_{17}Cl_2N \cdot HCl$ = 342.7

CAS—79559-97-0

IUPAC Name (1*S*,4*S*)-4-(3,4-Dichlorophenyl)-1,2,3,4-tetrahydro-*N*-methyl-1-naphthalenamine hydrochloride

Proprietary Names Altruline; Aremis; Besitrax; Gladem; Lustrax; Sealdin; Serad; Tatig; Zolof.

Chemical Properties A white to off-white crystalline powder (irritant). Mp onset of change 160° to 180° , partial melt 218° , total melt 243° to 245° . Solubility: water 3.8 mg/mL (pH 5.3), pH dependent. It is soluble in chloroform: methanol (1:1), methanolic 0.1 mol/L hydrochloric acid, dimethyl sulfoxide, ethanol and *N,N*-dimethylformamide; practically insoluble in aqueous 0.1 mol/L hydrochloric acid, aqueous 0.1 mol/L sodium hydroxide, propan-2-ol, acetone, ethyl acetate and acetonitrile. pK_a 9.48 ± 0.04 (water), 8.5 (ethanol: water (1:1)) and 8.6 (methanol: water (40:60)). Log *P* (octanol/water), 5.29.

Thin-layer Chromatography System TB— R_f 0.46; system TE— R_f 0.72; system TAE— R_f 0.25.

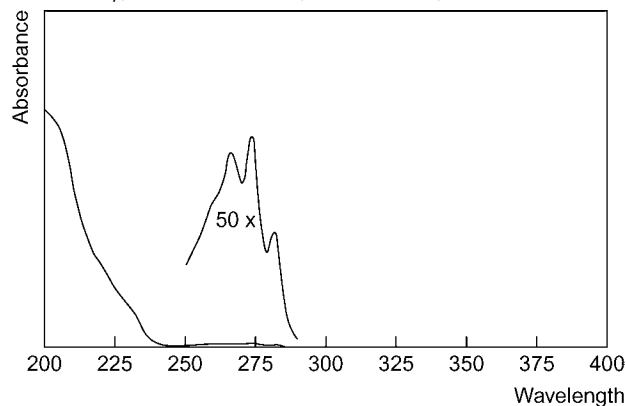
Gas Chromatography System GB—sertraline RI 2481, M (nor-) RI 2468, M (ketone) RI 2496; system GM—sertraline RRT 1.166, M (nor-) RRT 1.218. (both relative to iprindole).

Gas Chromatography-Mass Spectrometry Column: HP1 methyl silicone (0.2 mm i.d., 0.33 μ m). Temperature: 250° . Carrier gas: He, 0.9 mL/min. RI: 2401 [Mills, Roberson 1993].

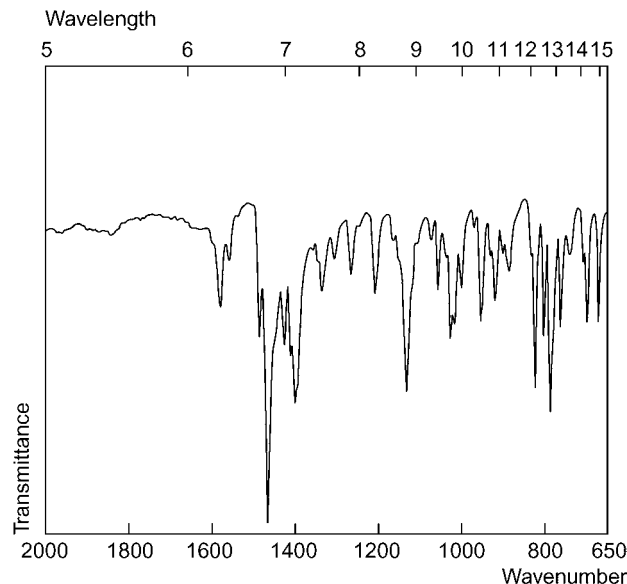
Column: 3% Silar 10C on Gas Chrom Q 80-100 mesh (0.8 m $\times$ 0.2 mm i.d.). Temperature: 255° . Injector temperature: 200° . Carrier gas: He, 40 mL/min. EI ionisation, SIM acquisition mode at m/z : 274. RT: 3.9 min. [Fouda *et al.* 1987].

High Performance Liquid Chromatography System HX—RI 460; system HZ—sertraline RT 8.2 min, M (desmethyl-) RT 7.0 min; system HAX—RT 14.5 min; system HAY—RT 7.7 min.

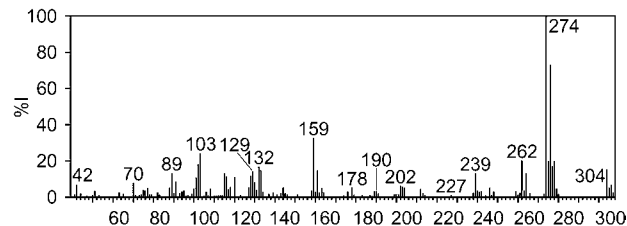
Ultraviolet Spectrum Aqueous acid (0.01 mol/L hydrochloric acid/methanol or methanol only)—265, 273, 281 nm (no alkaline shift).



Infrared Spectrum Principal peaks at wavenumber 1470, 1135, 825, 800, 790, 760 cm^{-1} (KBr pellet).



Mass Spectrum Principal ions at m/z 274, 159, 276, 103, 262, 132, 129, 161.



Quantification

Plasma HPLC Column: Ultraspher octyl C_8 (250 $\times$ 4.6 mm, 5 μ m). Mobile phase: acetonitrile: 0.086% w/v 1-octane sulfonic acid and 0.01% v/v *N,N,N,N*-tetramethylethylenediamine (pH 2.5, 43:57), flow rate 1.6 mL/min. UV detection (λ = 210 nm). Limit of detection, 5 μ g/L [Kristensen *et al.* 1998].

GC Column: SE-54 (12 m $\times$ 0.32 mm i.d., 0.25 μ m). Carrier gas: He, 3.0 mL/min. Temperature programme: 165° for 0.5 min to 210° at $50^\circ/\text{min}$ for 12 min. Limit of quantification, 1 μ g/L [Tremaine, Joerg 1989].

Serum HPLC Column: LC-8 (150 $\times$ 4.6 mm i.d., 5 μ m). Mobile phase: acetonitrile: methanol: water: diethylamine (1000:800:1200:2), flow rate 1.5 mL/min. UV detection (λ = 600 nm for 2 min followed by 214 nm). Limit of detection, 0.01 mg/L [Rogowsky *et al.* 1994].

GC-MS Column: RTX-5 (15 m $\times$ 0.25 mm i.d., 0.25 μ m). Temperature programme: 100° to 210° at $50^\circ/\text{min}$. Limit of detection, 10 μ g/L [Rogowsky *et al.* 1994].

Milk HPLC See Plasma [Kristensen *et al.* 1998].

Disposition in the Body Sertraline is absorbed slowly after oral administration, with peak plasma concentrations being reached within 4.5–8.4 h. Steady-state levels are achieved after 1 week of daily dosing. It is widely distributed throughout the body. Sertraline and its metabolite *N*-desmethylsertraline (inactive in *in vivo* models of depression) are extensively metabolised. Sertraline is metabolised by *N*-demethylation, oxidative deamination and subsequent reduction, hydroxylation and glucuronide conjugation; *N*-desmethylsertraline is metabolised by oxidative deamination and subsequent reduction, hydroxylation and glucuronide conjugation. The metabolites are excreted in urine and faeces; 40–45% of an administered dose is excreted in urine over 9 days with less than 0.2% as the unchanged drug. It is distributed in breast milk.

Therapeutic Concentration The serum therapeutic range is 50–250 μ g/L.

Forty-four healthy volunteers (male and female), in 2 groups (young aged between 18 and 45 years and older aged >65 years) were administered with a single 50 mg daily dose of sertraline. The initial dose was titrated to 200 mg over 30 days, which is the maximum dose. On day 30, peak sertraline plasma concentrations were determined as 118 μ g/L for the young males and 166 μ g/L for the young females. Concentrations were reached in 6.9 and 6.7 h, respectively. Peak plasma concentrations of the metabolite *N*-desmethylsertraline were 156 μ g/L in 9.1 h and 244 μ g/L in 5.9 h for the males and females, respectively. For the older patients, peak plasma concentrations of the drug were 135 μ g/L and 147 μ g/L for the males and females, respectively, and were reached in 7.8 and 6.4 h, respectively. For the metabolite, these values were 237 μ g/L and 274 μ g/L, respectively, attained in 13.8 and 8.2 h, respectively [Ronfeld *et al.* 1997].

Ten patients, aged 35–69 years, with stable chronic hepatic insufficiency caused by cirrhosis (mild to severe) and 10 equally matched healthy individuals were administered with a single 100 mg sertraline base dose after an overnight fast. For patients, mean peak plasma concentration for the drug was 35.2 µg/L (range, 20.0–40.8), observed at 4 h (range, 2–6); mean peak plasma concentration for *N*-demethylsertraline was 7.9 µg/L (range, 5.5–9.7), observed at 168 h (range, 96–216). For the healthy volunteers, mean peak drug concentrations were 20.6 µg/L (range 12.6–27.5) at 6 h (4–6) for the drug and 11.2 µg/L (10.6–13.0) at 6 h (6–10) for the metabolite [Demolis *et al.* 1996].

Toxicity

A 51-year-old white woman was admitted into hospital after ingesting 8000 mg sertraline. She had been treated 3 days before for acute alcohol intoxication. Symptoms included somnolence, nausea, tachycardia, anxiety and dilated pupils. On admission, serum concentrations were 2930 µg/L and 1679 µg/L for sertraline and desmethylsertraline, respectively. Other drugs detected included norpropoxyphene (268 µg/L) and nortriptylene (26 µg/L). No ethanol was found. On day 4, sertraline and desmethylsertraline serum concentrations were 453 µg/L and 1132 µg/L, respectively. The drug concentration reduced further to 305 µg/L on day 5 and the patient was discharged on day 6 after receiving supportive measures [Brendel *et al.* 2000].

Half-life Sertraline, approximately 26 h (mean for young and elderly 22–36 h); *N*-desmethylsertraline, 62–104 h; prolonged in hepatic impairment.

Volume of Distribution 20 L/kg.

Protein Binding ≈98%.

Dose 50 to 200 mg daily.

Brendel DH *et al.* (2000). Massive sertraline overdose. *Ann EmergMed* 36: 524–526.

Demme U *et al.* (2005). Systematic evaluation of 1-chlorobutane for liquid–liquid extraction of drugs. *Proceedings of the 12th TIAFT*, Seoul: 481–486.

Demolis JL *et al.* (1996). Influence of liver cirrhosis on sertraline pharmacokinetics. *Br J Clin Pharmacol* 42: 394–397.

Fouda HG *et al.* (1987). Gas chromatographic-mass spectrometric analysis and preliminary human pharmacokinetics of sertraline, a new antidepressant drug. *J Chromatogr* 417: 197–202.

Kristensen JH *et al.* (1998). Distribution and excretion of sertraline and *N*-desmethylsertraline in human milk. *Br J Clin Pharmacol* 45: 453–457.

Läer S *et al.* (1996). Determination of sotalol in human cardiac tissue by high-performance liquid chromatography. *J Chromatogr B Biomed Appl* 681: 291–298.

Mills TI, Roberson JC. *AT Instrumental Data for Drug Analysis*, 2nd edn, Vol. 4-5. Boca Raton FL: CRC Press 222–223.

Rogowsky D *et al.* (1994). Determination of sertraline and desmethylsertraline in human serum using copolymeric bonded-phase extraction, liquid chromatography and gas chromatography–mass spectrometry. *J Chromatogr B Biomed Appl* 655: 138–141.

Ronfeld RA *et al.* (1997). Pharmacokinetics of sertraline and its *N*-demethyl metabolite in elderly and young male and female volunteers. *Clin Pharmacokinet* 32(Suppl 1): 22–30.

Tremaine LM, Joerg EA (1989). Automated gas chromatographic-electron-capture assay for the selective serotonin uptake blocker sertraline. *J Chromatogr* 496: 423–429.

Setiptiline

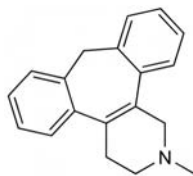
Antidepressant

C₁₉H₁₉N = 261.4

CAS—57262-94-9

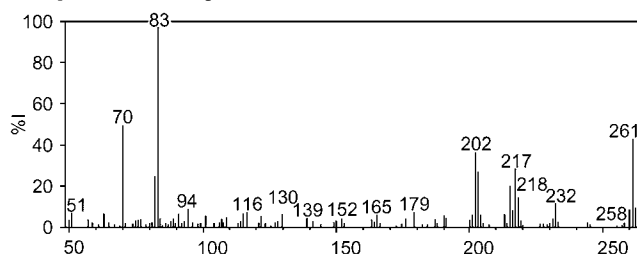
Synonym 2,3,4,9-Tetrahydro-2-methyl-1*H*-dibenzo[3,4,6,7]cyclohepta[1,2-*c*]pyridine

Proprietary Name *Tecipul*.



Gas Chromatography System GU—RT 14.1 min.

Mass Spectrum Principal ions at *m/z* 83, 70, 261, 202, 217, 203, 82, 232.



Quantification

Blood GC-MS Column: Supelco SPB-1 fused silica (30 m × 0.32 mm i.d., 0.25 µm). Temperature programme: 100° for 5 min, to 280° at 20°/min. Carrier gas: He, 0.8 mL/min. Internal standard (IS): imipramine. SIM acquisition mode (*m/z* 202, 217, 261 for setiptiline; 243 for IS). RT: setiptiline, 14.1 min; IS, 13.7 min. Limit of detection, 0.005 µg/g [Namera *et al.* 1998].

Plasma LC-MS Column: Inertsil C₈ (150 × 2.0 mm i.d., 5 µm). Mobile phase: methanol: 10 mmol/L ammonium acetate (pH 5.0): acetonitrile (70:20:10), flow rate 0.1 mL/min. SSI, positive ion mode, full scan mode. Retention time: 7.1 min. Limit of quantification, 0.15 mg/L, limit of detection, 0.1 mg/L [Shinozuka *et al.* 2006].

Disposition in the Body Setiptiline is metabolised via glucuronide conjugation, demethylation and hydroxylation in the liver. About 10.7% of a dose is excreted in urine as the glucuronide conjugate and ~9% as the other metabolites, in 48 h, resulting from aromatisation and hydroxylation following demethylation.

Therapeutic Concentration

Forty-five outpatients, aged between 22 and 86 years (mean for males, 48.8 years; mean for females, 59.3 years), being treated for depression were administered with setiptiline at doses and the frequency determined by the attending physicians according to symptoms and illness. The mean dose administered to the female volunteers was 0.08 mg/kg daily (range, 0.04–0.16 mg/kg) and to the male volunteers was 0.07 mg/kg daily (0.04–0.12 mg/kg). The mean plasma concentration observed was 3.54 (0.05–1.12) µg/L for the females and 2.24 (0.08–6.40) µg/L for the males. In terms of age, the mean plasma concentrations for mean ages of 23.8, 33.4, 44.0, 54.8, 65.6, 74.8 and 83.1 years were 1.92, 2.00, 3.06, 2.17, 2.61, 3.32 and 5.63 µg/L, respectively. The daily doses for these ages were 0.051, 0.057, 0.079, 0.062, 0.094, 0.086 and 0.064 mg/kg, respectively [Kamimura *et al.* 1994].

Toxicity Heart blood samples were taken from a subject whose death was suspected to be caused by acute setiptiline poisoning. A concentration of setiptiline of 1.77 µg/g was determined in blood taken from the left hand side of the heart and 0.78 µg/g from the right [Namera *et al.* 1998].

Half-life 11 h.

Kamimura M *et al.* (1994). The effect of age on plasma level of setiptiline maleate in depressed patients. *Prog Neuropsychopharmacol Biol Psychiatry* 18: 1015–1026.

Namera A *et al.* (1998). Simple analysis of tetracyclic antidepressants in blood using headspace-solid-phase microextraction and GC-MS. *J Anal Toxicol* 22: 396–400.

Shinozuka T *et al.* (2006). Solid-phase extraction and analysis of 20 antidepressant drugs in human plasma by LC/MS with SSI method. *Forensic Sci Int* 162: 108–112.

Sevoflurane

Anaesthetic

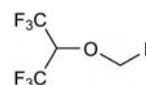
C₄H₇F₇O = 200.1

CAS—28523-86-6

IUPAC Name 1,1,1,3,3,3-Hexafluoro-2-(fluoromethoxy)propane

Synonyms BAX-3084; MR-654.

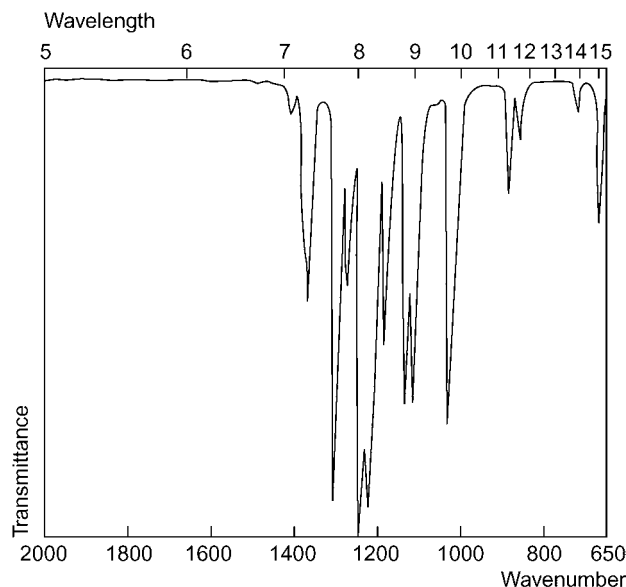
Proprietary Names *Sevofrane*; *Sevorane*; *Ultane*.



Chemical Properties A clear, colourless, volatile liquid with a non-pungent light odour. Non-flammable. Mp 25°. Bp 58.5°. Slightly soluble in water and miscible with ethanol, ether, chloroform and petroleum benzene. Log *P* (octanol/water), 1.75.

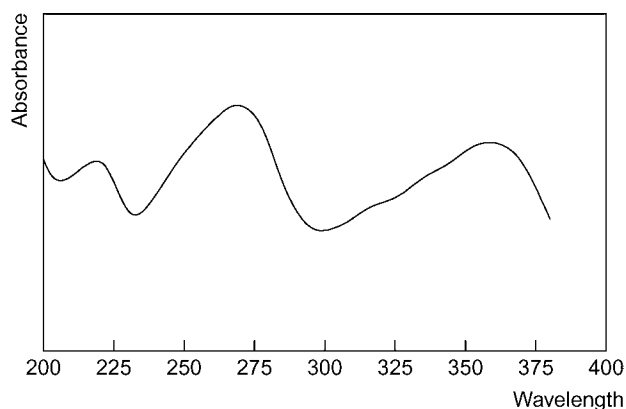
Gas Chromatography System GR—retention time 2.8 min.

Column: PoraPLOT Q (25 m × 0.32 mm i.d., 10 µm). Temperature programme: 30° for 2 min, to 250° at 15°/min for 5 min. Carrier gas: He, flow rate 3.8 mL/min. Detection: flame ionisation/FT-IR. IS: diethyl ketone. Retention time: (relative to IS) 0.758 [Ojanpera *et al.* 1998].

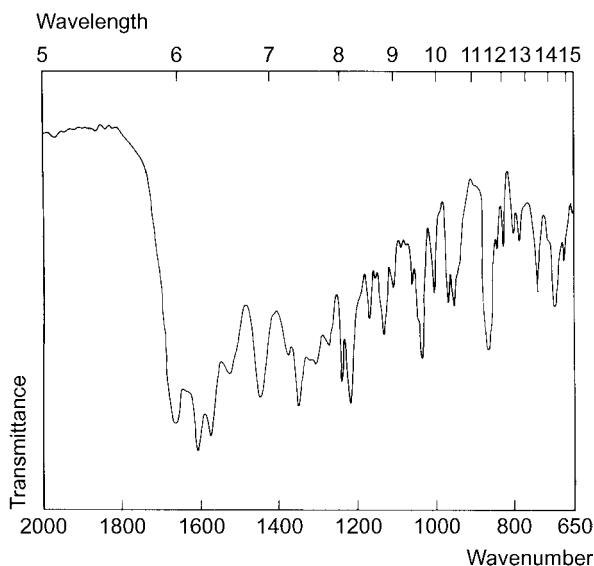


High Performance Liquid Chromatography System HX—RI 314; system HY—RI 265; system HAA—retention time 9.9 min.

Ultraviolet Spectrum Aqueous acid—270 ($A_1=417a$), 356 nm.



Infrared Spectrum Principal peaks at wavenumbers 1612, 1580, 1660, 1226, 1248, 1530 cm^{-1} (tetracycline hydrochloride, KCl disk).



Quantification

Plasma HPLC UV detection. Limit of detection, 100 $\mu\text{g/L}$ [Hermansson 1982].

Radioimmunoassay Limit of detection, about 20 $\mu\text{g/L}$ [Faraj, Ali 1981].

Urine HPLC See Plasma [Hermansson 1982].

Radioimmunoassay See Plasma [Faraj, Ali 1981].

Disposition in the Body Irregularly and incompletely absorbed after oral administration; widely distributed throughout the body. Up to about 60% of an IV dose and 20 to 50% of an oral dose is excreted in the urine unchanged in 48 h. Excreted in the bile and undergoes some enterohepatic circulation.

Therapeutic Concentration In plasma, usually in the range 1 to 5 mg/L .

Six patients with symptomatic pleural effusions and 1 patient with recurrent pneumothorax were administered 150 mg lidocaine, immediately followed by 20 mg/kg tetracycline instilled in the pleural space through a chest tube. Mean peak plasma concentrations for lidocaine and tetracycline were 1.3 mg/L and 3.6 mg/L , respectively, and these were reached after 86 and 96 min, respectively [Wooten *et al.* 1988].

Half-life Plasma half-life, about 9 h.

Volume of Distribution About 1.3 L/kg .

Clearance Plasma clearance, about 2 mL/min/kg .

Distribution in Blood Plasma : whole blood ratio, about 0.97.

Protein Binding 25 to 65%.

Dose 1 to 2 g daily.

Faraj BA, Ali FM (1981). Development and application of a radioimmunoassay for tetracycline. *J Pharmacol Exp Ther* 217: 10–14.

Hermansson J (1982). Rapid determination of tetracycline and lumecycline in human plasma and urine using high-performance liquid chromatography. *J Chromatogr* 232: 385–393.

Sweetman SC (2002). *Martindale: The complete drug reference*, 33rd edn. London: Pharmaceutical Press.

Wooten SA *et al.* (1988). Systemic absorption of tetracycline and lidocaine following intrapleural instillation. *Chest* 94: 960–963.

Tetraethylammonium Bromide

Antihypertensive

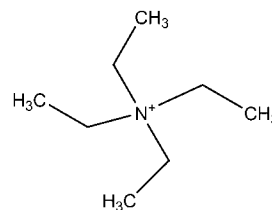
$\text{C}_8\text{H}_{20}\text{BrN}$ = 210.2

IUPAC Name Tetraethylazanium hydrobromide

Synonyms TEAB; tetramone bromide; tetrylammonium bromide; TMD 10.

Proprietary Names Beparon; Bromethyl; Etambro; Etamon(in); Sympatektoman; Teamon; Tetramon.

Br — H



Chemical Properties Tetraethylammonium bromide is a quaternary ammonium compound. White crystalline powder. Soluble 1 in less than 1 of water, 1 in 1 of ethanol and 1 in 4 of chloroform; freely soluble in acetone.

Thin-layer Chromatography System T1— R_f 0.03 (location reagent acidified iodoplatinate spray, positive reaction).

Dose Up to 500 mg IV .

Δ^9 -Tetrahydrocannabinol

Antiemetic, Appetite Stimulant, Hallucinogen

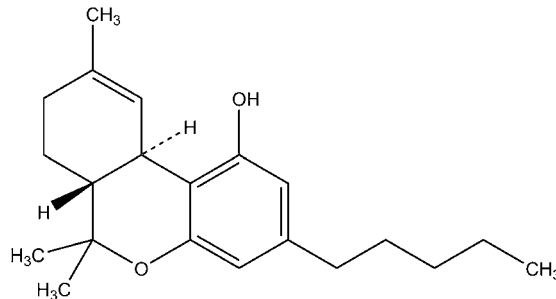
$\text{C}_{21}\text{H}_{30}\text{O}_2$ = 314.5

CAS—1972-08-3

IUPAC Name (6aR,10aR)-6,6,9-Trimethyl-3-pentyl-6a,7,8,10a-tetrahydrobenzo[c]chromen-1-ol

Synonyms Dronabinol; QCD-84924; (–)- Δ^1 -3,4-*trans*-tetrahydrocannabinol; (6aR,10aR)-6a,7,8,10a-tetrahydro-6,6,9-trimethyl-3-pentyl-6H-dibenzo[b,d]pyran-1-ol; Δ^9 -THC; Δ^1 -THC.

Proprietary Name Marinol



Chemical Properties Light yellow resinous oil that hardens upon refrigeration. Bp 200°. Essentially insoluble in water; soluble 1 in 1 in ethanol and acetone; readily soluble in chloroform and light petroleum. pK_a 10.6 [O'Neil *et al.* 2006]. Log *P* (octanol/water), 7.6. Binds readily to glass and plastic, limiting recoveries during analytical procedures. Absorption can be minimised by storing solutions in amber silylated glassware and by maintenance of the compound in a basic solution or organic solvent. For a study on the adsorptive loss of 11-nor- Δ^9 -THC-9-carboxylic acid, see Jamerson *et al.* [2005a]. Stability of Δ^9 -THC, 11-hydroxy- Δ^9 -THC, and 11-nor- Δ^9 -THC-9-carboxylic acid in urine was established after 3 freeze-thaw cycles, 72 h at 4°, and 24 h at room temperature [Abraham *et al.* 2007]. Δ^9 -THC concentrations showed a decrease of 26% in urine at –70° in silanised glass. The concentration of Δ^9 -THC in EDTA-plasma decreased steadily, with a decrease of 30% after 21 days and 43% after 41 days [Grauwiler *et al.* 2007].

Colour Test *p*-Dimethylaminobenzaldehyde—red/violet; Duquenois reagent—violet-blue/violet.

Thin-layer Chromatography System TA— R_f 11; system TE— R_f 31; system TI— R_f 30; system TJ— R_f 29; system TAH— R_f 50 (Fast blue B solution—red colour); system TAJ— R_f 00; system TAK— R_f 01; system TAL— R_f 31.

Plates: Merck 60F 254 silica gel. Solvent system: hexane : chloroform : dioxane (89 : 8.75 : 2.25). UV detection (λ = 220 nm). R_f 0.27. Limit of detection not reported [Debruyne *et al.* 1994].

Gas Chromatography System GA—RI 2473; system GB—RI 2350.

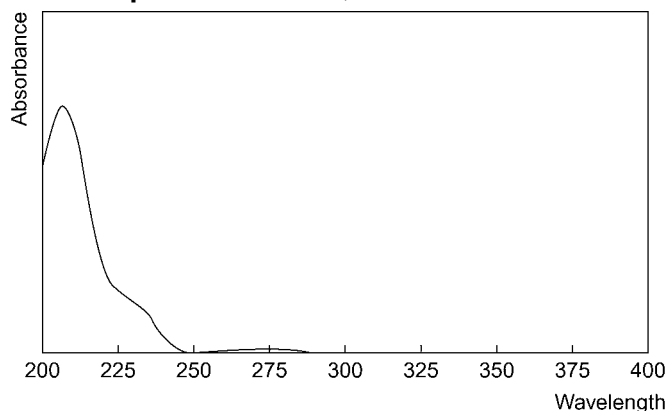
Gas Chromatography-Mass Spectrometry Column: methyl silicone capillary (25 m $\times$ 0.32 mm i.d., 0.25 μm). Carrier gas: N_2 . Temperature: 240°. FID. Retention time: 4.6 min. Limit of detection not reported [Debruyne *et al.* 1994].

High Performance Liquid Chromatography System HL—*k* 13.35.

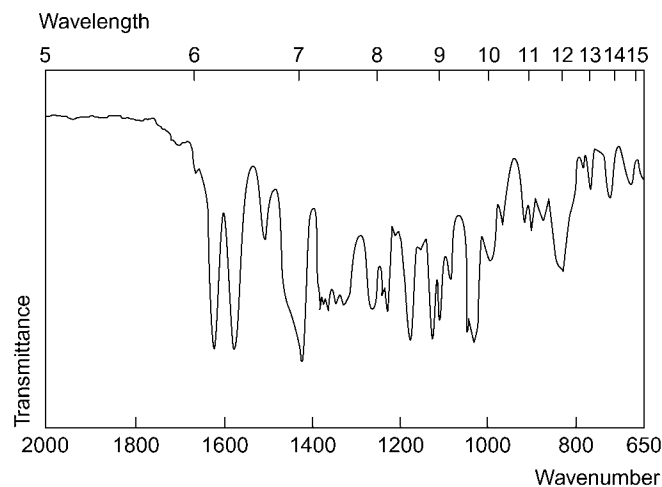
Column: Waters μ Porasil (150 $\times$ 4.6 mm i.d.). Mobile phase: hexane:chloroform (90:10), flow rate 2 mL/min. UV detection (λ = 220 nm). Retention time: 4.9 min. Limit of detection not reported [Debruyne *et al.* 1994].

Supercritical Fluid Chromatography Column: Cyanopropyl silica (250 $\times$ 4.6 mm i.d.). Mobile phase: methanol in CO₂ (2% to 7% at 15 min), flow rate 2 mL/min. APCI, positive ion mode, full scan and SIM acquisition mode. Retention time: 5.19 min. Limit of detection, 0.69 ng [Backstrom *et al.* 1997].

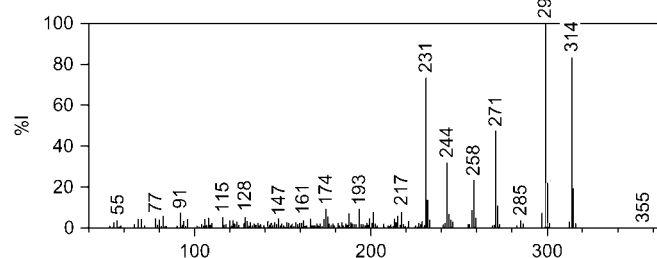
Ultraviolet Spectrum Ethanol—283, 276 nm.



Infrared Spectrum Principal peaks at wavenumbers 1580, 1040, 1620, 1180, 1130, 1050 cm⁻¹ (KBr disc).



Mass Spectrum Principal ions at *m/z* 299, 231, 314, 43, 41, 295, 55, 271.

**Quantification**

Blood GC-MS Column: DB-5MS fused silica capillary (15 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1 mL/min. Temperature programme: 70° for 1 min to 190° at 25°/min to 210° at 25°/min to 290° at 30°/min for 2.5 min. CI, negative ion mode, SRM acquisition mode. Limit of quantification, 0.5 μ g/L, limit of detection, 0.1 to 0.2 μ g/L [Thomas *et al.* 2007]. Column: Restek Rtx 5-MS capillary (15 m $\times$ 0.25 mm i.d., 0.25 μ m). Temperature programme: 70° for 1 min to 260° at 15°/min for 10 min. Limit of quantification, 1 ng/L; limit of detection, 0.5 ng/L [Ondra *et al.* 2006]. Column: rtx-200 (20 m $\times$ 0.18 mm i.d., 0.2 μ m) and DB-17 (15 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He. Temperature programme: 120° to 200° at 20°/min to 250° at 10°/min to 300° at 25°/min. SIM acquisition mode. Limit of quantification, 1.0 μ g/L for both Δ^9 -THC and 11-nor- Δ^9 -THC-9-carboxylic acid [Scurlock *et al.* 2006]. Column: 5% phenylmethyl silicone capillary (12 m $\times$ 0.32 mm i.d., 0.33 μ m). Carrier gas: He, 1.5 mL/min. Temperature programme: 120° for 2 min to 200° at 50°/min to 280° at 15°/min for 5 min. EI ionisation. Limit of

detection, $\approx$ 1 ng/L [Chiarotti, Costamagna 2000]. Column: HP-5MS 5% phenyl 95% methyl siloxane capillary (30 m $\times$ 0.25 mm i.d.). Carrier gas: He, 1.0 mL/min. Temperature programme: 60° for 1 min to 290° at 30°/min for 5 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of detection, 0.4 μ g/L Δ^9 -THC, 0.2 μ g/L 11-nor- Δ^9 -THC-9-carboxylic acid [Kintz, Cirimele 1997]. See also Bergman *et al.* [1981]; McCallum *et al.* [1978] and McCallum, Shaw [1981].

LC-MS Column: Symmetry C₁₈ (150 $\times$ 1.5 mm i.d., 5 μ m). Mobile phase: acetonitrile:0.1% formic acid (70:30), flow rate 0.3 mL/min. ESI, positive and negative ion modes. Limit of quantification, 2 μ g/L, limit of detection, 0.5 μ g/L [Teixeira *et al.* 2007].

Plasma GC-MS Column: DB-1MS capillary (15 m $\times$ 0.25 mm i.d., 0.25 μ m). EI ionisation, SIM acquisition mode. Limit of quantification and detection, 0.125 μ g/L [Lowe *et al.* 2007]. Column: HP-5MS 95% dimethylsiloxane 5% diphenylsiloxane capillary (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Temperature programme: 130° for 2 min to 300° at 20°/min for 5 min. SIM acquisition mode. Limit of quantification, 0.80 μ g/L Δ^9 -THC, 0.51 μ g/L 11-hydroxy- Δ^9 -THC, 0.88 μ g/L 11-nor-9-carboxy- Δ^9 -THC, limit of detection, 0.24 μ g/L Δ^9 -THC, 0.15 μ g/L 11-hydroxy- Δ^9 -THC, 0.26 μ g/L 11-nor- Δ^9 -THC-9-carboxylic acid [Nadulski *et al.* 2005a].

HPLC Column: Nucleosil C₁₈ (125 $\times$ 4.6 mm i.d., 3 μ m). Mobile phase: methanol:acetonitrile:5 mmol/L tetrabutylammonium perchlorate solution (pH 3.2; 50:25:25), flow rate 1.2 mL/min. UV detection (λ = 215 nm). Limit of quantification, 5 μ g/L [Abbara *et al.* 2006]. Column: LiChroCart Superspher 60 RP select B (250 $\times$ 4 mm i.d., 5 μ m). Mobile phase: 5.6 mmol/L tetrabutylammonium hydrogen sulfate (pH 2.3):acetonitrile:tetrahydrofuran (44:46:10) with 160 μ mol/L Na₂EDTA, flow rate 850 μ L/min. Electrochemical detection. Limit of quantification, 5 μ g/L for Δ^9 -THC and 11-nor- Δ^9 -THC-9-carboxylic acid [Kramer, Kovar 1999]. Column: Zorbax C₈ (250 $\times$ 4.6 mm i.d.). Mobile phase: acetonitrile:methanol:1 mol/L sulfuric acid (35:15:50), flow rate 1.8 mL/min. Electrochemical detection. Limit of detection, <0.5 μ g/L [Nakahara *et al.* 1989].

LC-MS Column: Synergi MAX-RP 80A C₁₂ (75 $\times$ 2 mm i.d., 4 μ m). Mobile phase: 10 mmol/L ammonium formate (pH 3.0):acetonitrile-10 mmol/L ammonium formate (90:10; 50:50 for 1 min to 21:79 at 12 min to 5:95 at 12.5 min for 2.5 min to 50:50 at 25 min), flow rate 400 μ L/min. APCI, SIM acquisition mode. Limit of quantification, 0.2 μ g/L for Δ^9 -THC, 11-hydroxy- Δ^9 -THC, and 11-nor- Δ^9 -THC-9-carboxylic acid, limit of detection, 0.1 μ g/L [Grauwiler *et al.* 2007]. Column: Phenomenex Luna phenylhexyl (50 $\times$ 2 mm i.d., 3 μ m). Mobile phase: 5 mmol/L ammonium acetate (pH 6.5):acetonitrile (70:30 for 0.4 min to 10:90 at 7.8 min for 0.4 min to 70:30 at 8.5 min for 2.5 min), flow rate 0.25 mL/min. ESI, positive ion mode, MRM acquisition mode. Limit of quantification, 0.8 μ g/L for Δ^9 -THC, 11-hydroxy- Δ^9 -THC and 4.3 μ g/L for 11-nor-9-carboxy- Δ^9 -THC, limit of detection, 0.2 μ g/L for Δ^9 -THC and 11-hydroxy- Δ^9 -THC, 1.6 μ g/L for 11-nor- Δ^9 -THC-9-carboxylic acid [Maralikova, Weinmann 2004].

Serum GC-MS Column: VF-5MS capillary (30 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He, 1 mL/min. EI ionisation at 70 eV. Limit of quantification, 0.5 μ g/L Δ^9 -THC, 0.3 μ g/L 11-hydroxy- Δ^9 -THC, 6.2 μ g/L 11-nor- Δ^9 -THC-9-carboxylic acid, respectively, limit of detection, 0.2 μ g/L Δ^9 -THC, 0.1 μ g/L 11-hydroxy- Δ^9 -THC, 1.9 μ g/L 11-nor- Δ^9 -THC-9-carboxylic acid [Kauert *et al.* 2007].

LC-MS Column: Luna phenylhexyl (50 $\times$ 2 mm i.d., 3 μ m). Mobile phase: 5 mmol/L ammonium acetate (pH 6.5):acetonitrile (70:30 for 1 min to 10:90 at 4 min for 1 min to 70:30 at 6 min for 4 min), flow rate 0.25 mL. ESI, negative ion mode, MRM acquisition mode. Limit of quantification, 7.5 μ g/L, limit of detection, 2.5 μ g/L [Jung *et al.* 2007].

Urine TLC Plates: TOXI-GRAMS MS (THC) biphasic with alkyl-silica extraction layer (14 $\times$ 5.5 cm). Solvent system: *n*-heptane:acetone:glacial acetic acid (50:50:1). R_f 0.78 for 11-nor- Δ^9 -THC-9-carboxylic acid. Limit of detection, 20 μ g/L for 11-nor- Δ^9 -THC-9-carboxylic acid [King *et al.* 1989].

GC-MS Column: DB-35MS capillary (15 m $\times$ 0.32 mm i.d., 0.25 μ m). Carrier gas: He. Temperature programme: 100° for 1 min to 200° at 35°/min to 250° at 23°/min to 325° at 20°/min for 1 min. SIM acquisition mode. Limit of quantification, 1.0 μ g/L Δ^9 -THC, 2.5 μ g/L 11-hydroxy- Δ^9 -THC, 2.5 μ g/L 11-nor- Δ^9 -THC-9-carboxylic acid, limit of detection, 2.5 μ g/L Δ^9 -THC, 2.5 μ g/L 11-hydroxy- Δ^9 -THC, 2.5 μ g/L 11-nor- Δ^9 -THC-9-carboxylic acid [Abraham *et al.* 2007]. Column: CP Sil fused silica capillary (12.5 m $\times$ 0.25 mm i.d., 0.4 μ m). Temperature programme: 160° for 2 min to 320° at 20°/min for 2 min. SIM acquisition mode. Limit of quantification, 2.6 μ g/L for 11-nor- Δ^9 -THC-9-carboxylic acid; limit of detection, 0.8 μ g/L, for 11-nor- Δ^9 -THC-9-carboxylic acid [Dietz *et al.* 2007]. Column: Zebtron ZB-5MS Phenomenex capillary (30 m $\times$ 0.25 mm, 0.25 μ m). Carrier gas: He, 1.4 mL/min. Temperature programme: 120° to 235 at 20°/min to 250° at 3°/min to 340° at 25°/min. SIM acquisition mode. Limit of detection, 1 ng [Huq *et al.* 2005]. Column: Supelco 5% phenylmethyl silicone capillary (17 m $\times$ 0.2 mm i.d., 0.33 μ m) or HP1 methyl silicone capillary (30 m $\times$ 0.2 mm i.d., 0.11 μ m). Carrier gas: He, 1.1 mL/min. Temperature programme: 160° for 0.5 min to 280° at 15°/min to 320° at 50°/min for 3 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of detection, 5 μ g/L for 11-nor- Δ^9 -THC-9-carboxylic acid [Strano-Rossi *et al.* 2005]. Column: DB-5MS bonded phase capillary (15 m $\times$ 0.25 mm i.d., 0.25 μ m). Carrier gas: He. Temperature programme: 230° to 290° at 55°/min for 1.0 min. SIM acquisition mode. Retention time: 1.8 min. Limit of quantification, 3.8 μ g/L for 11-nor- Δ^9 -THC-9-carboxylic acid, limit of detection, 3.8 μ g/L for 11-nor- Δ^9 -THC-9-carboxylic acid [Jamerson *et al.* 2005a]. See also Chiarotti, Costamagna [2000]; Jamerson *et al.* [2005b] and Kintz *et al.* [1995].

HPLC Column: LiChroCart Superspher 60 RP select B (250 $\times$ 4 mm i.d., 5 μ m). Mobile phase: 5.6 mmol/L tetrabutylammonium hydrogen sulfate (pH 2.3):acetonitrile:tetrahydrofuran (44:46:10) with 160 μ mol/L Na₂EDTA, flow rate

850 $\mu\text{L}/\text{min}$. Electrochemical detection. Limit of quantification, 15 $\mu\text{g}/\text{L}$ for 11-nor-9-carboxy- Δ^9 -THC [Kramer, Kovar 1999]. Column: RP-8 reversed phase (250 $\times$ 4.6 mm i.d., 5 μm). Mobile phase: acetonitrile:0.125 mol/L phosphate buffer (55:45), flow rate 1 mL/min. DAD (λ = 212 nm). Retention time: $\approx$ 12 min. Limit of detection, 10 $\mu\text{g}/\text{L}$ for 11-nor- Δ^9 -THC-9-carboxylic acid [Bianchi, Donzelli 1996]. Column: ODS (100 $\times$ 3.2 mm i.d., 3 μm). Mobile phase: 100 mL 0.5 mol/L monochloroacetic acid (pH 3.0) and 1.4 mL TEA to 1 L with water-methanol (30:70): 100 mL 0.5 mol/L monochloroacetic acid (pH 3.0) and 1.4 mL TEA to 1 L with water-tetrahydrofuran-methanol (5:15:80; 90:10 to 70:30 at 15 min to 40:60 at 30 min for 5 min), flow rate 1.0 mL/min. Electrochemical detection. Limit of quantification, 12 $\mu\text{g}/\text{L}$ for 11-nor- Δ^9 -THC-9-carboxylic acid [Fisher *et al.* 1996]. Column: Zorbax C₈ (250 $\times$ 4.6 mm i.d.). Mobile phase: acetonitrile: methanol: 0.1 mol/L sulfuric acid (35:15:50), flow rate 1.8 mL/min. Electrochemical detection. Limit of detection, <0.5 $\mu\text{g}/\text{L}$ [Nakahara *et al.* 1989].

LC-MS Column: Synergi MAX-RP 80A C12 (75 $\times$ 2 mm i.d., 4 μm). Mobile phase: 10 mmol/L ammonium formate (pH 3.0): acetonitrile:10 mmol/L ammonium formate (90:10; 50:50 for 1 min to 21:79 at 12 min to 5:95 at 12.5 min for 2.5 min to 50:50 at 25 min), flow rate 400 $\mu\text{L}/\text{min}$. APCL, SIM acquisition mode. Limit of quantification, 1 $\mu\text{g}/\text{L}$ for Δ^9 -THC and 11-nor- Δ^9 -THC-9-carboxylic acid, 2 $\mu\text{g}/\text{L}$ for 11-hydroxy- Δ^9 -THC; limit of detection, 0.5 $\mu\text{g}/\text{L}$ for Δ^9 -THC and 11-nor- Δ^9 -THC-9-carboxylic acid, 1 $\mu\text{g}/\text{L}$ for 11-hydroxy- Δ^9 -THC [Grauwiler *et al.* 2007]. Column: Luna phenylhexyl (50 $\times$ 2 mm i.d., 3 μm). Mobile phase: 5 mmol/L ammonium acetate (pH 6.5): acetonitrile (70:30 for 1 min to 10:90 at 4 min for 1 min to 70:30 at 6 min for 4 min), flow rate 0.25 mL. ESI, negative ion mode, MRM acquisition mode. Limit of quantification, 5 $\mu\text{g}/\text{L}$, limit of detection, 2.5 $\mu\text{g}/\text{L}$ [Jung *et al.* 2007]. Column: Symmetry C₁₈ (150 $\times$ 1.5 mm i.d., 5 μm). Mobile phase: acetonitrile:0.1% formic acid (70:30), flow rate 0.3 mL/min. ESI, positive and negative ion modes. Limit of quantification, 2 $\mu\text{g}/\text{L}$; limit of detection, 0.5 $\mu\text{g}/\text{L}$ [Teixeira *et al.* 2007]. Column: Waters 2695 HPLC-Micromass Quattro II reversed phase C₁₈ (200 $\times$ 4.6 mm i.d.). Mobile phase: methanol: water (85:15). APCL, positive ion mode, SIM acquisition mode. Limit of detection not reported [Yang, Xie 2006]. Column: RP-C8-select B (125 $\times$ 2 mm i.d., 5 μm). Mobile phase: 5 mmol/L ammonium formate-0.1% formic acid (pH 3): acetonitrile-0.1% formic acid (90:10 for 6.6 min to 70:30 at 26.6 min to 30:70 at 33.3 min to 10:90 for 4 min). ESI, positive ion mode, MRM acquisition mode. Limit of detection, <10 $\mu\text{g}/\text{L}$ for 11-nor- Δ^9 -THC-9-carboxylic acid [Weinmann *et al.* 2000]. Column: Zorbax Eclipse XDB-C₈ reversed phase (150 $\times$ 3.0 mm i.d., 5 μm). Mobile phase: 4 mmol/L formic acid in water: 4 mmol/L formic acid in acetonitrile (40:60 for 0.5 min to 20:80 at 3 min for 4 min to 40:60 over 3 min), flow rate 0.5 mL/min. API, ESI, positive ion mode, SIM acquisition mode. Limit of detection, 2 $\mu\text{g}/\text{L}$ [Breindahl, Andreasen 1999].

Meconium GC-MS Column: Agilent DB-5 MS (15 m $\times$ 0.25 mm i.d., 0.25 μm) and Agilent DB-17 MS (15 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He. Temperature programme: 200° for 0.2 min to 280° at 15°/min for 3.8 min. EI ionisation, FID, SIM acquisition mode. Limit of quantification, 10 $\mu\text{g}/\text{kg}$, limit of detection, 5 $\mu\text{g}/\text{kg}$ [Marin *et al.* 2007]. Column: DB-5MS (15 m $\times$ 0.2 mm i.d., 0.25 μm). Temperature programme: 200° for 0.2 min to 275° at 60°/min for 3 min to 300° at 20°/min to 325° at 60°/min for 1.5 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of quantification, 10 $\mu\text{g}/\text{kg}$; limit of detection, 5 $\mu\text{g}/\text{kg}$ [Coles *et al.* 2005].

Oral Fluid GC-MS Column: Ultra 1 methyl silicone capillary (16.5 m $\times$ 0.2 mm i.d., 0.11 μm). Carrier gas: He, 0.8 mL/min. Temperature programme: 70° for 2 min to 160° at 30°/min to 170° at 5°/min to 200° at 20°/min to 220° at 10°/min to 300° at 30°/min. EI ionisation at 70 eV, SIM acquisition mode. Limit of quantification, 1.9 $\mu\text{g}/\text{L}$, limit of detection, 0.6 $\mu\text{g}/\text{L}$ [Pujadas *et al.* 2007]. Column: HP-5 capillary (30 m $\times$ 0.32 mm i.d., 0.25 μm). Carrier gas: He, 1.8 mL/min. Temperature programme: 70° for 1 min to 300° at 20°/min. EI ionisation at 70 eV, SIM acquisition mode. Limit of quantification, 2 $\mu\text{g}/\text{L}$, limit of detection, 1 $\mu\text{g}/\text{L}$ [Drummer *et al.* 2007]. Column: VF-1MS capillary (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1 mL/min. EI ionisation at 70 eV, SIM acquisition mode. Limit of quantification, 2.4 $\mu\text{g}/\text{L}$, limit of detection, 0.5 $\mu\text{g}/\text{L}$ [Kauert *et al.* 2007]. Column: DB-5MS (15 m $\times$ 0.25 mm i.d., 0.25 μm). Temperature programme: 125° for 5 min to 250° at 40°/min for 1.3 min to 300° at 70°/min. SIM acquisition mode. Limit of quantification, 0.5 $\mu\text{g}/\text{L}$ [Moore *et al.* 2007]. Column: J and W DB-5 (15 m $\times$ 0.25 mm i.d., 1.0 μm). Temperature programme: 130° to 300° at 30°/min. Negative ion mode. Limit of quantification, 10 ng/L [Day *et al.* 2006]. See also Gunnar *et al.* [2005].

HPLC Column: LiChrospher Select B RP-8 (125 $\times$ 4 mm i.d., 5 μm). Mobile phase: methanol-0.15 mol/L aqueous sodium chloride (20:80): methanol-0.15 mol/L aqueous sodium chloride (82:18; 100:0 for 5 min to 0:100 at 5.1 min until 25 min to 100:0 at 25.1 min for 10 min), flow rate 0.8 mL/min. UV detection (λ = 220 nm). Limit of detection, 2 mg/L [Kircher, Parlar 1996].

LC-MS Column: Xterra C₁₈ MS reversed phase (150 $\times$ 2.1 mm i.d., 3.5 μm). Mobile phase: 10 mmol/L aqueous ammonium formate (pH 3.5): methanol (77:23 for 0.2 min to 4:96 at 6 min for 3 min), flow rate 200 $\mu\text{L}/\text{min}$. TOF-ESI, positive ion mode. Limit of quantification, 0.1 $\mu\text{g}/\text{L}$, limit of detection, 0.05 $\mu\text{g}/\text{L}$ [Quintela *et al.* 2007]. Column: Symmetry C₁₈ (150 $\times$ 1.5 mm i.d., 5 μm). Mobile phase: acetonitrile:0.1% formic acid (70:30), flow rate 0.3 mL/min. ESI, positive and negative ion modes. Limit of quantification, 5 $\mu\text{g}/\text{L}$, limit of detection, 2 $\mu\text{g}/\text{L}$ [Teixeira *et al.* 2007]. Column: Atlantis dC₁₈ (50 $\times$ 2.1 mm i.d., 3.5 μm). Mobile phase: acetonitrile: 5 mmol/L aqueous ammonium acetate (pH 5; 10:90 to 40:60 at 4 min to 90:10 at 4.1 min for 3.9 min to 10:90 at 8.1 min), flow rate 0.3 mL/min. ESI, positive ion mode, MRM acquisition mode. Limit of quantification, <0.16 $\mu\text{g}/\text{L}$ [Oiestad *et al.* 2007]. Column: Xterra MS C₁₈ (50 $\times$ 2.1 mm i.d., 3.5 μm). Mobile

phase: acetonitrile:0.05% ammonia (70:30), flow rate 0.3 mL/min. ESI, positive ion mode, single ion recording. Limit of quantification, 2 $\mu\text{g}/\text{L}$, limit of detection, 1 $\mu\text{g}/\text{L}$ [Teixeira *et al.* 2004, 2005]. Column: Xterra MS C₁₈ (100 $\times$ 2.1 mm i.d., 3.5 μm). Mobile phase: 0.1% formic acid: acetonitrile (15:85), flow rate 0.25 mL/min. ESI, positive ion mode. Limit of quantification, 2 $\mu\text{g}/\text{L}$ [Concheiro *et al.* 2004]. **Sweat GC-MS** CI, negative ion mode. Limit of quantification, 0.4 ng [Huestis *et al.* 2008].

Brain GC-MS Column: 6% OV-1 or Hi-EFF 9BP. FPD. Limit of detection, 10 $\mu\text{g}/\text{kg}$ [McCallum *et al.* 1978].

Hair GC-MS Column: DB-5MS capillary (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He, 1.0 mL/min. Temperature programme: 100° for 1 min to 275° at 35°/min for 3 min to 300° at 25°/min for 3.5 min. CI, negative ion mode, MRM acquisition mode. Limit of quantification, 0.05 ng/kg for 11-nor- Δ^9 -THC-9-carboxylic acid, limit of detection, 0.02 ng/kg for 11-nor- Δ^9 -THC-9-carboxylic acid [Kim, In 2007]. Column: DB-5MS capillary (15 m $\times$ 0.25 mm i.d., 1.0 μm). Temperature programme: 180° to 278° at 30°/min to 282° at 5°/min to 300° at 35°/min for 2.5 min. CI, positive ion mode for Δ^9 -THC, negative ion mode for 11-nor- Δ^9 -THC-9-carboxylic acid, SRM acquisition mode. Limit of quantification, 1 ng/kg for Δ^9 -THC, 0.1 ng/kg for 11-nor- Δ^9 -THC-9-carboxylic acid [Huestis *et al.* 2007]. Column: HP-5MS capillary. Temperature programme: 130° for 2 min to 300° at 20°/min. SIM acquisition mode. Limit of quantification, 37 ng/kg, limit of detection, 12 ng/kg [Nadulski, Pragst 2007]. Column: DB-35 MS (40 m $\times$ 0.18 mm i.d., 0.18 μm) and DB-1 (15 m $\times$ m i.d., 0.25 mm). Carrier gas: He, 1.6 mL/min. Temperature programme: 100° for 0.5 min to 200° at 50°/min to 260° at 10°/min for 2.9 min to 220° at 120°/min. CI, negative ion mode, SIM acquisition mode. Limit of quantification, 0.05 ng/kg [Moore *et al.* 2006]. Column: Supelco Equity 1 capillary (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He. Temperature programme: 70° for 1 min to 240° at 40°/min for 1 min to 280° at 20°/min for 5 min. CI, negative ion mode. Limit of quantification, 100 ng/kg, limit of detection, 50 ng/kg [Marsili *et al.* 2005]. See also Baptista *et al.* [2002]; Chiarotti and Costamagna [2000]; Deveaux *et al.* [2000]; Kim *et al.* [2005]; Sachs and Dressler [2000] and Strano-Rossi and Chiarotti [1999].

Nail GC-MS Column: HP-1 cross linked dimethyl silicone capillary (30 m $\times$ 0.25 mm i.d., 0.25 μm). Carrier gas: He. Temperature programme: 150° to 300° at 10°/min for 10 min. EI ionisation at 70 eV, SIM acquisition mode. Limit of detection, <0.1 $\mu\text{g}/\text{kg}$ for Δ^9 -THC and 11-nor- Δ^9 -THC-9-carboxylic acid [Lemos *et al.* 1999].

Other GC-MS Indoor Air. Column: HP-5MS fused silica capillary (25 m $\times$ 0.2 mm i.d., 0.33 μm). Carrier gas: He, 1 mL/min. Temperature programme: 40° for 1 min to 250° at 25°/min for 25 min. EI ionisation at 70 eV, CI, positive ion mode. Retention time, 13.7 min. Limit of quantification, 125 ng/m³, limit of detection, 100 ng/m³ [Chou *et al.* 2007].

HPLC Foods Containing Hemp. Column: Nucleosil 120-3 C₁₈ (125 $\times$ 2 mm i.d.). Mobile phase: acetonitrile:8.6 g/L 85% phosphoric acid (55:45 to 80:20 at 25 min to 90:10 at 26 min for 4 min to 55:45 at 31 min for 9 min). UV detection (λ = 210 nm). Limit of detection, 0.1 ng [Zoller *et al.* 2000]. Rat Cortex and Corpus Striatum. Column: ODS μ Bondapak C₁₈ reversed phase (300 $\times$ 4 mm i.d., 10 μm). Mobile phase: methanol:acetonitrile: 0.01 mol/L sulfuric acid (21:24:55), flow rate 3.0 mL/min. Electrochemical detection. Limit of detection, 1.5 ng [Nyoni *et al.* 1996].

LC-MS Mouse Brain and Blood. Column: Waters 2695 HPLC-Micromass Quattro II reversed phase C₁₈ (200 $\times$ 4.6 mm i.d.). Mobile phase: methanol: water (85:15). APCL, positive ion mode, SIM acquisition mode. Limit of detection not reported [Yang, Xie 2006]. Wastewater. Column: Xterra MS C₁₈ (100 $\times$ 2.1 mm i.d., 3.5 μm). Mobile phase: 0.05% TEA: acetonitrile (100:0 to 55:45 in 12 min to 0:100 in 2 min for 2 min to 100:0 in 2 min for 8 min), flow rate 200 $\mu\text{L}/\text{min}$. ESI, negative ion mode, MRM acquisition mode. Limit of quantification, 1.75 and 0.94 $\mu\text{g}/\text{L}$ for 11-nor- Δ^9 -THC-9-carboxylic acid in influent and effluent waste water, respectively, limit of detection, 307 pg [Castiglioni *et al.* 2006].

Note For a clinical evaluation of the Toxi Prep system for the screening of THC metabolites in urine, see Steinberg *et al.* [1997].

Disposition in the Body Δ^9 -THC is absorbed from the gastrointestinal tract but absorption is slow and irregular. However, Δ^9 -THC can be measured in plasma within seconds of inhalation of marijuana smoke. Δ^9 -THC is lipophilic and is widely distributed in the body. It is hydroxylated to the active metabolite 11-hydroxy- Δ^9 -THC by the hepatic CYP2C9 enzyme, and this is further oxidised to the intermediate 11-oxo-THC, followed by oxidation to 11-nor- Δ^9 -THC-9-carboxylic acid. Many other polar mono- and dicarboxylic acids are formed, which are conjugated with glucuronic acid to a variable extent. The inactive substances 8 α -hydroxy- Δ^9 -THC and 8 α ,11-dihydroxy- Δ^9 -THC are also formed by the action of CYP3A4. 11-nor- Δ^9 -THC-9-carboxylic acid undergoes glucuronidation of the carboxy group and this is excreted in the urine and can be detected up to 4 days after smoking one marijuana cigarette and up to 4 weeks after frequent use. Levels of THC increase rapidly after smoking, peak before the end of smoking, and quickly dissipate. Mean peak levels of 11-hydroxy- Δ^9 -THC are substantially lower than THC levels and occur directly after the cessation of smoking. 11-nor- Δ^9 -THC-9-carboxylic acid levels increase slowly and plateau for a long period of time. 11-nor- Δ^9 -THC-9-carboxylic acid glucuronide is present in blood at higher concentrations than the unconjugated acid and is unstable to hydrolysis, a common cause of misinterpretation for the prediction of cannabis exposure [Maurer *et al.* 2006]. Enterohaptic recirculation of metabolites may occur. Up to $\approx$ 25% of a dose is excreted in the urine in 3 days, mainly as 11-nor- Δ^9 -THC-9-carboxylic acid glucuronide, together with the other carboxylic acids in free and conjugated form. The Δ^9 -THC-*O*-glucuronide has been detected in urine. The major excretion route is via

the faeces, with up to ~65% of a dose being excreted in 5 days, mainly as 11-hydroxy- Δ^9 -THC and the carboxylic acids in conjugated form. Δ^9 -THC metabolites have been detected in urine for up to 12 days following a single oral dose. Δ^9 -THC crosses the placenta and is distributed into breast milk.

Blood Concentrations

Six volunteers with histories of marijuana use ingested liquid hemp oil (0.39 and 14.8 mg Δ^9 -THC/day), hemp oil in capsules (0.47 mg Δ^9 -THC/day), dronabinol capsules (7.5 mg Δ^9 -THC/day), or placebo. Plasma Δ^9 -THC and 11-hydroxy- Δ^9 -THC levels were low and never exceeded 6.1 $\mu\text{g/L}$. Analytes were detectable 1.5 h after the initial dosing with the 7.5 mg Δ^9 -THC/day regimen and 4.5 h after the start of the 14.8 mg Δ^9 -THC/day protocol. 11-nor- Δ^9 -THC-9-carboxylic acid levels peaked at 3.1 $\mu\text{g/L}$ during dosing with the low dose hemp oils. Fifteen and a half hours following the last THC dose, all plasma samples were negative for Δ^9 -THC and 11-hydroxy-THC, although plasma 11-nor- Δ^9 -THC-9-carboxylic acid levels persisted for at least 39.5 h after the end of dosing [Goodwin *et al.* 2006].

The distribution of 11-nor- Δ^9 -THC-9-carboxylic acid was investigated in postmortem specimens of 50 deaths in traffic accidents in New Jersey, USA. The mean concentration in heart blood, urine and bile was 0.081 (0.016 to 0.33), 0.314 (0.044 to 2.33) and 12.9 (1.03 to 43.7) mg/L, respectively. Vitreous humour samples were negative or below the limit of detection, [Lin, Lin 2005].

Twenty-four volunteers (12 male, 12 female; aged 18 to 45 years) were administered one of 10 mg Δ^9 -THC, cannabis extract (10 mg Δ^9 -THC + 5.4 mg cannabidiol), or water, and blood samples were obtained. Mean peak concentrations of Δ^9 -THC were 3.19 and 4.05 $\mu\text{g/L}$, respectively, attained at 63.6 and 56 min, respectively, for the dosed groups, although there was some evidence that cannabidiol partially inhibited the CYP2C-catalysed hydroxylation of Δ^9 -THC to 11-hydroxy- Δ^9 -THC [Nadulski *et al.* 2005b].

Ingestion of 20 mg Δ^9 -THC resulted in mean peak plasma concentrations of 6 $\mu\text{g/L}$, with a range from 4.4 to 121 $\mu\text{g/L}$. Higher plasma concentrations of 11-hydroxy- Δ^9 -THC were observed, possibly as a result of first-pass metabolism [Kochanowski, Kala 2005]. Passive inhalation of marijuana smoke has resulted in plasma Δ^9 -THC levels of 1 to 7 $\mu\text{g/L}$ and urine 11-nor- Δ^9 -THC-9-carboxylic acid levels as high as 39 $\mu\text{g/L}$ [Baselt 2004].

Seven healthy participants with histories of marijuana use received 0, 0.39, 0.47, 7.5 and 14.8 mg Δ^9 -THC/day orally for 5 days in a double-blind randomised study. There were no differences in mean time of maximum excretion rate, mean maximum excretion rate, and mean terminal half-life between the 4 Δ^9 -THC doses, with respective ranges of 67.4 to 94.9 h, 0.9 to 16.3 $\mu\text{g/h}$ and 44.2 to 64.0 h [Gustafson *et al.* 2004].

Six healthy men with histories of marijuana use smoked a single marijuana cigarette (placebo, 1.75% or 3.55% Δ^9 -THC). The smoking protocol consisted of a 2 s inhalation, a 10 s hold period, and a 72 s exhalation and rest period. A total of eight puffs were inhaled in 11.2 min. Mean plasma levels of 7.0 and 18.1 $\mu\text{g/L}$ Δ^9 -THC were observed after the first inhalation of the 1.75% and 3.55% Δ^9 -THC cigarettes, respectively. Levels continued to increase rapidly to mean peak plasma concentrations of 79.0 and 152.0 $\mu\text{g/L}$ for the 1.75% and 3.55% Δ^9 -THC cigarettes, respectively, attained after 9 min (before inhalation of the last puff sequence at 9.8 min) [Huestis *et al.* 1992].

After 11 subjects had smoked 11.6 to 15.6 mg (mean 13.0) of Δ^9 -THC from a cigarette over a period of 5 to 7 min, peak plasma concentrations of 0.03 to 0.12 mg/L (mean 0.08) Δ^9 -THC were observed within 3 min of termination of smoking; the concentration fell rapidly to 0.003 to 0.01 mg/L (mean 0.007) at 1 h and to 0.0006 to 0.003 mg/L (mean 0.0016) at 4 h, after smoking. When 20 mg Δ^9 -THC was given orally to the same subjects, peak plasma concentrations of 0.004 to 0.01 mg/L Δ^9 -THC were produced between 60 and 300 min after ingestion [Ohlsson *et al.* 1980].

Note For information on vaporisation as a smokeless cannabis delivery system, see Abrams *et al.* [2007]. For a trial investigating the efficacy of Δ^9 -THC in the treatment of spasticity in spinal cord injury, see Hagenbach *et al.* [2007], and for its efficacy in Tourette's syndrome, see Muller-Vahl *et al.* [2002].

Toxicity Δ^9 -THC intoxication may result in loss of consciousness, or even death, but reports of fatalities are rare. The following Δ^9 -THC levels were reported following 6 cardiovascular fatalities after cannabis use: 22, 4, 2, 5, 3 and 7 $\mu\text{g/L}$; no other drugs were detected apart from 400 mg/L ethanol in the fourth case [Bachs, Morland 2001].

The following postmortem levels were reported following a case of fatal poisoning by Δ^9 -THC: liver 37.5 $\mu\text{g/g}$, kidney 42 $\mu\text{g/g}$, and spleen 12 $\mu\text{g/g}$ [Tewari, Sharma 1980].

Bioavailability After ingestion, ≈ 6 to 20%. During smoking, ≈ 18 to 50%.

Half-life In plasma of frequent users ≈ 2 h.

Volume of Distribution Approximately 10 L/kg.

Distribution in Blood Plasma: whole blood ratio 1.8.

Protein Binding 94 to 99%.

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